CITY COUNCIL

SPECIAL MEETING AGENDA
City Hall – Council Chambers*
December 7, 2021 – 5:00 p.m.

***Masks are required inside City Hall and are to be worn at all times when attending in-person meetings.**

Notice: Due COVID-19 and pursuant to Governor Inslee's Proclamation 20-28 the Mayor and Council encourage you use one of the following ways to participate in the meeting remotely if you are unable to attend in person:

- **Call in** and listen to the live meeting: (888) 788-0099 or 253-215-8782
- **Zoom meeting code: 9877 860 2371 and passcode: 760617**

1. CALL MEETING TO ORDER

2. STUDY SESSION

a. A Tech Hub Action Plan

- Staff Presentation: Tim Johnson, Economic Development Director
- Council Discussion/Questions

3. EXECUTIVE SESSION

a. Collective Bargaining pursuant to RCW 42.30.140(4)(b)

4. ADJOURNMENT

COUNCIL MEETING DATE: December 7, 2021 **ITEM #:**_____

CITY OF FEDERAL WAY
CITY COUNCIL
AGENDA BILL

SUBJECT: PRESENTATION OF "A TECH HUB ACTION PLAN"

POLICY QUESTION: Should the Council accept receipt of the administrative draft of the "Tech Hub Action Plan."

COMMITTEE: **MEETING DATE:**

CATEGORY:

☐ **Consent**	☐ **Ordinance**	☐ **Public Hearing**
☐ **City Council Business**	☐ **Resolution**	☒ **Other**

STAFF REPORT BY: Tim Johnson **DEPT:** Economic Development

Attachments: **1. A Tech Hub Action Plan**

2. Addendum

Options Considered:

N/A

MAYOR'S RECOMMENDATION: N/A

MAYOR APPROVAL: 12/1/21 (Committee Initial/Date) 12/1/21 (Council Initial/Date) **DIRECTOR APPROVAL:** 12-01-21 (Initial/Date)

COMMITTEE RECOMMENDATION: NA.

Committee Chair	Committee Member	Committee Member

PROPOSED COUNCIL MOTION: N/A

(BELOW TO BE COMPLETED BY CITY CLERK'S OFFICE)

COUNCIL ACTION:

- ☐ **APPROVED**
- ☐ **DENIED**
- ☐ **TABLED/DEFERRED/NO ACTION**
- ☐ **MOVED TO SECOND READING** *(ordinances only)*

REVISED - 4/2019

COUNCIL BILL # _____
First reading _____
Enactment reading _____
ORDINANCE # _____
RESOLUTION # _____

A TECH HUB ACTION PLAN

2021

CITY OF
Federal Way
Tech
Centered on Opportunity

A Tech Hub Action Plan

City of Federal Way, Washington

Administrative Draft

October 2021

City of Federal Way

Jim Ferrell
Mayor

Susan Honda
Council President

Council Members
(By Position)

1. Lydia Assefa-Dawson
2. Hoang Tran
3. Leandra Craft
5. Greg Baruso
6. Martin Moore
7. Linda Kochmar

Tim Johnson
Director, Economic Development

Mary Cho
Alex Forte
Dominic Minichillo
Quinn Zilly
Maya Stieben
Ankit Bhat
Jason Golliet
Taylor Jackson

October 2021

TABLE OF CONTENTS

Acknowledgements
Message from the Mayor
Executive Summary i
Introduction 1
Federal Way Assessment 5
- Overview 5
- Assessment 5
- Summary 23

Case Studies 25
- Overview 25
- Greensville, South Carolina 27
- Petersburg, Virginia 33
- San Leandro, California 37
- Lancaster, California 48
- Alpharetta, Georgia 56
- Richardson, Texas 64
- South San Francisco, California 73
- Songdo, Incheon, South Korea 83
- Milton Keynes, England 93
- Summary 100

Transformative Functions 107
- Overview 107
- Highlights 107
- Summary 114

Actions 117
- Overview 117
- Action 1: Organization and Initiative 119
- Action 2: Infrastructure / Broadband Development 121
- Action 3: Entrepreneurial Ecosystem 124
- Action 4: University, College & School District Relations 126
- Action 5: Diversity, Equity, and Inclusion 129
- Action 6: Use of Technology and Data Analytics 131
- Action 7: Business Attraction 133
- Action 8: Workforce Development 135
- Summary 149

Conclusion..... 151
A Picture of Success..... 151
Ambition Without Pretense..... 152
Glossary..... 153
Sources..... 165

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Special acknowledgement and thanks to the University of Washington students (2021 Economic Development Intern Crew) who were instrumental in this effort: Alex Forte, Ankit Bhat, Dominic Minichillo, Jason Golliet, Quinn Zilly and Taylor Jackson. They continue the tradition of the program that comprises the largest and oldest economic development internship in the state with the University of Washington.

Pictured above from left to right: Ankit Bhat, Dominic Minichillo, Quinn Zilly, Alex Forte, and Jason Golliet. (Not pictured: Taylor Jackson).

Specifically, great thanks to Alex and Jason for their effort on the configuration of the Action Plan.

Lastly, warm thanks to former interns, Mary Cho, now Economic Development Specialist with the City of Federal Way for her research on Songdo and Smart Cities. Also, thanks to Maya Stieben, also a former intern, for her work on the preliminary framework and outline for this report.

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Message from Mayor Jim Ferrell

Federal Way's ambition is to become a tech hub. The City Council and I agreed to this priority at our annual retreat in early February 2020. Shortly after this action, I directed City staff to begin the research.

Unfortunately, and without question, the COVID-19 global pandemic caused the redirection of personnel to address the immediate needs of the City's residents and businesses.

However, not to be deterred, and as things stabilized, City staff returned to pursue the effort again in early 2021.

So, the purpose of this report is to identify how to become a tech hub. As such, the intent of this report is to provide answers to the questions on what is and how to become a tech hub.

This report incorporates a series of case studies that analyzes and compares how other cities of similar size and corresponding community characteristics to Federal Way achieved their success as a tech hub. The research identifies the resources needed, along with the outcomes and timelines for their accomplishment. This insight illustrates that they pursued similar if not exact actions to accomplish their efforts. Thus, the identification and understanding of their use of these actions are invaluable to our efforts towards becoming a tech hub.

Federal Way's Ambition - Tech Hub

For example, the cities built a strong partnership with their private sector along with a research university. These two actions alone were significant enough in achieving their efforts.

However, the case studies also illustrated that the cities achieved success through enhanced broadband services. This included either the deployment through a private sector, public sector or a public-private partnership model to provide last mile service by providing reliable, quick, accessible and affordable service along with the capacity and symmetry of broadband for their residents and businesses.

Additionally, research identified that their combined efforts help build entrepreneurial ecosystems through a variety of services and tools. These included private sector and university operated business incubators, accelerators, startup studios, and business foundries along with the business attraction and recruitment of innovation centers, co-working and makerspaces together with hosting meet-ups, mash-ups and hackathons to drive their success.

The research further identified their outreach to angel investors, venture capitalists, equity and institutional investors along with Community Development Financial Institutions requesting their financial assistance to help innovation focused entrepreneurs by providing access to and availability of capital.

A Work in Progress

However, while this report appears complete, it is still a work in progress. City staff is working in concert with the University of Washington – Tacoma in performing a workforce development analysis on the supply and demand of the experience and education of the city's labor force to determine opportunities. It is expected to be completed in the second quarter of 2022. While this element is important and it will be achieved, time is of the essence. Thus, we need to and can push forward with all of the other actions identified coupled with their programs and projects.

Meanwhile, input is still needed, specifically from Federal Way's entrepreneurs. The City will enlist the input of its entrepreneurs and community stakeholders in the effort to ascertain further opportunities and corresponding resources to implement to become a tech hub.

Beyond this endeavor, the City will also focus on balancing the effort of becoming a tech hub through the attraction of mature tech companies. The purpose of this action is to ensure diversification and sustainability of Federal Way's economy and hopefully provide the immediacy of employment opportunities for our residents beyond the organic growth of entrepreneurial development.

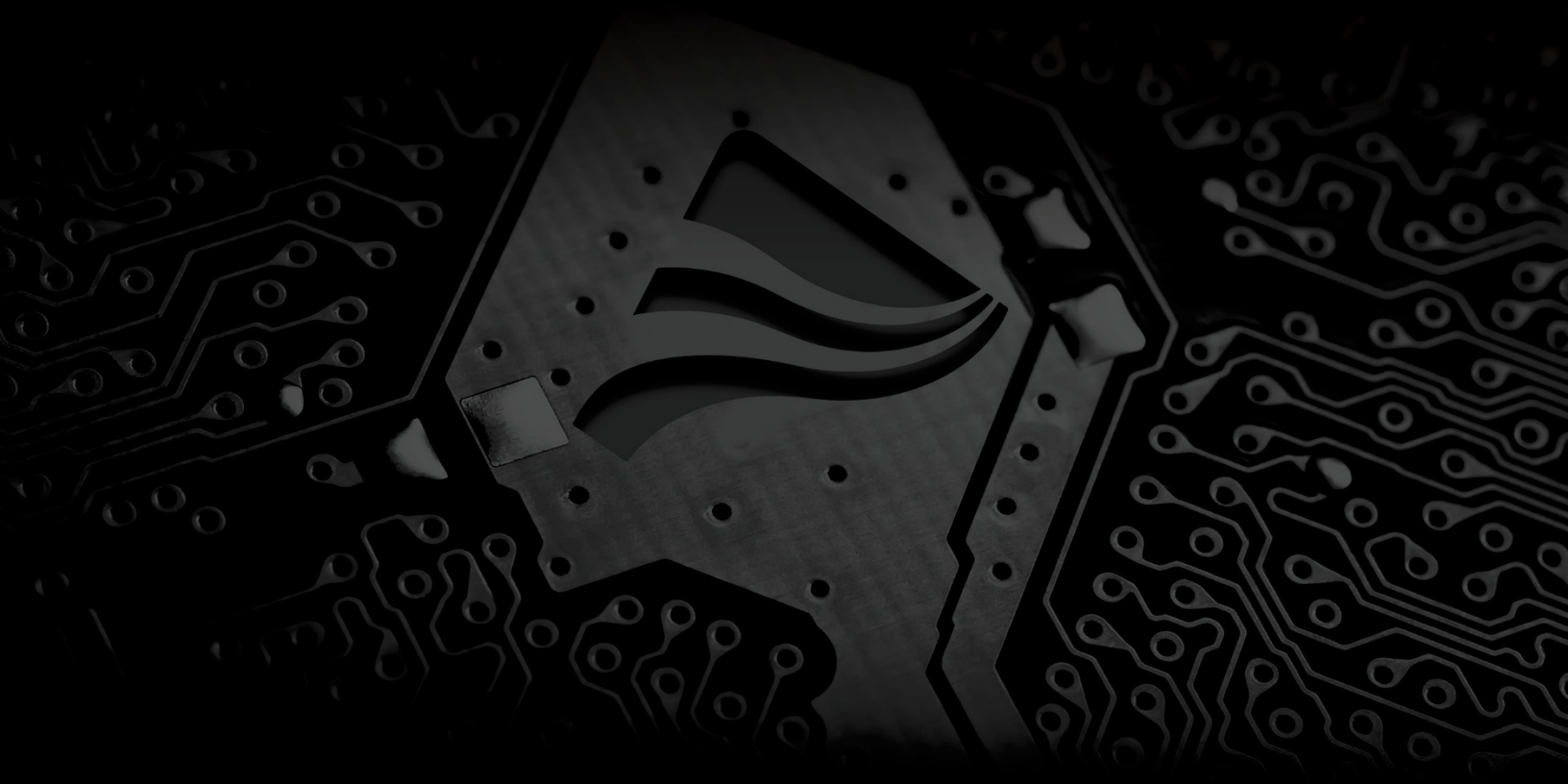
However, in either effort, research identifies that it takes time to scale up to become a tech hub. The cities reviewed in the case studies achieved success on average in a ten-year period. So, per my direction this "Action Plan," is on an expedited mission to reach success in less than a decade.

Federal Way's Mission: Reach Success of a Tech Hub in Less than a Decade.

The 8 actions identified comprise the backbone of the Plan. Within each of these, there are numerous corresponding programs and projects. Of importance, these actions also compliment our efforts to realize our economic development goals for our pending economic development strategy, "Residents, Community and Prosperity," which recommends how to create a strong, diversified, inclusive and sustainable Federal Way economy.

This report, "A Tech Hub Action Plan," is the culmination of over nine months of research by City staff supported by a group of passionate and gracious professors who provided their time along with economic students (interns) from the University of Washington dedicated to its publication and subsequent use. I thank them and all those recognized in the Acknowledgements. I specifically want to thank my fellow and former colleagues who served on the City Council for this vision and for all of their work in helping us launch our ambition of becoming a tech hub.

As such, I extend an invitation to you. Join the City Council and myself and other elected officials at the state and federal level on this journey. . . our community's journey, the ambition of becoming a "Tech Hub."



EXECUTIVE SUMMARY

EXECUTIVE SUMMARY

PURPOSE

Ambition- Tech Hub

The goal of the Mayor/City Council is to become a tech hub. The Mayor directed economic development staff to research and examine questions such as what is and how to become a tech hub? As such, Economic Development initiated the research and resources, subsequently leading to the publication of this report "A Tech Hub Action Plan."

Tech Hub-Defined

The research illustrates that a tech hub is a location where things happen faster. It is where ideation and product development can be streamlined. Experts identify that it is a community – informal or otherwise, that fosters innovation for technology, be it for mature and or startup companies.

Experts illustrate that a tech hub can occur in a city, a suburb or even in a building or office campus. A tech hub can cover a large or small geographic area. But whatever size or location, it can offer the activity which increases the potential for meetings, collaborations and general camaraderie among tech minded people working various technology-oriented businesses and products.

ASSESSMENT- FEDERAL WAY

Background

Federal Way, Washington (100,000 in population) is located in the Seattle-Tacoma metropolitan area of nearly 3 million in population. Federal Way has a very diverse ethnic population that is forecasted to see continued growth.

Federal Way is located along two major ground transportation corridors; Interstate 5 and the Pacific Coast Highway. It is also located along the Puget Sound and is mid-way between the Ports of Seattle and Tacoma. Federal Way is also located 15 minutes south of the Seattle-Tacoma International Airport.

Factors Affecting Federal Way

To help understand Federal Way and the opportunities and challenges it faces in becoming a tech hub, the research incorporated a PESTEL analysis. PESTEL is an acronym for Political, Economic, Social, Technological, Environmental, and Legal. The analysis evaluates outside factors that affect or have the potential to affect achieving a goal or mission. In this case, it is used to understand how outside factors affect or have the potential to affect Federal Way becoming a tech hub.

<u>Political</u>

Research has identified several key political items that could have a positive effect on Federal Way's efforts in becoming a tech hub.

Federal Way has a stable political environment coupled with an overall very healthy economy and tax base consistent with the City's Aa2 rating according to Moody's Investment Advisors.

Moody's identified that Federal Way is similar to other cities across the state in that it is likely to experience steady economic growth over the near-term. Sales tax and other economically sensitive revenues should improve as unemployment remains low, particularly in the Puget Sound. Strong improvement in the housing market will continue to boost assessed valuation growth and property tax collections. Pension costs are not a significant source of credit weakness, but remain a longer-term challenge.

Lastly, the research through a series of case studies identified that it takes on average a decade of concentrated efforts by political leadership in concert with a community's private sector and a research university to achieve an ecosystem for a tech hub. Thus, it is important to ensure the continuity and consistent focus of the local elected officials to remain on becoming a tech hub. In addition, a series of transformative functions need to occur as well, such as enhanced broadband services; an entrepreneurial ecosystem including access to and availability of capital; a ready, available skilled and experienced labor force for tech growth.

<u>Economic</u>

The research has also identified several key economic issues that could impact Federal Way's efforts in becoming a tech hub.

First, there are nearly 5,000 businesses located in storefronts in Federal Way. Anecdotal information identifies there could be an additional 3,000 home-based businesses (not all licensed) in the city as well. Thus, there could be a significant number of entrepreneurs in hiding that may be engaged in innovation. As such, further research is needed.

Second, because the City is positioned between multiple institutes of higher education such as the University of Washington in Seattle and its Tacoma campus, along with nearby Highline College, Federal Way has the capacity to act as a magnet to attract a young, diverse, and educated talent pool to meet the needs of tech-based businesses and entrepreneurs.

Thirdly, businesses locate to Federal Way because of the following locational advantages:

- Located between the Ports of Seattle and Tacoma.
- Located between University of Washington – Seattle and Tacoma campuses.
- Over 9 million consumers within a 2 ½ hour drive time. 50% of the state's population is located within a 2-hour drive either north or south from Federal Way.
- Located on the Interstate 5 corridor which is also 20 minutes away from SeaTac International Airport, the busiest airport within Washington. Prior to the pandemic it served over 50 million passengers a year, through 91 domestic and 27 international destinations. Federal Way's location to the airport provides ease of business access to the world.
- City government is proactive, and is able to provide expedited permitting, planning, and licensing for businesses that are looking to relocate or expand their business in the city.
- The City has a minimal tax on business such as no gross receipts or B&O taxes, unitary, inventory, or state corporate income tax.

The City has a diverse labor force with a deep base of requisite skills that are able to meet current employer's needs, however, this labor force leaves daily. Research identified that more than 85% go elsewhere to work according to the Census Bureau.

Also, while the startup ecosystem for entrepreneurs has been growing steadily in the Puget Sound region, Federal Way does not have either an accelerator or incubator, or even a startup studio or business foundry to help facilitate growth let alone anything that resembles an entrepreneurial ecosystem. That is not to say that entrepreneurs in Federal Way are not supported by a variety of programs, but none are physically located in the city. For example, the Small Business Development Center (SBDC) and the StartZone Microenterprise Program are both located at Highline College in Des Moines.

Social

Research has identified several key social issues specifically changing demographics that could affect Federal Way's efforts in becoming a tech hub.

The median age has continued to rise in Federal Way. In 2010, the median age was 36.2. In 2020, it was 37.6 and the median age is projected to increase to 38.5 in 2025. While many assumptions can be made, at least one key observation is that an older stable workforce may be valued by tech companies.

Research data from the demographic firm ESRI forecasts that there will be changes in the city's ethnic composition over the next 5 years. Data shows the following forecasted changes:

The Caucasian population was identified to be 87,972 in 2020 and forecasted to be 87,267 in 2025; a reduction of 705 in the population.

The Asian population was estimated to be 27,350 in 2020 and forecasted to grow to 32,562 in 2025; an increase of 5,212 in the population.

The Hispanic population was estimated to be 23,366 in 2020 and forecasted to grow to 26,145 in 2025; an increase of 2,779.

The Black population in 2020 was estimated to be 14,073 and forecasted to grow to 15,273 in 2025; an increase of 1200.

Another demographic change forecasted in Federal Way involves the change with the median household income. From 2020 to 2025 income is projected to have an annual growth rate of 2.25%, which is faster than the state and national rates. As an example, 40% of the current households make over $100,000. Given the projected growth rate, it is forecasted that by 2025 the number of households making over $100,000 will increase to 46%.

Technology

Research has identified several key technology issues that could impact Federal Way's efforts in becoming a tech hub.

As identified in the Introduction, the magnitude of the region's tech sector is captured in a recent report by the commercial real estate firm CBRE, which identified that the Seattle region added more than 48,000 tech jobs from 2016 to 2020. This is a 35% increase in employment over this period, representing the largest increase of any metro area in the nation.

Federal Way targets the attraction and growth of Information & Communication Technology along with Biotech/Life Sciences.

This is important as the life sciences sector showed a 23.5% increase in jobs from 2015 to 2019. More than 39,000 people are employed in the 1,100 firms across the state at an average wage of $106,122. This is valuable for Federal Way as the digital health and IT growth in the life sciences are areas of opportunities where the City has focused, as growth in these areas are growing and account for almost a quarter of the state's firms.

Additionally, the Seattle region was named the top emerging biotech market in 2019 by the CBRE and in 2020 was ranked as a top-ten life sciences cluster – though it's still far behind the major biotech/life science clusters of Boston and San Francisco.

Environmental

Research has identified several key environmental issues that could impact Federal Way's efforts in becoming a tech hub.

Climate change for example poses a variety of opportunities and threats in Federal Way's efforts to become a technology hub.

A key component of the City's vision should be the alignment of the "Tech Hub Action Plan" with its yet to be initiated Climate Action Plan. Specifically, the vision should drive sustainable outcomes for the residents and businesses (current and future), the environment and its businesses and their customers.

Climate scientists identify that a warming planet creates a wide range of business risks for businesses both large and small. Businesses risks include disrupted supply chains to rising insurance costs to labor challenges, which results in a higher business expenses and a higher cost of consumer goods and services. According to a consulting firm Deloitte, climate change and extreme weather routinely and directly impacts 70% of all economic sectors worldwide.
The changing environmental landscape presents great opportunity for tech-based climate change innovation. According to the United Nations Technology Executive Committee, "Tech-entrepreneurs," play a central role in the innovation process required to address climate change.

Federal Way needs to identify and use technologies, business models, and services that can be used to achieve a resilient and sustainable economy and should include a strong entrepreneurial ecosystem to support this action.

Legal
Research has identified several key legal or legislative items that could impact Federal Way's efforts in becoming a tech hub. These include both national and state regulations and goals.

The Federal Communications Commission (FCC) defines *broadband* as an internet connection of at least 25 megabits per second (Mbps) download and 3 Mbps upload. These have been the standards since 2015, but the standards change regularly as technology and connectivity improve. There is currently a call for a further increase in the benchmarks.

Meanwhile, the FCC has a plan illustrating four ways that the government can influence change within the broadband ecosystem.

Four Recommendations of the FCC National Broadband Plan:

1. Design and Establish Competition Policies.
2. Ensure Efficient Allocation and use of Government-Owned and Government-Influenced Assets.
3. Create Incentives for Universal Availability and Adoption of Broadband.
4. Update Policies, Set Standards, and Align Incentives to Maximize Use for National Priorities.

In addition to these four recommendations, the FCC Plan covers six recommended goals for the nation to use as a roadmap. These include:

1. At least 100 million U.S. homes having affordable access to actual download/upload speeds of at least 100mbps/50mbps.

2. The U.S. should become the leader of mobile innovation, with the fastest and most extensive wireless networks of any nation.

3. Bring affordable access to broadband services as well as the means and skills to subscribe if they choose.

4. Bring affordable access to at least 1 gigabit per second (Gbps) broadband service to anchor institutions (schools, hospitals, and government buildings).

5. Every first responder should have access to a nationwide, wireless, interoperable broadband public safety network to ensure community safety.

6. The ability for Americans to use their broadband service to track and manage their energy consumption to lead towards a clean energy economy.

Over the last decade, there have been many updates and changes to the Plan to ensure that regulations stay up to date. As of February 5th, 2021, with the introduction of the National Broadband Plan for the Future Act 2021, the FCC is required to update the National Broadband Plan to assess the progress of original goals; examine effects of the COVID-19 pandemic on the way people learn, work, receive medical information and treatment, and participate in civic communications; and analyze the change in reliance people will have on services enabled by broadband internet access services.

Meanwhile, in Washington, state laws have allowed some municipalities to offer communications services to residents but barred public entities from providing broadband service directly to customers. Recently, state legislation was passed allowing municipalities to provide this service.

ESHB 1336, (passed on July 25th, 2021), allows PUDs, ports, cities, and other public entities in the state to directly provide unrestricted retail broadband service to consumers. Under previous law, these entities could only offer wholesale services to private telecommunications providers. This legislation may become the catalyst to bring high-speed broadband access to underserved and unserved areas of Washington state.

Furthermore, the state has created specific goals for broadband development. These include the following:

- By 2024, all Washington businesses and residences have access to high-speed broadband that provides minimum download speeds of 25 Mbps and minimum upload speeds of 3 Mbps. This State standard aligns with the Federal Communication Commission's (FCC's) current minimum standard for broadband.

- By 2026, all Washington communities have access to at least 1 Gbps symmetrical broadband service at anchor institutions like schools, hospitals, libraries, and government buildings.

- By 2028, all Washington businesses and residences have access to at least one broadband provider with at least 150 Mbps symmetrical broadband service.

Of interest, identified in the legislation, "unserved area" means an area of Washington in which households and businesses lack access to broadband service at a minimum of 100 Mbps download speed and 20 Mbps upload speed.

CASE STUDIES

So, to better understand how to create a tech hub, the research incorporated case studies that analyzed cities of similar size and community characteristics to Federal Way that were successful in becoming a tech hub.

Dozens of cities were examined. However, the following cities were selected based upon a series of filters. First, they were similar in population size and in community characteristics to Federal Way. For example, they had to be part of an urban environment. They also needed to be located along an interstate or major highway/transportation corridor. There was no university or college located in the city, (at least when they started their efforts). Lastly, they had to become a tech hub (however, type of cluster or sector was not considered).

The following nine cities were studied, of which seven are domestic and two are international.

CITY	TECH HUB CLUSTER
1. Greensville, South Carolina	(Advanced Manufacturing)
2. Petersburg, Virginia	(Advanced Manufacturing)
3. San Leandro, California	(Broadband initiated Technologies)
4. Lancaster, California	(Green Tech)
5. Alpharetta, Georgia	(Financial Technologies
6. Richardson, Texas	(Information Communication Technology)
7. South San Francisco, California	(Bio-Tech/Life Sciences)
8. Songdo, Incheon, South Korea	(Smart Cities Tech)
9. Milton Keynes, England	(Mobility Technologies)

Elements for Success

The case studies identified the actors or individuals, along with the entities and organizations that initiated their tech hub efforts and actions for success. There are three organizations that combined create the entities for success: Local elected Officials, the Private Sector and a University.

Entities Needed for Success
Local Elected Officials, Private Sector & University

Local Elected Officials

Research identified through the case studies that the effort to become a tech hub was predominately initiated and sustained by local political leadership. Research also illustrated that, by virtue of their higher visibility and social profiles, local elected officials often played an effective role in convening different groups of stakeholders, also acting as boosters and, in broader terms, salespersons for their cities.

Tech Hubs Predominately Initiated by Local Political Leadership

The research from the case studies also discovered that 66% of the time, it was local elected officials that lead the effort. However, it was a combination of an active private sector and support from a university or universities that achieved the success for the cities to become tech hubs.

The main ways local elected officials and the public sector achieved a tech hub include the following:

- Convening Stakeholders, Acting as Boosters and Salespeople for their Cities
- Developing Infrastructure such as facilities, broadband, water, sewer, and energy
- Business Incentives including tax credits and expediated permitting
- Hosting Events for Entrepreneurs

- Convening Investors to identify opportunities to provide Access to and Availability to Capital
- Initiating and Engaging in Public-Private Partnerships

Private Sector

In addition, the research also ascertained the importance of the private sector in achieving a tech hub. Multiple examples where identified how the private sector can provide a combination of critical components. These include creating entrepreneurial tools from business accelerators to incubators. It also provided the needed capital from angel investors, venture capitalists, equity capital, and institutional investment for creating dynamic ecosystems for entrepreneurs. For example, as it pertains to Richardson, Texas, the private sector did most of the heavy lifting early in its history of becoming the Telecom Corridor. It started with the attraction of major company Collins Radio along with Texas Instruments and continued with a diversity of companies. However, following this, local elected officials and the private sector attracted the University of Texas to locate its Dallas campus to Richardson, a key component in driving the future of its entrepreneurial ecosystem.

The main ways the private sector works to develop a tech hub include the following:

- R&D, Innovation
- Creating jobs by establishing/expanding
- Creating offices and hubs
- Creating and fueling entrepreneurial tools
- Real Estate Investment
- Marketing/Promotion
- Networking

University

Research also identified that universities play a crucial role in a city becoming a tech hub. As previously noted, none of the cities reviewed in the case studies had a university when they began their effort to become a tech hub. The research discerned that each reached out to a university for help and assistance. For example, in Greenville, South Carolina, they were 35 miles away from Clemson University. Milton-Keynes is located between the campuses of Oxford and Cambridge. South San Francisco was also midway from the University of California at San Francisco and Stanford. No community had a prior gown-to-town relationship. Thus, they all reached out to create opportunities to a university in providing training, specialized workforce education in business and technology, and applied and advanced research that was located in these cities.

Accelerating the Approach

Research further identified that once the effort is initiated between all three; local elected officials, the private sector and a university, the effort accelerates rapidly in transitioning and growing the local economy into a tech hub. Over time, the efforts of the local elected officials, and private sector, coupled with the university are often equally shared.

The combined efforts of Local Elected Officials, Private Sector and a University Accelerate the Growth of a City becoming a Tech Hub

So, the largest takeaway from this analysis is that the local government through its elected officials are often the entity that initiates the effort to become a tech-based economy. Once the initiative is deployed

by and engaging the private sector and universities, the effort accelerates rapidly in transitioning and growing the local economy. Over time, the efforts of the public and private sector, coupled with the university are often equal.

As it pertains to the type of technology cluster(s) incorporated by cities, it appears that each city focused on sectors where they already had a locational advantage. These include infrastructure; a skilled labor force. Cooperation with nearby research and educational facilities and collaboration with existing tech-firms. These vital assets were identified and further reviewed before any time and resources were poured into sector development initiatives. None of these cities jumped on technology or economic development fads or chased a specific industry because it was fashionable or in vogue. Rather, they targeted a sector because analysis suggested that their city stood a realistic chance of succeeding in growing in that sector.

TRANSFORMATIVE ACTIONS

Research illustrated, that beyond the entities another elemtn for success is the incorporation of what is called transformative functions. These inlcude four elements.

- **Analytics and Promotion**
  These activities consist of feasibility and competitiveness analysis, planning, and market research, along with branding and marketing.
- **Investment Facilitation**
  These activities include infrastructure such as broadband/internet, zoning, permitting, and business facilities; customized workforce training; business recruitment, expansion, and retention; incentives; and site selection and retention services.
- **Entrepreneurial Assistance**
  This includes activities such as availability and access to capital, technology transfer, business accelerators, incubators, startup studios/foundries, innovation centers, makerspaces, coworking spaces, provision of market information, and export facilitation.
- **Convening Events**
  This function involves engaging stakeholders, fostering industry-academia links, developing partnerships and networks, establishing or leading growth coalitions through meetups, mashups, and hackathons as well as community meetings.

Research identified that there were stark differences in the approaches used by the cities to become a tech hub. However, asit pertained to Transformative Functions, success continued where the collaboration amongst the groups focused immediately on the development and deployment of broadband followed by, if not coinciding with, the formulation of a strong entrepreneurial ecosystem.

The ecosystem elements included:

- Business Accelerators
- Business Incubators
- Startup Studios
- Business Foundries
- Innovation Centers
- Tech Transfer Centers
- Meetups, Mashups, and Hackathons
- Coworking Spaces and Makerspaces

Lastly, while there appears to be no one specific use over another that leads to a Tech Hub, the use of Transformative Functions appears to be required in some percentage of use in order to achieve a tech hub.

ACTIONS FOR SUCCESS -THE PLAN
The Plan identifies 8 actions. They include the following and are highlighted below.

A. Organization & Initiative
B. Broadband Development
C. Entrepreneurial Ecosystem
D. University, College, and School District Relations
E. Diversity, Equity, and Inclusion
F. Technology and Data Analytics
G. Business Attraction
H. Workforce Development

Organization and Initiative
Adoption of city policies that will have catalytic effects on the number and quality of programs, projects and services on the and the potential impact on Federal Way residents, businesses, i.e., startups to mature growth companies.

Broadband Development
Examine citywide internet infrastructure and potential to enhance services to residents, business and visitors. Broadband's key components – Reliability, Speeds, Accessibility, Affordability, Bandwidth, and Symmetry – are what the City must aim for to become a tech hub. It must meet state policy standards, while enhancing education, develop the local economy for its businesses, and improve diversity and inclusion for its residents.

Entrepreneurial Ecosystem
Work to grow an entrepreneurial ecosystem from within the City. An entrepreneurial ecosystem's key drivers include supportive government policies and regulations; strong relationships with local universities; access to financing/capital; quality human capital, and an entrepreneur and business-friendly culture – can all be improved by the public sector directly and indirectly. Under these conditions, business tools – Incubators, Accelerators, Startups Studios/Foundries, Innovation Centers, Tech Transfer Centers, Makerspaces, Coworking Spaces, Meetups, Mashups, and Hackathons – can be utilized to fuel business.

University, College, and School District Relations
Create a beneficial relationship with universities such as University of Washington, both Seattle and Tacoma campuses; community colleges such as Highline; Federal Way School District, and other educational institutions that support STEM education and training, research and innovation and supportive of entrepreneurial success through education, mentoring, and networking.

Diversity, Equity, and Inclusion
Become a community that offers tools and resources designed to help local elected leaders build safe places where people from all racial, ethnic and cultural backgrounds thrive socially, economically, academically and physically.

Technology and Data Analytics
Identification of use of technologies to enhance the organizational effectiveness, efficiency, accountability and transparency of operations of the City of Federal Way. Also, formulation of objectives that identify, analyze and examine the feasibility of use of data collected by the City through its processes of business and determine its use for organization effectiveness and efficiencies and its value to the marketplace potentially leading to subsequent monetization.

Business Attraction
Attract businesses of all sizes and maturities to the City, especially in target technology sectors. Potential businesses include elements of an entrepreneurial ecosystem such as business incubators and business accelerators. Attract businesses that further the City's other aspirations in the areas of smart city technology, connections to academia, broadband development, entrepreneurial ecosystems, and disruptive technologies.

Workforce Development
The Center for Applied Urban Research at the University of Washington Tacoma will work with Federal Way and South Puget Sound communities to understand the characteristics of the region's labor force and identify information to help guide and prepare for the region's economic future. It will conduct a regional study in four phases focused on the following activities for the participating jurisdictions (currently Pierce County, Tacoma, Lakewood, Sumner, and Federal Way). Areas will include all of Pierce County and South King County as a single region. Detailed data at the local level will be provided to participating jurisdictions only.

Highlights of Recommended Actions
Some of the key "Actions" crafted for review include the following:

- Enhance Broadband Development, specifically last mile deployment to residences and businesses Fulfilling State and federal guidelines for speeds,

- Through new FCC National Broadband Plan, become "National Demonstration Project City

- Focused meetings by Mayor/City Council (Meet-ups) with local entrepreneurs to ascertain opportunities and to understand needed resources.

- Identify and implement at least one new tech-based solution per year to city issues. Formulation of at least one new tech-based solution per year, such as Eyes on Federal Way, or new finance software to enhance dashboard for citizen insight to city budget.

- Implementation of National League of Cities and Kaufmann Foundation Community Entrepreneurial Initiative awarded grant towards the formulation of a digital resource guide for entrepreneurs and another specifically childcare operators.

- Produce annual review measuring City efforts to achieving and sustaining a "Tech Hub."

- Create Tech Hub Action Plan website to promote the effort and present annual tech hub progress reports.

- Refocus City Business Attraction efforts towards recruitment of business accelerators, business incubators, and innovation centers to create a more diversified and sustainable economy.

- Formulate a tech Mentor Program to help start-ups

- Formulate outreach to investors and host financial conferences in support of innovation-based entrepreneurs.
    - Provide outreach to angel investors, venture capitalists and equity investors and Community Development Financial Institutions in support of entrepreneurs for access to and availability of capital

- Enhanced outreach to UW & WSU to ascertain opportunities with their Centers, Facilities Incubators and Laboratories.

- Identify and analyze feasibility of how to use existing data collected by City to improve its decision making, customer service and organizational effectiveness and efficiency.

- Host the International Underwater Robotics Competition 2023 and 2025

What Does Success Look Like?
If Federal Way succeeds at becoming a tech hub, what does success look like?

A Picture of Success
The research identified that there are many cities of similar size and characteristics to Federal Way that have been successful in becoming a tech hub. Research illustrated a series of actions needed for success. These included the following:

- Local Political Initiative & Organization
- Strong Private Sector Input and Participation
- Active Engagement with a Research University

In addition, research identified that through the case studies, that other transformation functions needed to occur as well such as deployment of an enhanced broadband system along with building a strong entrepreneurial ecosystem.

Ambition without Pretense

Federal Way needs to think big without going overboard. However, beyond the actions city staff has crafted five aspirations to help motivate the City towards its ambition of becoming a tech hub. The aspirations include:

- ***Become the City with the "Fastest Internet Speeds" in the State of Washington.***
- ***Become the "Most Advanced Smart City" in the Pacific Northwest.***
- ***Produce the "Finest Entrepreneurial Ecosystem" in the Pacific Northwest***
  through partnerships in the creation of business programs, services and tools such as access to capital, through inclusion with the outcomes of diverse startups, in terms of ethnicity, gender, technology, business, and employment growth.
- ***Most Connected City to Higher Education in the Nation.***
- ***Become the "Global Center for Disruptive Technologies."***

These aspirations reflect Federal Way's desire for prosperity and well-being, for equity and inclusion, towards becoming a community that is "Centered on Tech Opportunity," where the full potential of its citizens can be realized.

These outcomes reflect Federal Way's desire for shared prosperity and well-being, for unity and inclusion, and to become a community that is *"Centered on Tech Opportunity,"* where the full potential of its citizens can be realized.

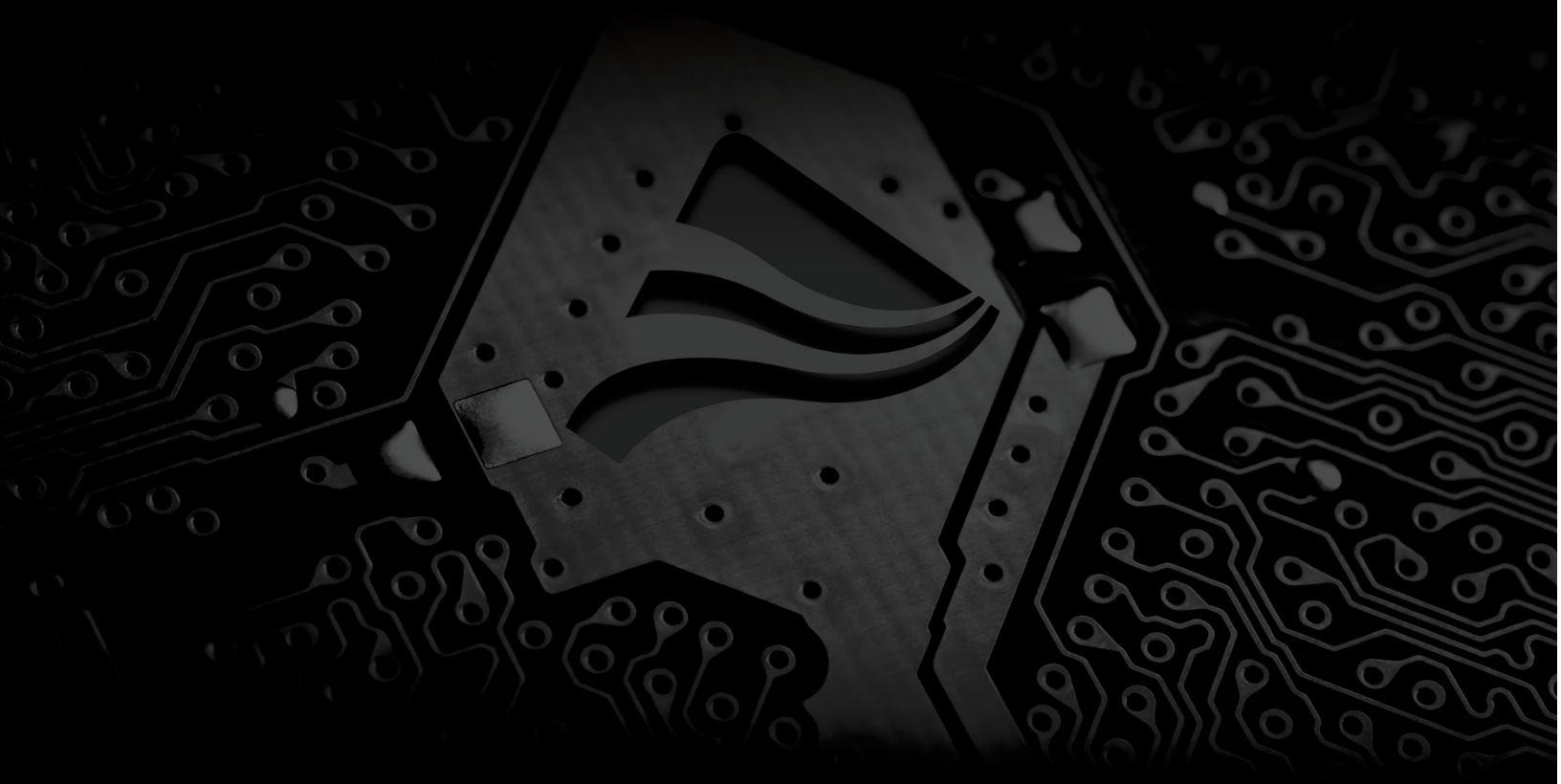
Lastly, the ability of Federal Way to achieve becoming a tech hub should be reached by three defining factors:

1. A Need to Develop its Own Tech Credentials
   Federal Way needs to create an unequalled entrepreneurial ecosystem that catches the attention of startups in the tech sector. The vibrancy of a strong entrepreneurial ecosystem should be a source of collective community pride. Federal Way should not hesitate to support the ambition of its entrepreneurs through education, and structured mentorship.

2. Entrepreneurial Parity
   The importance of entrepreneurs to Federal Way's economy plays a significant role in its economic future. Federal Way must focus its attention on supporting its entrepreneurs, not just large corporations.

3. Engineering our Own Center of Tech Gravity
   The City needs to develop a powerful lure and elevate its public image. Federal Way needs to illustrate that it is fashionable and cosmopolitan. Federal Way needs to become a foothold in the Puget Sound for tech- focused entrepreneurs in the effort to become a tech hub.

Becoming a tech hub is not a fairy tale but a plan of action based on realty, where its success lies in the journey not a destination.



INTRODUCTION

INTRODUCTION

Once upon a time there was a community in a large metropolitan area that was bestowed with worldly sized investments. Specifically, they included a corporate headquarters of a large international company employing thousands. It also included the construction of a major retail mall in its' downtown where millions of dollars were spent annually on merchandise and entertainment.

Surrounding cities were jealous of the community as it reaped the benefits from the investments such as a growing tax base, new jobs for its residents and a positive perception as a dynamic and exciting location for business.

Unfortunately, over time, the community changed. Sadly, the company moved its corporate headquarters and the mall saw multiple ownership changes constituting varying shifts in retail operations and consumer volume. The community also faced a variety of significant economic factors impacting it from a "Great Recession" to a global pandemic in just over a decade.

Regrettably, unlike a fairy tale, things don't always turn out to be happily ever after.

The reality of business development and growth is found in an analysis by the Small Business Administration which identifies that on average 10 to 12 percent of businesses close annually across the nation. The moral of this story is that Federal Way had a rich economy but it must always be a steward to its future. It must constantly work to ensure the well-being of its economy .

So, due to these economic realities, it must confront its future. To do so, its Mayor and City Council have prioritized their focus and commitment on becoming a tech hub.

So, how can a community once again achieve such a renowned economy? How can a city achieve a lofty goal of becoming a tech hub and specifically when the speed of technological change is so fast?

To answer these and other questions related to becoming a tech hub, this research pursued a variety of underlying issues.

For instance, technology is transforming society's habits and the routines of work and personal lives at an amazing rate. Given this rate, the change means technology companies are constantly innovating and creating to compete. As an example, it has been 13 years since the iPhone was introduced transforming the way people interact with technology and software that is now commonplace. Today, Apple is on the 12th edition of its iPhone in 13 years. So, how can a community interested in becoming a tech hub achieve success when the speed of business engaged in creating technology is moving faster than the community?

Federal Way's economic history had been concentrated on a natural resource-based economy, specifically timber and wood products. Federal Way was home to the corporate headquarters of the Fortune 500 firm Weyerhaeuser, one of the world's largest growers of timber and producers of wood products. Unfortunately, that came to an end after 40 plus years of operations in Federal Way when the company announced a major restructuring in 2014. Weyerhaeuser downsized and left Federal Way selling its 400 plus acre headquarters' campus.

Thus, to address the array of issues caused by this action, not the least, the loss of its largest employer and property taxpayer, the City's political leadership sought a policy direction to achieve a diverse and sustainable economic future.

The Mayor and City Council prioritized becoming a tech hub at their annual organizational retreat in February 2020. Through this action, the priority would immediately become the responsibility of City staff, specifically economic development to perform the research, review and illustrate opportunities, along with presenting actions for implementation.

Fortunately for Federal Way, it has several comparative advantages to help achieve this lofty ambition. For example, it is located in the center of a major metropolitan region comprised of over 3 million people. It is located along the Interstate 5 corridor between the urbanized region of the state capital of Olympia to the south and the industrial/military port in Everett located to the north. The region is strong in technology and innovation with companies like Boeing, Microsoft, Amazon and Blue Origin, along with Apple, E-Bay, Facebook, Google and Oculus. It has a large entrepreneurial ecosystem with business incubators and accelerators supported by angel investors, venture capitalists, equity and institutional investors as well. The region is also home to the University of Washington in Seattle and its campuses in Tacoma and Bothell. Additionally, Federal Way is located minutes from an international airport and located midway between the Ports of Seattle and Tacoma.

The magnitude of the region's tech sector is captured in a recent report by CBRE, which identified that the Seattle region added more than 48,000 tech jobs from 2016 to 2020. This is a 35% increase in employment over this period, representing the largest increase of any metro area in the nation.

Tech Hubs

So, the concept of becoming a tech hub if not simply through the attraction of technology-based business coupled with building a strong innovation led entrepreneurial ecosystem appears possible. However, what is a tech hub and what factor(s) does it take to become one?

Research illustrates that a tech hub is a location where things happen faster. It is where ideation and product development can be streamlined. Experts identify that it is a community – informal or otherwise, that fosters innovation for technology, be it for mature and or startup companies.

Experts illustrate that a tech hub can occur in a city, a suburb or even in a building or office campus. A tech hub can cover a large or small geographic area. But whatever size or location, it can offer the activity which increases the potential for meetings, collaborations and general camaraderie among tech minded people working various technology-oriented businesses and products.

A Tech Hub can occur in a city, a suburb, or even in a building or office campus.

Experts also claim that tech hubs develop slowly but surely and are accelerated by concentrated efforts supported by infrastructure, such as robust broadband coupled with fast internet speeds that provide growth and opportunities for entrepreneurs that are enhanced by business incubators, accelerators and innovations centers to move their efforts to the marketplace.

Additionally, there are developmental stages of a tech hub. For example, in 1992, the now defunct federal agency, The Office of Technology Assessment, (OTA) identified and analyzed a cross section of technology

development initiatives undertaken by communities across the nation to determine what types of programs have been attempted, how well they worked, and the factors that impacted their effectiveness. Over the years, state and local efforts to achieve technology -based economies have been driven by the political philosophies to increase employment and revenue for local jurisdictions. Communities are generally influenced by the rapid growth associated with technology-based industry. OTA found that many local initiatives attempt to emulate the characteristics of the preeminent technology communities in the nation such as the Silicon Valley and the Research Triangle by creating specific strategies geared toward achieving their goal.

The OTA identified that there are five distinct types of tech-based communities. These include the categories list below and are subsequently followed by their definitions:

1. High Technology Center
2. Diluted Technology Centers
3. Spillover Communities
4. Technology Installation Centers
5. Bootstrap Communities

<u>High Technology Centers</u>
Typified by the cities in the Silicon Valley, specifically those in Santa Clara County, such as Sunnyvale, Palo Alto, Mountain View, Santa Clara and San Jose have a high concentration of research-oriented companies and major research-oriented universities that identify them as High Technology Centers. The large companies present in this type of Center continually spin off other technology-oriented companies. The universities located in this type of Center continue to provide technology to industry located in this type of Center. Combined with available capital, it provides for the synergy which creates this type of center.

<u>Diluted Technology Center</u>
This type of Center can be found in cities that have a large base of technology-oriented companies, skilled work forces, along with research universities and venture capital firms. But this type of Center, the high technology or biotech orientation of the area is diluted in larger, broader and more mature economies. Metropolitan areas such as New York, Los Angeles and Chicago illustrate this type of Center.

<u>Spillover Communities</u>
These communities are typically located adjacent to a High Technology Center or Diluted Centers. While these communities usually lack most or all of the ingredients that comprise a high technology or tech hub, they are close enough to such an area to take advantage of its resources. For example, a technology-based company located in such a community can exploit the search capabilities at nearby universities, use venture capital firms located in the region and obtain engineers and scientists from the large technical workforce around the region. Meanwhile, employees seeking graduate degrees or courses in their field can attend local universities. Typically, the objective of such communities is to capture the spillover of companies from the center which are looking for lower cost land and a less competitive labor market. Three examples of this type of communities are Lowell, Massachusetts to Boston as it pertains to biotech. Another example is Naperville, Illinois to Chicago as it pertains to technology and research with companies like Bell Labs, Western Electric and Nokia. Yet another is the unincorporated area of Montgomery County, Maryland to Baltimore as it pertains to medical research and services.

Technology Installation Centers

These types of Centers are the home of a major research or technology-based institution, but they lack most of all of the other ingredients of a Tech Hub. The institution creates a local base of researchers and skilled workers and in some cases, this has led to extensive spinoff activities in the local economy. In other cases, however, the technical base created by the research institution produces few new firms and often remains unavailable to new employers coming into the area. This is because pay scales are usually quite high at such operations. Additionally, there is no technology transfer opportunities or they are heavily regulated making it difficult for research staff to initiate entrepreneurial activities of their own. This group had historically been typified by the policies associated by national laboratories and even NASA. Examples of these types of communities include Sunnyvale, California adjacent to NASA's Ames Research Center; Pasadena, California and it proximity to the NASA's Jet Propulsion Laboratory; Princeton Plasma Physics Laboratory in Princeton, New Jersey.

Bootstrap Communities

A number of communities began their development efforts possessing none of the characteristics of a tech hub. They have depended instead on low operating costs and attractive living environments to allure the expansion of technology facilities to their communities because of those factors. These facilities generally manufacture products which no long have a high technological input and at this stage in their life cycle competitive operation costs are far more important than the research capabilities of a high technology center. However, when several of these facilities have located in an area, their combined work forces create a pool of skilled labor that a more sophisticated operation can build upon. Additionally, the combined engineering work forces at such facilities create enough demand to merit the addition to or improvement of engineering and science courses at local universities. Of interest in the 1980's Austin, Texas; Colorado Springs, Colorado along with Orlando, Florida were identified in this category of communities by the OTA.

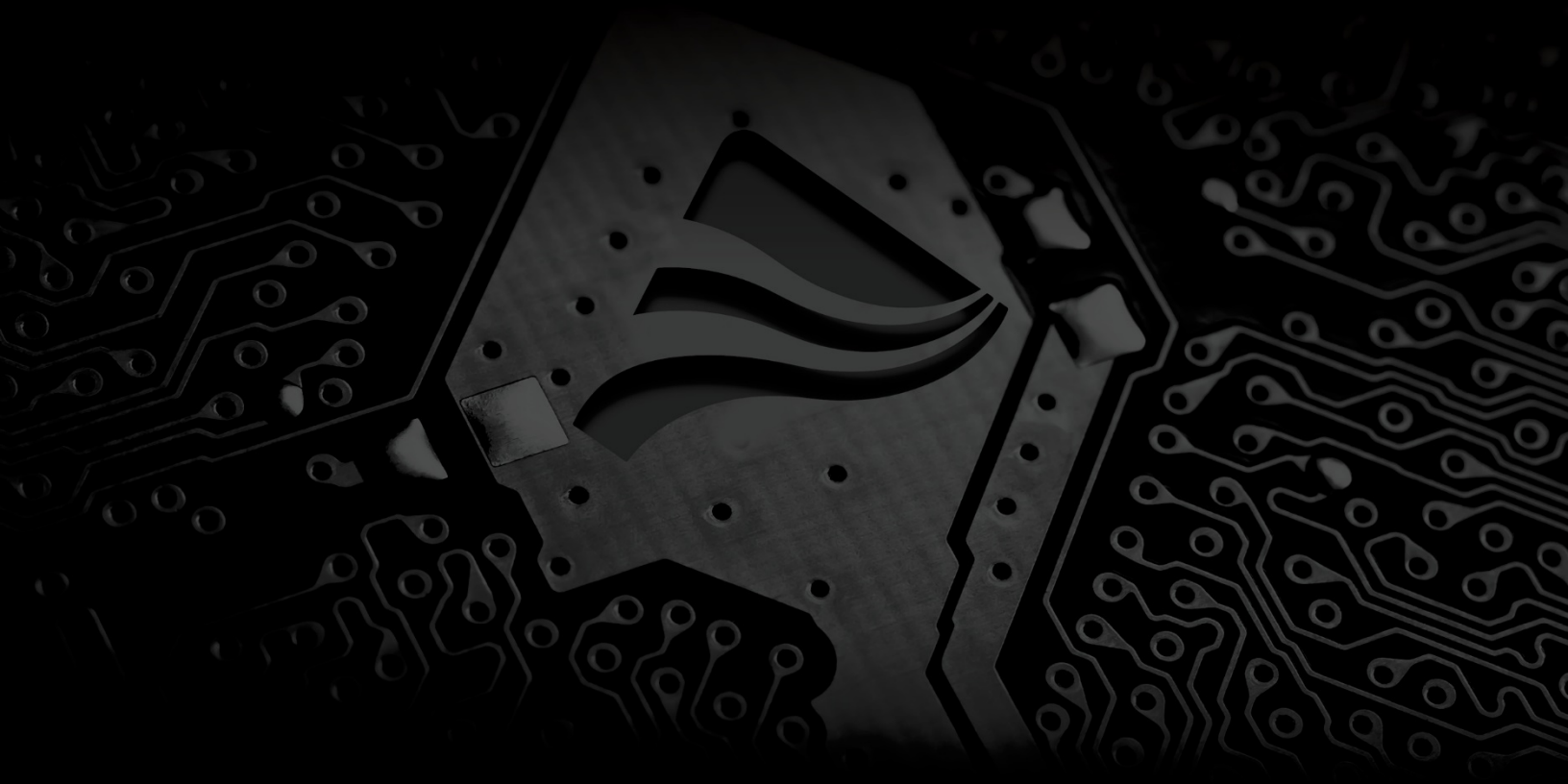
As previously mentioned, tech hubs develop slowly but surely and is the case some thirty years later. The categories may not have changed, but the major tech hubs in the categories have. For instance, tech hubs in the nation now include cities like Austin, Seattle, Boston, Chicago, Los Angeles, New York and San Francisco. Others taking shape include Minneapolis, Pittsburgh and Miami.

Ambition

Today, experts and analysts alike identify that it takes a series of locational advantages to become a successful tech hub. However, this research also identified that a community needs transformative actions such as a subculture of like-minded people, fully engaged public and private sectors, along with the influence and participation of a top tier university and/or a strong entrepreneurial ecosystem, specifically with robust broadband services and access to and availability of capital to achieve a tech hub.

This report provides an exhaustive examination and actions needed along with the programs and projects to become a tech hub

However, this is a work in progress. Through this report will come the opportunity to leverage and embrace further input in the ways of concepts and ideas from the community i.e., through residents and entrepreneurs and stakeholders. This is not a fairy tale, but a plan of action based on realty, where its success lies in the journey not a destination, to live happily ever after.



FEDERAL WAY: ASSESSMENT

FEDERAL WAY: ASSESSMENT

Overview

The purpose of this section is to illustrate the issues or factors that impact or could impact Federal Way in becoming a tech hub.

Assessment

To help understand Federal Way and the opportunities and challenges it faces in becoming a Tech Hub, the research incorporated a PESTEL analysis. PESTEL is an acronym for Political, Economic, Social, Technological, Environmental, and Legal. This Chapter evaluates outside factors that affect or have the potential to affect an organization in achieving its goal or mission. In this case, it is used to understand how outside factors affect or have the potential to affect Federal Way becoming a Tech Hub.

Political: Determines how the current political climate may influence development of policies and programs around technology.

Economic: Examine the effects income levels, wage rates, housing costs and other economic factors may affect the issue of achieving a tech hub.

Social: Acknowledgement of perceptions of Federal Way on becoming a tech hub.

Technology: Examine technology sector and entrepreneurial ecosystem in Federal Way to determine how it may affect achieving a tech hub.

Environment: Identify the environmental factors affecting technology and the impact of technology on the environment.

Legal: Anticipate the impacts of current or proposed laws and regulations which could impact the issues surrounding

The following provides a summary of Federal Way.

Federal Way – Background

The City of Federal Way is located in the Seattle-Tacoma region of over 3 million people. It is approximately 100,000 in population size with nearly a 300,000 - Trade Market Area. It is located 20 miles south of Seattle in the southern end of King County, just north of Tacoma which is located in Pierce County. It is located along Interstate 5 and the Puget Sound, a salt water inlet from the Pacific Ocean. Federal Way is geographically not large, as it is approximately 23.7 mi^2 in size. Neighboring communities include Auburn, Des Moines, Fife Heights, Kent, Milton and Tukwila.

Of significant opportunity, Federal Way is located less than 20 minutes to the SeaTac International Airport. In 2024, light rail is scheduled for operation and will connect Federal Way to the international airport- by way of a 20-minute trip.

Political

Research has identified several key political factors that could impact Federal Way's efforts in becoming a tech hub.

The City of Federal Way was incorporated in 1990. It operates as a strong mayor form of government. The Mayor is elected citywide as are its seven city councilmembers. City Council members are elected to four-year staggered terms. The Mayor operates the City on a day-to-day basis.

The annual budget for the City of Federal Way and all of its funds is just over $110 million. As it pertains to its fiscal operations, the Moody's Investor Services in its annual comment in May 2021, identified that the credit position for the City of Federal Way is of "very high quality," and its Aa2 rating is a little stronger than the median rating of Aa3 for US cities. The notable credit factors include a robust financial position, a negligible debt burden and a moderate pension liability. It also reflects a considerable tax base and a healthy wealth and income profile.

The City's economy and tax base are characterized by Moody's as very healthy overall and is consistent with the City's Aa2 rating.

Federal Way is very healthy overall and consistent with the City's Aa2 bond rating

Moody's

Moody's identified that Federal Way is similar to other cities across the state in that it is likely to experience steady economic growth over the near-term. Sales tax and other economically sensitive revenues should improve as unemployment remains low, particularly in the Puget Sound. Strong improvement in the housing market will continue to boost assessed valuation growth and property tax collections. Pension costs are not a significant source of credit weakness, but remain a longer-term challenge.

Lastly, research identified that through the case studies it takes on average a decade of concentrated efforts to achieve an ecosystem for a tech hub. This includes enhanced broadband services, entrepreneurial ecosystem including access to and availability of capital, a ready, available skilled and experienced labor force for tech growth and the overarching business infrastructure (broadband) to support a tech hub. Thus, it is important to ensure the continuity and consistent focus of the local elected officials to remain on becoming a tech hub.

Economic

Research has identified that several key economic factors that could impact Federal Way's efforts in becoming a tech hub.

First, there are nearly 5,000 businesses located in storefronts in Federal Way. Recently anecdotal information identifies there could be nearly an additional 3,000 home-based businesses in the City as well. This could be a significant number of entrepreneurs in hiding that may be engaged in innovation.

Second, because the City is positioned between multiple institutes of higher education such as the University of Washington, Seattle and its Tacoma campus, along with nearby Highline College, Federal Way has the capacity to act as a magnet and attract a young, diverse, and educated talent pool to meet the needs of businesses.

Thirdly, businesses locate to Federal Way because of the following advantages:

LOCATIONAL ADVANTAGES FEDERAL WAY

- Located between the Ports of Seattle and Tacoma.
- Located between University of Washington – Seattle and Tacoma campuses.
- Over 9 million consumers within a 2 ½ hour drive time. 50% of the state's population is located within a 2-hour drive either north or south from Federal Way.
- Located on the Interstate 5 corridor which is also 20 minutes away from SeaTac International Airport, the busiest airport within Washington. Prior to the pandemic it served over 50 million passengers a year, through 91 domestic and 27 international destinations. Federal Way's location to the airport provides ease of business access to the world.
- City government is proactive, and is able to provide expedited permitting, planning, and licensing for businesses that are looking to relocate or expand their business in the city.
- The City has a minimal taxes on business such as no gross receipts or B&O taxes, unitary, inventory, or state corporate income tax.

While, the City already has a diverse labor force with a deep base of requisite skills that are able to meet current employer's needs, this labor force leaves daily. Research identified that more than 85% go elsewhere to work according to the Census Bureau.

The startup ecosystem for entrepreneurs has been growing steadily in the region. Seattle and Tacoma are home for business accelerators and incubators along with startup studios. Unfortunately, Federal Way does not have either an accelerator or incubator. However, the City is striving to bring in more technology companies including attracting an incubator, accelerator, or even a startup studio to help facilitate growth and to improve if not create an entrepreneurial ecosystem.

Entrepreneurs in Federal Way are supported by a variety of regional programs however none are physically located in city.

The Small Business Development Center (SBDC) is at Highline College. It provides support to established small businesses and startups. Support by the SBDC comes in many forms such as providing referrals to lending partners, finalizing and packaging a business plan, marketing & sales assistance, and classes to help understand how to start a business.

There are also other resources such as the StartZone Microenterprise Program at Highline College which offers no-cost classes to help teach rising entrepreneurs about business. There is the Seattle SCORE office which provides volunteer advisors that assist entrepreneurs in finalizing their idea and begin to develop a business plan.

Additionally, while there are a variety of micro to small office spaces in Federal Way, Federal Way has never been home to either a business accelerator or incubator. However, there are other opportunities for Federal Way to develop its entrepreneurial ecosystem such as the University Initiative.

The University Initiative was started in 2014, when the City actively started recruiting a branch or satellite campus of a university and/or college. The purpose is to help improve workforce development of Federal Way residents, but also act as a catalyst for investment within the City. Highline College in concert with The University of Washington-Tacoma will open a branch campus within Federal Way, and possibly Highline as well. There have been many proposed names for this education center as well as locations, and educational pathway opportunities that may be offered by this program.

Social

Research has identified several key social factors that could impact Federal Way's efforts in becoming a tech hub.

Changing Demography

The median age has continuously risen. Since 2010, the median age was 36.2, in 2020 it was 37.6 and the median is projected to increase to 38.5 in 2025.

Population by Ethnicity

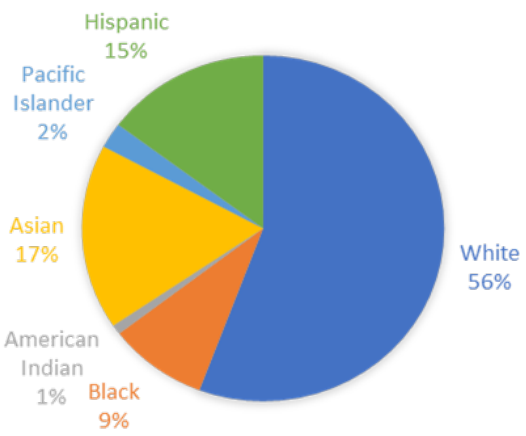
Research data from ESRI illustrates that there will be changes in the city's ethnic composition over the next 5 years. Data shows changes in reduction in the white population (4%) and growth in Asian (3%) and Hispanic population (1%).

This forecast illustrates that the two largest increases by ethnicity will be with the Asian and Hispanic communities. The Asian community is forecasted to going from 27,350 to an estimated 32,562 in 2025 and the Hispanic population from 23,366 to an estimated 26,145 in 2025. The Black population in 2020 is estimated at 14,073. It is forecasted to increase to 15,273 in 2025. This in turn could manifest in the way of the future ethnicity of entrepreneurs in Federal Way.

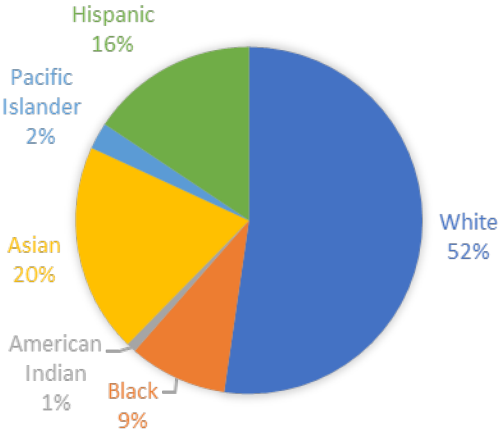
While the ethnic composition is forecasted to change, so is the median household income. From 2020 to 2025 income is projected to have an annual growth rate of 2.25%, which is faster than the state and national rates. However, as shown below one of the key issues that Federal Way is facing is that over 40% of current households make over $100,000. This means that there is over 60% of households that make less than $100,000. Fortunately, by 2025 the projected number of households making over $100,000 will increase to about 46%, but that still leaves 54% of households making less than $100,000.

Below is a chart that shows the comparison between 2020 and 2025 household income differences.

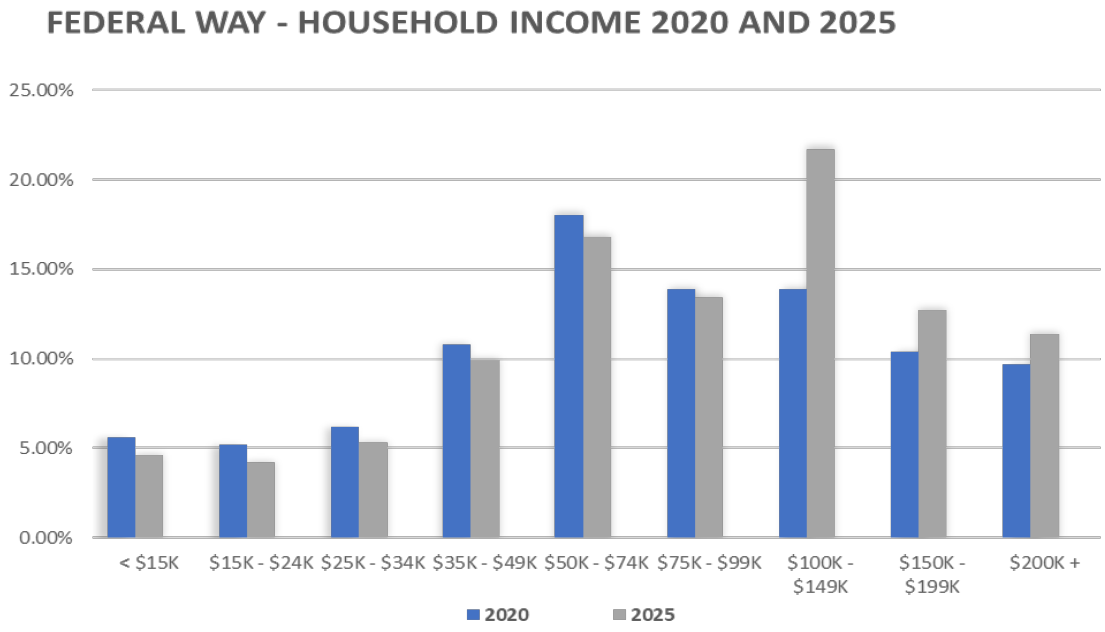
Ethnic Composition 2020 versus 2025



2020



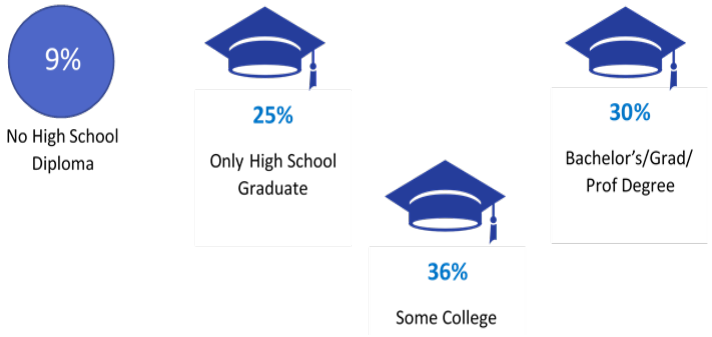
2025



According to the Census Bureau, 30% of Federal Way residents have a bachelor's degree or higher. A City goal has been to elevate this number to improve educational and skill levels necessary to propel its residents in the current and future tech market. Along with providing a strong K-12 education for the city's youth, the goal is to work with the University of Washington and Highline College in the creation of a satellite campus in Federal Way (opening in 2022) to assist its citizens in pursuing higher education.

Highline College and the University of Washington to open satellite of classrooms in Federal Way 2022

Federal Way Educational Statistics



Through developing community and university relations, the City is expecting to attract a satellite campus of higher education but also technical and coding schools to help bring more options for our citizens to receive an education and achieve their academic goals.

Federal Way needs to be active in pursuit of relationships with both Highline College and specifically the University of Washington to grow its entrepreneurial ecosystem

The University of Washington is one of the most significant research - based institutes in the in the Pacific Northwest let alone in the nation

UNIVERSITY OF WASHINGTON

#1 Most innovative Public University in the US, Reuters
#5 among universities based upon their total R&D spending at $1.4 billion in 2019
#7 nationally for Graduate Entrepreneurship , Princeton Review 2021
#7 Best University for Technology Transfer, Milken Institute
#8 best for global universities, US News and World Report 2020
#9 in the Nation for the Number of Startups Launched AUTM 2017
#9 in the nation amongst all US universities for computer Science program, 2020
#11 in the nation amongst all US universities for engineering

Of interest in the 2010 Census, the percentage of the population that had a Bachelor's/Graduate including professional degree comprised 24 percent. Estimates currently show 30%. This population appears to represent the growth in the median household income to 40 percent. The forecast of 46 percent could indicate increasing educational obtainment as it correlates to income.

Technology

Research has identified several key technology factors that could impact Federal Way's efforts in becoming a tech hub.

Size and Scale

As previously identified in the Introduction, the magnitude of the region's tech sector is captured in a recent report by the commercial real estate firm CBRE, which identified that the Seattle region added more than 48,000 tech jobs from 2016 to 2020. This is a 35% increase in employment over this period, representing the largest increase of any metro area in the nation.

Federal Way in its targeted industries/clusters analysis of its economic development strategy includes Information & Communication Technology sector,

Additionally, the life sciences sector showed a 23.5% increase in jobs from 2015 to 2019. More than 39,000 people are employed in the 1,100 firms across the state at an average wage of $106,122. Valuable for Federal Way is that the digital health and IT sector where opportunities have been focused are growing and account for almost a quarter of the state's firms which are located in 136 cities across the state.

The Seattle region was named the top emerging biotech market in 2019 by the CBRE and in 2020 was ranked as a top-ten life sciences cluster – though it's still far behind the major biotech/life science clusters of Boston and San Francisco.

While the Seattle region is home to Microsoft and Amazon, unfortunately little if any direct impact is felt from these companies on the Federal Way economy. Additionally, Federal Way has a small array of tech companies.

According to SourceLink at the University of Missouri, on average, 1% of a city's entrepreneurs are engaged in innovation. Based upon that ratio, Federal Way would need to have at least three to four time the number of businesses engaged in innovation than its current inventory. Thus, the opportunity to grow the sector is omnipresent to achieving a tech hub status.

Technology-based Companies; Federal Way

- Applexus Technologies
- Epicor Software
- Texas Instruments
- Diagnostechs
- MedChem Source
- QS/I
- McObject
- International Paper: Global Cellulose Fiber Innovation Center
- Cogent
- Rockwell Collins
- PlanLED
- Neumeier Engineering (Advanced Manufacturing)
- 7th Power Inc.
- Sagemax Bioceramics
- Infotech Systems Inc
- ITSource
- ISS Galaxy Technologies and Services (home-based business/international focused company)
- Pape Kenworth, sales/distribution of electric trucks (pending)

Tech Attraction

Research examined a variety of industry clusters specifically those that were the tech clusters from the case studies. These include:

- Advance Manufacturing
- Biotech/Life Sciences
- Clean Technology
- Financial Technologies
- Information and Communications Technology
- Smart Cities Technologies
- Transportation/Mobility Technologies

Broadband and Entrepreneurial Ecosystem

Research through the case studies illustrates that broadband and a strong entrepreneurial ecosystem are two key elements necessary to create a strong and sustaining tech-based economy. Internet access facilitates data analysis, collaboration, information storage, and a whole host of other functions crucial to business. Entrepreneurial ecosystems are what create business ventures in the first place and what grow startups into mature companies. These two tools enhance each other, snowballing business attraction, growth, and retention in a community.

There are, of course, a multitude of other factors that influence an economy's development. These include proximity to airports, highways, and major cities; access to and availability of capital; the business-friendliness of local, state, and federal government policies; and much more. However, research has

determined that there are no tools more fundamental to success than broadband and a strong entrepreneurial ecosystem. Without these two tools, nothing can grow, and with these two tools, everything else can be built.

Broadband
If capital is the lifeblood of business, then broadband is its nervous system.

Fast, reliable, and affordable broadband in combination with a strong entrepreneurial ecosystem initiates and turbo-charges the economic development of a community, fostering both entrepreneurship and fueling its growth. A city with ambitions to develop its community and economy can succeed by focusing on these two factors.

Broadband has been identified as an essential utility for many businesses, schools, and households in recent years. Today, reliable, fast, and affordable internet is becoming fundamental to society as electricity was in the 20th century. Although the necessity for broadband is apparent, our nation's digital divide (the persistent gap between American's who have internet access and those who do not) has been increasing, exposing issues of inequality facing underserved communities across the country.

The internet has changed many aspects of many American's lives, such as how communities conduct business, provide education, deliver healthcare, ensure public safety, organize knowledge and data, and so much more. Many cities within the nation and in Washington have examined broadband infrastructure as a necessary utility for their citizens and recognized economic benefits along the way. However, there have been limitations to what local governments can provide. Although broadband has evolved rapidly throughout the last decade, millions of Americans still do not have access to reliable internet.

Without access to high-speed internet, many rural communities – and even parts of urban areas – are left behind. A Microsoft study in 2018 concluded that roughly 19 million rural Americans do not use broadband, primarily due to a lack of access. Furthermore, this lack of access and reliability is a significant barrier to socioeconomic opportunity, education, health, and overall quality of life. Broadband can serve as a lifeline for these small communities, connecting students to online degrees and connecting sick patients to medical consultation that is locally unavailable.

Yet, broadband is also consistently identified as a top challenge facing small businesses in rural America and stifles entrepreneurship by limiting the ability of individuals to take on work. In today's economy, broadband is critical to building resilient and future-ready communities.

According to Pew Research Center, 93% of adults are using the internet as of 2021, and it has become a priority for federal and state governments to provide internet access to their citizens. During this 2021 legislative session, 47 states, the District of Columbia, and Puerto Rico have pending legislation addressing broadband in issue areas such as educational institutions and schools, municipal-run broadband networks, and underserved communities.

When trying to create a successful broadband infrastructure, there are key factors to consider, such as:

- Reliability
  Internet connection must be reliable to minimize outages and maintain a consistent connection to support residents, businesses, and public entities.

- Speed
  The available internet needs to provide a fast-enough connection to accommodate the current and future demands of video conferencing, file sharing, and increased telework.

- Accessibility
  The network needs to be able to reach unserved and underserved areas of the community. These efforts will close the digital divide and ensure equal opportunity to those within the community.

- Affordability
  Access to high-speed and reliable internet needs to be affordable to access more residents and close the digital divide. This is also a necessity for established and startup businesses that rely on the internet.

- Bandwidth
  Within populated communities, buildings, and residences, more people will be connected to the same internet service at a given time. Having a high enough capacity to handle the vast number of residents, students, and businesses is essential to maintain efficient work conditions and optimal internet capabilities.

- Symmetry
  Symmetrical internet connection means the connection provides the same download and upload speeds. A symmetrical connection enables fast data speeds in both directions, making it ideal for large or small offices, apartment complexes, public schools, and other communities where multiple people connect and transfer data simultaneously to the internet.

Many cities have developed their broadband infrastructure to achieve economic success, such as San Leandro, California. Research found that the enhancement and expansion of broadband helped facilitate San Leandro's economic growth, and their efforts have resulted in some of the fastest internet speeds available to businesses in the nation at up to 100 Gigabits per second.

Through these critical factors, communities like Federal Way will be able to provide equal opportunity and relative advantage over other communities by providing high-speed, reliable, accessible, and affordable internet that will facilitate the needs of a tech-forward community. However, many communities either do not fully understand the importance of broadband or do not have supportive state regulations that allow them to create and expand their broadband network. Organizations throughout the nation have made efforts to highlight the nation’s digital divide and make aware the importance of public policy when investing in broadband infrastructure.

Infrastructure Bill

President Biden has proposed a $1 trillion bipartisan infrastructure bill, called the Infrastructure Investment and Jobs Act (IIJA), which focuses on investments in roads, railways, bridges, and broadband internet. Specifically, $65 billion has been allocated to broadband efforts that the White House says will "deliver reliable, affordable, high-speed internet to every household." Currently, the FCC says that roughly 14 million Americans do not have access to broadband at the speeds necessary to work or learn online (25mbps up/3mbps down).

Additionally, within this proposal, $14 billion will be aimed at helping low-income Americans pay for internet service. The digital divide has become apparent as our society has transitioned to utilizing the internet more for our daily lives over the last two years. This bill hopes to reduce that divide by providing a $30 monthly benefit to help low-income citizens pay for internet service.

However, as of the printing of this document, the infrastructure bill has not yet been enacted into law. If the bill becomes law, it will be a boon for every American to access this necessary utility.

Broadband and Federal Way

Considering the information provided, it is apparent that the City of Federal Way has numerous opportunities to provide a reliable, accessible, and high-speed broadband network to its residents and facilitate economic growth.

Previously identified, the current broadband environment in Federal Way cannot support its goal of becoming a tech hub. However, Federal and State regulations support enhancing broadband at the local level, presenting an opportunity for the City.

The City is in an advantageous position to utilize these regulations and create a thriving environment that attracts new businesses, entrepreneurs, and a talented workforce while simultaneously supporting current businesses and residents within the community.

Entrepreneurial Ecosystem

An entrepreneurial ecosystems is a blend of social, economic, cultural, and political components within a region that grow businesses, create jobs, and develop economies. New products and services created by entrepreneurs can produce a positive cascading effect, where they stimulate related businesses or sectors that need to support new ventures, furthering economic development.

An entrepreneurial ecosystem can be started anywhere with a strong enough stimulus, but they tend to grow under certain similar conditions. It takes a combination of some or all of these conditions to achieve success. Conditions important for a successful entrepreneurial ecosystem include the following:

- Supportive Government Policies and Regulations
  Components influenced or controlled by local, state, national, and international governments include ease of starting a business, tax incentives, and ease of commerce. Physical components influenced by these governments include access to infrastructure, telecommunications, and transport.

- Strong Relationships with Local Universities
  Universities promote a culture of respect for entrepreneurship, play a key role in idea-formation for new companies, and serve as career pipelines to almost every industry. Universities may also have entrepreneurship programs, events, or partnerships that contribute to the ecosystem.

- Access to Financing/Capital
  Depth in the availability of financing allows more ideas to be tested and enables early-stage companies to scale quickly. Types of finance include angel investors, private equity, venture capital, and access to debt.

- Quality Human Capital
  Regions with a skilled workforce and many students supplying workers to established companies and help build early-stage companies. Types of human capital include management talent, technical talent, those with entrepreneurial experience, and students.

- Entrepreneurial and Business-Friendly Culture
  Some aspects of cultural support are openness to networking, risk tolerance, preference for self-employment, local success stories, mentorship ecosystems, celebration of innovation, and a research and innovation culture.

Federal Way needs to ensure these conditions for success**.**

Entrepreneurial Ecosystem
Along with these general conditions, there are many specific elements of business infrastructure that boost the growth of an ecosystem. Key elements of an entrepreneurial ecosystem include the following:

- Incubators
- Accelerators
- Startup Studios/Foundries
- Innovation Centers
- Tech Transfer Centers
- Makerspaces and Coworking Spaces
- Meetups, Mashups, and Hackathons

Incubators are multi-month to multi-year collaborative programs for startup companies, often physically located in one central workspace, designed to help startups succeed by providing workspace, funding, mentoring, and training. A city that attracts an incubator can expect new employment opportunities for residents and, specifically, youth through participation in startups. Along with increased entrepreneurialism, incubators can also bring new products and services to an area.

Accelerators are holistic business advisory services that provide an intense, rapid, and immersive education aimed at accelerating the life cycle of young, innovative companies, providing funding and compressing years' worth of learning-by-doing into just a few months. Accelerators are beneficial to cities because they operate as business multipliers, strengthening every cohort of companies. Areas with accelerators see participating local businesses grow quickly, creating jobs and attracting skilled workers.
Startup studios/foundries are a team of engineers and entrepreneurs working on multiple startups at one time. Operators of studios or foundries are real co-founders on projects they pursue. They mentor members of the team to become the future CEOs and CTOs of the companies built with the hope of their projects turning into scalable, full-fledged companies. Establishing a startup studio/foundry in a city concentrates entrepreneurial and technological effort locally, thus producing startups that could become mature local businesses.

Innovation Centers are facilities where a specified team of employees is located and dedicated to innovation, allowing companies to experiment with disruptive technologies, new business models, and new products. Centers tend to be physically separated from the parent company and are set up more like a startup than a traditional R&D department. Innovation centers improve the economy and overall quality

of life of a region by upgrading its technological infrastructure and cultivating a startup ecosystem through investment and collaboration.

Tech transfer centers are hubs or areas where scientific findings are transferred from one organization to another for further development with the intention of connecting to inventors and helping commercialize new products and innovations. Tech transfer centers are essential tools for any city near a university, as students with business ideas need a way to commercialize. The human capital created by educational institutions only has economic effects if business ideas can become a reality.

Makerspaces are collaborative workspaces inside a school, library, or separate public/private facility for making, learning, exploring, and sharing tools of all kinds. These spaces may be open to everyone or just to entrepreneurs and have various maker equipment, including 3D printers, laser cutters, CNC machines, soldering irons, and even sewing machines. Bringing a makerspace to a city encourages local innovation and strengthens entrepreneurial culture, providing a place for ideas to become prototypes, the first step towards creating a successful business.

Coworking spaces are shared professional-style working spaces where that you may pay a monthly fee to use. Coworking spaces often foster community, hold networking events, and provide ideal spaces for small companies, startups, or remote workers. Many large companies operate facilities across the nation and the world, including Impact Hub, Spaces, and WeWork, though coworking can happen anywhere. This presents an opportunity for a city to attract or create a coworking space to strengthen local entrepreneurial work and collaboration.

Meetups are events where a group of like-minded people can listen to experts, share experiences, and simply communicate on a wide variety of subjects. Networking is how business is done and ideas are exchanged, so a city interested in creating an entrepreneurial ecosystem would see considerable benefits from encouraging or hosting meetups.

A mashup is a web page or application that integrates complementary elements from two or more sources. Mashups represent the intersection of different industries of technology and business to create new, innovative products. Examples of mashups include smart street lights, cell phones, and applications built on top of Google Maps. To encourage mashups, entrepreneurs can be brought together to exchange and create ideas using the eight other elements.

Hackathons are competitions that typically occur within a 48-hour time limit in which computer programmers and others involved in software development collaborate intensively on software projects. Hackathons engage the technical minds of a community by testing their skills and bringing them together to collaborate. Hackathons lead to prizes, recruitment, new business ideas, and much more.

These nine elements are central to the creation, growth, and retention of companies in a region to maintain and grow an entrepreneurial ecosystem.

Research identified that Federal Way has no incubators, accelerators, startup studios or business foundries. It does have an innovation center, operated by International Paper focused on cellulose. However, beyond that, Federal Way needs to build an ecosystem.

Environmental
Research has identified that several key environmental issues that could impact Federal Way's efforts in becoming a tech hub.

Climate change poses a variety of factors in Federal Way's efforts to become a technology hub.

A key component of the city's vision should be its alignment of the Tech Action Plan with its yet to be developed Climate Action Plan. Specifically, the vision should drive sustainable outcomes for the residents and businesses (current and future), the environment and it businesses and their customers. The City should ensure commitment to sustainability, creating value for its residents and business while protecting the planet and improving people's lives.

Challenges and Threats
A warming planet creates a wide range of business risks for businesses both large and small, both of which are essential to creating an entrepreneurial ecosystem. Businesses risks include disrupted supply chains to rising insurance costs to labor challenges, which results in a higher business expenses and a higher cost of consumer goods and services. According to a consulting firm Deloitte, climate change and extreme weather directly impacts 70% of all economic sectors worldwide routinely.

Opportunity
The changing environmental landscape presents great opportunity for tech-based climate change innovation. According to the United Nations Technology Executive Committee Technology, "Tech-entrepreneurs," play a central role in the innovation process required to address climate change.

Through the development of technologies, business models, and services that can be used to achieve low-emission and climate resilient sustainable efforts to innovative climate technologies, Federal Way needs to develop a strategy through its Climate Change Plan including a strong entrepreneurial ecosystem to support this sector.

As it pertains to Clean Technologies, research illustrates a variety of opportunities from recruiting private research institutes and research & development organizations, to companies engaged in energy generation, storage, efficiency, and infrastructure. Other companies that could be recruited include those that are engaged in water and waster water research, air and environmental research, and recycling and waste research. Additionally, the opportunity could include housing a major university or even global research center on clean technology. As former U.S. Assistant Secretary of Energy Dan Reicher said, " Clean Tech is growing up."

Legal
Research has identified several key legal or legislative factors that could impact Federal Way's efforts in becoming a tech hub.

National Regulations
The Federal Communications Commission (FCC) defines *broadband* as an internet connection of at least 25 megabits per second (Mbps) download and 3 Mbps upload. These have been the standards since 2015, but the standards change regularly as technology and connectivity improve. There is currently a call for a further increase in the benchmarks.

Previously, in 2009, the FCC began developing a National Broadband Plan to ensure that every American has "access to broadband capability." The FCC has outlined a detailed strategy for achieving affordability and maximizing the use of broadband to advance "consumer welfare, civic participation, public safety, homeland security, community development, health care delivery, energy independence and efficiency, education, employee training, private sector investment, entrepreneurial activity, job creation and economic growth, and other national purposes."

The FCC has a plan illustrating four ways that the government can influence change within the broadband ecosystem.

Four Recommendations of the FCC National Broadband Plan:

1. Design and Establish Competition Policies.
2. Ensure Efficient Allocation and use of Government-Owned and Government-Influenced Assets.
3. Create Incentives for Universal Availability and Adoption of Broadband.
4. Update Policies, Set Standards, and Align Incentives to Maximize Use for National Priorities.

In addition to these four recommendations, the plan covers six recommended goals for the nation to use as a roadmap. These include:

1. At Least 100 million U.S. homes having affordable access to actual download/upload speeds of at least 100mbps/50mbps.

2. The U.S. should become the leader of mobile innovation, with the fastest and most extensive wireless networks of any nation.

3. Bring affordable access to broadband services as well as the means and skills to subscribe if they choose.

4. Bring affordable access to at least 1 gigabit per second (Gbps) broadband service to anchor institutions (schools, hospitals, and government buildings).

5. Every first responder should have access to a nationwide, wireless, interoperable broadband public safety network to ensure community safety.

6. The ability for Americans to use their broadband service to track and manage their energy consumption to lead towards a clean energy economy.

Over the last decade, there have been many updates and changes to the plan to ensure that regulations stay up to date. As of February 5th, 2021, with the introduction of the National Broadband Plan for the Future Act 2021, the FCC is required to update the national broadband plan to assess the progress of original goals; examine effects of the COVID-19 pandemic on the way people learn, work, receive medical information and treatment, and participate in civic communications; and analyze the change in reliance people will have on services enabled by broadband internet access services.

State Regulations

Currently, in Washington, state laws have allowed some municipalities to offer communications services to residents but barred public entities from providing broadband service directly to customers. Recently, state legislation was passed allowing municipalities to provide this service.

ESHB 1336, (passed on July 25th, 2021), allows PUDs, ports, cities, and other public entities in the state to directly provide unrestricted retail broadband service to consumers. Under previous law, these entities could only offer wholesale services to private telecommunications providers. This legislation may become the catalyst to bring high-speed broadband access to underserved and unserved areas of Washington state.

Furthermore, the state has created specific goals for broadband development. These include the following:

- By 2024, all Washington businesses and residences have access to high-speed broadband that provides minimum download speeds of 25 Mbps and minimum upload speeds of 3 Mbps. This State standard aligns with the Federal Communication Commission's (FCC's) current minimum standard for broadband.

- By 2026, all Washington communities have access to at least 1 Gbps symmetrical broadband service at anchor institutions like schools, hospitals, libraries, and government buildings.

- By 2028, all Washington businesses and residences have access to at least one broadband provider with at least 150 Mbps symmetrical broadband service.

Of interest, identified in the legislation, "unserved area" means an area of Washington in which households and businesses lack access to broadband service at a minimum of 100 Mbps download speed and 20 Mbps upload speed.

American Rescue Plan Act (ARPA)

State, local, and tribal governments across America have been under an unprecedented amount of strain due to the COVID-19 pandemic. The need for services has increased quickly, and the crisis also illuminated critical shortcomings and inequalities within American infrastructure. The current public health crisis has devastated the health and economic wellbeing of millions, but ARPA is hoping to revitalize the nation’s economy.

ARPA is a $1.88 trillion stimulus bill, and it will provide a total of $350 billion specifically to eligible state, local, territorial, and tribal governments for their continued response to the pandemic. Washington state itself has received approximately $4.4 billion from the ARPA and has appropriated roughly $400 million in the Capital Budget. Within this budget, $260 million will be used to provide broadband infrastructure grants.

Infrastructure Bill

President Biden has proposed a $1 trillion bipartisan infrastructure bill, called the Infrastructure Investment and Jobs Act (IIJA), which focuses on investments in roads, railways, bridges, and broadband internet. Specifically, $65 billion has been allocated to broadband efforts that the White House says will

"deliver reliable, affordable, high-speed internet to every household." Currently, the FCC says that roughly 14 million Americans do not have access to broadband at the speeds necessary to work or learn online (25mbps up/3mbps down).

Additionally, within this proposal, $14 billion will be aimed at helping low-income Americans pay for internet service. The digital divide has become apparent as our society has transitioned to utilizing the internet more for our daily lives over the last two years. This bill hopes to reduce that divide by providing a $30 monthly benefit to help low-income citizens pay for internet service.

(NOTE, as of the publication of this report, the infrastructure bill has not yet been passed by Congress).

Broadband and Federal Way

The City of Federal Way has numerous opportunities to provide a reliable, accessible, and high-speed broadband network to its residents and facilitate economic growth.

Previously identified, the current broadband environment in Federal Way cannot support its goal of becoming a tech hub. However, Federal and State regulations support enhancing broadband at the local level, presents an opportunity for the City.

The City is in an advantageous position to utilize these regulations and create a thriving business climate that attracts new businesses, entrepreneurs, and a talented workforce while simultaneously supporting current businesses and residents within the community.

PESTEL Analysis

Political Factors	Determines how the current political climate may influence development of policies and programs around technology.	• Strong Mayoral Government • Stable Political Environment • Healthy Tax Base and Economy
Economic Factors	Examines how economic factors may affect the goal of achieving a Tech Hub.	• Locational Advantages • Low Taxes and overall Cost of Doing Business
Social Factors	Examines how current and future demographics may affect becoming a Tech Hub.	• Changing Demography • Promote higher education
Technological Factors	Evaluate and examine technology sectors and entrepreneurial ecosystem in Federal Way to determine how it may affect achieving a Tech Hub.	• A small number of established tech companies and smaller tech companies • Broadband infrastructure • Entrepreneurial Ecosystem
Environmental Factors	Identifies the environmental factors affecting technology and the impact of technology on the environment.	• Alignment with yet to be developed Climate Change Plan • Need to ensure sustainability • Attraction and recruitment of
Legal Factors	Anticipate the impacts of current or proposed laws and regulations that could impact the issues surrounding achieving a Tech Hub.	• Align City Efforts with State & Federal Broadband Goals • Utilization of ARPA dollars to achieve goals • Infrastructure Bill to achieve goals • Build Back Better to achieve goals

Summary

The research identified a broad array of items that could affect and/or enhance Federal Way's ability to become a tech hub.

Federal Way has a strong and stable political environment. It has a healthy tax base and strong financial position.

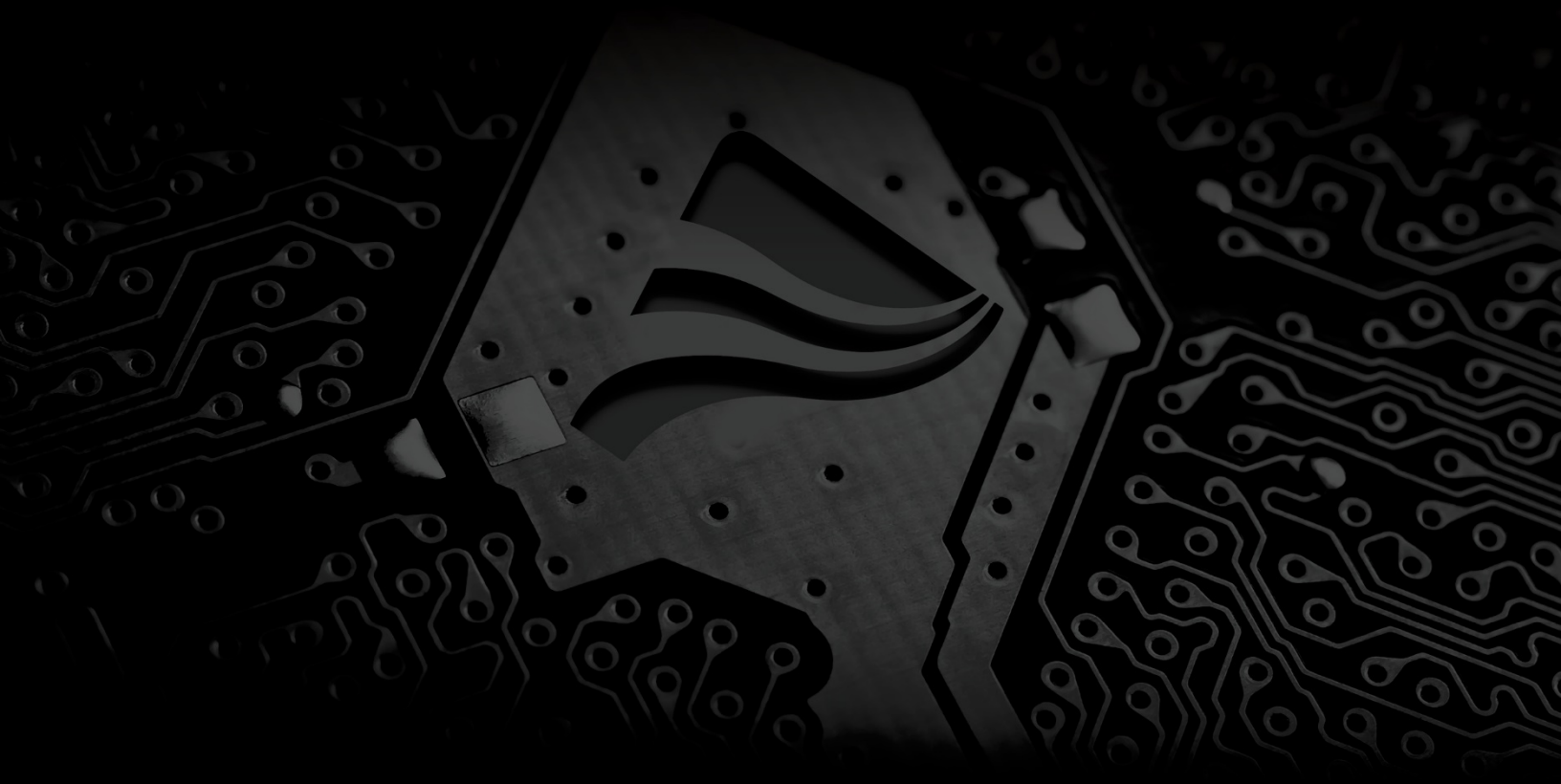
Additionally, economically, Federal Way has several locational advantages such as being centrally located between Seattle and Tacoma, located along the major Interstate on the west coast, (I-5) coupled with a low cost of doing business.

Meanwhile, the social factors illustrate a forecast of a growing population and along with changes in demography, i.e. increase in ethnic composition and income. Thus, these changes could manifest themselves in the number of businesses of color. The City needs to be strategically aware of their requirements in the context of access to and availability of capital to assist their growth and success.

Key to technological factors is the small number of established tech companies ion Federal Way. The opportunity to grow this sector is very relevant to the ambition of becoming a tech hub. Additionally, there is need to enhance broadband service, specifically the last mile of service. Also, the formulation and building of an entrepreneurial system is critical.

There are a variety of environmental opportunities but specifically making sure that the tech hub Action Plan aligns with the city's yet to be formulated Climate Change Plan.

Lastly, there are a variety of legal issues including alignment of the City's efforts with state and federal broadband goals, utilization of American Recovery Plan Act, Infrastructure Bill and the Build Back Better legislation with the actions and corresponding programs and projects identified in the Plan.



CASE STUDIES

CASE STUDIES

Overview

To understand how to become a tech hub, researchers developed an approach to acquire insight to the question what is the key factor(s) and/or action(s) in becoming a tech hub?

To acquire the answer to that question, research incorporated several approaches. These included inductive research including personal interviews and secondary data collection, accountability and legitimacy of the use of data and heuristics, "rule of thumb" to illustrate findings.

The research began with a deep dive in identifying cities that have successfully become tech hubs. Research examined dozens of cities across the nation and the globe that have been successful. In addition, a screening process was used to select similar cities to the population and characteristics of Federal Way. This included cities of similar size, such as 100,000 to 250,000 in population. Additionally, screening included the following: A suburb of a metro area; adjacent to an interstate or major highway; no university when they started their effort, however close proximity to one; age of city when they initiated effort and maturity of the technology and entrepreneurial ecosystem along with identifying the jobs created and investments made.

In the effort to identify successful factor(s) and/or action(s), nine case studies were completed on cities that succeeded at becoming tech hubs.

Research involved analyzing the similarities and differences among these cities, as it provided the opportunity to identify common and cross-cutting themes. The intent was to understand the following:

- What stakeholder organization(s) and individual(s) did cities rely on to spur tech-based development?
- What strategies did the communities use to achieve success?
- Under what type of economic conditions did such success occur, and
- What lessons from this experience might be valuable to Federal Way.

The cities reviewed include:

1. Greenville, South Carolina
2. Petersburg, Virginia
3. San Leandro, California
4. Lancaster, California
5. Alpharetta, Georgia
6. Richardson, Texas
7. South San Francisco, California
8. District of Songdo, Incheon, South Korea
9. Milton Keynes, England

It is important to note that the research did not seek similar types of technologies. Research was not focused on the technology as it was the methodology used to achieve becoming a tech hub. Suffice it to say, each city that was analyzed achieved success through a different type of technology. For example, Greenville and Petersburg focused on advanced manufacturing while San Leandro concentrated on

broadband. Lancaster incorporated clean technologies and Alpharetta focused on financial technologies. Richardson committed itself to information and communication technology (ICT), while South San Francisco focused on life sciences and biotech. Songdo concentrated on Smart City technologies and Milton Keynes focused on transportation and mobility technologies.

The value of the case studies is not in identifying a type of technology that acts as a panacea or a silver bullet for success. Rather, it is to ascertain a common theme for success and to understand who the actors were and what actions, initiatives, and tools they used to achieve the development of becoming a tech hub.

The following presents the case studies.

GREENVILLE, SOUTH CAROLINA

Overview

Is it possible for history to rewrite itself? Cities in the southeastern part of the nation's oldest manufacturing corridor are rewriting manufacturing history.

Advanced manufacturing is the use of innovative technology to improve products or processes. It can include production activities that depend on information, automation, computation, software, sensing, and networking. However, as these technologies have become more widely used in the manufacturing processes, the line between traditional and advanced manufacturing has become blurred, rendering the definition less relevant. While others have framed the definition, the diversity of the sector is difficult to define. Think tanks like the Brookings Institute, consultancies such as Deloitte or even the federal government through the Federal Reserve all have similar but varying definitions.

The reason for the definitions may be due to the speed of evolution moved by the disruptive technologies being used in this sector. They are creating vast changes. For example, many experts believe there is something identified as the 4th industrial revolution, or Industry 4.0. This involves the newest innovations in Advanced Manufacturing being identified as "Smart Manufacturing." The National Institute of Standards and Technology defines Smart Manufacturing as fully-integrated, collaborative manufacturing systems that respond in real time to meet changing demands and conditions in the factory, in the supply network, and in customer needs.

A report by the tax and consultancy firm Deloitte has identified the ten most promising Industry 4.0 technologies, which include predictive analytics, smart factories, advanced materials, connected products, digital design, simulation and integration, high performance computing, advanced robotics, additive manufacturing, open source design, and augmented reality. So, the framework is Advanced Manufacturing, and its detailed inventions and innovations are called smart manufacturing.

While headlines have identified that manufacturing has less of an impact on the economy than it did post WWII, the manufacturing industry still employs a significant number. Today, over 12 million people are employed in the sector nationally. More than 305,000 in Washington State are employed in the manufacturing sector, and in King County there is an estimated 105,000 people employed in the manufacturing sector.

However, based on the definition of Advanced Manufacturing by the Brookings Institution, the subsector of advanced manufacturing would employ an estimated 6.3 million nationally; 154,000 in Washington State, and 60,000 in King County.

According to Grandview Research, some of the key global players in the development of smart manufacturing technologies are Siemens AG, General Electric, Rockwell Automation, Schneider Electric, Honeywell International, Emerson Electric, and The Fanuc Corporation.

The five largest manufacturers in the U.S. based on revenue are Exxon Mobil, Apple, General Motors, Ford, and General Electric.

This research has identified major cities across the globe and the nation that have created successful business environments for growing advanced manufacturing ecosystems. These cities are presented in the following chart:

GLOBAL ADVANCED MANUFACTURING HUBS

Copenhagen, Denmark	**Istanbul, Turkey**
Queensland, Australia	**Ulsan, South Korea**
Lombardi, Italy	**Tamil Nadu, India**

Source: World Economic Forum

However, screening involved similar characteristics to Federal Way. These included similar population size (100,000- 250,000); a suburb-location in a metropolitan area; no university within the city limits although in close proximity to a major research university; age of the city; successful entrepreneurial ecosystem; and metrics of success such as jobs created and investments made.

The research identified two communities, Greenville, South Carolina; and Petersburg, Virginia. These cities were chosen as case studies for advanced manufacturing due to the unique growth conditions and resulting state and local public policies that created an environment for growth of the sector. Additionally, they were selected based upon similarities to Federal Way, such as population, location along an interstate, and proximity to a major research university.

SCREENING PROCESS

- **Population Size (100,000-250,000)**
- **Suburb to Metropolitan Area**
- **Adjacent to Interstate**
- **No University, yet close proximity to one**
- **Age of City compared to Federal Way**
- **Maturity of Technology & Entrepreneurial Ecosystem**
- **Jobs Created & Investments Made**

Taking these individual filters into account, the most notable similarities to Federal Way is the distance to a major metropolitan Area (Seattle and Richmond). This proximity allows the cities to access the talent pools within these major cities and benefit from nearby business and technology ecosystems. Also, proximity to major research universities (UW, Virginia Commonwealth, and Clemson). All are located along an Interstate. However, when it comes to age, Petersburg was incorporated in the 1749, 241 years before Federal Way.

Meanwhile, the differences are stark. Greenville and Petersburg both have extensive entrepreneurial ecosystems that have created the growth of their advanced manufacturing technology sectors. Although each approached their efforts slightly different. Another major difference is that both Greenville and Petersburg have been heavily supported by state resources to enhance their growth of technology sector as compared to Federal Way.

Additionally, Petersburg has an institute of higher education – Virginia State University, which is an all-black student body of 3,000 scholars. Also, it has nearby Bland College, a satellite campus of William and Mary.

Greenville, South Carolina

Greenville is a city in upstate South Carolina that is similar to Federal Way in several respects including population, proximity to a major research university, and access to an international airport. The city has a population of about 70,000, is about 30 miles east of Clemson University, and Greenville-Spartanburg International Airport (GSP) is only 13 miles from downtown Greenville. The city is located midway between Atlanta and Charlotte on Interstate 85. Greenville has a strong advanced manufacturing presence, with companies like General Electric, Lockheed-Martin, and the Michelin Tires North American Headquarters.

Greenville is located in the northwest corner of South Carolina along Interstate-85 (I-85), one of the busiest interstates in the nation. Dubbed the center of the "Charlanta" mega-region, Greenville is situated approximately halfway between two of the largest cities in the Southeast—Atlanta, Georgia; and Charlotte, North Carolina. Interstate-85, the backbone of the manufacturing region, stretches from Richmond, Virginia; through Atlanta, Georgia; to southern Alabama.

Economic History

Greenville's growth as an advanced manufacturing hub began in the late 1960's with the arrival of several large manufacturing firms, and the beginning of Clemson University's presence in the city. According to *Greenville News*, the transition to manufacturing came at a pivotal time when the city's historic textile industry was beginning to struggle, and mills were laying off workers.

In 1967, General Electric announced that it would be building a 300,000 square foot gas turbine plant in Greenville. Although the company already had a plant in the nearby city of Irmo, Greenville was chosen for several reasons. This included available and reasonably priced land, water, educational programs at Greenville Technical College, and the city's leadership. In 1970, GE invested another $60 million and increased the plant's employment to 1,500. Today, the plant employs more than 3,000 workers.

General Electric has continued to invest in Greenville by opening an advanced manufacturing R&D facility on the campus of their turbine plant called Advanced Manufacturing Works (AMW). This facility, which opened in 2017, represents a $400 million investment. The purpose of the facility is to research technologies to accelerate improvements in every aspect of the manufacturing process including design, engineering, product development, production, supply chain, distribution, and service and will unlock new productivity and growth across GE's power portfolio. Some of these new technologies include materials science, 3-D printing (additive manufacturing), automation, advanced software platforms, and robotics.

In 1975, Michelin opened its radial tire plant in Greenville. Then, in 1988, Michelin moved its North American Headquarters to the city. According to Greenville News, Michelin chose Greenville because of its conservative labor environment, availability of qualified labor, location, and because they were given a five-year tax holiday. Michelin now employs more than 4,300 workers in Greenville County.

Michelin is credited with much of Greenville's success in becoming an advanced manufacturing hub. Michelin has

invested over a billion dollars in Greenville County. By establishing its headquarters in the city, it transformed the county "from a fading textile industry to modern, diversified manufacturing."

In 1984, Lockheed began operations in Greenville as the Lockheed Aeromod Center, the production line for the F-16 fighter aircraft. The facility employs over 500 workers, contributing an annual economic impact of more than $100 million to the surrounding area.

Another important step in the development of Greenville as an advanced manufacturing hub was their ability to develop a university presence in the city. In 1981, Clemson University began offering graduate-level courses at the Greenville Technical College campus. This partnership would eventually develop into what is now called University Center Greenville (UCG). According to the UCG website, in 1987, Clemson University with Greenville Tech's cooperation invited other state-supported institutions to form a consortium for the Greenville area. The proposed new Greenville Higher Education Center was chartered by the South Carolina Commission on Higher Education and officially began its operations on the Greenville Tech campus. Currently, UCG partners with ten different institutions of higher learning to further the education of the residents of Greenville.

The reasoning behind the establishment of the Center was to bring Bachelor and Graduate (Masters and Doctoral) degrees from many of the State's most respected and highest ranked universities to the citizens and economic community of Greenville. The Greenville region is the largest metropolitan area in the state that does not have a public university within its city limits. Although Clemson University and the University of South Carolina are within an hour commuting distance of downtown Greenville, neither is accessible to working professionals who cannot afford a two-hour round trip to take day or evening classes.

However, Clemson doubled down on their presence in Greenville. In 2007, Clemson opened the Clemson University International Center for Automotive Research (CU-ICAR) in Greenville. The CU-ICAR is an advanced-technology research campus where education, research, and economic development collaborate to create a global venue for the automotive industry; an automotive ecosystem that helps companies make connections and build relationships. With the addition of the Center for Emerging Technologies in 2012, CU-ICAR currently has 20 Campus Partners from around the world. These on-site residents work in close quarters with our students and faculty. The CU-ICAR was made possible thanks to early investment by industry partners including AT&T, BMW, JTEKT Koyo, and Michelin.

Advanced Manufacturing

The City of Greenville and the surrounding area contains a concentrated cluster of advanced manufacturing industry. Along with the companies already listed, other manufacturers include Sealed Air Corp, Magna/Drive Automotive, Bosch Rexroth, Mitsubishi Polyester, and the Hubbell Lighting corporate headquarters. BMW also has yet a second facility in the region with a manufacturing plant in the nearby city of Spartanburg.

Greenville has continued to target advanced manufacturing in their business attraction efforts. According to the Greenville Area Economic Development Corporation, a private non-profit developed by Greenville County, while they have focused on advance manufacturing, their targeted industries for the area also include advanced materials, automotive, aviation/aerospace, biosciences, data centers, headquarters, and office space.

Tech and Entrepreneurial Ecosystem

Greenville has also been successful as a tech hub in developing and growing an entrepreneurial ecosystem, thanks in large part to a group of local entrepreneurs and founders who created the support network, NEXT.

NEXT began with a small group of entrepreneurs in the mid 2000's who saw the need for an organization to help the growing startup community in upstate South Carolina. They created a plan to build an organization that could become a peer network, foster fellowship amongst innovators, and help provide entrepreneurs access facilities, talent, and capital. With the financial support of the Greenville Chamber of Commerce and eventually the City of Greenville, NEXT began offering entrepreneur support and networking services to the local startup community in 2006.

Since its inception, NEXT has grown and continues to add value to the city's entrepreneurial ecosystem. The organization has achieved a variety of actions such as the formulation of the Upstate Carolina Angel Network, the opening of three coworking spaces, the launch of a venture mentoring service and virtual incubator, and holds an annual conference to connect entrepreneurs with venture capitalists. Through these efforts NEXT has been able to raise over $300 million for startup businesses.

Over the past 15 years, NEXT has opened three different spaces. These spaces, NEXT Innovation Center, NEXT on Main, and NEXT Manufacturing, provide entrepreneurs, startups, and small technology businesses with flexible lease options and a collaborative work environment. Together, the spaces total over 100,000 square feet and currently support more than 110 companies.

Within NEXT is the non-profit makerspace facility known as Synergy Mill. This facility provides a shared workshop with woodworking tools, metalworking tools, electronics tools,

welding equipment, 3d printers, a laser cutter, workbenches, and more.

Yet another element is the minority business accelerator program in Greenville. It was established to serve as a catalyst to drive a positive community economic impact by increasing the overall capacity of locally-based high potential minority business enterprises located within the market area. The accelerator program prepares minority-owned businesses for growth and expansion to create a competitive advantage for minority owned businesses located in Upstate South Carolina to become the most inclusive business community in the world. The program grows the economic success of disadvantaged organizations through business development services, mentoring, technical assistance, and access to large corporations for partnership opportunities.

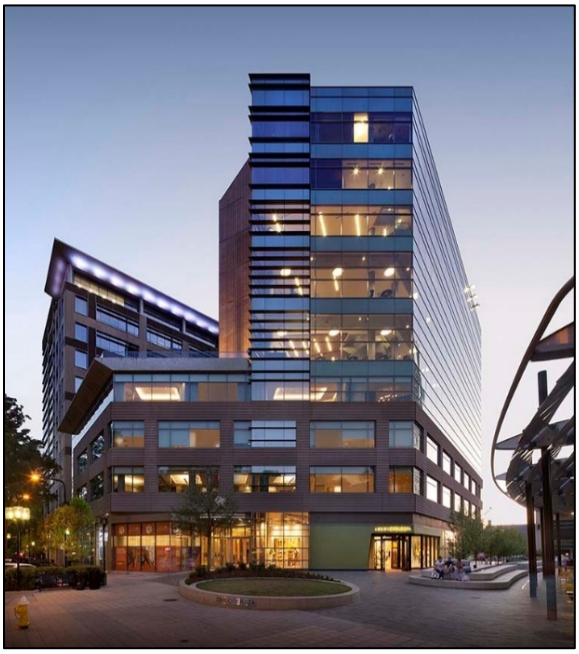
NEXT Facility in Greenville
Source: NEXT & Hughes Development

The Greenville Technical College Center for Manufacturing Innovation Business Incubator is intended to help firms make use of the nearly $7 million in advanced manufacturing equipment and hardware at the Center for Manufacturing Innovation. The Center provides opportunities for businesses to collaborate with students and faculty to eventually grow into their own innovation through prototyping and/or start-up phase.

In 2014, Clemson moved their MBA program for corporate entrepreneurship and innovation, and their business analytics to the Greenville One building.

Greenville One is also home to several other of Clemson's graduate-level programs, such as accountancy, real estate development, and management, as well as Clemson's Phyfer Innovation Hub.

Another element of Greenville's technology and entrepreneurial ecosystem is the Phyfer Innovation Hub. It provides MBA students from Clemson along with faculty, and the upstate community a place to meet and share innovative thought and practice. This includes mentoring promising startups and providing ongoing support for workplace innovation. The Phyfer Innovation Hub is located at the office complex known as Greenville One.



Greenville One Building
Source: Clemson University

Greenville is also home to the International Transportation Innovation Center (ITIC). ITIC is an independent non-profit proving ground for automotive OEMs, suppliers, technology providers, and research universities to test and validate mobility innovations. In addition to offering multi-configuration test tracks, we provide a variety of services for developing and testing transportation megatrends such as vehicle automation, connectivity, and electrification.

ITIC is headquartered in Greenville, SC, at the S.C. Technology & Aviation Center (SCTAC), a sophisticated business park that is home to more than 110 companies in key market segments, such as advanced manufacturing, trade, technology, and automotive/aerospace innovation. SCTAC is also home to Donaldson Airfield, South Carolina's largest general aviation airport with an 8,000-sq. ft. runway capable of accommodating large aircraft. The airport is particularly convenient for companies that need to transport components for testing and for customers who wish to fly directly to the testbed.

The ITIC serves an important purpose for a growing tech hub like Greenville. Along with research and testing, the mission of the ITIC is also to help the area attract nationally renowned companies and researchers, attract knowledge-based economy jobs, accelerate economic and community development, and enhance the educational and scientific initiatives of South Carolina.

In addition, Clemson University's International Center for Automotive Research (CU-ICAR) has also contributed a great deal to Greenville as a growing advanced manufacturing hub. Since the location was opened, the Center has invested over $250 million and created 770 on site jobs. CU-ICAR focuses specifically on seven major areas of research in the automotive field based on the current innovation trends in the field. These areas are advanced powertrain systems, automotive systems integration, human factors, manufacturing and materials, vehicle performance, vehicle-to-vehicle connectivity, and vehicular electronics.

CU-ICAR works on this research with a variety of industry partners, including BMW, Bosch, Chrysler, GE, General Motors, Honda, and Toyota.

South Carolina Technology Clusters

- **Advanced Materials**
- **Chemical and Engineering Process**
- **Defense**
- **Energy**
- **Information Technology**
- **Life Sciences**

Access and Availability of Capital

The entrepreneurial ecosystem in Greenville is served in large part by a variety of angel and venture capital investors including accredited investors in Greenville.

Labor Force

The City of Greenville has an employed population of 37,000, with nearly 46% of the labor force completing a bachelor's degree or higher. According to the City of Greenville, almost 12% of workforce is in the manufacturing industry. The city and surrounding area have an abundance of talent to pull from thanks to Greenville Technical College's Center for Manufacturing Innovation and Clemson.

Public Policy

The City of Greenville, Greenville County, and the State of South Carolina all have public policies designed to attract targeted industries. These policies target companies that establish or expand corporate offices, manufacturing companies, R&D, and other high-technology companies.

In addition, the South Carolina Research Authority (SCRA) Launch Program provides investments and support to early stage, start-up technology companies in the following technology clusters

Some of these policies include:

Supportive Industry Policies

- **City of Greenville Business License Abatement**
- **Greenville County Fee-In-Lieu of Property Taxes**
- **Greenville County Property Tax Abatement**
- **State Job Tax Credits**
- **State Job Development Credits**

Summary

Greenville provides an example of a city that has found success as a growing tech hub, especially in the area of advanced manufacturing. This success has had a positive impact on Greenville's economy.

While, Greenville is a much older community than Federal Way, it too experienced the need to transition its economy to one that is more sustainable and future focused. Historically, the City focused its attention on attracting manufacturers. Impacted by changes in the economy and off-shoring of business activity, it subsequently amended its economic development plan to formulate an ecosystem around advance manufacturing.

This effort has paid off. Today, the City of Greenville outpaces U.S. per capita personal income by 36.9% and domestic product growth from 2010 to 2018 by 44.9%.

Ranked, Rated & Graded

Greenville has been ranked by CNBC as one of the top ten cities for people under 35. It has also been ranked as one of America's top three strongest job markets by Bloomberg/BusinessWeek, and it was also ranked as a top ten engineering hub with the greatest capacity for innovation by Forbes.

According to Ken Brower, Interim CEO of NEXT, "There is still a great deal of potential for more growth, especially for the entrepreneurial ecosystem. The next step for the hub is to create a stronger collaboration between the entrepreneurial ecosystem, the local advanced manufacturing industry, and institutions of higher learning." "By working together, this hub can create a virtuous cycle of talent, innovation, and investment to further the city's growth as a tech hub," according to Brower.

Research also identified that they also were able to build a relationship with Clemson University to bring education, investment, and a stronger pool of talent to the City. In the past two decades, innovation and entrepreneurial activities have exploded thanks to investment by the City in advanced manufacturing due to its relationship with Clemson University.

Lastly, the Wall Street Journal identified Greenville as one of several breakout cities on the forefront of America's economic recovery, stating that Greenville may actually be the best example of such a city. The article describes a dramatic transformation of a community from a textile-based economy to one focused on manufacturing and technology. This transformation relied on attracting automotive and manufacturing firms and investing in its downtown area. The diversification to advanced manufacturing has enhanced the economy despite the effects from the pandemic. Apparently, its future looks bright in a post pandemic world.

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PETERSBURG, VIRGINIA

Petersburg, VA, is part of the Richmond Metropolitan Statistical Area which is comprised of an estimated 1.3 million people. Petersburg has a population of 32,000. The city is 23.1 square-miles in size and is one of 13 jurisdictions that comprise the Richmond-Petersburg Metropolitan Statistical Area. It is located 23 miles south of downtown Richmond and is 25 miles from the Richmond International airport

It is located along two major interstates, I-95 and I-85. It is also at the juncture of two mainline railroads, Norfolk Southern and the CSX.

Overview

Similar to Greenville, Petersburg is focused on Advanced Manufacturing. The previous overview summarizes the marketplace and screening process.

Economic History

Located along the Appomattox River as a settlement in the 1700's, the City was focused on tobacco, trade, and transportation. Historically, it was an agricultural area and a series of mills. It grew as a hub from river transportation and railroads. Thus, the concept of networks is well understood in community planning.

For Petersburg it was linked historically by water, later by rail, and then highway to adjacent communities such as Richmond. It is also served by the Richmond International Airport. However, it was the development of highways that lead to the departure from the core of the city that aided to its subsequent economic decline.

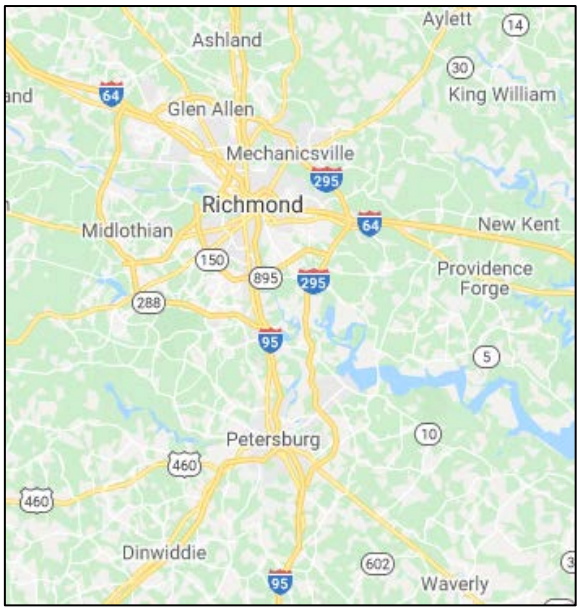
Petersburg is known for its rich historical background, which shines through its Postbellum and Edwardian architecture widely present in the city's downtown area. While local leaders have been listless to respond, over the decades, the state government, universities and the private sector have led efforts to resuscitate the economy.

Common Wealth Center for Advanced Manufacturing Statistics

- **Employs over 60 Employees including**
- **40 researchers, including 13 PhDs from around the world and 4 graduate research assistants**
- **9 interns**
- **7 business staff**
- **34 Members *(academia, government & industry)***

For example, in recent years the Petersburg has been the backdrop for many films and movies such as Steven Spielberg's *Lincoln* and AMC's *Turn.*

Petersburg's downtown continues to revitalize, spurred by growth of nearby community of Fort Lee and the unique housing opportunities created by adaptive reuse of renovated warehouses and historic homes along with growth of local small businesses and entrepreneurs. Additionally, the City is home to festivals and monthly arts events increasing tourism throughout the year.



Tech and Entrepreneur Ecosystem

Advance Manufacturing has a significant impact on the economy of the Petersburg/Richmond metropolitan area. There are over 1,100 manufacturing facilities in the region that employ nearly 40,000.

Petersburg is part of the Virginia 's Gateway Region. It is known for its game-changing research and innovation centers specializing in advanced manufacturing. Some of the major employers serving the region and Petersburg include DuPont, General Electric and Rolls-Royce North America.

"Game Changing Research and Innovation Centers"

Self-proclaimed: Petersburg, Virginia

There are a variety of organizations that assist businesses in the area of advanced manufacturing.

Completed in 2012, the $17 million Commonwealth Center for Advanced Manufacturing (CCAM) is a facility that works in concert with Virginia's major universities including University of Virginia, Virginia Tech, Commonwealth of Virginia, and Virginia State University for research and development labs for manufacturers and university researchers.

CCAM engages in a variety of areas. These include, surface engineering and manufacturing systems to 3D additive manufacturing and composite materials development. However, these areas are determined by the members and as such, are subject to change as technologies and member needs evolve.

Seven manufacturers including Siemens, Rolls-Royce, and Canon located units to the Center. It brings together some of Virginia's biggest companies and brightest minds to solve challenges in advanced manufacturing technologies critical to Virginia's economy.

CCAM is a 61,998 sq. ft. state-of-the-art building with 16,000-square foot high bay area and labs on a 20-acre campus.

CCAM has also secured a combination of state and federal funding to build a $12.6 million Advanced Manufacturing Apprentice Academy next to its main facility.

The Commonwealth Center for Advanced Manufacturing
Source: Perkins and Will

The purpose of the Advanced Manufacturing Apprentice Academy is to train high school and college students, specifically older students that are returning for retraining in advanced technology fields such as advanced manufacturing, data analytics, bioscience, and information technology. The Academy opened in 2018. Its purpose is to deliver a truly sustainable advanced manufacturing workforce pipeline for the region.

The Manufacturing Skills Certification is designed to facilitate the workforce needs of advanced manufacturing and technology employers in Virginia. After training and assessment delivered on-site or at an education institution, the MSI awards certifications for Manufacturing Technicians or Specialists, indicating readiness for employment in advanced manufacturing and building skills in precision machining, systems and processes, machine programming, CAD, precision measurement, machine troubleshooting and maintenance, and more.

Meanwhile in addition to these technology and entrepreneurial programs, the Community College Workforce Alliance supports local companies with job fairs, customizable employee assessments, specialized trainings, and convenient access to many locations. Working collaboratively with the Apprentice and Employee Development Center, companies create individualized programs of study or fast-tracked certificate programs that feature classes tooled directly to company needs.

Also, the Precision Machining Program at John Tyler Community College is accredited by the National Institute for Metalworking Skills. It has come to play an important role in the community. For example, in 2011, Rolls-Royce was the first of several companies to donate specialized machinery to the program to train local employment candidates directly on modern company machinery. The Institute allows students to develop hands-on expertise that is directly applicable to advanced manufacturing systems across many different crafts or trades.

Also, as previously identified, Virginia State University, a small black college located in Petersburg, offers a large number of sciences, technology, engineering, and mathematics degrees and advanced degrees. These include computer science and engineering, manufacturing engineering, electrical and electronic engineering, mathematics, biology, and chemistry. Success Squared (S2) program through Virginia State University provides interns involved in science, technology, engineering and math that provide valuable skills for employers.

Virginia Commonwealth University (VCU) has undergraduate and graduate engineering programs with heavy research emphasis, especially in biomedical, chemicals, life science computer, electrical, and mechanical engineering, and computer science. There are over 1,700 undergraduate and 280 graduate engineering students enrolled each year. VCU also coordinates a statewide partnership that allows students to take graduate courses in 14 additional engineering fields offered on other campuses.

The University of Virginia (UVA) is located an hour away in Charlottesville, is an important workforce source for the region due to R&D and corporate partnerships in the area. The top four national public university has more than 2,000 undergraduate and close to 1,000 graduate engineering students with 11 different engineering programs and four off-campus collaborative programs, including Mechanical and Aerospace Engineering and Science, Technology, and Engineering.

Virginia Tech, while located outside the Petersburg/Richmond region, provides students to area employers. For example, the University's Industrial and Systems Engineering, Environmental Engineering, Biological Systems Engineering, and Civil and Environmental Engineering programs consistently rank in the top ten nationally. Aerospace and Science and Mechanics are

additional programs offered among the 15 total undergraduate engineering degrees.

Additionally, the program *GO Virginia,* a business led initiative that fosters private-sector growth and job creation through state incentives for regional collaboration serves the needs of startups and entrepreneurs through access to capital be it debt or equity.

The Virginia Economic Development Partnership, (VEDP) works together with the state and the state's Community College Workforce Alliance to help approve grants for building development and capacity building for cities like Petersburg to attract advanced manufacturers. An example is the Virginia Talent Accelerator Program which is a workforce initiative by the VEDP in collaboration with the Virginia Community College System. They receive financial assistance through the governor's office and state legislature. Launched in 2019, the program accelerates new facility start-ups through the direct delivery of recruitment and training services that are fully customized to a company's unique products, processes, equipment, standards, and culture. All program services are provided at no cost to qualified new and expanding companies as an incentive for job creation. The Federation of Advanced Manufacturing Education, (FAME) at Richard Bland College of William and Mary is a significant player in workforce development. The program is a network of employers that sponsor students through a work/learn model. The Advanced Manufacturing Technician program produces entry level, multiskilled technicians for advanced manufacturers.

The state of Virginia also provides, The Commonwealth's Opportunity Fund. It is a state program available to assist small and medium sized businesses secure a site for development or expansion of facilities.

City Efforts

The City of Petersburg is economically competitive through innovative strategies for business attraction, expansion, and retention. The development and implementation of these strategies establish the City's prominence as a premier location of commerce and industry in Central Virginia. Improvements to the City's infrastructure are strategically planned to expand water, wastewater, natural gas, broadband, and other utility systems to accommodate new development. The City works collaboratively with the Commonwealth of Virginia, regional organizations, economic development agencies, and other jurisdictions, local businesses, and its citizens to enhance investment and job creation.

Incentives

As previously mentioned, the City has not played a significant role in formulating a technology or entrepreneurial ecosystem. However, they do offer a variety of incentives for businesses, big and small.

The City of Petersburg provides several incentives through its Technology Zone and its Enterprise Zone along with its federal Opportunity Zone.

Also, the city provides assistance with infrastructure with its Tap Fee Reduction–Fees associated with connecting to the city's water and sewer system. The program can reduce fees by fifty percent.

The City Assessor offers an Enhancement to Real Estate Rehabilitation Tax Exemption. It is yet another program. This is a partial exemption of property taxes on commercial real estate over a five-year period.

The City of Petersburg also provides a rebate of 50% on taxes for machinery and tools collected by the City from new industries locating in the City's Enterprise Zone. The rebate starts the year that the industry begins operations, and it is available for five consecutive years.

Additionally, the City provides discounted acquisition prices on City-owned industrial sites. Industrial sites owned by the City of Petersburg are sold at below market prices to targeted businesses that meet the wage and benefit requirements of the State Jobs Grant incentive at the $500 grant level (minimum of five full-time positions earning at least 150% of the federal minimum wage with health benefits,

Summary

As previously identified, Petersburg started as an agricultural town focused on tobacco and trade. It continued its growth from river transportation to railroad followed by highway development and now by air.

Petersburg has also experienced economic declines. Most recently it started in the 1980s because the City specialized in only a few sectors making the City vulnerable to economic cycles. This resulted in a population decrease for 30 consecutive years. So, like so many American cities, it too suffered through the Great Recession. However, a resurgence has begun. State government and university efforts in the advanced manufacturing sector have begun to transform Petersburg as a hub in advanced manufacturing.

Petersburg found success in growing as an advanced manufacturing tech hub for several reasons:

- **Proximity to educational and research institutions supporting advanced manufacturing.**
- **A talented but not necessarily a high-priced educated workforce.**
- **A variety of Research Centers.**
- **State and University investment.**

So, how did the Petersburg succeed in establishing itself in advanced manufacturing?

It appears that Petersburg's path has some similarities to Greenville's.

The state government working with organizations such as Virginia Economic Development Partnership, which partnered with Petersburg, provides resources and funding to the City. For example, the state has been proactive in funding university engineering education, incentives for manufacturing companies, and assistance for cities like Petersburg to attract and grow advanced manufacturers. As such, Virginia's investments have paid off. There are now 1,100 manufacturing facilities in the Richmond/Petersburg Metropolitan Area that employ nearly 40,000.

Petersburg has grown in advanced manufacturing during the decade of 2010 through both public and private action. The transformation was led primarily by the state government and universities which invested heavily in the area and with it has come $100's of millions in private investment.

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SAN LEANDRO, CALIFORNIA

Overview

What role does broadband play in creating and growing an economy, specifically a city focused on achieving a tech-based economy? According to the City of San Leandro, "access to high speed, reliable links to the Internet and internal networks is a basic, 21st Century utility, as vital to economic development as electricity or water."

Broadband is defined as a high-speed internet connection. The performance of internet is measured as bandwidth, the maximum amount of data an internet connection can transmit in a given amount of time.

The Federal Communications Commission (FCC) defines broadband as an internet connection that meets their benchmark of 25 megabits per second (Mbps) download and 3 Mbps upload. These have been the standards since 2015, but the standards change regularly as technology and connectivity improve. For example, before 2015, the benchmark was only 4 Mbps download / 1 Mbps upload. There is currently a call for another increase in the benchmarks.

The standard unit for bandwidth is megabits per second (Mbps) with high-end broadband performing at over 1,000 Mbps or 1 gigabit per second (Gbps). For reference, at 25 Mbps, it would take about 23 minutes to download a 4 GB movie, while at 1 Gbps it would take 34 seconds. In 2019, the global broadband internet services market size was $264 billion according to Research and Markets.

Broadband is delivered in five main ways: Fiber-Optic, Wireless, Digital Subscriber Line (DSL), Cable, and Satellite.

- Fiber-optic is capable of the fastest speeds of all the methods, using pulses of light through fiber strands to transmit data. The top fiber provider in the nation is Google Fiber with an average download speed of 164 Mbps.
- Wireless broadband, or Wi-Fi (wireless fidelity), connects homes and businesses to the internet using radio signals. The top wireless provider in the nation is Starry Internet with an average download speed of 106 Mbps.
- DSL, which transmits data over copper phone lines, was once ubiquitous but is no longer used by most Americans. The top DSL provider in the nation is Frontier with an average download speed of 71 Mbps.
- Cable uses the same copper cables that deliver television signals to deliver internet data. The top cable provider in the nation is RCN with an average download speed of 133 Mbps.
- Satellite uses networks of satellites to deliver internet, a popular option for rural subscribers. The top satellite provider in the nation is HughesNet with an average download speed of 10 Mbps.

Fiber optic networks have revolutionized internet speeds and connectivity. Fiber can transmit data far faster than copper cables because it uses light instead of electrical pulses to carry data from one point to another. Light can also keep signals strong as they travel, which isn't always the case with electricity through copper cables. By the time copper cables bring electrical pulses to their destination, they have often weakened significantly. This makes fiber superior to both DSL and cable methods of broadband delivery.

Wi-Fi and satellite, both being wireless methods of broadband delivery, cannot practically achieve the speed and efficiency of fiber. Wireless networks become congested, as multiple users crowd, and thus slow, the network. Fiber, on the other hand, delivers internet directly to a single user, allowing that user to fully and exclusively utilize the broadband.

Considering these advantages of fiber and the increasing recognition of the power of municipal fiber networks, this case study will focus on fiber networks and the impact of bringing them to a city.

Broadband Speeds

Ookla, a web testing and network diagnostics firm, listed the top five countries for internet speeds in their ranked 2021 Speedtest Global Index for fixed broadband (in contrast to mobile broadband).

TOP 5 NATIONS FOR INTERNET SPEEDS

1. **Singapore**
2. **Hong Kong**
3. **Monaco**
4. **South Korea**
5. **Thailand**

Average speeds ranged from 250 to 214 Mbps. The United States placed 11th in the index with a 195 Mbps average download speed.

HighSpeedInternet.com, an independent researcher on Internet Service Providers (ISPs), identified the top five metro areas in the United States for internet speeds.

TOP 5 METRO AREAS FOR INTERNET SPEED

1. **Washington DC**
2. **Philadelphia**
3. **San Francisco**
4. **New York City**
5. **Boston**

These metros had median download speeds of 74 to 66 Mbps. The Seattle Metro Area has a median download speed of 58 Mbps.

BroadbandNow, an independent researcher on broadband, ranked the following US cities as the top five for download speeds (list filtered for cities over 15,000 people), ranging from 927 Mbps to 902 Mbps.

TOP 5 US CITIES FOR INTERNET SPEEDS

1. **Harbor City, CA (Los Angeles Metro)**
2. **Hoffman Estates, IL (Chicago Metro)**
3. **Villa Rica, GA (Atlanta Metro)**
4. **Dracut, MA (Boston Metro)**
5. **Randolph, MA (Boston Metro)**

Hartford, Connecticut, while only being 17th on the list at 878 Mbps, is notable because of its population of over 122,000.

Comparing the top-five lists of cities and metros, cities between 15,000 and 50,000 residents are capable of achieving much higher average speeds than the metro areas of between 4 million and 20 million people because of their relative sizes.

There are three main dimensions to measure a company's fiber network: average speed, number of subscribers, and miles of fiber.

HighSpeedInternet.com ranked the top fiber internet providers in the nation to be Google Fiber, Verizon, RCN, Metronet, and Xfinity. These providers average from 164 to 128 Mbps average download speeds with advertised maximum speeds of either 940 Mbps or 1 Gbps.

BroadbandNow reports that AT&T is the national leader in number of fiber subscribers at 37.8 million followed by Verizon, Earthlink, and Century Link at 25 million subscribers.

For total miles of fiber, SP Global, a financial information and analytics company, reports that AT&T is also the leader with 1.3 million fiber route miles followed by Verizon, Century Link, and Earthlink at 145,000 miles.

Regulatory Environment

BroadbandNow has determined that nationally there are 18 states that have explicit restrictions on municipal broadband. These laws include requiring municipal broadband systems to be self-sustaining, inhibiting the common practice of bundling internet with voice and data, requiring broadband to price-match incumbent Internet Service Provider (ISP) services, and a large variety of other laws that make creating a competitive municipal broadband network difficult.

Specifically, in Washington, public utilities are allowed only to sell or lease broadband wholesale and barred from providing broadband service directly to customers. Since 2018, public-private partnerships have been allowed to provide broadband though these partnerships are restricted to a single telecom provider.

However, on July 25th of 2021, Washington State House Bill 1336 will come into effect that will "create and expand unrestricted authority for public entities to provide telecommunications services to end users." States overall are removing broadband restrictions, opening the door for increased internet access and speed for the nation's people and businesses. The removal of broadband restrictions will finally make it feasible for municipalities to bring broadband to rural areas and faster internet to places already covered.

Screening Process

This research has identified leading cities across the globe and the nation that have successfully invested in broadband infrastructure.

However, research screened cities based upon similarities to Federal Way. For example, the screening involved the following: population size (100,000 - 250,000), a suburb - located in a metropolitan area, no university within the city limits yet close proximity to a major research university, age of the city, successful entrepreneurial ecosystem, and metrics of success such as jobs created and investments made.

SCREENING PROCESS

1. **Population Size (100,000 – 250,000)**
2. **In or Near Major Metro Area**
3. **Adjacent to an Interstate**
4. **No University, however close proximity to one**
5. **Age of City**
6. **Maturity of Technology and Entrepreneurial Ecosystem**
7. **Jobs Created & Investment Made**

The research identifies the City of San Leandro, CA, was chosen as a case study for several reasons. San Leandro has shown continued political leadership to achieve its goal as a "Center of Innovation" by creating a city-wide fiber network and deploying public Wi-Fi and smart city technologies.

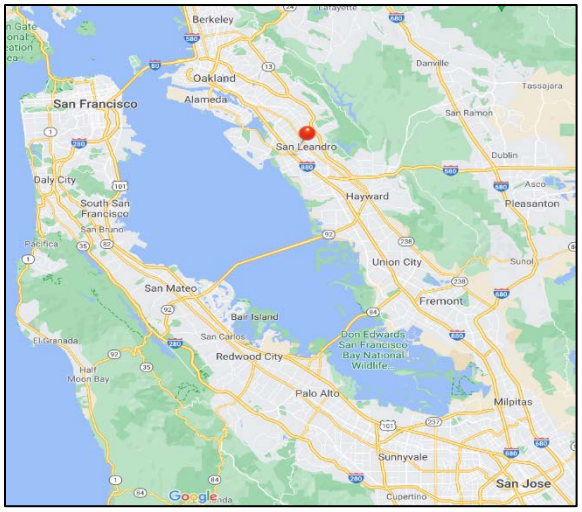
There are several similarities between San Leandro and Federal Way. First, it is a suburb of Oakland, like Federal Way is to Seattle. Second, San Leandro is served by interstates I-580 and I-880 and is adjacent to Oakland International Airport and the Port of Oakland. Third, like

Federal Way, San Leandro does not have a major research institute or university. Fourth, the cities have similar-sized populations.

Though San Leandro has about 10,000 fewer residents than Federal Way, it is a considerably smaller city by area, 13 square miles versus 22. However, San Leandro has more than a 50% higher population density than Federal Way, at about 6,700 people per square mile versus 4,300.

The cities are marginally different in education and income. In 2019, San Leandro residents had a median income of $78,000 in contrast to Federal Way's $67,000. 32% of-age residents have Bachelor's Degrees in San Leandro while 27% do in Federal Way.

San Leandro, California



San Leandro, California, is a city located on the east side of San Francisco Bay in Alameda County.

It is 10 miles south of Oakland and 35 miles north of San Jose. It is 13 square miles and has a population of 90,000. In 2019, 40,800 residents were employed in San Leandro.

There are no universities in San Leandro, though the University of California, Berkeley is located 15 miles northwest. Santa Clara University and San Jose State University are 35 miles to the southeast, and Stanford University is 25 miles south, all across the Bay. California State University, East Bay is 25 miles to the northeast.
As of 2019, San Leandro had 20.9 million square feet of industrial space, 6.2 million square feet of retail space, and 2.3 million square feet of office space. The top manufacturing employers in the City are Acco Engineering, Coca Cola, Columbia Cosmetics, Ghirardelli, Kennerly-Spratling (plastic injection molding), OSIsoft, Mi Rancho (tortillas), and PCC Structurals (structural engineering).
The City brands itself as the "Prototype City of the Future." They define their brand in today's innovation economy measuring success by the speed at which ideas and information can be transferred across the nation or the globe. They recognized that they are located in the globally renowned tech hub of the San Francisco Bay Area. To compete with neighboring cities, they deployed a fiber-optic loop that offers speeds far faster than the average internet connection in the nation.

San Leandro, "The Prototype City of the Future."

City of San Leandro

Economic History

San Leandro was a small, edge community to Oakland before WWII. Prior to defense buildup due to WWII in the San Francisco Bay area, specifically Oakland and Alameda, there were just a few major employers. They included C.L. Best Tractor Company (later Caterpillar Tractor), Friden Calculating Machine Company, and the Hudson Lumber Company.

In the postwar period, corporations such as Chevrolet, Chrysler, Dodge, and General Foods opened west coast branch plants in and around San Leandro.

In 1947, the City Council understood the need to grow its tax base through attracting manufacturing companies. They agreed to provide full municipal services to the Motor Parts division of Chrysler even though its plant was located outside the city limits and paid no taxes. Within two years, taxpayer opposition overturned the political leadership that had made this deal. So, the City Council quickly annexed the Chrysler plant site and its surrounding area into the city. The actions illustrated a business-friendly environment that created at its peak, over 20,000 jobs in San Leandro's manufacturing sector. San Leandro became a center for manufacturing with hundreds of firms of all sizes taking root.

However, by the mid-1970s, San Leandro's era of expansion slowed. Manufacturing began shifting to lower-cost areas across the country and abroad. A generation-long "de-industrialization" befell the City with the loss of many of its large manufacturing companies. Sadly, it spelled the end of manufacturing in San Leandro.

Broadband Network

San Leandro was not recognized as a center of any type of technology compared to its neighbors in the Bay area such as Sunnyvale, Santa Clara, Palo Alto, and South San Francisco. However, an opportunity presented itself to the community and political leadership of the community.

The founder and CEO of a local software company, OSIsoft, was looking to build a data center in San Leandro but was disappointed by available network speeds provided by the marketplace. So, the CEO approached the City with the idea of bringing enhanced broadband to San Leandro.

Thus, in early 2011, the process of a decade-long effort began to achieve the goal of enhanced broadband. The Mayor and OSIsoft' s CEO started the discussion of a Public-Private- Partnership (PPP) to achieve this activity.

Each entity had different perspectives. Obviously, for the private sector, it was about speed and constant and consistent service. The City of San Leandro was interested in a fiber network to enhance its economic development goals of creating an innovation-based economy.

Additionally, the City approached the opportunity as an infrastructure project to be used by next-generation manufacturers and commercial operations.

Six months into the discussion, the parties agreed to enter into an exclusive negotiating rights agreement. The City, in concert with the newly formed San Leandro Dark Fiber ("SL Dark Fiber"), the private entity created for this partnership began outlining the process for the installation of a public-private partnership for deploying fiber in the City.

SL Dark Fiber proposed to initially install and operate a fiber-optic network of 11-miles. The plan was to use City-owned conduit that was already in existence being used to control traffic signals. The agreement identified that SL Dark Fiber would be responsible for all project costs, including design, development, construction, and improvement cost.

Additionally, the City agreed not to solicit or accept any bids, offers, or proposals by any other person or entity, with respect to the acquisition of any interest in City-owned conduit or the development of the conduit.

SL Dark Fiber was required to provide the City confirmation that it had sufficient financial resources and committed to funding the project.

After nearly a year, the City and SL Dark Fiber agreed to the framework and business negotiations and subsequently entered into a license agreement.

The license granted SL Dark Fiber non-exclusive, non-revocable access to and use of City conduits and vaults ("property") and the public right-of-way for the creation of the fiber network in exchange for an allotment of fiber strands for the City. SL Dark Fiber's responsibilities included the installation and operation of a fiber-optic cable system consisting of not more than 288 strands.

The license provided the City of San Leandro with unrestricted ownership of 30 strands or a minimum of 10% of all the strands if space constraints limited SL Dark Fiber to install fewer than 288 strands. The initial term of the agreement was 20 years, with SL Dark Fiber having the right to renew the lease for up to 20 more years.

For the first 10 years of the license, SL Dark Fiber would only pay a fee of $1 per year to the City for the use of the City's conduit. This was, in part, to enhance the cash flow for the private entity of the PPP during the initial capital improvement phase.

However, according to the license, beginning on year eleven, the fee would be increased. SL Dark Fiber would be required to pay an "agreed-upon amount" based upon one of two factors. The first being based upon a per linear foot of fiber cable per year, or the second factor, a prevailing market rate, whichever is lower, for all property.

The license specifies that SL Dark Fiber may enter into agreements with affiliates or third parties to provide telecommunication services using the fiber strands SL Dark Fiber owns. At the time of the signing of the original license, two vendors (Internet Service Providers) were identified as potential telecommunication service contractors with SL Dark Fiber. They included Phase 3 Communications and Lit San Leandro.

The license does not provide SL Dark Fiber exclusive use of the conduit, but gives the City the right to permit other providers of telecommunication or other entities to install equipment or devices in the property. The City did agree, however, that it would not make or allow changes to the conduit that would affect SL Dark Fiber's ability to use the property, including reducing the capacity of conduit specified for use or services provided by SL Dark Fiber.

Finally, SL Dark Fiber was required to provide a confidential annual report to the City, detailing information regarding the number of subscribers, capacity used, capacity available, and revenue received from the fiber network.
For the City of San Leandro, the license provides that after five years into the term of the license, the City could permit others access to unused conduit space as long as such access was not considered to constitute a reduction in capacity of conduit specified for use by SL Dark Fiber.

Additionally, as part of the agreement, the City agreed to assist SL Dark Fiber in obtaining any necessary city permits or entitlements from the City such as encroachment permits or environmental reviews.
Now, while all agreements to create the fiber network were between the City of San Leandro and SL Dark Fiber, another company was involved - Lit San Leandro ("LSL"), also founded by the CEO of OSIsoft.

Together, SL Dark Fiber and LSL comprised the partnership that worked with and continues to work with the City to create the fiber loop.

LSL is a local exchange carrier utility company that maintains the fiber network, creates last-mile connections, and offers internet services from over 100 Internet Service Providers (ISPs). The fiber network is open access, meaning

LSL partners with many ISPs, so customers can lease fiber from LSL and choose whichever ISP they would like services from.

The common services that LSL provides are business internet solutions, diverse fiber paths, and fiber leases between buildings. In conjunction with their contractors, LSL determines how it will get fiber from the network to a given building, sometimes involving construction. Once the given building is physically connected to the network (the "last mile" is completed), the ISP of choice will install their equipment and turn on their service.

After finding success in the creation of the initial network, the City of San Leandro applied and was awarded $2.1 million in a matching grant totaling $4.2 million from the US Department of Commerce, Economic Development Administration (EDA) in September 2012.

This investment facilitated the underground construction of a 7.5-mile extension of 2-inch conduit to the initial 11-mile fiber loop, bringing the length of the fiber network to 18.5 miles.

While unconfirmed, it appears that this was the first-ever grant awarded by EDA for a community broadband project.

Meanwhile, the City also launched the Broadband Connection Program in 2013. The Program was designed to reduce the cost of connecting a commercial or industrial building to the broadband infrastructure within the City of San Leandro. The Broadband Connection Program serves to stimulate innovation and high-tech business activity in order to make commercial and industrial districts in the City more competitive.

For approved projects, the City will contribute up to half of the business or property owner's out-of-pocket expenditures, to a maximum of $10,000, as a loan secured through a loan agreement and a promissory note. The funds can only be used to connect a property or business to the broadband network.

The loans are structured as interest-bearing, forgivable, one-year loans. The project is evaluated annually by City staff to determine if it continues to meet the terms of the agreement. If no default occurs within the one-year term of the agreement, the loan is forgiven. If a default occurs as defined in the Agreement, the loan becomes due and payable immediately. City staff has the discretion to work with the participant to cure the default and structure a repayment schedule. Upon notice of a change in ownership during the term of the agreement, the loan may be assigned to the new participant at the sole discretion of City staff.

If an applicant is a tenant, they must include a copy of the lease agreement and an affidavit from the property owner allowing them to make the proposed building and site modifications. The tenant will be required to verify that they have at least a five-year lease on the property. If the tenant has less than a five-year lease, the property owner is required to co-sign the loan agreement. Loans are designed to reimburse the participant for completed improvements. The City will reimburse the applicant for the agreed-upon list of improvements upon written request specifying the amount and providing copies of bills and invoices for the completed work.

Eligible applicants must own the commercial property to be improved and be tenants of the commercial property or hold a master lease for the commercial property. The property must be located within the City of San Leandro.

Applications to the Broadband Connection Program were evaluated on the applicant's ability to stimulate high-tech business activity, innovation, and job creation.

The City of San Leandro Community Development Department reported that five years after its inception, the program has awarded over $82,000 to local businesses.

Meanwhile, in 2014, the City began leveraging its portion of the fiber network for the benefit of its citizens. During the year, the City of San Leandro and the San Leandro Unified School District entered into a Fiber License Agreement. It allowed 10 City-owned fiber strands to be accessed and used by the San Leandro Unified School District. This effort enabled access and use of the fiber for $1 per year for the 20-year term of the Agreement.

This free access to the City's fiber did not mean the School District made no investments, however. This new connection to fiber was met with $5 million in investment from a federal technology bond program. The District used the money to upgrade switches and Wi-Fi, train teachers, and add computers, going from a student-to-computer ratio of 40-to-1 to 2-to-1.

By 2015, all 17 San Leandro Unified School Districts sites were connected to the internet with a dedicated, synchronous 10 Gbps connection through City fibers.

Also, in 2015, the License between the City and SL Dark Fiber LLC had its first amendment. The main purpose of the amendment was to create terms for the 7.5-mile extension of the fiber network. The amendment provided the City with unrestricted, exclusive access to 25% of the new fiber created, significantly more than the 10% the City had previous rights to in the original 11-mile fiber loop. This meant the City had exclusive rights to 30 strands in the 11-mile loop and 72 strands in the extensions.

However, before any construction of the network extension could begin, SL Dark Fiber was required to, at its own cost, procure and provide the City complete electronic construction drawings and maps of any additions made to the original fiber network by SL Dark Fiber. Any additions required written consent from the City per the original license.

Also, the amendment added a requirement for SL Dark Fiber to provide the City electronic GIS and mapping data for the fiber network project and related facilities, including all lateral conduit, third party conduit and ducts, access and termination enclosures, and complete fiber, cable, logical and physical network data, including fiber count and identification of the dedicated strands for the City.

The amendment also made the relationship between the City and SL Dark Fiber a more transparent one. In addition to the previously required elements of the annual reports by SL Dark Fiber, the future annual reports were subsequently required to specify current and historical performance data of the fiber network.

Additionally, at the time of the amendment, a total of four vendors were identified as potential telecommunication service contractors with SL Dark Fiber, only one of which (LSL) had been listed in the original license agreement: Crosslink Networks, PAXIO, Unwired LTD, and Lit San Leandro.

Outcomes

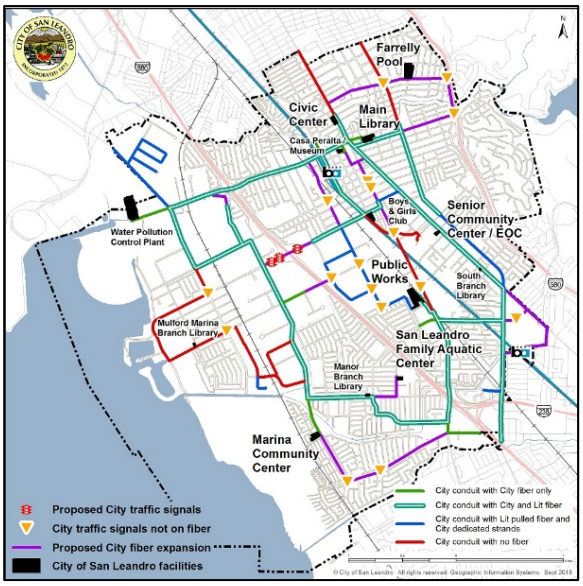
Ten years after its inception, the fiber network is connected to over 70 buildings and 250 businesses in the City through Lit San Leandro, and the City now has two tech campuses connected to fiber: Gate 510 and the San Leandro Tech Campus. With these two sites connected to fiber internet, businesses located in them have a competitive advantage over firms almost anywhere else in the nation.

Through the fibers exclusively controlled by the City, the fiber network has been powering public Wi-Fi since 2015.

Public Wi-Fi is now present downtown and in all of the City's libraries, community centers, city facilities, and in many of its public parks. The Wi-Fi infrastructure and deployment initially cost the city $70,000.

When planning the Wi-Fi network, the City recognized that it did not own any facilities in the downtown core, and there were no commercial buildings with direct access to the Fiber loop. However, City staff determined that the ideal location for a system was on a historic 50-foot sign in the Pelton Plaza shopping center in the center of downtown. Additional mesh access points could then be deployed on the City's streetlights and buildings to expand the Wi-Fi network.

To achieve this, the City formed a public-private partnership with the commercial property owners of the Pelton Plaza to



gain exclusive access to the sign in exchange for building and managing a Wi-Fi system on the property. The historic sign was connected directly to the City's fiber loop, and mesh access points were deployed around the area. Each point creates a four-block radius of Wi-Fi.

Along with the public Wi-Fi, the City of San Leandro implemented smart-city technologies including a city data analytics dashboard, a smart streetlight network, smart irrigation clocks, and smart traffic signals.

A Wi-Fi Access Point on One of the City's Streetlights
Source: PR Newswire

Tech and Entrepreneurial Ecosystems

The San Francisco Bay Area is home to world-class universities and research institutions. These universities and institutions attract people and companies from across the globe. The concentration of people has provided the environment for innovation and opportunities.

For example, the University of California, Berkeley and Stanford University are two of the top universities in the world and are both in the Bay Area. Two of the United States Department of Energy's 17 national laboratories are

in the Bay Area: the Lawrence Berkeley National Laboratory and the Lawrence Livermore National Laboratory.

California is a state with no restrictions or roadblocks on municipal broadband. California was already leading the nation in number of broadband lines before executive orders and a multitude of legislation strengthened the state's broadband position even more. The first executive order came in 2006, creating a Broadband Task Force to "remove barriers to broadband access, identify opportunities for increased broadband adoption, and enable the creation and deployment of new advanced communication technologies." Most recently, a 2020 executive order directed five California agencies to "regularly convene private-sector companies in an effort to understand and predict current and future demand for broadband" with the goal of increasing access to broadband across the state. At the end of 2018, nearly 95% of California households had access to 100 Mbps or greater advertised download speeds.

Thus, San Leandro is in a state and region of innovation. Coupled with the region's strengths, San Leandro has two major business technology campuses connected to fiber. These two facilities illustrate the growth that has occurred because of the development of the fiber network: Gate510 and the San Leandro Tech Campus.

Gate510

One of the former manufacturing facilities, the former Chrysler Dodge plant, a 720,000 square foot, two-building complex found a new use. In 2014, Lit San Leandro partnered with the building's owners to connect the property to fiber. The connection created the opportunity to redevelop and rebrand into an innovative tech campus known as Gate510.

It provides flex space for scientists, entrepreneurs, makers, creators, and visionaries to collaborate, create, invent, and work. There are labs, offices, industrial, retail, and production spaces available from 1,000 to 10,000 square feet. There are flexible suite options, high ceilings, and fiber internet. The Town Center has flex space for meetups and coworking as well as events, performances, and socializing.

Upon its creation, Gate510 had tenants primarily involved in advanced manufacturing, robotics, and drones. In 2014, Type A Machines, a 3D printer manufacturer, set up the Bay Area Manufacturing Hub a 3D-printing cluster comprised of 12 companies in Gate510. The Hub was formed to create technical and collaborative efficiencies between these companies of the same industry to decrease costs and increase innovation in the field of 3D printing. Over the years, the tenant makeup of Gate510 has shifted to biotech.

Gate510 tenants are primarily founded and run by people from surrounding universities like UC Berkeley and Stanford. These companies come to Gate510 for its space and relatively inexpensive costs in the context of the Bay Area.

Gate510*: From Car Plant to Tech Campus*

While Gate510 does not directly offer business resources, the concentration of companies does create unique opportunities. For example, ten biotech companies in Gate510 were able to bundle themselves to reduce their individual hazardous material off-haul costs.

San Leandro Tech Campus (SLTC)

Constructed from 2014 to 2016, the San Leandro Tech Campus (SLTC) is a 1 million square foot, technology-focused, mixed-use development adjacent to the Downtown San Leandro BART Station. The primary tenant of the first phase of development was OSIsoft in the creation of a 500,000 square foot global headquarters, and the major tenant of the second phase was Ghirardelli Chocolate Company.

The campus is continuing to expand, and an apartment complex is on the way: a seven-story, mixed-use, residential building with ground-floor retail. The campus is connected to Lit San Leandro, offering any business on the campus extremely fast internet speeds.

In addition to these campuses, the City launched the Founders Circle in 2014, a group of San Leandro business leaders, partners, and potential partners that came together to create connections, improve supply chains, and improve communication among San Leandro businesses and the City. The City of San Leandro hoped this group

would attract the workforce critical to growing businesses related to sectors such as software and hardware, smart city, big data, energy technologies, as well as advanced manufacturing.
Today, the Founders Circle is a growth-stage investment firm. In the past eight years, Founders Circle has raised three funds totaling nearly $1B under management. To date, they've invested in 73 companies, 12 of which are now publicly listed.

In addition to the creation of the City's fiber network, the public-private partnership also created and funded the position of Chief Innovation Officer. This goal of this position has been to drive tech-based economic development in the City and manage the City's role in the fiber network.

Summary

The City of San Leandro was not actively searching for an opportunity regarding broadband. However, a local businessman elevated the conversation nearly a decade ago to modernize the City's communications infrastructure.

Together, the public-private partnership built an 18.5-mile fiber-optic loop in an effort to attract a new generation of companies to San Leandro. The fiber loop was also connected to the outside world via BART's fiber network, allowing businesses in San Leandro to tap into direct, long-haul fiber.

While San Leandro claims to have the fastest internet speeds in the nation at 10 Gbps, its efforts are still a work in progress. Research illustrates that marketplace competition was possibly taken off guard or embarrassed by the focus and deployment of the network by the public-private partnership. However, they too provide similar speeds, although research has not been able to ascertain the differences or similarities in pricing.

Additionally, while research identifies a simple deployment program, it was the capacity of existing city assets (underground conduit) that provided the opportunity. A license was created that allows SL Dark Fiber to own the fiber optic cable that runs through the City's underground conduit. Meanwhile, LSL, a sister organization to SL Dark Fiber owns and operates the switch and routing facilities that bring Internet service to the community.

Due to confidentiality, research was unable to determine the pricing, let alone the methodology, used to ascertain a revenue model. However, the obligation by a City Council ten years ago to an arrangement for 20 years is significant. The continued commitment beyond 20 years with renewals every five years for a total of another 20 years illustrates a devotion, if not promise, to building the City's economy. It is simply an impressive community commitment.

Today, the City of San Leandro is evaluating how to maximize the value of its broadband and fiber optic assets. Through this process, City staff has formulated a Smart City Strategy utilizing the internet connectivity services available to it to deliver technology services. These include the following:

- Digital Transformation: the process of creating a digital city government experience, often called "e-Government," which includes digital services and open data
- Smart Buildings & Facilities connecting facilities and using integrated management systems for climate control, HVAC, and energy
- Smart Street Lights, Small Cells, and preparing for the deployment of 5G
- Intelligent Traffic Signal Systems that adapt to real-time traffic conditions, making San Leandro streets more efficient
- Using technology as a force multiplier for data and intelligence to improve public safety
- Digital Inclusion which entails connecting the unconnected and helping community members engage in the digital economy
- Public Wi-Fi and infrastructure to drive economic development and bridge the digital divide

In addition, the City, through a consultant, has combined a market assessment, a needs assessment, and a future broadband network design into a series of recommendations to continue to build and serve the City's broadband network. San Leandro is now in the process of implementing a series of recommendations to continue to build its broadband network. These include the following:

1. Develop Fiber-Friendly Public Policies
2. Enhance Data Center and Network Interconnects
3. Work with Community Anchor Institutions on meeting their broadband needs
4. Implement Smart City Strategy applications
5. Formalize an internal Broadband Initiative Program (BIP)
6. Release RFPs for Construction and Operations
7. Incorporate broadband governance strategies into the daily operations of all City departments and work with local and regional public agencies and private investors

The City hopes to implement fiber management systems, standardized agreements, pricing policies, published rates and terms, as well as create an enterprise fund to maintain proper budgets, cost accounting, and revenues.

Research identified that the commitment by the City Council in 2011 to obligate future councils to the future development of broadband has proven to be a valued decision. While still not recognized as a center of technology like their neighbors, the ability to claim the fastest internet speeds in the nation is not lost on the economic development opportunities as an attraction for the next generation of technology-based companies.

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A BRIEF BROADBAND HISTORY OF SAN LEANDRO, CALIFORNIA

"THE PROTOTYPE CITY OF THE FUTURE"
CITY OF SAN LEANDRO

2011
Work on Public-Private Partnership begins

Exclusive Negotiating Agreement signed between the City and San Leandro Dark Fiber

2012
Lit San Leandro goes live, providing up to 10 Gbps speeds and connecting its first building to the fiber-optic network

Bidding opens for fiber loop improvements

City enters agreement with HP Communications for fiber loop improvements

San Leandro Dark Fiber LLC and Lit San Leandro LLC are created

Licensing Agreement signed between the City and San Leandro Dark Fiber

Construction of 11-mile fiber network begins

City is awarded $2.1 million as part of a matching $4.2 million grant from the EDA

2013
11-mile fiber network is finished

Forgivable Loan Program begins for business fiber

2014
City enters fiber agreement with the San Leandro Unified School District

2015
License Agreement First Amendment signed

Creation of the 7.5-mile fiber network extension begins

City begins deploying public Wi-Fi

All 17 schools in the School District are connected to fiber

2016
City begins smart city upgrades

2017
7.5-mile fiber network extension is finished

2018
City finishes smart city upgrades

LANCASTER, CALIFORNIA

Overview

There is an old adage - "What do you do with lemons? Make lemonade!" In Lancaster, California - what do you do with over 300 days of sunshine?

The City of Lancaster was stunned by not one but two noteworthy impacts. First, it was confronted by the Great Recession and a 20 percent unemployment rate caused by it. Simultaneously, it was confronted by a new and significant landmark state environmental regulation - Assembly Bill 32.

Political leadership used the city's locational advantage of plentiful sunshine as the solution for both by embracing clean technology. But what is Clean Technology?

Research identifies clean technology as any technology that reduces or eliminates pollution, waste, or use of natural resources while seeking to be competitive in cost and performance.

Now, there is some conversational confusion. Though "cleantech" is now the more widely used term, "Greentech" was once popular to describe the same industry. The two have become interchangeable, and you often hear of "green and clean technologies" in the same conversations.

"Climate tech" is yet another term often used. Although it is a relatively new phrase that has come into popularity only over the past few years, Climate tech encompasses any technology seeking to address the climate crisis through decarbonization. This means anything that is climate tech is also under the broader umbrella of cleantech. There is also the term "the green economy." It is the broadest term of them all, including cleantech and non-tech subsectors that contribute to environmental business efforts such as environmental consulting and carbon management accounting.

While certain terms may come in and out of popularity, the underlying sectors of technology are the most important to define. Cleantech spans many subsectors, including energy generation, storage, infrastructure, and efficiency; transportation, water and wastewater, air and environment; materials, manufacturing/industrial, agriculture, and recycling and waste.

CLEANTECH SUBSECTORS

- **Energy Generation**
- **Energy Storage**
- **Energy Infrastructure**
- **Energy Efficiency**
- **Transportation**
- **Water & Wastewater**
- **Air & Environment**
- **Materials**
- **Manufacturing/Industrial**
- **Agriculture**
- **Recycling & Waste**

Nationally, cleantech, specifically in the energy sector, employs more than 3.3 million people in the United States and more than 85,000 people in Washington State. The median wage for the sector in the nation is $23.89 per hour and slightly higher in Washington at $25.39 per hour.

In 2020, it was estimated by the College of London professors, Georgeson and Maslin that the green economy produced a revenue of $7.9 trillion globally with $1.3 trillion of that in the United States.

Some of the largest cleantech companies in the nation include GE (energy), Iberdrola (energy), Vestas (wind), NextEra Energy (energy), Renewable Energy Group (biodiesel), First Solar (solar), Turner Construction (green construction), Tesla (electric vehicles), GM (electric vehicles), Waste Management (waste), and Republic Services (waste).

In the United States, the San Francisco Bay Area reigns as the top cleantech hub followed by Los Angeles and Boston. Worldwide, outside of the Bay Area, Shanghai and Beijing, China, are the next two largest hubs for cleantech.

However, research screened cities based upon similarities to Federal Way. The screening involved the following: Population size (100,000 - 250,000), suburb of a metropolitan area, no university within the city limits yet close to a major research university, adjacent to an interstate, age of the city, successful entrepreneurial ecosystem, and metrics of success such as jobs created and investments made.

SCREENING PROCESS

- **Population Size (100,000-250,000)**
- **Suburb to metropolitan Area**
- **Adjacent to Interstate**
- **No University, however, close proximity to one**
- **Age of City compared to Federal Way**
- **Maturity of Technology & Entrepreneurial Ecosystem**
- **Jobs Created & Investments made**

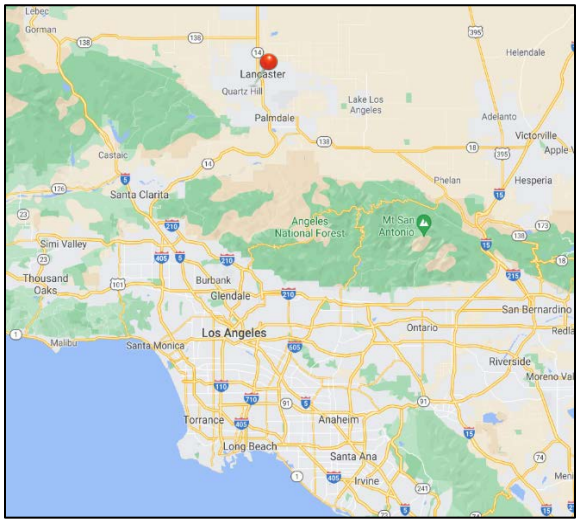
From the research, the City of Lancaster, California, was selected as a case study for the cleantech sector because of its success in cleantech and similarities to Federal Way. For example, Lancaster is without a major university. It has a similar-sized population, and has shown leadership and success in attracting and helping clean technology companies grow.

Taking these screening filters into account, the most notable similarity between Lancaster and Federal Way is their proximity to a major metropolitan area (Seattle and Los Angeles). Being outside the second most populous city

in the nation has allowed Lancaster to leverage its relatively low costs and vast land while still benefiting from LA's resources.

An important difference between Lancaster and Federal Way is access to natural resources. Lancaster has used its land and days of sun to develop, while Federal Way is built out and is about a quarter the area. Lancaster may have the University of Antelope Valley, but the University is very small and is not analogous to the University of Washington.

Lancaster, California



Lancaster is a city of 160,000 people located in southern California, specifically in northern Los Angeles County in the Antelope Valley of the Western Mojave Desert. Lancaster is about 70 miles north of Los Angeles and is neighbored to the south by its twin city Palmdale, a city of 155,000. Lancaster is 94 sq. miles in size.

Lancaster is home to Antelope Valley College, the University of Antelope Valley, a satellite campus of California State University, Bakersfield, and a satellite engineering program of California State University, Long Beach.

Lancaster is served by the Antelope Valley Transit Authority (AVTA) local and commuter bus services as well as the Metrolink commuter rail. The Palmdale Regional Airport is directly south of Lancaster. The Bob Hope/Hollywood Burbank Airport is 60 miles south of Lancaster, and the Los Angeles International Airport is 75 miles south.

Economic History

Antelope Valley has a rich history of aerospace. Edwards Air Force base is about 30 miles northeast of Lancaster. Although the primary space shuttle landing site was at Kennedy Space Center in Florida with 78 landings, Edwards Air Force Base was used as a backup and saw 54 landings. NASA's Neil A. Armstrong Flight Research Center (formerly Drydan) is at the base and is known for its X-15 experimental rocket ship program which set speed and altitude records in the 1960s. The research center continues to be NASA's primary center for high-risk, atmospheric flight research and test projects.

In adjacent Palmdale, home to the classified aircraft manufacturing plant US Air Force Plant 42, major aerospace companies such as Lockheed Martin, Boeing and Northrop Grumman are engaged in the development of sophisticated aircraft. These include the Space Shuttle orbiters, B-1 Lancer Bomber, B-2 Spirit Bomber, F-117 Nighthawk Fighter, and F-35 Joint Strike Fighter along with the fully autonomous aircraft the Global Hawk.

Additionally, 30 miles north of Lancaster is Mojave Air & Space Port, where everything from fighter jets to airliners to spacecraft are tested, and airliners are stored.

Clean Tech – A New Opportunity

Lancaster had not even considered cleantech development until the call to action that was the Great Recession and California's Assembly Bill 32.

As an indicator of how impactful the Great Recession was to Lancaster, unemployment reached 17.5% in July of 2009, significantly up from 7.4% in 2007. Another indicator was housing starts. In 2007, there were 2,800 new single-family homes built. However, little if any growth was seen again until 2012 when permits were issued for 200 new homes.

The Mayor then, who still serves in that role today, stated, "The recession really acted as a crucible that forced city governments everywhere to ask what works as a business policy and what doesn't."

In his first year in office, the Mayor and the City Council dealt with impacts from the financial crisis of the Great Recession specifically as it pertained to housing.

However, the need for an economic strategy and its implementation could not wait. Additionally, and simultaneously, cities in California like Lancaster were also forced to address a significant landmark piece of environmental legislation- AB 32. California's Global Warming Solutions Act of 2006 (AB 32), set an absolute statewide limit on greenhouse gas emissions, and confirmed California's commitment to transition to a sustainable, clean energy economy.

It appeared that for the City to address these, the solution needed to be multifold: Formulate a strategy to ensure an economic recovery that would create a sustainable economy while addressing the state environmental requirements.

The City needed to work with a wide range of innovative companies and regional partners to find alternative sources for economic support to generate affordable renewable

energy as well as new opportunities to create desirable skilled jobs in the community.

Lancaster's challenge focused on implementing an operation to attract reputable private firms to design and develop new affordable solar solutions which would generate not only plentiful clean energy, but new revenues and employment opportunities from which all residents would benefit.

Recognizing the 300 days of sunshine, the Mayor identified a strategy that was focused on becoming the solar capital of the world.

The Mayor also understood that to convince the community to accept this vision, it needed to make a commitment to solar as the basis of its economic rebirth.

The effort began less than a year later. The City initiated an effort aimed to harness Lancaster's sunshine to generate solar energy. The goal was to produce as much power as the community utilizes making Lancaster one of the world's first net-zero cities.

The strategy began with the City of Lancaster working with the solar energy firm eSolar to bring its 20-acre, 5-megawatt Sierra SunTower solar project from conception to completion. In the summer of 2008, planning began on the concentrated solar power (CSP) plant. CSP is a system that uses mirrors to aim incoming sunlight at a central receiver that uses this light to generate steam that drives a turbine. The City had "no permitting process or codes" for such a project at the start. But, through political will and availability of land located close to high-power transmission lines, the City was successful in landing the project. In August 2009, just 14 months later, eSolar held the grand opening of the nation's only operating solar thermal power tower plant.

Next, to catalyze the net-zero campaign, the Mayor led the 2010 launch of the "Solar Lancaster" program, one of the most expansive solar community programs undertaken at the time. The program was one of the nation's first combined public-private partnerships to offer more affordable ways to adopt solar power.

The private partner was the solar firm SolarCity. Through the partnership, the company enabled local residents and businesses to install solar energy systems at significant cost savings over their existing energy bills.

SolarCity, the company started by Elon Musk's cousins (subsequently purchased by Tesla), partnered with the City to install rooftop solar systems on virtually all municipal buildings and schools. Lancaster also changed its building code to require that new homes include rooftop solar, the first city in America to take that bold step.

SolarCity helped create a local solar training program in partnership with the University of Antelope Valley. The City also worked with SolarCity to install 2.5 megawatts of new solar capacity across six different City sites including City Hall, Clear Channel Stadium, and the Lancaster Performing Arts Center.

This was one of the most expansive solar community programs ever undertaken and included clients such as Sierra Toyota Car Dealership, Lancaster School District (19 sites), Desert Christian School (3 sites), Eastside School District (2 sites), Lancaster Business Park, and Lancaster Baptist Church.

In addition, the City worked with SolarCity to install 2.5 megawatts of new solar capacity across five City facilities, most notable, the city's Municipal Stadium, City Hall, and the Lancaster Performing Arts Center. The stadium, home of the Lancaster JetHawks, was the first Minor League Baseball Stadium in the nation to go green. Its solar system offsets close to 100 percent of its energy use.

In 2011, the City of Lancaster partnered with SolarCity and US Topco Energy, a renewable energy developer, to develop and construct several solar energy facilities. This included a 4-megawatt solar farm and another 2.3-megawatt farm.

Meanwhile, the Mayor also led an initiative to streamline and fast-track renewable energy developments of all kinds, from small single-home installations to major developments which provided hundreds of megawatts of renewable energy. The permitting process to install solar energy was reconfigured with the approval process performed over the counter. Furthermore, zoning adjustments were made to allow various types of renewable energy in additional areas throughout the City.

The Sierra SunTower in 2010
Source: Bryant Solar Thermal

By March 2013, the Lancaster City Council unanimously approved the adoption of changes to their zoning code that adopted a new ordinance requiring new single-family residential units built within Lancaster on or after January 1, 2014, to provide solar-generated power at a minimum average of 1 kilowatt per unit.

With this adoption, the Mayor's previous adage about being the solar capital of the world was replaced with "The Solar Capital of the Universe."

"The Solar Capital of the Universe"

- Rex Parris, Mayor of Lancaster

Additionally, as previously mentioned, the project to install solar energy systems in 2014 on six municipal facilities throughout the City was forecasted to generate 1.5 megawatts of renewable energy. In addition, the City's Power Authority, which was formed in 2011, partnered with Eastside and Lancaster School Districts to install 7.5 megawatts of solar energy-producing parking shade structures at 25 school sites throughout Lancaster. These projects not only provided the school districts with substantial energy savings and shaded parking but will also generate an estimated taxpayer savings of $6 million and City revenue of $20 million over 15 years.

By 2015, Lancaster was named the solar energy capital of California, with 132 watts of solar energy production capacity per capita. This is more than three times the per capita capacity for the runner-up, San Jose with 40 watts per capita, and over five times that of third-place San Diego at 24 watts. In addition, Lancaster boasts three of the state's top five solar ZIP codes.

At the end of 2016, the Western Antelope Dry Ranch utility-scale solar power facility became operational. This was a partnership between Lancaster and sPower. The 10-megawatt facility provides enough renewable energy to power more than 1,800 homes in Lancaster, ensuring low energy costs for residents and businesses while it continues to advance the City's net-zero goals. Continuing the development, sPower has since added to the facility, which now has 130-megawatts of solar, the largest community choice aggregation-feeding solar power plant in California.

Meanwhile, university researchers and private industry contacted the City. In 2011, the University of Ottawa in concert with a Canadian solar company approached the City for a testing facility. The City used existing space at a City recreation and park site for the University of Ottawa's SUNLAB project, testing concentrator photovoltaic systems (CPV) for commercial viability.

The 85 MW sPower Solverde 1 Solar Farm in Lancaster
Source: Global Power Journal

Some of the largest solar farms in the area are sPower's 85-megawatt Solverde 1 Solar Farm pictured above, Exelon Generation's 242-megawatt Antelope Valley Solar Ranch, and SunPower Services' 579-megawatt Solar Star farm.

Public Policy

The City had made significant progress with transition to solar power, addressing the needed economic transformation, but it still needed to address the issue of the state's environmental legislation. To achieve that, the City embarked on several initiatives.

Clean Energy Leadership

Confronted with being able to address the issues of entering into energy agreements with other public agencies and private firms to advance the solar power generation, distribution, sales and marketing; the City needed to create a legal organization authorized to achieve these elements. The challenge was met through the formation of the new joint powers agency, called the Lancaster Power Authority (LPA) and its affiliates. Combined with the LPA, the City and its Redevelopment Agency set the stage to transform not only how it generated electricity, but also the revenue generation to support the City's many programs.

Lancaster Choice Energy

In addition, Lancaster created an initial feasibility report in 2013 to become a Community Choice Aggregator (CCA) for energy in the area. A CCA is an alternative to investor-owned systems where local organizations aggregate individual electricity customers into a jurisdiction that can create large contracts with electricity producers. There are only eight states in the nation that allow it: Massachusetts, Ohio, Illinois, New Jersey, New York, Rhode Island, Virginia, and California. CCA has been legal in California since 2002.

By May 2014, an ordinance to establish the CCA was adopted, and in February 2015, an implementation plan was finalized. The CCA, otherwise known as Lancaster Choice Energy (LCE), began serving municipal accounts in May 2015 and began serving residential and commercial accounts in October 2015.

LCE offers ratepayers in the City of Lancaster three options: Clear Choice, Smart Choice, and Personal Choice.

- Clear Choice rate plan provides electricity purchased and generated from 35% renewable energy sources.
- Smart Choice rate plan provides electricity from 100% renewable energy sources.
- Personal Choice rate plan is geared towards those who generate their own solar or wind power as they may sell excess energy they generate to LCE for an offset in their power cost.

With Lancaster Choice Energy's default program, Clear Choice, customers save on average 3% to 15% over Southern California Edison's (SCE) current rates. LCE is in business to find the best sources of clean energy at the lowest possible price and procure it for customers.

LCE rates are set annually by the Lancaster City Council, so consumers have a steady, predictable outlook on rates and the satisfaction of working with a local provider, governed by local leaders whose interests are aligned with those

consumers. Since 2015, LCE has saved the Lancaster community more than $2.8 million.

High Desert Power Authority

In 2012, the High Desert Power Authority was formed through the partnership of the City of Lancaster and the City of Pittsburg, California, as a Joint Powers Authority.

The project was created to open extensive areas for future alternative energy development in Los Angeles, Kern, and San Bernardino counties, while being capable of delivering more than two gigawatts of renewable generation to Antelope Valley and eastern Kern County.

CalChoice

Emphasizing the "community" in community choice aggregation even more, the Cities of Lancaster and San Jacinto partnered to create CalChoice as a Joint Power Authority (JPA) in 2017 to help cities in Southern California Edison territory participate in community choice aggregation (CCA) without having to sacrifice control or any of the benefits associated with a traditional JPA.

CalChoice is governed by the Lancaster City Council with each city joining as an associate member of the JPA. Through aggregation, CalChoice cuts costs for each city through the benefits of economy of scale: as more cities are added, each city's share of the fixed costs goes down. This allows smaller cities to start a CCA that might have been too small to successfully operate one on their own. Each city also has access to experienced staff who have been through the implementation and operational process of a CCA.

Advanced Energy Community

In 2019, the California Energy Commission awarded Lancaster a $5 million grant to lay the groundwork for its Advanced Energy Community. The Advanced Energy Community encompasses several different components, including a virtual power plant integrating five megawatts of solar with ten megawatt-hours of storage, community microgrids, affordable microgrid housing developments, microgrid community resiliency centers, commercial energy storage, and innovative energy data management and rate design.

Diversifying the Strategy

While solar generation has been Lancaster's claim to fame, the City has developed in other areas of clean technology, including transportation and, in the future, hydrogen.

Business Recruitment

The Chinese battery and automobile manufacturer BYD (Build Your Dreams) had just opened its new North American headquarters in Los Angeles as it began looking for a place to build electric buses.

The Mayor of Lancaster knowing this, made four trips in quick fashion to BYD's China headquarters presenting tax breaks and permits to secure the bus plant in 2010.

Fortuitous for BYD, it was able to take advantage of an existing facility that met its needs when a local RV plant moved out of Lancaster. This meant a ready-made, fully permitted facility with personnel experienced in transportation manufacturing.

In 2014, BYD unveiled the world's largest electric vehicle, a 60-foot articulated bus that carries 120 passengers over 170 miles per charge. The bus was designed and manufactured in Lancaster. The Antelope Valley Transit Authority (AVTA), which serves the Lancaster-Palmdale area, also implemented the first all-electric bus fleet in the nation thanks to BYD's proximity. By 2019, BYD had produced its 400th electric bus at its Lancaster manufacturing plant, and in May 2021, AVTA's electric busses crossed the aggregate five-million-mile mark.

BYD's plant has since expanded to nearly 500,000 square feet, making it the largest electric bus manufacturing facility in the nation. Construction of an on-site, 2-megawatt solar power plant with a microgrid also began in 2020. This plant is not just about manufacturing; it is also about innovation and use of clean technology inventions.

In collaboration with the Chinese-based global energy company SGH2, Lancaster will soon be home to the world's largest green hydrogen production facility. Lancaster will be co-owner of the facility. They have plans to break ground in 2021 with full operations anticipated to begin in 2023. The facility will use recycled mixed paper waste to produce hydrogen that emits a half to a third the carbon of hydrogen produced using electrolysis or renewable energy and at a fifth to a seventh the cost. Lancaster will supply a guaranteed feedstock of recyclables, saving between $50 and $75 per ton in landfilling and landfill space costs.

Many leading global companies and top institutions have joined SGH2 and the City of Lancaster to develop and implement the project, including Berkeley Livermore Lab, University of California, Berkeley, Fluor, Thermosolv, Integrity Engineers, Millenium, HyetHydrogen, and Hexagon.

Tech & Entrepreneurship Ecosystem

Lancaster Small Business Incubator

The Lancaster Small Business Incubator hosts a variety of companies from tech to non-tech start-ups. The incubator is a 30,000 square foot, multi-tenant building with ten separate spaces varying from 1,600 to 5,000 square feet. The building is suitable for office work, R&D, light manufacturing, and distribution. It boasts high ceilings, doors, and bays. Lease options of the space are flexible, with most companies staying for multiple years. There is SBA financing available along with access to the resources of their Small Business Development Center.

Small Business Development Center (SBDC)

The SBDC is a technical resource partner of the US Small Business Administration and works with Antelope Valley, Santa Clarita Valley, and San Fernando Valley businesses.

The SBDC has 20 professional business advisors who provide one-on-one advising to businesses at any stage of maturity. SBDC advisors are also flexible in their business hours, accommodating the schedules of business owners. There is no cost to work with SBDC, as they are sponsored by SBA, College of the Canyon (their host), various cities, and various other sponsors.

Lancaster; Dealing with the State

The State of California has been engaged in cleantech for decades, formulating some of the most progressive economic incentives for cleantech; and Lancaster has worked diligently to fulfill the policies of the state, specifically its legislative mandates. Listed below are some of the cleantech incentives the state offers that were used by businesses and investors in Lancaster's achievements.

California Cleantech Business Incentives

- **Alternative Energy and Advanced Transportation Sales and Tax Exclusion**
- **California Climate Investments Initiative**
- **California Sustainable Energy Entrepreneur Development Initiative**
- **Clean Transportation / Alternative and Renewable Fuel and Vehicle Technology Program**
- **Electric Program Investment Charge (EPIC) Program**
- **Electric Vehicle Charging Station Program**
- **Hybrid and Zero-Emission Truck and Bus Vouchers**
- **Self-Generation Incentive Program**

Summary: Solar and Beyond

The story of Lancaster's resolve is an example of how a community through its political leadership can quickly respond to a recession while addressing environmental regulations in its efforts to diversify and sustain its economy in just over a decade.

Lancaster was known for an economy based upon aerospace. However, when impacted by the Great Recession, its single focused economy was horribly impacted. It was confronted by nearly a 20 percent unemployment rate.

Additionally, and simultaneously, it was also confronted by mandatory state environmental legislation. It felt the weight of the world on its shoulders in trying to respond.

Who said government can't move quick?

The City worked quickly towards a resolution of both items. Its leadership created an elegant strategy using its locational advantage of 300 plus days of sunshine to transition to solar energy coupled with a comprehensive approach from the innovations of clean technology to solve its economic plight.

The City solved for both short-term issues created by the collapse of its economy by shifting its economic strategy to achieve solutions to also answer the requirements of the regulatory environmental legislation. Additionally, while achieving short term solutions, it simultaneously created a long-term plan for the diversity and sustainability of its economy.

Lancaster is a community of firsts through these efforts. From solar towers to solar mandates and electric buses, so, it is no surprise that the City's success has been celebrated through prestigious honors in clean technology development such as the 2021 Energy Globe Award, the 2019 Green Power Leadership Award from the EPA, and three Eddy Awards as the Most Business-Friendly City in Los Angeles County in 2007, 2013, and 2019.

The City's success is due to a variety of factors. They include customer service, partnerships and leadership.

For example, it has led to the business attraction of companies engaged in clean energy innovations involved in transportation, mobility and energy generation like BYD, eSolar, and SolarCity.

Additionally, the City invited an opportunity with the University of Ottawa in solar energy research with its deployment of SUNLAB.

Lancaster has been instrumental if not relentless in formulating new city codes and ordinances along with a host of new entities by incorporating new opportunities to address the needs of environmental regulations regarding the effectiveness and efficiency of energy generation, distribution, sales, and marketing.

Through all of this, Lancaster has demonstrated how local government can respond and quickly implement projects and partnerships to fuel long-term agendas that benefit its residents and businesses for the benefit of a region, if not a state.

Lancaster has seized upon its natural resource and existing assets to turn itself into an alternative energy hub. Its vision of becoming the Solar Capital of the Universe is certainly a noteworthy effort along its road towards its achievement of solving the two major issues it confronted.

Combined, the activities have not only brought jobs and improved the City's tax base but also enhanced the image of the City as a forward-thinking community.

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Catherine Grooms, Director of Small Business Development Center, Los Angeles Network

Organizations

BYD

CalChoice

California Governor's Office of Business and Economic Development

City of Lancaster

Federal Reserve Bank of St. Louis Economic Data

KBHome

NASA Armstrong Flight Research Center

Rocket Homes

SG H2 Energy

University of Ottawa SUNLAB

U.S. Energy Information Administration

A BRIEF ECONOMIC HISTORY OF LANCASTER, CALIFORNIA

"THE SOLAR CAPITAL OF THE UNIVERSE."
MAYOR REX PARRIS, CITY OF LANCASTER

2008
Established goal of becoming "the alternative energy capital of the world."

2009
Partnered with eSolar to open a twenty-acre, 5 megawatt (MW) solar thermal power tower plant

2010
Brought Build Your Dreams and KB Home together to build net-zero homes.

Launched Solar Lancaster program

2011
Lancaster Power Authority (LPA) created

LPA partnered with Lancaster and Eastside school districts and SolarCity

Partnered with First Solar to develop a 230 MW utility-scale solar project

2012
Partnered with Beautiful Earth Group

Partnered with The University of Ottawa's SUNLAB and Canadian solar firm Morgan Solar

High Desert Power Authority created

California Clean Energy Authority established

2013
Residential solar requirment adopted

2014
Solar Mandate comes into effect

2015
All new homes required to have greywater "recycle-ready" plumbing

Lancaster Choice Energy begins serving residential, commercial, and municipal accounts as a community choice aggregator

Sreetlights acquired from Southern California Edison

California Clean Energy Authority created

2017
CalChoice created

ALPHARETTA, GEORGIA

Overview

How has a small community in the state of Georgia become such a major player in technology, specifically Fintech?

The accounting and consulting firm Deloitte has one of the best definitions of Fintech. Their definition of financial technology refers to computer programs and other modern technologies used by businesses that provide automated and improved financial services.

However, people are confused by the term - Fintech. According to the Sloan School of Management at M.I.T., it is a portmanteau of "financial services" and "technology" and its function(s) have enabled people, specifically young people, to perform financial transactions not otherwise available through traditional banking systems. These include day trading stocks from their mobile devices, splitting checks through a payment app, or updating car insurance through a mobile application.

Fintech has achieved the moniker of being mobile-first, customer-centric, and disruptive. However, the industry is a center of opportunity brought upon by the internet and unfortunately pushed on by the pandemic and a post pandemic world that may have less face-to-face contact that was once a cornerstone for financial services.

FINTECH SERVICES

- **Personal Finance**
- **Payments & Billing**
- **Insurtech**
- **Capital Markets**
- **Money Transfer & Remittances**
- **RegTech**
- **Blockchain**
- **Mortgage and Real Estate**
- **Wealth Management**
- **Lending**

Source: Deloitte

In 2018, the US accounted for 57% of the global Fintech market. Nationally there are over 6.4 million people directly employed in the sector. In Washington there are nearly 95,000 employed, and in the Seattle region there are roughly 75,300 employed.

Some of the major companies in the nation include, Robinhood, Coinbase, Square, and SoFi. In Washington, notable companies that are located but also serve the nation include Remitly, Avalara, and PitchBook. Those that specifically are known in the Seattle area are companies such as Snap! Raise, Gravity Payments, and Transparent Systems.

As of 2021, there are about 8,770 Fintech startups in North America according to the research firm Statista. Fintech is global and has been projected to reach a market value of $305 billion by 2025.

This research has identified major cities across the globe and the nation that have created successful business environments for growing Fintech ecosystems. These cities are presented in the following chart:

MAJOR FINTECH HUBS

San Francisco	**Tokyo**	**Singapore**
Atlanta	**New York**	**Hong Kong**
Paris	**Chicago**	**Shanghai**
Beijing	**Seattle**	**Sydney**
London	**Seoul**	**Norway**

Source: Deloitte

However, research screened cities based upon similarities to Federal Way. For example, the screening involved the following: Population size, (100,000 - 250,000); a suburb - located in a metropolitan area; no university within the city limits yet close proximity to a major research university; age of the city, successful entrepreneurial ecosystem; metrics of success such as jobs created and investments made.

The research identified the City of Alpharetta, Georgia, in the Atlanta Metropolitan Statistical Area. Alpharetta was chosen as a case study for Fintech due to their infrastructure investments to support technology businesses and the support of its residents by creating a sustainable and technology-based future focused economy. Additionally, it was selected based upon similarities to Federal Way, such as population, location in a metropolitan area, and proximity to a major research university.

For many tech hubs to grow and become successful, they need the basics of a talented and educated workforce, access to and availability of capital, excellent infrastructure, and education system(s). That is the same needed for Fintech organizations, as well as the effectiveness of pro-active public policies designed to enable Fintech growth

SCREENING PROCESS

- **Population Size (100,000-250,000)**
- **Suburb to Metropolitan Area**
- **Adjacent to Interstate**
- **No University, however close proximity to one**
- **Age of City compared to Federal Way**
- **Maturity of Technology & Entrepreneurial Ecosystem**
- **Jobs Created & Investments Made**

Taking these individual filters into account, the most notable similarities to Federal Way and Alpharetta is their distance to a major metropolitan area (Seattle and Atlanta) which will allow both to access the talent pools within these major cities as well as benefit from nearby business and technology ecosystems.

However, there is a slight difference. Alpharetta has both university and college satellite campuses. These include Georgia State University and Gwinnett Technical College. Although it is noteworthy to point out that these two satellite campuses are relatively new; opening doors in 2000, and 2016 respectively. However, the similarities are close as Federal Way will become home to a satellite campus of the University Washington in the fall of 2021.

Alpharetta, Georgia

Alpharetta is a suburb of Atlanta located in northern Fulton County. It is about 25 miles north of Atlanta. The City is within the Atlanta Metropolitan Statistical Area of just over 6 million in population.

Alpharetta has a population of 65,590 as of 2019. It is 27 sq. miles in size.

It is served by State Route 400 which is concurrent with US Route 19. It is a multi-lane highway to and from Atlanta.

The City had humble beginnings in the early 1800's known for its agricultural products and trading posts. It remained rural until 1980's when suburbanization found its way from Atlanta. Its population went from 3,000 to nearly 70,000 in less than 30 years.

Its daily work population swells to over 100,000 -120,000 due to the commercial and business growth that has occurred.

The City has self-proclaimed itself as the "Technology City of the South," because of the 700 plus technology companies located there including some 45 Fintech companies.

"Technology City of the South"

Self-proclaimed: Alpharetta, GA

Today, this concentration of Fintech businesses within Alpharetta continues to rapidly grow. According to the Technology Association of Georgia, the state's Fintech firms generate $72 billion a year in revenue, third highest only behind California and New York.

Alpharetta is home to Fintech companies such as the Fintech giant - FISERV, to companies such as BitPay, Ingo Money Inc., Paymetric, and 43 more Fintech firms.

Economic History: Two Waves of Growth

The economic success of Alpharetta appears to have come in two waves. The first, via the marketplace that found Alpharetta as an alternative location to higher priced Atlanta and the second coming much later after the influx of growth and the corresponding formulation of a municipal organization. The results from the second wave were initiated by elected officials of Alpharetta, and it came fast with a well-oiled plan to boost the economic activity of the City.

Again, the population growth and corresponding building boom of the 1980's created by the marketplace looking for a cheaper alternative to Atlanta was the first wave. However, and unbeknownst to anyone including city officials, it provided the foundation for the future opportunity for Fintech growth.

According to the City's economic development organization, *GrowAlpharetta,* the City installed hundreds of miles of fiber optic cables to address the future growth of the city. Meanwhile, Georgia's largest electricity

provider, Georgia Power, established a series of redundant electric networks throughout the city.

Together this foresight made Alpharetta an ideal environment for the growth of business, specifically tech. With internet download speeds 60% higher than US average, along with 25 million sq. ft. of office space and available and affordable land for development, it made for an ideal location, and, as such, many companies made Alpharetta their home.

Meanwhile, in 2012, some 30 years after the first wave of economic activity, Alpharetta's elected officials recognized that even with the arrival of businesses, the City struggled to have a "sense of community" and had for some time, specifically regarding the tech sector.

They knew they needed a spark to change the community.

The efforts of branding the City as the "Technology City of the South," came from a former mayor. Then Mayor, David Belle Isle, said, "I guess you could say this tagline and subsequent strategy was our community's moonshot." The Mayor and City Council recognized that all of the basic components were in place. These included essential infrastructure, a highly educated workforce, nearby research universities, specifically Georgia Tech, and a handful of influential technology companies that wanted to help grow the sector.

The Mayor and City Council looked for a spark to ignite it. The spark came in the way of the formulation of the Alpharetta Technology Commission, and an announcement to the world that Alpharetta would be known as the "Technology City of the South." The City formulated a strategy and immediately began implementation.

Through the City's elected officials' leadership, Alpharetta landed some of the largest economic development projects looking for sites in the metro Atlanta area. An example includes the relocation of the Fintech giant – FISERV in 2014. It is a global Fintech and payments company focused on ecommerce. Its relocation brought over 1,800 jobs.

Alpharetta's Fintech Success

According to a 2018 report by Technology Association of Georgia, large Fintech companies such as *CheckFree* and *Equifax* had been in the state for decades and were part of the thriving Financial, Insurance, and Real Estate sector of the state's economy.

CheckFree as a company had made numerous acquisitions and continued to grow in the state throughout the years. Additionally, former CheckFree employees that gravitated on to other firms specifically in executive positions have help grow countless Fintech companies across the state. But in 2007, FISERV acquired CheckFree.

However, and again, unbeknownst to city officials, the acquisition of CheckFree would create an opportunity to attract a major Fintech business.

FISERV's announcement in 2014, began the growth of FINTECH to permeate Alpharetta's economy. So, with this action, yet another major opportunity continued building the foundation for Fintech in the City.

The strategy that included branding, marketing, business recruitment, and incentives, along with customer service related to permits, served the community well. The relocation of FISERV became a significant change and launched Alpharetta as a major host city for Fintech companies.

Tech and Entrepreneurial Ecosystem

Again, coupled with significant built communication and energy infrastructure and a large market player – Fiserv, the following identifies how Alpharetta continued to grow the resources and created the tools used to become the "Technology City of the South."

Associations and Organizations

There are a variety of young organizations that promote growth and the development of the tech industry within Alpharetta. The most notable contributors are the Technology Association of Georgia (TAG) and Tech Alpharetta.

Alpharetta Technology Commission

The Alpharetta Technology Commission started in 2012 as an advisory organization of local tech leaders that was created to help the City with innovation, national branding, and growth of the technology sector. It was established as a strategic advisory board of local tech leaders by the Mayor/City Council. However, by 2017, it had matured and changed its name to Tech Alpharetta. Additionally, that same year they also opened an innovation center. The Center provides a unique supportive space for tech entrepreneurs, mature businesses, startups, and venture capitalists to meet, network, and innovate. Research identifies that it is more of a co-working space. Membership includes office space or desk space, internet, parking, educational & mentoring programs, investor connections, and events.

The Innovation Center is supported financially by the City at $100,000 per year. It has helped Fintech companies like DataSeers software, a company that solves fraud related to money laundering with connections to the industry and coaching. Currently the Center is home to more than 50 tech startups focused on Fintech, biotech, cybersecurity, smart city, data analytics, med device, healthcare IT, voice recognition tech, and many other technologies. Georgia Tech is co-located at the Center with its Alpharetta Startup Circle program.

In addition, a corresponding ecosystem activity is the Greater Alpharetta Tech Network whose mission is to help connect the technology community by hosting locally-based tech thought leadership along with educational and networking events for tech professionals.

Tech Alpharetta
Source: The Atlanta Journal-Constitution

The Technology Association of Georgia, (TAG)

Alpharetta figuratively won the lottery when the state-wide organization, the Technology Association of Georgia, moved their main offices to the city in 2018.

TAG which was established in 1999 provides tech companies with local access to events, networking, and opportunities to connect and engage with TAG's statewide membership of over 2,000 technology companies consisting of more than 35,000 tech professionals.

TAG has helped create a thriving tech community in the state but has been fortuitous for Alpharetta as well. For example, while TAG has many societies and sector chapters, one focuses on Fintech aptly named TAG Fintech which has provided the host city with numerous opportunities due to their presence in Alpharetta.

The Association's goal is to build recognition of Georgia as the hub of choice for the world's leading financial technology providers. They have been able to do this by leveraging major sponsors in Fintech such as Elavon, FISERV, and e-Merchant as well as fostering a strong business environment within the state due to the ongoing success of the Association's endeavors.
TAG conducts an annual Fintech conference as well as a world-class Fintech summit called Fintech South, in Atlanta. This event hosts 2,000+ leaders from around the world for 3 days of content to help drive business success and keep the community up-to-date on topics and trends.

TAG has had significant global and statewide success through marketing over the years, but their face-to-face contact at their main office in Alpharetta illustrates the value of brick and mortar operations. Before the pandemic it had nearly 1,400 visitors in 2018, that lead some 40 plus tech events such as networking, luncheons, and forums on tech brands and companies.

Georgia Tech Financial Services Innovation lab
Source: Georgia Tech

In addition to Tech Alpharetta and TAG, Georgia Tech, and Fintech Atlanta have also helped Alpharetta achieve its goals of becoming a more prominent Fintech hub as well as an established tech hub.

Accelerators, Incubators, and Tech Transfer

There are a wide variety of pre-accelerators and accelerators along with incubators in the Atlanta region. They include some familiar players found also in the Seattle Region such as TechStars. But there are a variety of independent locally operated enterprises focused on Fintech.

One such entity is located in Atlanta, at the Georgia Institute of Technology, aka Georgia Tech, and its Tech's Advanced Technology Development Center (ATDC). The center is home to the financial services innovation lab. This program creates a unique collaboration between a nonprofit, university-based incubator and major financial services firms to help launch and scale Fintech companies across the state. USBank/Elavon, a global payments provider and subsidiary of U.S. Bancorp is the current lead sponsor of the program. USBank invites all early stage Fintech companies to join in the effort to drive innovation across financial services and payments technologies.

This program is open for students of Georgia Tech but also members and companies within the ATDC portfolio. ATDC also has a place in Alpharetta, GA, and provides mentorship in many different topics related to Fintech such as finance, sales, business development, marketing, technology, etc.

The Financial Services Innovation Lab aims to be a hub for financial education, research, and industry in the Southeast. The Lab acts as a platform to connect and bring together faculty and students across Georgia Tech with the financial services industry and Fintech entrepreneurs.

When suburbanization occurred, Alpharetta was the recipient of a labor force that possessed a high preponderance of bachelor's degrees. The US Census Bureau identifies over 60 percent of the population has a bachelor's degree. Alpharetta had a talented labor force that could support the needs of a growing tech-based economy let alone a Fintech. However, in addition, there are a variety of other educational programs supporting the sector. They include the Georgia FinTech Academy, a collaboration between the region's Fintech industry and the 26 public institutions of the University System of Georgia, is a talent development initiative ensuring the region has a strong pipeline of Fintech talent. The Academy is the first of its kind in the nation and allows students to access specialized educational experiences necessary to enter the FinTech sector.

Access to Capital

There appears to be a coordinated coalition of venture capitalists in metro Atlanta, and investment is alive and well even during the pandemic.

Funding for venture-backed startups based in metro Atlanta area hit a five-year high in 2020, reaching nearly $2 billion across 188 deals, according to preliminary data from Crunchbase.

This is due to the efforts of many to enhance the ecosystem. For example, Venture Atlanta is the largest technology innovation event in the southeastern part of the nation. It is proclaimed as where the region's most promising tech companies meet the country's top-tier investors. As the Southeast's largest investor showcase helping launch more than 500 companies and raise nearly $6 billion in funding to date, the event connects the region's top entrepreneurs with local and national investors and others in the technology ecosystem who can help them raise the capital they need to grow their businesses. The annual nonprofit event is a collaboration of the Atlanta CEO Council, Metro Atlanta Chamber, and the Technology Association of Georgia (TAG).

Policy Regulation – Local Incentives

The City provides local incentives through their economic development toolkit that has allowed for new or growing businesses to further their economic impact.

These local incentives include expedited permitting and inspections, discounted permitting and inspection fees, local job creation credits, tax abatements, and revenue bonds.

For example, Alpharetta's expedited permitting and inspections allows for companies to have 24/7 access to the city's Development Ombudsman, providing applicants pre-submittal meetings for their architects, engineers, and contractors before starting a project. The city also has same day inspections without any extra charge to the company and or developer.

Discounted permitting and inspection fees allowed by the *High Impact Permit and Inspection Grant* which was created by Alpharetta Development Authority provides the City to further incentivize new or expanding businesses. Projects that exceed $5 million in real property capital investment would qualify for a *Base Permit Fee* reduction of 20%-45% with a cap of $250,000.

The City of Alpharetta has also established a Local Jobs Creation Grant Program to encourage new and expanding businesses to hire from the local workforce. For each job created that qualifies for the Georgia Job Tax Credit Program and for which a current resident of the City of Alpharetta is hired, the City will pay a direct grant of $250-$500 per new employee, up to a maximum of $80,000.

The City's tax abatement program, coordinated through the Alpharetta Development Authority, is designed to lower the real and personal tax liability on the value of new investments made in Alpharetta and can provide significant savings for companies. To qualify for tax abatement, businesses must meet minimum thresholds in the areas of job creation, average salary, and fiscal impact.

The Alpharetta Development Authority is the agency that issues revenue bonds for financing projects that will have a significant impact on the City of Alpharetta in terms of job creation and capital investment. These are tax-free bonds that can provide significant savings to qualifying projects over more traditional financing methods. The final authorization to issue bonds rests with the Alpharetta Development Authority board of directors.

State Incentives

The state of Georgia also provides tax incentives and credits to businesses through their Department of Economic Development (GDECD). One such incentive is the opportunity to minimize or even eliminate state corporate income tax. This may apply to companies large and small, preexisting companies, or new to the state companies.

Additionally, the state provides incentives such as Tax Credits, Quality Jobs Tax Credits, Mega Project Tax Credits, Sales and Use Tax Exemptions, Investment Tax Credits, Research and Development Tax Credits, and Retraining Tax Credits.

Outcome: Georgia's Transaction Alley;
Source: American Transaction Processors Coalition

Transaction Alley

Much of Alpharetta's success may be attributed to the large cluster of Fintech companies that permeate the region, (see map below). Within metro Atlanta, including Alpharetta, Fintech companies have evolved into a cluster to become what is known as "Transaction Alley." This area encompasses Fintech and supports the technological needs of the financial services industries within the area and, specifically, the payment-processing infrastructure of the economy. Throughout this area alone, more than $208 million has been invested in Georgia's Fintech industry since 2013, according to the state of Georgia.

More than 60 percent of the companies in this U.S. industry sector call Georgia home, and 70 percent of all U.S. payments processed annually run through Georgia. The global payment processing (merchant acquiring) industry processes approximately 369 billion card transactions annually with projected growth to 854 billion by 2028.

City officials are quick to point out that along State Route 400 highway there is also an abundance of Fintech companies that makeup what is known as the "Payments Corridor of the Nation." Many new and growing Fintech companies have originated here such as Kabbage and Greenlight.

According to TAG, Georgia first began to attract financial technology companies in 1987, when state lawmakers lifted caps on credit card interest rates and annual fees of 18% and $12, respectively.

Other organizations such as the American Transaction Processors Association (ATPC), which was founded in 2014, were created to promote, protect, and preserve the interests of Georgia's transaction processing industry through proactive public relations and government affairs activities. This organization is also the first of its kind in the payment processing industry in the U.S. according to a 2020 Fintech Ecosystem Report by TAG. The ATPC is also the first of its kind at both the state and federal levels to be able to handle business interests of the Fintech community.

According to the event conference, Fintech South, there are more than 170 Fintech companies in metro Atlanta that generate more than $100 billion in revenues annually of which nearly 30 percent are in Alpharetta. It appears that there was a combination of activities that helped catapult Alpharetta into the thin atmosphere of success as it pertains to Fintech.

Summary

As previously identified, Alpharetta was a sleepy agricultural town. With people looking for a cheaper alternative to Atlanta, the suburbanization occurred 70 years later. But after 30 years of growth as identified, the City struggled to find an identity. It was recognized by the Mayor and City Council that they needed to step up and take the initiative. However, they said they needed a spark. So, what was the spark, and how did they get to success in such a short amount of time from 2012 to the current day?

First, the City's corresponding response to the stunning growth of burying communication fiber during the development was a significant factor in creating the foundation for its tech future.

Second, with the rapid suburbanization came a highly educated labor force. 64% of the population has a Bachelor's degree, a luxury for future business.

The third element was Alpharetta's ability to attract one of the largest Fintech companies in the nation. Fiserv selected the City as a new location for their expansion in metro Atlanta ensuring a supply chain and corresponding interest by others in Fintech.

In addition, Atlanta public and private leaders created a coalition of companies ranging from Fortune 500s to startups and other organizations to cement the metro Atlanta area as the recognized global capital of financial technology. The coalition is funding and fueling efforts to recruit, retain, and expand businesses and jobs across the metro Atlanta.

Also, metro Atlanta is home to a host of startup incubators dedicated to creating environments for early-stage Fintech companies to grow. Founded in 1980, the Advance Technology Development Center (ATDC) at Georgia Tech boasts a 90 percent success rate for its startups after their first five years. Much of ATDC's success in helping Fintech companies over the years is tied to the level of access it provides to Georgia Tech resources, mentorship by seasoned entrepreneurs, and the support of established corporate partners. Additionally, the Atlanta Tech Village is the fourth largest tech hub in the U.S. and stands as a great resource for startups, promoting faster connections between talent, ideas, and capital.

The whole ecosystem is local, as metro Atlanta is not only home to incubators but also processing companies, payments companies, plus a strong venture capital community. Many local venture capital firms focus solely on Fintech for their investments, which has strengthened metro Atlanta's economy, including Alpharetta's, and increased technology developments within the financial services industry.

Technology sector talent is a key factor. According to The Atlanta Journal-Constitution, Atlanta is ranked 10th out of 100 metro areas for STEM professionals in terms of career opportunities, and students interested in a career in the Fintech industry are able to find comprehensive programs at many local academic institutions. For example, Georgia

State University offers one of the first university Fintech-specific labs through its J. Mack Robinson College of Business. The University of Georgia offers an MBA with a concentration in financial technology, and Georgia Tech's engineering programs are all ranked in the top ten in the nation, with programs available that focus specifically on innovation within the Fintech industry.

Atlanta and Alpharetta specifically have built a great reputation as one of the biggest tech hubs in the Southeast. Local support of tech and the city of Alpharetta's commitment to economic development, startups are thriving all over the city.

In the 2016 report "State of Georgia's Fintech Ecosystem," it identified that the state was ready to take off in Fintech. Capital Investment was exploding during this time globally as Fintech transitioned to become more mainstream.

According to Crunchbase, VC funding has reached an average of 1.35 billion in Atlanta annually since 2016, with an annual average of 184 deals as of January 2021 in the Greater Atlanta area alone, and Alpharetta is near the center of it all.

The infrastructure investments have had economic pay-off. The City's efforts to achieve their goal of being the Technology City of the South, were working, but it took 30 years and a recognition and committed political leadership to make it happen.

While Tech Alpharetta addresses everything tech, there does not appear to be an organization devoted strictly to the Fintech cluster in Alpharetta. However, maybe they do not need to have a Fintech cluster since the Technology Association of Georgia focuses on Fintech as well as the Metro Atlanta Chamber and Fintech Atlanta.

So, was it a spark or just simply blind luck? It appears it was a combination of things that all occurred within less than a decade. But it was the direction of a mayor and council that enabled the community
to succeed, without that leadership, it probably would not have occurred to the level reported.

Resources

Advanced Technology Development Center (ATDC)

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Fintech Atlanta

Georgia Department of Economic Development

Georgia Tech Fintech Lab

Grow Alpharetta

Mayor, T. (2021, February 4), *Fintech, explained*; MIT Sloan School of Management.

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Tech Alpharetta

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RICHARDSON, TEXAS

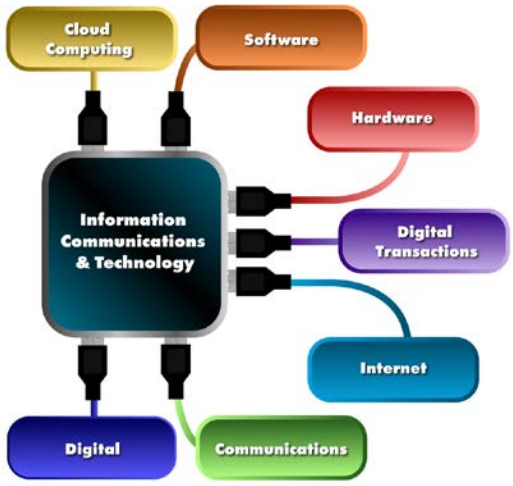
Overview

Is it possible for a City to have and achieve a vision for 70 years? Apparently, that is the case with Richardson, Texas. The community saw an opportunity and has remained constant and consistent to their cause of building an electronics and telecommunications technology-based economy.

Information and Communication Technology (ICT) is defined as a diverse set of technological tools and resources used to communicate and to create, disseminate, store, and manage information. Additionally, it is similarly defined as a diverse set of technological tools and resources to transmit, store, create, share, or exchange information.

ICT is comprised of a multitude of elements from cloud computing, software, hardware, digital transactions, internet, communications, and data.

ELEMENTS OF ICT



Source: City of Federal Way

Nationally, there are over 55 million people directly employed in the sector. In Washington State, ICT is among the state's largest sectors. It is fast growing, pays strong wages, and is an increasingly important driver of other industries in the state. According to the State's Employment Security Department, there are 313,100 people employed in this sector. In King County there are over 188,000 employed. Average annual wages, including benefits are $225,000 per worker. Even more importantly, every job In ICT creates several more jobs, with a jobs multiplier of 3.7 in the state.

Nationally, some of the major information and communication-based companies include Apple, Microsoft, Amazon, Alphabet (Google), Dell, and IBM.

In Washington, specifically the Seattle area, major companies include Amazon, Microsoft, Tableau, Boeing, T-Mobile, and Nintendo. In the Greater Seattle area (if not Federal Way itself) many ICT giants have campuses. Amazon, Microsoft, T-Mobile and Boeing all have large facilities within the Greater Seattle Area.

Research has identified cities across the globe and in the nation that are ICT hubs. They include Los Angeles, San Jose, and San Francisco, California; Austin, Texas; New York, New York; Tokyo, Japan; Beijing, China; London, England; and of course, Seattle.

GLOBAL ICT HUB CITIES

Los Angeles	**San Jose**
San Francisco	**Austin**
New York	**Tokyo**
Beijing	**London**

Source: Tech Monitor

However, research screened cities based upon similarities to Federal Way. For example, the screening involved the following: Population size (100,000-250,000); a suburb - location in a metropolitan area; adjacent to an interstate and no university within the city limits let alone proximity to a major research university.

SCREENING PROCESS

- **Population Size (100,000-250,000)**
- **Suburb to Metropolitan Area**
- **Adjacent to an Interstate**
- **No University, however close proximity to one**
- **Age of City compared to Federal Way**
- **Maturity of Technology & Entrepreneurial Ecosystem**
- **Jobs Created and Investments Made**

Given the similarities to Federal Way, the research identified the "Telecom Corridor" in Richardson, Texas, as a city for analysis due to its rich history of maintaining and attracting business in the ICT sector. Originally coined the "Electronic Suburb" in the late 1950's, it grew into a telecommunications hub without a university, let alone a formal economic development strategy.

Richardson is a vibrant and diversified business community. It is located in one of the largest economic regions in the world, the Dallas-Fort Worth metropolitan area. It is home to companies seeking and producing the latest technological breakthroughs.

Today, the Telecom Corridor area accounts for over 150,000 jobs in over 600 technology companies and contains over 25 million square feet of office space.

The most similar characteristic between the two cities of Richardson and Federal Way is the proximity to a major established tech hub (Dallas and Seattle). Both Federal Way and Richardson are located in major tech hubs in the United States with multiple Fortune 500 companies. This proximity also allows for access to talent pools through universities serving the regions. In addition, both cities are located along highways if not interstates.

However, the cities differ in multiple aspects. In 1969, the City of Richardson became the location of the University of Texas-Dallas. As such, over 50 percent of the population has a bachelor's degree or higher.

Another key difference between the two cities is the physical layout of the cities. Of Richardson's 28.6 square miles of land, approximately 12 square miles are dedicated to commercial and tech space, with the majority of the city being dedicated to office space. Federal Way is slightly smaller at 23.7 square miles of land area with only a small portion dedicated to office and commercial space.

Another distinction is that Federal Way is served by not one but two major port facilities - Seattle and Tacoma. Not so in the Dallas-Fort Worth metropolitan area.

Richardson, Texas

Richardson is located in the greater Dallas/Fort Worth metropolitan area specifically, northeast of Dallas.

Originally settled in 1873 by settlers from Kentucky and Tennessee, Richardson now has a population of over 116,000. The Dallas/Fort Worth metro area has a population of over 7.5 million.

The City is 28.6 square miles. Of these 28.6 square miles, approximately 12 square miles is dedicated to the city's established and growing ICT Hub: The Telecom Corridor.

Richardson is a 20-minute drive from Dallas and is within 30 minutes of 44 Fortune 500 corporate headquarters.

Economic History

Like the history of some many other communities, Richardson was located along a railroad line. Such was the case for Richardson, although it was home to the Atchison, Topeka and Santa Fe, the Southern Pacific, and the St. Louis and Southwestern railroads. Richardson was also an agricultural oriented economy. However, it remained a sleepy farming community until the 1950s as Post-World War II brought major highways and interstates.

Texas in the mid 1950's was an area with plentiful surplus of land. So local property owners and businessmen encouraged economic activity and with it came the precursors of tech companies. Thus, Richardson was known as the "electronic suburb."

Basically, companies not satisfied with the current state of their communities sought out Richardson. As an example, in 1956, Texas Instruments selected the east side of U.S. 75 just north of Interstate 635, for its corporate campus. Shortly after, in 1957, Collins Radio, a well-established electronics company from Cedar Rapids, Iowa, built the first of what would be a multi-building campus on a 400-acre site just north of Texas Instruments. Both companies attracted a variety of engineering talent, which further paved the way for a number of other technology-based businesses to locate to Richardson.

Richardson was a popular suburb for upper income, college-educated professionals. Education was a focus for the community, and the Southwest Center for Advanced Studies opened there in 1964. It would become the University of Texas at Dallas in 1969.

The pace of growth accelerated even further in the 1980's. In part, this was due to the deregulation of the telecommunications industry. Mature companies involved in ICT continued their location to Richardson. For example, MCI located to the area because their microwave network relied on the products and technical expertise of Collins Radio. During this period, the concept of the Telecom Corridor would become a market-driven brand with communication and telephone businesses in the Texas area and the Midwest region creating and relocating campuses and headquarters to the city.

During this period local government became engaged in economic development. The Richardson Chamber of Commerce and the City of Richardson collaborated to form a joint economic initiative named, "The Richardson Economic Development Partnership." The main purpose of this initiative was to build a vibrant and thriving local economy by attracting new investment and jobs, through

marketing and recruitment efforts and by working with existing employers.

In 1989 with the hopes of creating a telecom and technology corridor, local property owners set aside over 11 square miles dedicated to office and tech space toward the effort of creating what would be named and is now known as the "Telecom Corridor," in Richardson.

By 1990 Richardson had 102 manufacturing firms, with an emphasis on electronics and telecommunications.

In 1994, a new collaborative approach was needed to truly allow Richardson to transform from a business center to a large-scale telecom corridor. In August of that year, four past Chairs of the Richardson Chamber of Commerce formed the Tech Titans (formerly known as the MetroPlex Technology Business Council) with the purpose of providing leadership and promoting collaboration among area companies. The first meeting was held in 1995 and the founding leaders invited 80 high tech companies to become charter members, which all of them subsequently joined. Since its founding over 25 years ago, the organization has become the largest technology industry association in the southwestern United States. The organization provides, networking, advocacy, workforce development, and innovation partners program.

Major companies that have moved into the Telecom Corridor include: AT&T, Hitachi, Huawei, Juniper Networks, L-3 Communications, Verizon, Ericsson, Rockwell Collins, Cisco Systems, Raytheon, MetroPCS, Honeywell, Motorola, Texas Instruments, and Samsung Wireless.

Of significance, Richardson has access to Dallas' talent pool due to the close proximity of the cities. This has enabled Richardson to grow becoming an employment center. With over 25 million square feet of office space and over 150,000 jobs, Richardson is considered to be one of the larger telecom hubs in the United States. However, it is also diversified with other tech subsectors from software, electronic equipment, data centers, cloud computing, and medical devices.

Richardson, Texas:
"The Best Place in the U.S. to Locate your Technology Company"

Self-Proclaimed, City of Richardson

In 2013, The Richardson Economic Development Partnership, announced that Richardson's Telecom Corridor® area had earned the prestigious "Area of Innovation" membership status in the International Association of Science Parks (IASP). The Telecom Corridor is only the 8th U.S. area accepted to IASP membership and is one of the first to earn the "Area of Innovation" designation. IASP defines areas of innovation as places designed and curated to attract entrepreneurial-minded people, skilled talent, knowledge-intensive businesses and investments, by developing and combining a set of infrastructural, institutional, scientific, technological, educational and social assets, together with value added services, thus enhancing sustainable economic development and prosperity with and for the community.

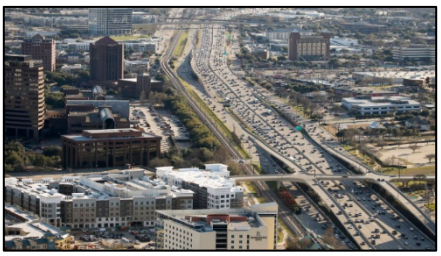
Today, the Dallas/Fort Worth region is one of the largest technology centers in the nation. A variety of high-tech companies in the region employ about 230,000 people. Telecom, IT, Biomedical, Semiconductor, and Nanotechnology are all represented in the area of Dallas/Fort Worth.

But it is Richardson that has come to be known as the Telecom Corridor. It is 11 square miles in size, has over 12 million square feet of office space, 13 million square feet of tech space, and estimated 150,000 plus jobs, in over 600 tech companies, of which 50 are foreign firms such as Ericsson, Samsung, and Hitachi, to name just a few.

Tech and Entrepreneurial Ecosystem

Richardson was originally able to succeed as a center of economic development for electronics and communication without having any formal branding or business recruitment efforts or entrepreneurship tools. It relies on several factors. These include its geographic location being conveniently located near Dallas along with its location along the networks of highways, interstates, and railroads. It was also close to a well-trained labor force along with the ability to attract businesses with available and inexpensive land for its growth.

The Dallas/Fort Worth metropolitan area is home to dozens of accelerators and incubators. However, for Richardson as time progressed, more attention was needed to creating and supporting aspects of the tech and entrepreneurial ecosystem by local leaders. Thus, success came to Richardson through the University of Texas.



Telecom Corridor, Richardson, Texas
Source: Vernon Bryant

University of Texas- Dallas

Since 1969, the University has played a role for the growth of the Telecom Corridor. UT-Dallas has an estimated 29,000 students. About a third of the students are enrolled in masters and doctorates in engineering, business, and the sciences.

The University provides a variety of accelerators, incubators, innovative centers, and networks to help grow start-ups. The Institute for Innovation and Entrepreneurship was started in 2006 and is the center of activity for these programs.

CometX

CometX is an eight-week, experiential accelerator program that pulls student entrepreneurs into an environment that incubates ideas and engages prospective customers in preparation for the launch of a new business concept. This 8-week program covers the following topics:

- Getting started and minimum viable product (MVP)
- Startup legal basics
- Building a team
- Business model canvas
- Launch and marketing
- Fundraising, Investment, and KPI's
- Building a pitch deck and how to pitch

Guest speakers, subject matter experts, and mentors are included throughout the program. The program concludes with a pitch competition.

GalXc

The GalXc Women's accelerator provides education, training, networking, and mentoring to a cohort of women. Participants develop skills and competencies to become effective business leaders and founders of high-achieving startups. This 8-week program covers the following topics:

- Getting started and customer discovery
- Startup legal basics
- Financial foundations
- Founders stories
- Launch and marketing
- Building & scaling
- Building a pitch deck and how to pitch

Guest speakers, subject matter experts, and mentors are included throughout the program. The program concludes with a pitch competition.

Innovate(her)

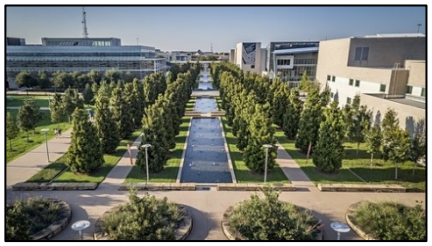
Since 2018, the Institute for Innovation and Entrepreneurship has partnered with Capital One to reach more than 630 middle school girls in North Texas with the Innovate(her) program, a day-long conference that equips middle school girls with science, technology, engineering, arts, and math (STEAM) skills for higher education and future careers. The program gives the middle schoolers an introduction to several topics, including technology, finance, innovation, and entrepreneurship.

The Innovate(her) curriculum has multiple dimensions:

- A free one-day conference for middle school girls, held at The University of Texas at Dallas;
- A workbook of modules on technology, design thinking, building confidence, and personal finances created by UT Dallas and Capital One;
- A network of women entrepreneurs, Capital One corporate professionals, and UTD students mentoring and guiding the students through a full-day of interactive, hands-on activities

Participants spend the day with mentors from the community who work in STEAM careers or have started their own companies. Students work alongside them throughout the workshops. The middle schoolers get to hear first-hand how women have overcome challenges and developed their careers or started their businesses. According to Million Women Mentors, women are 50% of the workforce but only 24% of the STEM workforce. Women who are mentored are 2.5 times more likely to be confident in their ability to succeed in school and their careers. UT-Dallas believes this interaction is essential to inspire the girls to think about STEAM careers.

Mentors are recruited through the UT Dallas network of mentors, Capital One's Business Resource Groups for Women and Women in Technology, and professional organizations in the community. A wide variety of careers and interests are represented so the middle schoolers have an opportunity to meet someone who inspires them.



University of Texas, Dallas in Richardson

Blackstone Launchpad

This program helps students at The University of Texas at Dallas succeed in entrepreneurship and in their careers. Open to all students and recent alumni in all majors, the campus-based Blackstone LaunchPad & Techstars enables participants to access mentoring, grow their network, and access the resources to accelerate the success of their business.

The Blackstone LaunchPad & Techstars network helps students succeed in entrepreneurship and in their careers. Accessible to over 700,000 college students per year, LaunchPad is designed to complement existing entrepreneurship on-campus resources, coordinate off-campus opportunities for schools and students to expand their reach, and convene students and staff at world-class

events leveraging our industry-leading mentors, networks, and brands.

Blackstone LaunchPad & Techstars was developed as part of the Blackstone Charitable Foundation's Entrepreneurship Initiative, modeled after a successful program at the University of Miami, and is powered by Techstars, the worldwide network that helps entrepreneurs succeed.

UT Dallas Seed Fund

The UT Dallas Seed Fund is an academic program established by the Institute for Innovation and Entrepreneurship in 2016 and has supported startups with early stage funding typically $50,000 for a startup once a year.

UT Dallas Design Makerspace

The University of Texas-Dallas campus holds a maker space where faculty and students are able to collaborate on ideas and share knowledge. The maker space contains wood working tools such as miter saws, scroll saws, and more. The maker space also has fabrication tools, a 3D printer, and a soldering station. This space is also used for workshops. Workshops are held at the venue often; but due to the covid-19 pandemic, it has dwindled. Meetups can occur at the maker space as well.

Venture Development Center

The University of Texas at Dallas Venture Development Center (VDC) is a leading incubator designed to help students, faculty, and alumni commercialize their ideas and inventions. It does this through targeted, one-to-one support enabling ventures to identify and build relationships with the key industry partners, investors, entrepreneurs, and others to help them succeed.

The VDC offers 20,000 square-feet of state-of-the-art dedicated office, lab, and meeting space, as well as shared business services and equipment. They help companies build out their teams, protect their Intellectual Property (IP), and connect with expert advisors. They work to get minimum viable products (MVPs) into customers' hands and prove a viable, scalable business model.

For startups, they help founders identify investor options, write and rehearse their company pitch, and navigate funding terms and other questions.
For **researchers** and **inventors** seeking industry connections for translational research and licensing, they help identify partners, make introductions, and identify commercialization paths, including licensing agreements, IP assignments, and other contracting alternatives.
While Richardson is the main campus for the University of Texas-Dallas, many of their programs are located outside of the city. For example: Backlot, located in Fort Worth is a space for independent creatives to work in a collaborative environment. It provides access during typical office hours, or a 24/7 dedicated space, access to studio space for video or photography, gear rental, and space to meet with partners or clients in a professional setting.

Collide Village in nearby Addison is a milestone-driven accelerator where we invest in startups, teach them about evidence-based entrepreneurship, and help founders brand and commercialize their innovations.

The University has also created an Office of Technology Commercialization in 2008 and also a master's degree in entrepreneurship that began in 2010.

Other Entrepreneurial Tools

While the University of Texas-Dallas provides a significant role in support of business and entrepreneurial ecosystem, there are dozens of private and public operated accelerators and incubators.

As previously identified, the City and the Chamber of Commerce created the initiative the Richardson Economic Development Partnership. It is dedicated to building a vibrant and thriving local economy by attracting new investment and jobs through marketing and recruitment efforts and by working with existing employers.

Tech Titans as a private non-profit organization continues to provide a network to encourage and promote the technology ecosystem.

In addition, as expected in a major metropolitan area, there are a variety of tools and programs accessible to both established businesses and entrepreneurial start-ups.

Access to Capital

There are well over 100 venture capitalists located in the Dallas Fort Worth Metropolitan Area according to CrunchBase and Tracxn. Additionally, there are several venture capitalists located in Richardson, such as Hexa Global Ventures. Thus, entrepreneurs can find angel, pre-seed, seed, private equity, corporate venture in the area.

State Incentives

Texas is considered a very "business friendly" environment. With no state personal income tax and a relatively low cost of living, it is a perfect environment for companies to grow and startups to flourish. Texas also has many economic development incentives, specifically the following:

- Texas Enterprise Fund
  The Texas Enterprise Fund is the largest such fund in the US. This fund awards "deal-closing" grants to companies considering a new project for which one Texas site is competing with other out-of-state sites. The fund serves as a financial incentive for those companies whose projects would contribute significant capital investment and new employment opportunities to the state's economy.

- GURI
  The Governor's University Research Initiative grant program (GURI) has a goal to bring the best and brightest researchers in the world to the State of Texas. This program is a matching grant program to assist eligible institutions of higher education in recruiting distinguished researchers.
- Texas Enterprise Zone
  The Texas Enterprise Zone Program is an economic development tool for local communities to partner with the State of Texas to promote job creation and significant private investment that will assist economically distressed areas of the state.
- PDSBI
  The Product Development and Small Business Incubator Fund (PDSBI) is a revolving loan program financed through original bond issuances. The primary objective of the program is to aid in the development, production, and commercialization of new or improved products and to foster and stimulate small business in the State of Texas.

Infrastructure

Broadband

There are nearly 418 linear miles of fiber. There are three major carrier fiber optic trunk lines that operate in the city, they include AT&T, Verizon, and now T-Mobile (previously Sprint). Richardson claims it has "fiber-to-the -curb" in all residential areas. However, speeds differ for business versus residential. Although it does not appear that even though it is coined the Telecom Corridor, there are not significant internet speeds that has provided a locational advantage to attract business.

Data Centers

There are a variety of data centers in Richardson. They include the following:

- Terremark (Verizon) Richardson (Alma)Regional DC-04 MFA. The data center is a SAS 70 Type II certified facility. The data center has a centrally located command center manned by security personnel 24x7x365.

- Terremark – Richardson Regional DC-04 MFA. The data center is a SAS 70 Type II certified facility. The data center has a centrally located command center manned by security personnel 24x7x365.

- Verizon Business – Premium Data Center. The data center is SAS 70 type II Audited facility. The data center has security and monitoring system including onsite security guards, and video surveillance 24x7x365. The facility offers fully scalable bandwidth connections between company and customers' equipment at 1.544 Mbps (DS-1/T-1) through 2.5 GB (OCx/GB) levels. The facility has direct access to the Verizon's IP Backbone via data center primary Internet connectivity. Verizon offers either tiered or burstable primary circuits. Tiered bandwidth offers network connectivity with a maximum bandwidth limit, while burstable bandwidth provides usage based flexible connectivity.

- Databank – North Dallas Data Center DFW1

- Flexential- Richardson Data Center. This is the physically largest data center in Richardson at 450,000 square feet.

Labor Force

The Richardson Department of Economic Development recorded in 2017 that the two counties that comprise the Richardson Trade Market Area has a workforce of 1.71 million people. Skills of the labor force range from accounting to IT. Additionally, with several different companies headquartered in Richardson, it provides a variety of career path opportunities. It is identified that 80% of the area's labor force has 1-2 years college education. Also, 50% of the adult workforce (25 & over) has an undergrad or advanced degree.

DART Light Rail (Commuter) and Public Transportation (Bus Service)

The Dallas Area Rapid Transit is a modern public transportation system that includes a network of buses and light-rail lines that connect Richardson to 12 other cities in the metropolitan area. There are 4 light-rail stops in the city, and it will have 2 additional stops by 2022; one of which be on a line that connects directly to the DFW International Airport.

DART has enabled Richardson to attract and attain large employers to the city as employees have the opportunity to enjoy an extensive public transportation system.

Additionally, Richardson is also served by the DART bus service.

Quality of Life

Richardson has a relatively low cost of living with housing expenses being 12% lower than the national average according to the State of Texas. It has highly rated school districts and a plethora of respected research universities within the state and Richardson. The quality of life is good as Richardson also has several sites for parks and recreational activities.

Summary

Richardson provides a great example of a city that has fully committed itself into becoming a community focused on a singular goal and sticking to it.

Driven by a group of property owners and businesses in the early 1950's, it has been dedicated to the focus for nearly 70 years.

Starting with the allocation of over 40% of the available land zoned for commercial development, it has allowed for available and inexpensive land in the effort to continuously attract new businesses.

With constant investment in infrastructure and proximity to Dallas, the City was able to grow. The growth took it from being known as the "Electric Suburb" to the "Telecom Corridor" and now with a new moniker, the "Innovation Quarter."

Continued Growth: The Innovation Quarter

The City in 2017 began a visioning and strategic planning process in an effort to rebrand the City. The City rezoned 1,200 acres in the heart of the City and entitled it "The Richardson Innovation Quarter," aka the IQ. It is being rezoned for the City's living laboratory. The purpose is to move its economy to new technological heights.

City officials call it a place where "tech companies large and small can develop both commercial and top-secret telecommunication technologies." The IQ has been given the tagline of "Live. Work. Invent," building upon the city's reputation as a center for innovation and entrepreneurship.

The IQ is best described as an innovation center that focuses on tech startups and collaborations. It is estimated that the 1,200 acres will employ over 19,000 engaged in innovations and inventions. The IQ is on pace to be fully open by February of 2022.

The IQ will have multiple maker spaces and components of a tech hub ecosystem. The city and UT Dallas announced the construction of a 27,500-square-foot facility. UT Dallas plans to take 10,000 square feet, and 3,000 square feet will be dedicated for programs, networking, and event space that will hopefully attract entrepreneurial and startup activity to the area's thriving tech business ecosystem.

UT-D will also place five newly created UT Dallas research centers plus an extension of the University's venture development center in the heart of the IQ. Those research centers will include:

- **The Center for Applied AI and Machine Learning**
  CAIML is a research and development center that partners with companies to apply cutting-edge artificial intelligence (AI) and machine-learning technologies to their products, services, and business processes.

- **The Center for Applied AI**
  CAIQUE *will showcase emergent applied AI research at the University to engage companies, win extramural grants, strengthen international partnerships, and provide a global network for affiliated students to excel as UT Dallas alumni.*

- **The Center for Imaging and Surgical Innovation**
  CISI brings together UT Dallas engineers and computer scientists and UT Southwestern investigators and clinicians to develop emergent technologies to improve patient care and human health. UT Dallas, UT Southwestern, and local companies will collaborate on translational research, clinical trials, and product commercialization.

- **The Multi-Scale Integrated Interactive Intelligent Sensing Center**
  MINTS will develop and deploy a cost-effective irrigation decision-support tool that uses super-resolution machine learning and remote sensing imagery from satellites, drones, and security cameras. The center will collaborate with municipalities, corporate campuses, and homeowners to save money by optimizing their irrigation water usage.

- **The Center for Smart and Connected Mobility**
  CSCM aims to provide advanced solutions for the engineering of smart, multimodal, cyber transportation infrastructures to improve safety and enhance mobility for drivers, bicyclists, and pedestrians. The center aims to pursue community engagement, strengthen industry collaboration, accelerate the transition of research into the real world, and advance education and outreach efforts.

The Centers will be under the umbrella of the University's new Centers for Emergent Novel Technology known as CENT-IQ.

The CENT-IQ will feature a series of co-working, research, and lab spaces. Each center will focus on solutions related to specific tech specialties, which will could include applied artificial intelligence, machine learning, imaging and surgical innovation, and smart mobility.

The remainder of CENT-IQ will be occupied by the City's Office of Strategic Initiatives which coordinates with partners to promote Richardson's 1,200-acre innovation district. Approximately 7,000 square feet will be set aside for future expansion or flexible partnering opportunities.

The innovation quarter will create upwards of 19,000 new jobs and attract more young talent from around Texas as Richardson has continued to live up to its reputation as a community built around its businesses.

So, is it possible for other cities follow Richardson's blueprint?

Richardson's economic success was rather unique. Prior to any public leadership or collaboration throughout the City, businesses relocated to Richardson due to the marketplace,

specifically because of its proximity to Dallas along with the availability and reasonable cost of land for development.

Additionally, it was fortuitous that the University of Texas system identified Richardson as the location for University of Texas-Dallas.

With the location and growth of the university has come the technology and entrepreneurial ecosystems such as accelerators, incubators, meetups, mashup, co-working, and makerspace. These are the very elements needed to grow, sustain, and continue the vision as a tech hub in Richardson.

Resources

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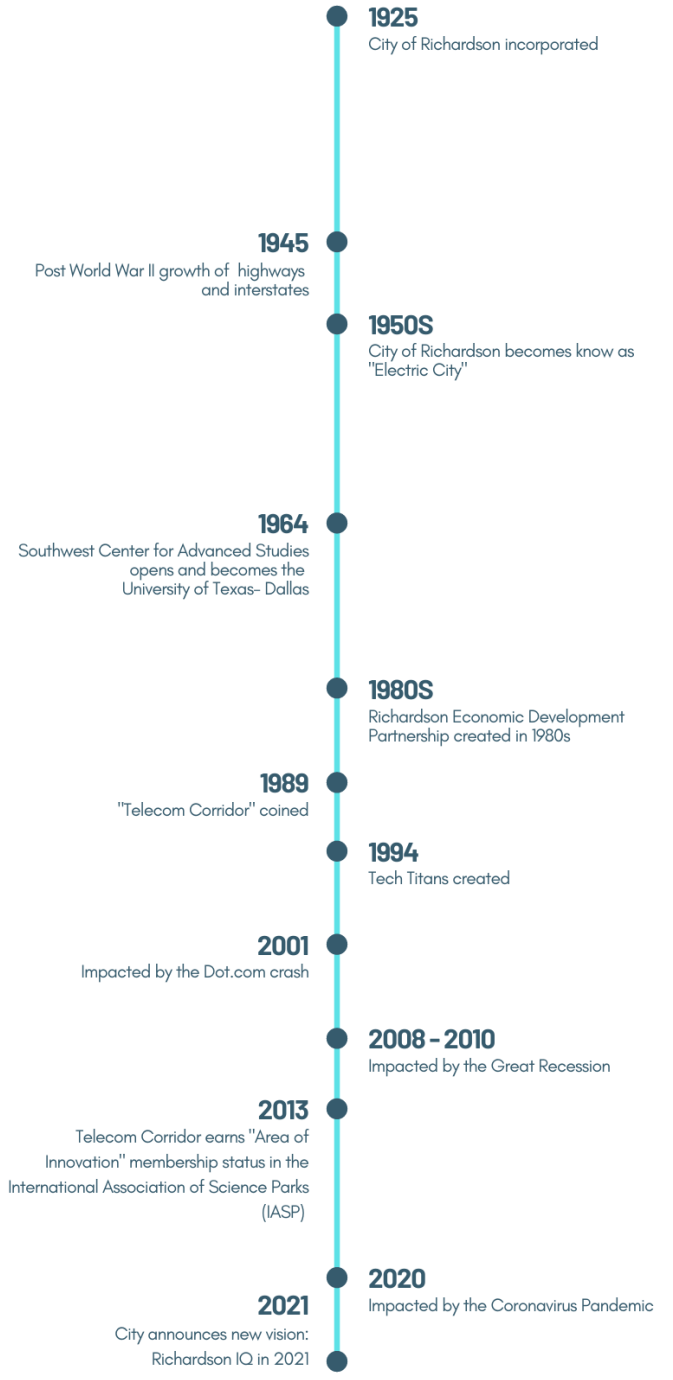
A BRIEF ECONOMIC HISTORY OF
RICHARDSON, TEXAS

"THE BEST PLACE IN TH U.S. TO LOCATE YOUR TECHNOLOGY COMPANY"
CITY OF RICHARDSON



SOUTH SAN FRANCISCO, CALIFORNIA

Overview

How does a city of old meat packing plants, rusting steel mills, and dusty truck depots become the birthplace for biotechnology?

Despite being used interchangeably in conversation, the terms biotechnology and life sciences have scientific differences. Life sciences is the broad study of living organisms and life processes while biotechnology is the understanding of natural science to develop technologies to help improve lives and the health of the planet. In other words, life sciences is a diverse sector that includes biotechnology. In this sector, biotechnology applies the broad fundamentals of life sciences to create processes resulting in products.

The life sciences sector includes:

- Research, testing, and medical laboratories which consist of firms engaged in the R&D of biotechnology and other life sciences.
- Drugs and pharmaceutical firms that develop and produce biological and medicinal products and manufacture pharmaceuticals and diagnostic substances.
- Medical devices and equipment firms that develop and manufacture surgical and medical instruments and supplies, laboratory equipment, electromedical apparatus including MRI and ultrasound equipment, dental equipment, and supplies.
- Agricultural and Industrial Biotechnology firms engaged in agricultural processing, organic chemical manufacturing, and fertilizer manufacturing. The sub-sector includes industry activity in the production of ethanol and other biofuels.
- Biotechnology Distribution companies that coordinate the delivery of related products spanning pharmaceuticals, medical devices, and agricultural biotechnology. Distribution is unique in its deployment of specialized technologies including cold storage, highly regulated monitoring and tracking, and automated drug distribution systems.

Understanding the dimensions of the sector, simply, it is large. In 2020, the United States accounted for 48.2 percent of the global life sciences market. Nationally, there are nearly two million people directly employed in the sector. In Washington there are nearly 36,000 employed, with nearly 70 percent of the total located in the Seattle region. In 2020, the Seattle metro area ranked #9 in a CBRE Life Sciences report for the largest life sciences market in the United States.

The table below illustrates the number of employees, firms, and wages in life sciences illustrated by the categories of national, state, and regional totals.

LIFE SCIENCE EMPLOYMENT, FIRMS & WAGES

	Employment	Firms	Avg. Salary
US	1.87M	101,143	$107,000
WA State	35,914	1,144	$93,146
Seattle	24,320	959	$95,981

Sources: US - BIO Economic Report 2020
WA State - Life Science Washington Economic Impact Report 2019
Seattle - City of Seattle

Nationally some of the major companies in life sciences and biotechnology include Amgen, AbbVie, Biogen, Bristol-Myers, Eli Lily, Gilead Sciences, Johnson & Johnson, Pfizer, and Vertex. Notable companies that have facilities in the Greater Seattle area include NovoNordisk, Celgene, and Bio-Rad Laboratories. Those that are well recognized in the Seattle area include Seagen, Juno Therapeutics, and Nanostring Tech.

As of 2020, there are about 2,650 biotechnology firms in the United States according to research firm IBIS World.

This research has identified major cities across the globe and the nation that have created successful business environments for growing life sciences and biotechnology ecosystems. These cities are presented in the following chart:

MAJOR LIFE SCIENCES & BIOTECHNOLOGY HUBS, UNITED STATES

Denver-Boulder	**San Francisco**
San Diego	**Washington, D.C.**
Raleigh-Durham	**New Jersey**
Philadelphia	**New York City**
Seattle	**Los Angeles**
Chicago	**Orange County, CA**
Boston-Cambridge	

Source: CBRE US

Globally, the life science & biotechnology industry concentrates heavily in a few cities, specifically those that are near to research universities, national laboratories, and

a significant concentration of venture capital firms. These include San Francisco, Boston, New York/New Jersey, San Diego, and London.

However, research screened cities based upon similarities to Federal Way. For example, the screening involved the following: Population size (100,000 - 250,000), a suburb located in a metropolitan area, no university within the city limits yet close proximity to a major research university, adjacent to an interstate, age of the city, successful entrepreneurial ecosystem, metrics of success such as jobs created, and investments made.

The research identified South San Francisco, California, which is located in the San Francisco Bay Area. South San Francisco was chosen as a case study for biotechnology due to its unique existing conditions and tech-focused public policies that created a business-friendly environment. This environment ultimately transformed the City into a host for global pioneers and leaders in the biotechnology sector. Additionally, it was selected based upon similarities to Federal Way such as population, location in a metropolitan area, and proximity to a major research university.

SCREENING PROCESS

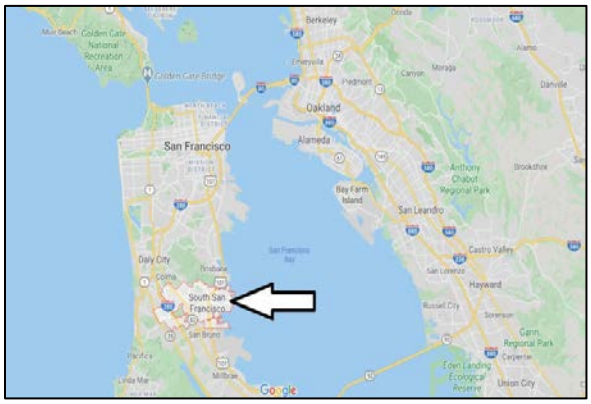
- **Population Size (100,000-250,000)**
- **Suburb to Metropolitan Area**
- **Adjacent to Interstate**
- **No University, however, close proximity to one**
- **Age of City compared to Federal Way**
- **Maturity of Technology & Entrepreneurial Ecosystem**
- **Jobs Created & Investments made**

Taking these individual filters into account, the most notable similarities to Federal Way and South San Francisco is their distance to a major metropolitan Area (Seattle and San Francisco) and major research universities (UW, Stanford/UC Berkley). This proximity allows the City to access the talent pools within these major cities and benefit from nearby business and technology ecosystems.

However, there is a slight difference between the two cities. South San Francisco has access to significantly more venture capital. According to the tech focused website CrunchBase, the state of California led the nation in venture capital investments with a total of $84.3 billion in 2020, while Washington had a total of 4.4 billion. Nonetheless, Washington remains a major state for venture capital investment. It is also worthy to note that Washington State has seen a high percentage of growth in venture capital investment. From 2016 to 2020, venture capital investment in Washington grew 170.2% compared to California's growth of 119.2%.

This case study details and examines actions undertaken by South San Francisco by examining its history to identify the policies and events implemented that accelerated life sciences and biotechnology business growth.

South San Francisco, California



South San Francisco is located ten miles south of San Francisco and 30 miles northwest of Silicon Valley. It is located on the western shore of the San Francisco Bay, in northern San Mateo County, and is within the San Francisco-Redwood City-South San Francisco Statistical Area of about 1.65 million in population.

The City has a population of 67,789 as of 2019 and is 30.17 square miles in size.

The San Francisco Bay Area is the second-largest life sciences hub in the world only behind Boston. The City hosts a significant segment of the hub as it was first to plan for biotechnology growth mainly due to the location of Genentech which acted as the pioneer and subsequent magnetic for others to grow there.

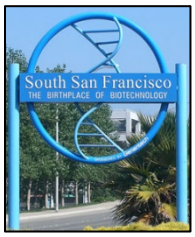
Today, there are 35,247 employees that work within the city limits.

MAJOR COMPANIES, SOUTH SAN FRANCISCO

- **Genentech**
- **Merck**
- **AstraZeneca**
- **Roche**
- **Pfizer**
- **Amgen**

The community prides itself on being at the forefront of the biotechnology industry. The City's Economic Development Office states, "We are the first city in the nation to plan for biotechnology growth." As such, they refer to themselves as "The Birthplace of Biotechnology," and the hub of the biotechnology industry. They have also branded themselves

using the tagline "94080: America's Biotechnology Zip Code." Suffice it to say, no matter how they brand themselves, it is a thriving biotechnology ecosystem.



"The Birthplace of Biotechnology"

94080: America's Biotechnology Zip Code

Source: City of South San Francisco

Source: Curbed SF

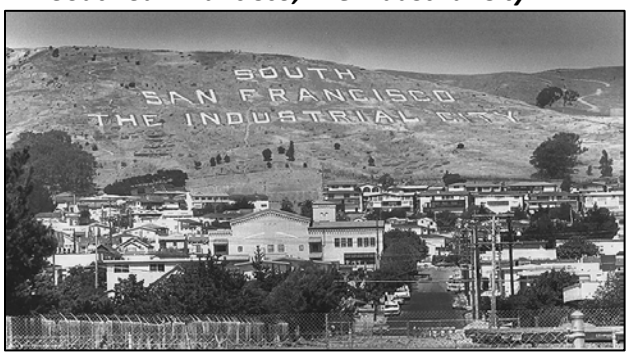
Today, with over 200 biotechnology companies, the City has a diverse group within the industry. For example, 61% of the city's biotechnology companies specialize in biopharmaceutical research and development (R&D), e.g., vaccines, allergenic products, gene therapies, etc.). Additionally, the City is home to the nation's largest cluster in R&D for biotechnology, with over 11.5 million square feet currently in operation and nearly four million square feet approved or under construction.

Additionally, the concentration of biotechnology firms in the City is rapidly growing, shown by a 70% increase in life science space in the last five years.

Economic History

Things were not always focused on biotechnology. The City began in 1908. Its economy was comprised of manufacturers including a large shipbuilding industry. Growth continued with a considerable portion of land primarily devoted for manufacturing. In a 1928 special election, voters approved to fund the construction of a sign on San Bruno Mountain proclaiming South San Francisco as the Industrial City.

"South San Francisco, The Industrial City"



Source: City of South San Francisco

However, even with the sign, it was not a warm and welcoming city. South San Francisco was simply known as an industrial area east of US 101. The newspaper, the *San Francisco Chronicle* described the City before biotechnology as "old meat packing plants, rusting steel mills and dusty truck depots."

But that all began to change 68 years later. In 1976, Professor of biochemistry Herbert Boyer from the University of California, San Francisco (UCSF) and venture capitalist Robert Swanson founded Genentech. They began in an inexpensive 2,000 square feet of rented office space. In doing so, they launched the biotechnology industry in the City.

Organic growth continued due to inexpensive land and buildings. The biotechnology industry capitalized on the location of the City being close to the research and development infrastructure at nearby research universities and federal research facilities.

The innovation in biotechnology's early years was done by a handful of brilliant scientists at the world-class research institutes such as Stanford and the University of California campuses of San Francisco and Berkeley (UCSF & UCB).

Growth for the industry exploded when the federal government declared a war on cancer in the early 70's, which funded researchers pushing the frontiers of genetic engineering.

Concurrently, during the 1970's and 1980's, biotechnology specifically in Northern California experienced tremendous growth. By 1987, the Bay Area's biotechnology industry had grown to 112 companies, supporting over 19,000 jobs and revenues of $2 billion. Between 1987-1990, 81 new biotechnology companies were formed alone. This rapid expansion was subsequently stimulated by the growth of research and laboratory facilities at Stanford. Through these investments the City worked to claimed its share of these expansions.

These efforts were captured in a city publication entitled, "25 Years of Achievement: Biotechnology Industry Report." The City had a few early advantages that helped it transform. Most importantly, company access to researchers and students from Stanford, UCSF, and UC Berkeley provided significant benefits. Then coupled with the presence of industry leader Genentech, it provided a major value-add component in identifying the City as the center of a cluster that encourages collaboration.

Additionally, the report memorialized the strong relationship with biotech companies specifically working with the city officials engaged in administration, planning, and building departments. Lastly, the report identified the proximity to a sophisticated group of venture capitalists in the Bay Area, familiar with biotechnology innovation that invested heavily in the industry sector as the key factors for success.

Even with those advantages, the City had challenges that limited the growth of the biotech companies. These included the lack of employees and affordable commercial space. Companies looking for bioprocessing and

manufacturing space were often out of luck. Additionally, business taxes and public infrastructure costs continued to be impediments to site selection in the area. The City also had congested streets due to heavy traffic.

However, strong public private relationships were formulated and sustained that ensured constant and consistent positive outcomes. The combination of the first biotechnology company- Genentech coupled with the world-renowned research performed at Stanford and UCSF, and UC Berkeley attracted companies looking to follow in Genentech's footsteps. Peter Pellerito, a consultant for the Bay Area Bioscience Center in 1998 stated, "In biotech, South San Francisco is a special place for this industry because that's where the biotech firm Genentech was born."

Today, because of efforts of both the public and the private sectors, South San Francisco is home to an unequaled biotechnology ecosystem. This ecosystem has propelled the City into becoming a global hub for biotechnology, illustrated by its current 200 biotechnology firms and 200,000 biotechnology employees.

Genentech and South San Francisco

As previously mentioned, in 1976 Genentech started its headquarters in a 2000 square foot warehouse with only five employees and belief that DNA technology could improve lives. Genentech transformed the City from "smokestack industries" to biotechnology. Its presence acted as a magnet to attract others to the City.

Genentech: South San Francisco

Today, the company is currently expanding and working closely with the City. In 2020, the City Council approved an expansion that would double the headquarters to 9 million square feet and add as many as 12,550 employees.

As part of the expansion, Genentech offered the City $30 million for an affordable housing program and $15 million for transportation initiatives.

Genentech presence has also provided valuable community support. For example, the company provides several scholarships to high school seniors - including two four-year scholarships of $50,000 per year to exceptional students to provide them the opportunity to pursue a degree in science, technology, engineering, and/or mathematics.

Genentech also funds local school district programs such as Futurelab which provides Science, Technology, Engineering, and Math (STEM) education for K-12 students in the South San Francisco Unified School. The curriculum improves college readiness and inspires students to pursue careers in STEM fields. Futurelab features unique programs for students such as: Gene Academy, Helix Cup, and Science Garage.

For example, the Science Garage is a fully functioning biotechnology laboratory complete with centrifuges, thermal cyclers, gel electrophoresis machines, and other equipment funded by Genentech Inc. The 6,900-square-foot lab, constructed with a $7.9 million investment from the Genentech Foundation, is being built at South San Francisco High School, but is also accessible to all students in the District.

Tech and Entrepreneurial Ecosystem: The Building of an Unequaled Ecosystem

The growth of the biotechnology sector was primarily stimulated by business locations. Companies slowly located to the City during the 1970's & 80's. However, by the 1990's, a series of private actions began with the location of business accelerators, incubators coupled with other entrepreneurial programs. Today, there are business accelerators, innovation centers, and co-working spaces all contributing to the ecosystem. The following provides an overview.

Associations

The California Life Sciences Association (CLSA) is located in South San Francisco. It is the premier trade association representing California life sciences and biotechnology. It is heavily involved with biotechnology in the City. For example, their work includes assisting biotechnology companies, research universities and institutes, and investors throughout the state, but also in South San Francisco. By acting as the custodian of the industry cluster, it cultivates the industry and helps facilitate innovation. The organization supports entrepreneurship, education, and career development.

Business Accelerators and Incubators

The impact of the incubators and accelerators in the City has been significant. Business incubators provide the services for startups and entrepreneurs needed for success. With more than 200 active biotechnology companies in the City, many are startups. The thriving environment has created companies like Calcio Life Sciences and Freenome. Freenome was founded in 2014. The company detects and

treats diseases using machine learning. The firm closed with $270 million in Series C funding for early cancer detection, financing a total of over $500 million since its launch. The startup's funding was made possible by being incubated by StartX.

StartX is a non-profit community of serial entrepreneurs, industry experts, tenured Stanford professors, and well-funded growth-stage startups. They help companies hire elite talent, secure funding, and tap into one of the most innovative networks in the world - the Stanford University Alumni Network. Their cohort of companies have a combined valuation of $26B and have started 700+ companies with an average funding of $11M. Although StartX is located in Palo Alto, its presence is vital to the local biotechnology ecosystem.

CARB-X is an accelerator network focused on antibiotic resistance and is one of the largest public-private partnerships dedicated to anti-bacterial innovation. The accelerator is investing over $500M in the development of antibiotics and has a close partnership with CSLI.

JLABS @ SSF is an incubator operated by Johnson & Johnson which opened in 2015. The incubator offers operational support, educational programs, and can accommodate 30-40 companies concurrently.

JLABS takes investment out of the equation, eliminating the setup time typically required of a startup. JLABS member benefits also include access to entrepreneurial programs, vital industry connections, partner network opportunities, and mentorship.

Life Sciences Conferences

The City is home to the South San Francisco Conference Center (built 1993), it is a prime location to host biotechnology events in the Bay Area.

Conveniently located minutes from San Francisco Airport (SFO), near transportation and hotels, and a few short miles from downtown, South San Francisco Conference Center is easily accessible. The conference center and event venue offer flexible meeting spaces for conferences, corporate events, and special meetings. The 20,500 square foot flexible space has 13 meeting rooms that can host meetings of up to 1200 attendees.

South San Francisco Conference Center
Source: City of South San Francisco

Innovation Centers

There are several innovation centers in the City. For example, there is Johnson & Johnson West NA Innovation Center. Its focus advances early-stage innovation, providing essential resources to meet the needs of life science entrepreneurs. The Center provides four benefits:

- A team of experts
- Unique collaboration
- Tools
- Guidance

The Center offers access to JLAB sites, business incubators, and training in business development.

Additionally, Eli Lilly has an innovation center or lab. In 2019, Eli Lilly unveiled its first shared innovation lab located in the City and designed to speed discovery of medicines through collaboration with local biotechnology companies. Lilly Gateway Labs offer companies direct access to Lilly's scientists, team members, and executives, as well as exposure to scientific and functional expertise. With over 65,000 square feet of space, Lilly Gateway Labs offers 32 state-of-the-art, flexibly designed private lab modules. The space was designed for collaboration and meant to create new companies in the City.

Collaborative and Co Working Space

Business infrastructure and co-working spaces are also vital to foster cross-fertilization between firms. JLABS @ SSF also offers a shared lab and co-working space. The shared space is a non-competitive cooperative climate. This state-of-the-art 30,000- square-foot facility provides startups and entrepreneurs a combination of shared and private specialized lab facilities.

Throughout the City entrepreneurs and startups will also find many different available forms of co-working spaces. For instance, Bonneville Labs, JLL, and Smart Labs also offer lab space to collaborate and create. For example, Airport Boulevard Business Park offers more than 52,000 square feet of commercial real estate space in four buildings in a prime South San Francisco location. Additionally, high quality lab space is widely available for biotechnology startups to utilize.

Diverse Group of Companies

The allure of collaboration has attracted many companies and expansions in the City over the last few years. As a result, the size of the cluster has increased, and with its numerous high paying jobs have been brought to the City.

For example, Horizon Therapeutics announced the opening of its new office in SSF, a 20,000 square-foot facility. A company official stated the location of the City and its ecosystem allows them the opportunity to connect to newer technologies, leading academic institutions, and the dynamic scientific area at large.

Recently, Amgen moved into a nine-story R&D facility. It provides 240,000 square feet of space. According to a company official the new location will foster even greater collaboration across our strong scientific team, accelerate the R&D process, and provide a venue for increased engagement with the Bay Area's abundant scientific and educational communities.

Merck's New Regional Hub
Source: City of South San Francisco

Merck built in 2019, employs 300 people at their new regional hub in the City, and there's room to grow – the space has capacity for nearly 500 in total. The site is a research center set up to foster collaborations across businesses, functions, and even industries. Scientists are working with business developers, and the company will work with collaborators from academia and industry.

Real Estate

The City has always been concerned about limited lab space and real estate given the City's geography.

Thus, the City partnered with four Real Estate Investment Trust Companies to address the supply issue, sparking the climate for more growth. Recently, the problem was finding space for start-up companies since vacancy rates were around two percent or less pre-COVID.

The Real Estate Trusts responded to the shortage by proposing new developments and receiving new entitlements for projects that would add five million square feet in the City. The recent South San Francisco projects include the following: Genentech's campus expansion, BioMed Realty Trust's 1.25 million square foot Pacific Gateway campus, 2.25 million square foot waterfront campus at Oyster Point Marina; HCP Investment's 884,000 square foot Britannia Cove campus (now under construction), and Alexandria's 291,000 square feet of office and lab space.

Oyster Point

The Cove at Oyster Point is a massive property owned by Kilroy Realty. The development will comprise 2.5 million square feet of lab and office space for biotechnology firms. The biotech- Cytokinetics Inc. was the first to announce its location at the site. It will pay $65.40 per square foot for 234,892 square feet of space.

Rendering of proposed Oyster Point
Source: Biz Journals

Astra Zeneca's new location—at the Cove at Oyster Point—is the new workplace for around 400 employees gathered from AZ's TIDE (Technology Innovation & Delivery Excellence) unit, as well as subsidiaries Acerta Pharma, MedImmune, and Pearl Therapeutics.
Oyster Point symbolizes the biotechnology boom in the City. In the last five years, life sciences space has grown more than 70 percent.

Gateway of Pacific

Near the entrance of the City, just off Highway 101, Gateway of Pacific aims to become a new landmark life sciences cluster in the biotechnology hotbed, according to real estate firm BioMed Realty. The Gateway of Pacific

campus is set to include 2.2 million square feet encompassing ten buildings, to be finished in late 2021.

The development includes an amenity referred to as "Central Park," an environment for connection and collaboration outside of the office. The Park is uninterrupted by vehicles and offers a variety of experiences on ±5.5 acres. It provides space for interaction to areas for quiet contemplation.

Location of The Gateway of Pacific Campus
Source: City of South San Francisco

AbbVie was the first big firm to reserve space at the Gateway of Pacific, claiming 480,000 square feet for to build a central spot for its oncology business. Several of its neighbors include Calico, CytomX and Alector. AbbVie renewed its $1.5 billion aging-focused research collaboration with nearby Calico, which has grown to over 150 employees in SSF since being backed by Google.

Labor Force and Educational Institutions
Biotechnology is by far the largest sector for employment in the city. In 2016, there were 35,247 people employed in South San Francisco with over 20,000 working directly in the biotechnology industry. The sector continues to grow as employment has increased by over 25% since 2013.

Given this growth, education plays an important part in the economy. It is deceiving as the numbers of residents of South San Francisco do not reflect the educational requirements of the workforce of a biotechnology-based economy. For example, only 36% of the residents in the workforce have a bachelor's degree or higher. This may be due to the transitory nature of the labor force or location of the City's labor force – meaning they may live in nearby communities such as Pacifica, San Mateo, Palo Alto, or Menlo Park.
Nonetheless, the City works closely with the region's smaller universities and community colleges to sustain and train a talented workforce. Nearby are San Francisco State University, University of San Francisco, and three 2-year colleges: City College of San Francisco, Skyline, and College of San Mateo.

For example, the College of Business at San Francisco State University, in partnership with the City of South San Francisco and the California Life Sciences Association (CLSA), launched an MBA program specifically for biotechnology professionals.

However, it maybe the proximity to prestigious research giants such as UC Berkeley and San Francisco along with Stanford University that make the difference. The City is 21.5 miles from UC Berkeley; 27.4 miles from Stanford University, and only 11.3 miles from UCSF. These attraction of graduates from these nationally rated institutions may provide the personnel needed to support the sector.

National Laboratories

There are 17 national research laboratories across the United States. Four of them are located in California, and all are within 50 miles of South San Francisco.

This proximity also provides a large, highly educated and experienced workforce. Opportunities for collaboration with top research scientists, and an influx of federal research dollars contribute to the growth of the sector.

The Lawrence Berkeley Lab has trained tens of thousands of university science and engineering students who are advancing technological innovations across the nation and globally. Sixty startups have been created using lab technology and 516 patents in the last ten years, helping to create 10,000+ jobs.

Lawrence Livermore National Laboratory has a mission of strengthening the United States' security through the development and application of world-class science and technology. In the fiscal year 2020, LLNL awarded more than $240 million in procurements to businesses in California.

The Stanford Linear Accelerator Center (SLAC)/National Accelerator Laboratory is a joint effort between U.S. Department of Energy and Stanford University. Over 2,700 scientists use the facilities annually, with four Nobel Prizes awarded to laureates using the SLAC resources.

Sandia National Laboratories generates nearly $1 billion in both direct and indirect economic output in the State of California with nearly half coming from the San Francisco Bay Area according to a report by the Center for Economic Development at California State University-Chico. Over 1,100 California employees work primarily at the two laboratory facilities; one at the Livermore site and the other at the Joint Bioenergy Institute (JBEI) in Emeryville.

Access to Capital

The Bay Area is a hotbed for venture capital, particularly for biotechnology. Life Sciences and biotechnology companies in from south San Francisco have raised 1.1B in venture capital over the last three years.

Additionally, the Bay Area has been ranked first for National Institute of Health (NIH) funding and in the number of patients. For example, the University of California San Francisco, the city's closest university partner, lead the state with $599M in NIH funding in 2018, reported by CLSA.

Local Government

The local government has played a vital role in facilitating growth. The City's economic development office stated that their top priority is to attract the largest and fastest-growing biotechnology companies.
Although this was not always the case. Former city manager Mike Wilson, recalls that in the early 70's, it took time for local officials to get accustomed to the new and growing industry.

One of the concerns was that biotechnology firms often used hazardous materials raising concerns and straining the relationship with the community. To address this issue, the City created industrial zones with no housing to ensure safety in hopes of mitigating any biohazard issues. The City also spent considerable time examining and amending zoning. Public policies helped create a clear distinction between industrial and residential areas. The City then discouraged non-biotechnology firms from locating in the permitted zones for biotechnology firms.

Additionally, the City had also been flexible with zoning laws doubling the maximum building height to accommodate biotechnology firms.

Currently, with a sustained life science/biotech ecosystem, the City's economic development team focuses on recruitment and retention activities. Recruitment is primarily focused thru digital campaigns using the internet and social media. The economic development team also assists businesses during the planning, permitting, construction, and opening of a new or expanding business. The unique identity of biotech provides a singular focus for the City.

Although singularly focused, the City works tirelessly to ensure a quality technology and entrepreneurial ecosystem. For example, the City partners with the San Mateo Small Business Development Center (SBDC) and the Bay Area Entrepreneur Center to make business resources and advising available and accessible.

These centers offer one-on-one advising with professionals, business training, and funding. The City implements a consistent and streamlined permitting process that provides ease and efficiency for companies that choose to locate their business to South San Francisco.

The City takes pride in having one of the fastest turnaround times for approving development projects in the state.

Finally, the City partners with companies, on a variety of activities to promote the industry, the city, and its residents.

Infrastructure & Transportation

One of the City's economic development goals is to serve the unique needs of the biotechnology cluster through city services and infrastructure.

The City understands the necessity of an effective transportation system and infrastructure for recruiting firms and employees. Therefore, the City announced they have invested over $300 million toward street improvements, water quality, and sewer delivery upgrades.
The City has also invested over $250 million dollars to build public improvements, from free shuttle service, to expanded bus routes, and partnerships with rideshare and bike rental programs.

The City is also currently embarking on an expanded Caltrain (heavy rail) Station. These improvements will benefit both the business climate and the quality of life for all residents and employees. Continuous improvements to the already existing infrastructure are prioritized providing plenty of options for travel.

Local transportation is superb to and from the City with numerous local options:

- SamTrans (bus)
- Caltrain (heavy rail)
- BART (light rail)
- San Francisco Bay Ferry
- Free Public Transportation in South San Francisco (shuttle)

SamTrans (San Mateo County Transit District) provides bus service throughout the San Mateo area with regular stops at San Francisco International Airport and downtown San Francisco. The service also operates BART (Bay Area Rapid Transit), a subway system that connects San Jose, San Francisco, and Oakland. Caltrain is a commuter heavy rail system that goes from Gilroy to San Francisco and runs through the Peninsula. There is also a free shuttle service which allows residents to move about the city free of charge or take them to a SamTrans.

Quality of Life

South San Francisco is an excellent place to live from a climate perspective. The City is often a preferred living location for many in the San Francisco Bay Area. The City has phenomenal weather with warmer temperatures and more sunny days than San Francisco but keeps the mild climate.

The area is also more affordable and family-friendly than most of the Bay Area and attracts many young families. It is close to world-class wineries, gorgeous beaches, and Olympic-style ski slopes.

However, the median home value is just north of $1 million and it also has a cost-of-living that is two times that of Federal Way.

Summary: Success Breeds Success

So, how did the City of South San Francisco succeed in establishing itself as the "Birthplace of Biotechnology," and as a global leader in this technology?

The San Francisco Bay Area has had a long history of producing innovative and forward-thinking entrepreneurs. In fact, the "Birthplace of Silicon Valley" is just 24 miles from the "Birthplace of Biotechnology," - South San Francisco.

The story of two Stanford graduates, Bill Hewlett and David Packard that founded the Hewlett-Packard Company (HP) in 1939 in a small one-car garage in Palo Alto, California making it the birthplace for the Silicon Valley.

That same spirit of entrepreneurship, but only for biotechnology, can be found in South San Francisco. Professor Herbert Boyer and venture capitalist Robert Swanson in the 1970's founded Genentech. Similar to the humble beginnings of the entrepreneurial spirit that birthed Silicon Valley, the entrepreneurial spirit that birthed the first biotechnology firm could not be contained in an inexpensive 2,000 square feet of rented office space in South San Francisco.

Thus, it appears that success was due to a combination of conditions and activities.

First and most notably, it was the founding of the biotechnology company – Genentech that acted as a magnetic helping attract others in the sector to South San Francisco.

Second, it was due to the region's long-standing spirit of entrepreneurship, producing innovation but also forward-thinking entrepreneurs. Again, people drawn to South San Francisco.

Third, the City's close proximity to educational and research institutions. This locational advantage specifically gave the City an early advantage in establishing itself as a global biotechnology hub. Proximity to world-class universities and federal research facilities that were early innovators in the biotechnology field, such as Stanford University, UC Berkley, and the UC San Francisco, provided access to a highly educated labor force. In fact, the founder of Genentech taught at the University of California, San Francisco, located just 11 miles from South San Francisco, highlighting the importance of a pipeline of educated workers plays in creating a Tech Hub.

Fourth, business-friendly public policies. The City of South San Francisco approved infrastructure investments needed to support the growing biotechnology industry. In addition, the City sustained the momentum from the founding of Genentech and the organic growth of companies by regularly engaging with companies. Today, the City actively recruits and sustains employees and firms through digital marketing campaigns that promote the City's services and continued investments into public infrastructure, transportation and quality of life.

The fifth and final factor is the City's proximity to a sophisticated group of venture capitalists. In 2020, California led the nation with a total of $84.3 billion in venture capital, according to Crunchbase. Today, Venture Capitalists continue to invest into mature biotechnology companies in South San Francisco.

The Bay Area's entrepreneurial spirit, powered by a highly educated workforce from nearby universities and research facilities, nurtured by sound local public policy and funded by nearby venture capital, created the perfect storm of economic activity that established South San Francisco as a global biotechnology hub.

Today, as a result of these investments and commitments, South San Francisco has over 200 biotechnology firms that employ over 20,000 in the sector.

Resources

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Karl Stickel, Director of Entr5epreneurship and Industry, Economic Development, City of Seattle

Marc Cummings, Life Science Washington

Didi Caldwell, President, Global Location Strategies

Bob Hess, Vice Chairman, Global Strategies, Newmark

Andy Shapiro, Biggins Lacy Shapiro

SONGDO, INCHEON, SOUTH KOREA

Overview

How did 1,500 acres of marshy tidal flats in the Yellow Sea develop into a growing smart city technology hub?

In recent years, the term "Smart City" has been a buzzword across the globe. While there is no exact definition of the term, there is a general consensus among industry leaders that smart cities are cities that leverage information communication technology (ICT) and data analysis to increase the efficiency and effectiveness of core systems that are central to a city's operation and development.

Cities can deploy smart city programs and services through individual projects, such as smart waste management systems, smart traffic control systems, and smart air quality sensory systems. Additionally, cities can scale up individual smart city programs and services by taking an integrated approach. In an integrated smart city approach, various projects are connected, and the majority of data and platforms are linked together.

Regardless of a city's approach, smart cities add digital intelligence to existing urban systems, making it possible to do more with less. Connected applications put real-time, transparent information into the hands of users to help them make better choices. These tools can save lives, prevent crime, and reduce the burden of disease. A smart city can save time, reduce waste, and even help boost social connectedness. When cities function more efficiently, they also become more productive places to do business and promote economic growth.

According to the consulting firm McKinsey, there are generally three elements to a successful smart city: A Technology Base; Smart Applications; and Adoption and Usage by Cities, Companies, and the Public.

First, the technology base includes a critical mass of sensors, or the Internet of Things (IoT), that are connected by a high-speed communication network. The IoT collects real-time data throughout the city that is then transmitted and analyzed to aid decision-making.

Second, smart applications act as essential tools that translate and analyze the collected data into alerts, insight, and action. Smart applications are primarily developed and maintained by private sector technology providers and app developers, making public-private partnership partnerships essential in a smart city's operations.

Thirdly, the widespread adoption and usage of applications by cities, companies, and the public is crucial. It is only through usage by the public that smart city technologies are able to influence changes in public behavior for increased efficiency.

THREE ELEMENTS TO A SUCCESSFUL SMART CITY

1. **Technology Base**
2. **Smart Applications**
3. **Adoption and Usage by Cities, Companies, and the Public**

Source: McKinsey Global Institute

While the concept of smart cities seems like a contemporary urbanism trend, the use of technology and data to improve city operations and development can be traced back as early as the 1970s specifically during an increase in urbanization.

The City of Los Angeles in 1974 created the first big urban data project. Shortly after, the City of Amsterdam launched De Digitale Stad, (DDS) in 1994. The DDS was an electronic social networking platform designed to bridge the gap between the City and its citizens. What started as a bulletin board system offering a menu with line numbers to continue to further pages, subsequently developed into a virtual digital city.

Despite these early uses of data and technology by cities, it was not until the early 2000's that the smart cities movement caught on and took off.

Supporting this growth were early innovators, including global companies like Cisco and IBM.

For example, Cisco dedicated $25 million to research smart cities, ultimately launching the Connected Urban Development program. The program included pilot programs with the cities of San Francisco, Amsterdam, and neighboring Seoul.

IBM formulated the Smarter Cities campaign, a comprehensive approach to helping cities run more efficiently, save money and resources, and improve the quality of life for citizens. Under this campaign, over 100 Smarter Cities Forums were held around the world.

Today, growth of smart cities continues around the world. The United Nations Department of Economic and Social Affairs projects that the present percentage of the world's urban population of 55.7% will increase to 68.4% by 2050. With increasing urbanization and populations, major cities are turning to innovative solutions in order to address urban challenges, such as traffic, crime, and air pollution.

Today, the global market for smart city technologies is valued at over $111 billion. Grand View Research forecasts that the global market size of technologies used for smart cities will reach $676 billion by 2028. The marketplace is estimated to achieve a compounded annual growth rate of 29.3% over the forecast period.

As the demand for smart cities and smart city services increases, it is projected that there will be an increased demand in several high-paying jobs, both in the public and private sectors. This includes tradition technical jobs, as well as new hybrid positions. According to the website TechRepublic, the following jobs are expected to see an increase in demand.

HIGH GROWTH JOBS IN SMART CITY SERVICES

Machine Learning Scientist
Data Scientist
Cybersecurity Analyst
Cloud Architect
Industrial Network Engineer
Alliance/Partnership Manager
Virtual Reality Specialist
Chief City Experience Officer
Network Reliability Engineer
Urban Informatics Analyst
Integration Engineer

Source: TechRepublic

While there may be a change in the marketplace, today, North America cities dominate the smart cities market and account for a revenue share of over 30%. This is in part due to the strong presence of key market players, as well as a well-developed information, communication, and technology infrastructure; the collaboration of federal and local governments with ICT vendors and start-ups; and the presence of prominent technology vendors.

For example, the Greater Seattle Area has companies engaged in smart city technologies such as Cisco, Ericsson, IBM, Itron, Microsoft, Oracle, Schneider Electric, Siemens AG, and Verizon. The region is also home to the Huawei Tech Research Center.

Moving forward, the Asia Pacific region is expected to emerge as the fastest-growing regional market from 2021 to 2028. Projected growth is attributed to the increased focus on the development of digital infrastructure in the region, especially in China and India. Growing urbanization and the rise in disposable income are also projected to further propel this market growth.

Screening Process

This research has identified leading cities across the globe and the nation that have successfully adopted smart city technology. The following city governments were ranked based on ten factors—vision of a smart city, leadership, budget, financial incentives, support programs, talent readiness, people centricity, innovations, smart policies, and track record. These cities are presented in the following:

2020 GLOBALLY LEADING SMART CITIES

Singapore	**Seoul**
London	**Barcelona**
Helsinki	**New York**
Montreal	**Shanghai**
Vienna	**Amsterdam**

Source: Statista

However, research screened cities and districts based upon similarities to Federal Way. For example, the screening involved the following: Population size (100,000 - 250,000); a suburb - located in a metropolitan area; no university within the city limits yet close proximity to a major research university; age of the city, successful entrepreneurial ecosystem; metrics of success such as jobs created and investments made.

The research identified the District of Songdo in Incheon, South Korea. Songdo was chosen as a case study for smart city technology due to their public infrastructure investments and strong public-private partnerships. Additionally, it was selected based upon similarities to Federal Way, such as location to a metropolitan area and proximity to major research universities.

SCREENING PROCESS

- **Population Size (100,000-250,000)**
- **Suburb to Metropolitan Area**
- **Adjacent to Interstate or Major Highway**
- **No University, however close proximity to one**
- **Age of City compared to Federal Way**
- **Maturity of Entrepreneurial Ecosystem**
- **Jobs Created & Investments Made**

Although Songdo is a district within a city, there are several similarities between Songdo and Federal Way. The most notable similarities between Federal Way and Songdo is their distance to a major metropolitan area (Seattle and Tacoma, and Seoul and Incheon) which allows both to access talent pools within these major cities as well as benefit from nearby business and technology ecosystems. Similarly, to Federal Way, Songdo is also within close proximity to an international airport—Songdo is located about 13 miles southeast from Incheon International Airport, while Federal Way is about 15 miles south of SeaTac International Airport. This allows multinational firms easy access to and from both cities.

However, there are a few differences. Songdo has the presence of university satellite campuses within city limits.

These include extended campuses of George Mason University, Ghent University, the University of Utah, and the State University of New York. Most recently, Stanford established the Stanford Center at the Incheon Global Campus, which will serve as a research center on smart, sustainable cities, and urban communities. In addition, Songdo is also home to satellite campuses of Yonsei University, Incheon National University, Inha University, Incheon Catholic University, and POSCO Global University. Despite these differences, the similarities are close as Federal Way will become home to a satellite campus of the University of Washington in the Fall of 2021.

Additionally, the District of Songdo has a larger population of 185,975 compared to the City of Federal Way's population of 98,340.

For the purposes of this case study, we will examine how the South Korean government successfully created a thriving smart city in the span of two short decades.

Songdo, Incheon, South Korea

Source: Gale International Press Kit (2015)

Songdo is a district of 185,975 people located in northwestern South Korea. The District is within the City of Incheon, South Korea's third largest metropolitan city by population. Songdo is located about 20 miles southwest of Seoul, a city of 9,588,711 people. Songdo is 2.2 square miles in size and is part of the Incheon Free Economic Zone (IFEZ).

The District was designed to have an "ideal mix of residential, cultural, business, retail, and recreational environments." There is 40 million square feet of commercial space, 25 million square feet of residential space, and ten million square feet of retail space. Additionally, there is five million square feet of hospitality space and ten million square feet of green public space.

Map of Songdo
Source: Songdo IBD

Population Growth of Songdo

YEAR	SONGDO POPULATION
2003	2,274
2004	2,215
2005	17,729
2006	21,114
2007	23,621
2008	27,395
2009	33,293
2010	41,425
2011	55,178
2012	61,608
2013	71,792
2014	86,002
2015	99,424
2016	112,814
2017	123,832
2018	139,580
2019	159,326
2020	185,975

Source: Incheon Free Economic Zone

The District is a $35 billion public-private development project was financed by the Gale International (61%), POSCO E&C (30%), and Morgan Stanley Real Estate (9%). The South Korean Government financed infrastructure that private sector partners built upon.

Songdo was developed on 1,500 acres of reclaimed land from the Yellow Sea off of the southwest coast of the City of Incheon. Prior to development, the area was home to a small fishing industry. The greater City of Incheon was a major industrial center primarily manufacturing iron, steel, coal, light, metals, plate glass, textiles, chemicals, and lumber. Today, Incheon's economy is led by manufacturing and transportation industries, accounting for 25.5% and 11.3% of the total economy, respectively.

The transformation of Incheon's economy started when 500 tons of sand was poured into the tidal flats in 2000. Shortly after reclamation, planning and development of Songdo began in 2001. Development was completed in 2015.

Land reclamation in Songdo.
Source: Dong-a Ilbo

Today, the District is a smart and sustainable city. The District is serviced by a ubiquitous sensor network, a fiber-optic communication network, and a Smart City Operation Center, all of which are essential elements to a smart city. Smart city services in Songdo are managed by the Incheon Smart City Corporation and the Incheon Free Economic Zone. Smart city programs in Songdo include transportation, crime prevention, disaster prevention, environment, and e-governance.

In terms of sustainability, Songdo is also the first urban development to earn pre-certification for the US Green Building Council's 'LEED for Cities' classification, with over 21 million square feet of LEED-certified space.

The District is served by the Incheon International Airport and the Port of Incheon. The Incheon Bridge, the world's tenth longest spanning cable-stayed bridge, connects the District to Incheon International Airport, located just 13 miles away.

Within the District, the Incheon Subway System connects to the Seoul Metropolitan Rapid Transit (SMRT) system and the National Railroad network, allowing for easy access to much of the Incheon and Seoul Metropolitan Area.

Today, Songdo is a home to multiple global smart city technology businesses, including Cisco and IBM. Songdo is also home to South Korea's largest startup support space, Incheon Startup Park, and the nation's leading startup accelerator, SparkLabs.

"Songdo, The Ultimate Smart and Sustainable City"
Source: NewCities

Additionally, Songdo is home to numerous international businesses and international agencies, including Daewoo International, Daifuku, Samsung Biologics, the United Nations Office for Sustainable Development, the World Bank Korea, Green Climate Fund, and the Global Green Growth Institute.

Songdo is also home to several university campuses. These include extended campuses of George Mason University, Ghent University, the University of Utah, and the State University of New York. Most recently, Stanford established the Stanford Center at the Incheon Global Campus, which will serve as a research center on smart, sustainable cities, and urban communities. In addition, Songdo is also home to satellite campuses of Yonsei University, Incheon National University, Inha University, Incheon Catholic University, and POSCO Global University.

Economic History: From Marshy Tidal Flats to the World's Smartest City

While the planning and development of Songdo primarily took place during a crucial time of smart city innovation in the early 2000s, Songdo's roots can be traced further back to the 1997 Asian Financial Crisis. In the aftermath of the global economic crises, the South Korean government began to explore ways to reduce the national economy's reliance on export-oriented manufacturing.

Meanwhile, the City of Incheon set out on a mission to develop and provide its citizens with a higher quality of life and more housing options through the development of a smart district. The District was initially to be an alternative to the outdated development of Incheon and neighboring Seoul. The City envisioned the ambitious development of Songdo on 1,500 acres of tidal flats along the Yellow Sea. Following the establishment of the globally-ranked Incheon International Airport in 2001, the City of Incheon began to orchestrate its development.

To bring this vision to life, in 2001, then Mayor Choi Ki-Sun recruited several US developers to visit the site. Of the visiting firms, New York City-based developer Gale International was selected to oversee the planning and development of Songdo, transforming the undeveloped marshy tidal flats into what is now claimed to be the world's "ultimate smart city."

In 2002, Gale International formed a joint venture with Korea-based construction firm POSCO E&C and recruited global practitioners KPF for planning and design, Arup for engineering and technology, and Cisco for smart city technology. Together, the private firms began to create the master plan that would bring Songdo to life.

Shortly after planning began, under the "Act on Designation and Operation of Free Economic Zones," the federal government established the nation's first free economic zone, the Incheon Free Economic Zone (IFEZ). In efforts to

transform the nation's economy, the IFEZ was established, consisting of Songdo and two other Districts in the Incheon Metropolitan Area. Also, the IFEZ includes the Incheon International Airport and the Port of Incheon. It was established to oversee the establishment and operationalization of a publicly operated ubiquitous smart "city," ultimately improving the business and residential environment within its three Districts. By doing so, the IFEZ aims to attract higher levels of foreign investment, specifically from the tech industry.

In accomplishing its mission, the IFEZ also provides a wide range of tax benefits, consulting services, and user-friendly government administrative services for businesses. International businesses are further incentivized by the relaxing of various regulations, such as labor laws.

Under the IFEZ, the law also established and designated the New Songdo International City, LLC. (NSIC) as a developer, marketer, and manager of residential projects, office buildings, commercial, educational, and public facilities in Songdo "International Business District."

As a result of this federal effort, the City's development plans for Songdo were incorporated into a larger regional strategy, ultimately coordinating and streamlining public efforts.

Soon, the federal and municipal government begin to heavily invest into the smart infrastructure of Songdo. This included the construction of the Incheon Bridge, the installation of a ubiquitous sensor network, and a fiber-optic communication network. Under the new establishment of the IFEZ, approximately $60 million was invested into the District's smart city infrastructure alone. These public investments laid the foundation for private sector developers to create the world's "ultimate smart city."

Sustainability and Green Growth

The development further benefitted from former President Lee Myung-Bak's effort to promote green and low-carbon growth in South Korea. In 2009, the National Assembly passed a $38 billion economic stimulus package for green investment. The investment amount increased to $83.6 billion spanning five years when the Framework Act for Low Carbon Green Growth was passed by Korea's National Assembly in 2010. Under the former President's initiative, Songdo was developed as a smart city with an increased emphasis on sustainability.

More than 40% of its area is reserved for green space, including the park of 40 hectares (100 acres), 26 kilometers (16 mi) of bicycling lanes, numerous charging stations for electric vehicles, and a waste collection system that eliminates the need for trash trucks. As identified previously, Songdo is also the first urban development to earn pre-certification for the US Green Building Council's 'LEED for Cities' classification, with over 21 million square feet of LEED-certified space.

The Incheon Smart City Corporation: IFEZ Smart City Operations

In 2011, the City Council of Incheon passed policy to establish the Incheon Smart City Corporation to operationalize the smart "city." The Corporation was jointly established with the City of Incheon and two private firms, KT and Cisco, in 2012.

The Corporation oversees the utilization of publicly mined data in the Incheon Free Economic Zone and combines it with the expertise and capital of the private sector to operate and manage the Incheon Free Economic Zone's smart city systems. This includes designing ICT development, operating smart city facilities, developing and providing private smart city services, and overseeing partnerships with the private sector.

Most importantly, the Corporation is tasked with developing smart city service models and exporting them overseas. For example, Colombian government officials worked with the Corporation in developing smart infrastructure in the cities of Valledupar and Villavicencio.

As the managing entity for the IFEZ' smart city systems, the Corporation provides services internally to the local governments in South Korea and externally to foreign governments and private firms looking to tap into South Korea's expertise and experience in developing smart cities. These services include:

- Consulting Services
- Ubiquitous System Design
- Ubiquitous System Implementation
- Ubiquitous System Operations
- Research and Development

The Corporation's Smart City Operations Center in Songdo is the heart of the IFEZ' integrated smart city system. The Center collects, stores, and processes various data and intelligent city infrastructure in order to provide smart city services. The Center has an integrated control system to check each corner of the city in real time. Real-time monitoring of data allows for the real-time decision-making in smart public safety, transportation, and environmental programs. The system also discovers and measures abnormal circumstances detected by the ubiquitous sensor network and processes civil complaints.

As of 2014, the Corporation employed 25 regular full-time employees and 33 contract workers.

The IFEZ' smart city technology system has been globally recognized for its cutting-edge technology. The International Data Corporation (IDC) recognized the IFEZ' smart city technology system as an excellent project in

urban administration. In 2017, the GeoSmart Asia Awards awarded the system with the grand prize in software selection using electronic maps and GIS.

Songdo's Smart City Technology Success

South Korea is not new to adopting smart city technology. In fact, the City of Seoul was one of the early adopters of smart city technology. Dating back to 2003, Seoul introduced advanced ICT technology, Bus Management Systems (BMS), and Global Positioning Systems (GPS). Today, Seoul is ranked the second leading smart city in the world, behind Singapore.

However, unlike the organic development of neighboring city Seoul and its adoption of smart city technology, it is important to recognize that Songdo was a centrally planned smart development. From the very beginning, the municipal and federal government in partnership with private-sector firms, planned and invested to make this new development a globally recognized smart city and high-technology hub.

But how did the local government attract these private sector firms? Wim Elfrink, former Cisco executive, stated that while selecting the company's next project, the IFEZ's unwavering support for the project in the form of investments into infrastructure and coordination of the elite team of developers sealed the deal. Following his praise of the efforts of the South Korean government, he further stated, "If you can't get it done there [Korea], we can't make it [a smart city] anywhere."

This investment into the District's infrastructure and strong public-private partnerships with industry leaders in planning, developing, and marketing Songdo has begun to pay off. Today, Songdo is a "living lab" for smart city technology companies and startups and is home to a young smart city technology ecosystem, consisting of industry leaders, startup accelerators, and research institutions.

Tech and Entrepreneurial Ecosystems

Coupled with significant investment into infrastructure and strong public-private partnerships, the following identifies how Songdo continued to grow resources and tools used to become a young smart city technology ecosystem.

Accelerators, Incubators, and Tech Transfer

SparkLab's Internet of Things Accelerator

The government's early investment into Songdo's technological infrastructure paid off quickly when Sparklabs, one of the leading incubators in Seoul's growing startup ecosystems, launched the Internet of Things (IoT) Accelerator in Songdo in 2015. Based out of Seoul and Songdo, the accelerator program is designed to leverage Songdo's technological infrastructure as a "test bed" for startups focused on enterprise and consumer connected devices, sensors, data analytics related to the Internet of Things space, platform plays, and wearables.

SparkLabs accelerator program provides a five-month mentorship program; funding; office space; a structure program; and access to a network of entrepreneurs, venture capitalists, angel investors, and executives.

When asked about the company's decision to expand into Songdo, SparkLabs co-founder HanJoo Lee responded, "Global leaders, such as Cisco and LG, and potential others are integrating cutting edge technologies into this smart city built from the ground up. It really provides a unique opportunity for companies to see how ubiquitous technology can be integrated into our lives without creating interference of distractions. Songdo is a giant petri dish for companies such as Google to new startups."

Today, there are 77 startups participating in the accelerator program.

Startup Coworking Space

Incheon Startup Park

In February 2021, South Korea's largest startup support space, Incheon Startup Park, opened in Songdo. Incheon Startup Park is a joint project between the Ministry of SMEs and Startups and Shinhan Financial Group. Shinhan Financial Group financed the project with over $46 million. Incheon Startup Park aims to foster startups by combining private experience, capabilities, and public support to create open innovation startup bases modeled after Silicon Valley in the United States and Zhongguancun in China.

Innovation Centers

Cisco Global Center for Intelligent Urbanization

The Cisco Global Center for Intelligent Urbanization was established in 2009 under a joint agreement between Cisco and Gale International. The company's center is focused solely on Intelligent Urbanization and drives innovation with new solutions. The innovation is part of Intelligent Urbanization, Cisco's global initiative to help cities use the network as the next utility to optimize their resources for long-term growth and quality of life. The Center also features a training facility that will utilize Cisco's Intelligent Urbanization Labs concept.

Rendering of Incheon Startup Park
Source: Incheon Startup Park

Research Centers
The Technology Research Center of Incheon Smart City Corporation conducts research and development projects related to smart cities. Research is focused on the Internet of Things (Iot), Cloud, Artificial Intelligence (AI), Big Data, and Unmanned Aerial Vehicle (UAV). In line with the mission of the Incheon Smart City Corporation, the Research Center works with local experts who are working on the expansion of smart cities, as well as national and international research and academic institutes.

Labor Force and Educational Institutions
Songdo is home extended campuses of George Mason University, Ghent University, the University of Utah, and the State University of New York. In addition, Songdo is also home to satellite campuses of Yonsei University, Incheon National University, Inha University, Incheon Catholic University, and POSCO Global University.

Most recently, in partnership with the IFEZ, Stanford University established its flagship research center in South Korea, the Stanford Center at Incheon Global Campus in Songdo. The Center conducts research in various fields, including smart city technology, urban environment, finance and economy, and smart city system integration and logistics. It also actively carries out mutual exchanges through joint projects with Korean universities, institutions, and businesses.

The presence of numerous prestigious educational institutions in Songdo continues to supply Songdo's tech ecosystem with a highly educated workforce.

Marketing Efforts: New Songdo International City Development, LLC: NSIC
As identified previously, the New Songdo International City Development, LLC. (NSIC) was established by Gale International in partnership with the Korea-based POSC E&C to develop, market, and manage residential projects, office buildings, commercial, educational, and public facilities in Songdo "International Business District." Today, NSIC continues to work with the IFEZ in promoting the District.

Public Policy – Local and Federal Incentives
The IFEZ, in tandem with the City of Incheon, provides tax benefits, business support, deregulation, and administrative support to incentivize foreign direct investment. This includes reductions in tariffs, financial and cash grants, exemption from certain labor regulations, efficient municipal administrative services, and consulting services.

Outcome: Smart City Technology Hub
Much of Songdo's success in creating a young smart city technology ecosystem appears to be attributed to the federal and municipal governments' early investments into the city's technological infrastructure. Considered one of the world's first specially designed smart cities, Songdo has attracted businesses, startups, accelerators, incubators, and research institutions with its unparalleled ubiquitous sensor network and fiber-optic communication network.

As a result of forward-thinking and with strategic investments in partnership with private-sector partners, Songdo has become a "living lab" for startups, businesses, and educational institutions in the District.

Legal Troubles
During the research, a legal dispute had been discovered between the public and private sector. Despite continued early indicators of economic success, the public-private partnership that brought the District of Songdo had run into rough waters, drawing negative media attention.

In June 2019, Gale International sued its partner, POSCO E&C, following years of project delays over conflict regarding profit and cost allocation.

The Board of Audit and Inspection of the South Korean government took issue with the joint venture's failure to attract overseas investment in 2010 and Gale's improper receipt of $179 million through dividends. Based on this, the Incheon Metropolitan Government ordered NSIC to sell its land back to the City. Last year, POSCO E&C split with Gale, taking on the entirety of the project financing loans.

Gale also filed arbitration seeking $2 billion in damages against the South Korean government. Gale's attorney asserted that "South Korea expropriated a substantial portion of Gale's investment in an unlawful manner," adding that "South Korea caused Gale more than $2 billion (US) in damages."

The Korean government believed that Gale had decided to take legal action against the Korean government as its dispute with POSCO E&C over profit sharing remained unresolved. This raises the question, has the District succeeded in what it set out to do?

Summary
Songdo's development was initiated by government, specifically the South Korean government in concert with the City of Incheon.

The outcome has received international recognition as a leading smart and sustainable "city."

In the effort to achieve this recognition, the federal and municipal governments set out several goals.

First, the City of Incheon embarked on the development project with the goal of providing a higher quality of life to its citizens. Ideally, Songdo was to be an alternative district to live and work in, in comparison to older developments in Incheon and neighboring Seoul.

Second, in the aftermath of the 1997 Asian Financial Crisis, the federal government incorporated the development of Songdo into its strategy to diversify the nation's economy. The short-term goal was to develop a successful smart "city" along with a development model that could be exported. The long-term goal was to leverage the smart "city" to attract businesses, ultimately developing a high-technology industry in Songdo.

Two decades later, it appears that the South Korean government has achieved its ambitious goals.

What started as 1,500 acres of undeveloped tidelands, is now home to over 185,000 residents, over 1,600 domestic and global companies, and sophisticated technology and entrepreneurial ecosystems.

In achieving these goals, it appears that there were a combination of conditions and activities that transformed the District.

It is important to first note Songdo's close proximity to the City of Seoul, and the important role it played in the District's development as a smart "city." Seoul has been an early adopter of smart city technology in the early 2000s. The early hands-on experience of Seoul in adopting and deploying smart city technologies was a strong foundation for Songdo to develop into a successful master planned smart city.

Correspondingly, there were activities that took place to attract residents, recruit businesses, and create a strong and sophisticated technological and entrepreneurial ecosystem.

First, it was the municipal and federal government's coordinated investment into Songdo's high-tech infrastructure which laid the foundation for the District's success. The City of Incheon and the IFEZ invested heavily into the smart city infrastructure that would create the envisioned smart city. Additionally, the federal government funded much of the District's sustainable development and green growth efforts. These investments appealed to potential residents looking for convenience and a high quality of life. Also, it appealed to businesses looking to leverage the District's smart city infrastructure for increased operational efficiency.

Second, strong public-private partnerships were consistently leveraged to create a technology startup ecosystem in Songdo to support South Korea's hotbed of technology startups.

The nearby Seoul Metropolitan Area has recently become home to a thriving technology startup scene, with over ten unicorn startups, each worth over $1 billion. Thus, watching what happened in Seoul through the power of startups to generate jobs, the South Korean government utilized public-private partnerships to build a technology startup ecosystem in nearby Songdo. In doing so, the government created the environment for technology startups to thrive, while continuing to diversify the national economy. Today, the federal government and IFEZ continuously works to create public-private partnerships with essential players in technology and entrepreneurial ecosystems.

Third, Songdo's numerous university and college campuses attract students who eventually supply a steady pipeline of high-skilled workers to the District's technology and startup industry. As identified, Songdo is home to satellite campuses of four international universities and numerous elite universities based out of South Korea. It is also worthy to note that businesses can also access highly-educated workers in neighboring Seoul, where the majority of the country's elite universities are located.

Lastly, New Songdo International City, LLC. (NSIC) aided in the District's development by marketing residential projects, office buildings, commercial, educational, and public facilities. NSIC's efforts in promoting Songdo as a thriving smart city where residents and businesses can enjoy a high quality of life has contributed to the successful recruitment of residents and businesses to the District.

As a result of these conditions and actions, Songdo has become home to over 185,000 people and a sophisticated technological and startup ecosystem. Songdo has numerous smart city technology business leaders, such as Cisco, IBM, and ADT; and the nation's largest co working space for startups, the Incheon Startup Park.

Additionally, numerous innovation and research centers focused on smart city technologies have opened, such as Cisco's Global Center for Intelligent Urbanization and the Stanford Center at the Incheon Global Campus.

So, has the District succeeded in what it set out to do?

Based upon indicators of economic success such as jobs, tax base, incomes, quality of life; it appears that the District has succeeded in meeting its goals.

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A BRIEF ECONOMIC HISTORY OF

SONGDO DISTRICT, INCHEON, SOUTH KOREA

"THE ULTIMATE SMART AND SUSTAINABLE CITY"
NEWCITIES

1997
Asian Financial Crisis

2001
Opening of Incheon International Airport

Incheon Mayor recruits US developers

Gale International selected to develop masterplan

2002
Act on Designation and Operation of Free Economic Zones establishes the IFEZ and New Songdo International, LLC.

Gale forms partnership with POSCO E&C and recruits KPF, Arup, and Cisco

Government begins land reclamation

2005
Land reclamation is complete

Private-sector partners complete masterplan

2009
$38 billion green investment economic stimulus package is passed by South Korean National Assembly

2010
Framework Act for low Carbon Green Growth is passed by South Korean National Assembly

Federal government completes Incheon Bridge construction

2012
Incheon Smart City Corporatino is established

2015
Construction is completed

2020
New Songdo International, LLC. continues to market and manage Songdo

인천광역시
Incheon Metropolitan City

MILTON KEYNES, ENGLAND

Overview

Current literature defines transportation and mobility technology as the process of making transportation of people and goods more seamless, sustainable, accessible, affordable, and safe through the use of technology. Additionally, according to the tech focused website BuiltIn, they similarly define it as, "the tools, machines, and general technologies used to solve problems or improve conditions related to the transportation of people or goods." What

this research identified is that transportation/mobility technology is very diverse. It comprises a variety of fascinating categories such as autonomous vehicles, aerospace transportation, electric vehicles, mobility-as-a-service, first and last mile transportation, and smart transportation.

Nationally, there are over 5,658,400 people directly employed in the sector. In Washington there are 100,012 people working in transportation and mobility and locally in King County there are 52,207 people employed.

Some of the major companies in the United States leading innovation in transportation and mobility include Tesla, Uber, Waymo, and Ford. In Washington, they include some of the largest companies in the nation as well such as Boeing, Expeditors, Alaska Airlines, and Blue Origin. Similar to the companies in the state they are also in the Greater Seattle area such as Convoy, ioCurrents, and Tideworks.

Research has identified cities across the globe and the nation that have had success in attracting and growing the expansion of companies engaged in transportation/mobility technology including Berlin, Germany; Chicago, Illinois; Dallas, Texas; and Los Angeles, California.

However, research screened cities based upon similarities to Federal Way. For example, the screening involved the following: Population size (100,000 - 250,000); a suburb - located in a metropolitan area; no university within the city limits yet close proximity to a major research university; age of the city; successful entrepreneurial ecosystem; metrics of success such as jobs created; and investments made.

The research identified the City of Milton Keynes, England, as one of two case studies on transportation/mobility technology. Milton Keynes was chosen as a case study due to their impressive investment and presence in the field of transportation/mobility technologies, as well as their future plans to further expand in this sector. This is shown through deploying various innovative mobility technologies, such as driverless pods and an autonomous robot fleet for deliveries.

SCREENING PROCESS

- **Population Size (100,000-250,000)**
- **Suburb to Metropolitan Area**
- **Adjacent to Interstate**
- **No University; however, close proximity to one**
- **Age of City compared to Federal Way**
- **Maturity of Technology & Entrepreneurial Ecosystem**
- **Jobs Created & Investments made**

Milton Keynes, England

City of Milton Keynes
Source: Financial Times

Milton Keynes is located in Buckinghamshire, England. It is located in south central England just north of London. More than 7.5 million people live within a one-hour drive from the city, making it a popular location for talented

employees and one of the most productive cities in the UK. Specifically, Milton Keynes is in what is entitled the "Golden Triangle," from which 90% of England's population can be reached in less than five hours such as London, Cambridge, Oxford, and Leicester.

Milton Keynes was established in 1967, as an answer to the increasing housing shortage in London. The City has a population estimated at 230,000 according to the 2011 British Census. Milton Keynes is also large at 34 square miles. It is aptly named the "City of Dreams" as it is home to a diverse number of technology companies, including those in the transportation /mobility tech field. This self-proclaimed label is part of their effort to recruit both established technology companies and cutting- edge startup businesses from all sectors, rather than focusing on just one. Despite this, the transportation and mobility tech field has seen some of the most significant growth thanks to the emphasis placed on artificial intelligence, autonomous robotics along with other technologies involved in the mobility tech field.

"The City of Dreams"

Self-Proclaimed Title: Milton Keynes

Although there is no research institute in the city, with its close proximity to the cities of Cambridge and Oxford, Milton Keynes has managed to develop relationships with various institutes of higher education such as the Universities of Oxford and Cambridge that have addressed diversity and inclusion in a community that is 70% white. British business journalists view Milton Keynes from a business and economic standpoint as an exciting up and coming city. It is home to more than 14,000 businesses (2019) and boasts the second highest startup rate outside of London. This is largely due to their ability upon its inception to establish the city as an emerging tech hub. They achieved this through a strategy of attracting several large domestic companies and international corporations along with a high number of startups.

As previously mentioned, Milton Keynes has been labeled as a high-tech growth city, thanks to a commitment by both local government and the private sector.

An example of this growth is the information provided by the British-based Centre for Cities which identified Milton Keynes as one of the top growth cities in all of England, specifically due to the large number of knowledge-based businesses and workers. Milton Keynes has demonstrated through these labor statistics that there are over 11,000 software engineers living in Milton Keynes, which is an estimated 8% of their labor force. An obvious site location advantage.

11,000 Software Engineers Live in Milton Keynes

Additionally, business growth in terms of annual gross value added (% growth) identifies that Milton Keynes is one of the top three cities in all of the UK, with an economy worth over £12 billion and growing. More than 3,000 companies have relocated there in the last decade, and half of the inquiries to the City's website "Invest Milton Keynes" identify themselves as companies looking to expand. In addition, Milton Keynes also had the 6th highest number of startups per 10,000 citizens in England as of 2019, showing a healthy startup ecosystem.

British business journalists also humorously referred to Milton Keynes as "Going Back to the Future." Journalists share the concept of stepping out onto the streets of Milton Keynes is akin to stepping into the pages of a sci-fi novel with robot deliveries, driverless cars, city-wide Internet of Things, full fiber or broadband, wireless bus charging and dock less bikes. The City has become a forerunner in tech and has the attraction of other innovative businesses with a winning reputation.

As previously identified, Milton Keynes also has the advantage of a great geographical location for businesses to consumers but is also matched by its reputation for enabling businesses to thrive.

Examples of these companies include national and international corporations, many that are well known including Daimler Chrysler, Aston Martin, WD40, Network Rail, Red Bull Racing, Marshall Amps, Mercedes, Ford, Domino's Pizza, and Xero UK. It's also welcoming Santander's £150 million Tech Hub and Goldman Sachs' first UK office outside of London.

Initial Steps

Much of the initial development was led by various business professionals deciding to bet on Milton Keynes as an emerging tech hub, along with the potential establishment of the local government's vision of a Cambridge-Milton-Keynes as a "growth corridor," in the early 1970's.

In concert with this interest by the private sector, the Milton Keynes City Council over the last 20 years have coalesced the energies of the public and private sectors combined with universities in the region to achieve amazing goals.

Their significant tech expansion was set into motion in 2006. Nearly 15 years later, Milton Keynes is now reaping the benefits of a long period of investment in several tech investments. This appears in many different forms throughout Milton Keynes, such as 2,500 traffic cameras all relaying data to each other or numerous sensors dedicated to managing transport routes. With these various sensors, Milton Keynes has become a smart city, gathering data on many different aspects of its infrastructure to then analyze and draw observations from, often utilizing AI systems to determine these observations.

For example, the City owes its success to many different factors. The most noticeable being the investment and implementation of cutting-edge high tech that enables innovation in many fields. These include Big Data as in the processing, analyzing, and extracting information and observations from massive sets of data. Robotics with the development of machines to perform tasks traditionally performed by humans. 5G, fifth generation mobile network with significantly greater bandwidth and download speeds, allowing for possible integration of IoT. Additionally, the investment in the Internet of things (IoT) formulating a network of sensors and other data-gathering tools that share this data with each other over the internet. Lastly, investment was made in Artificial Intelligence as in the simulation of human intelligence through machines that are then able to perform tasks that usually require human intelligence to accomplish.

MAJOR INVESTMENT MADE BY CITY COUNCIL IN MILTON KEYNES

- **Big Data**
- **Robotics**
- **5G**
- **Internet of things (IoT)**
- **Artificial Intelligence**

To bolster this effort, Milton Keynes recently developed MK:Smart, a citywide initiative in collaboration with stakeholders designed to create innovative tech-based solutions to support economic growth in Milton Keynes.

The focus of MK:Smart is on developing and maintaining a large data hub in Milton Keynes (MK Data Hub) to increase their data-collection capabilities throughout the city. By doing so, Milton Keynes then provides even more reason for small and medium sized companies to locate to the city, as this massive quantity of data is available and provided for analysis to enhance innovation. It has become a draw for high-tech companies, but especially those in the transportation and mobility field. Essentially, this unparalleled quantity of analysis-ready data is very attractive for any high-tech company as the data collected represents many different aspects of the city, and it appeals to many different types of tech companies.

In order to keep up with the large amount of data being collected by the City, it has also invested heavily in Artificial Intelligence (AI) endeavors to accommodate its growing AI sector. As such, Milton Keynes also holds many events demonstrating and celebrating AI, creating a tech-savvy culture that other tech hubs may not share.

In 2014, thanks to the city's impressive initial investment in artificial intelligence, Milton Keynes was chosen along with three other UK cities to be awarded £10 million each from the national government agency Innovate UK (part of UK Research and Innovation) for developing and testing autonomous cars in the real world. This funding is what jump-started Milton Keynes' endeavors into the mobility tech field, which is currently thriving. Two years after the initial funding was announced, Milton Keynes had its first self-driving car (courtesy of Transport Systems Catapult) traversing the streets of Milton Keynes.

Because this was the first ever instance of an autonomous vehicle on UK public streets, many other transportation and mobility tech companies and projects followed suit and located to Milton Keynes.

These new companies include an autonomous vehicle company that delivers products to residents to driverless shuttles.

Adding to the diverse tech scene, Milton Keynes was also the host city to the world's first robot competition integrating robots into real time smart city data. The event is known as SciRoc. It is a European Union funded and supported program. It is focused on how robots will integrate in the cities of the future as physical agents located in them serving the needs of consumers, professional and public and private emergencies.

Additionally, the city government also facilitates knowledge transfer partnerships between local companies such as Santander, Mercedes Benz, Red Bull Racing, and Volkswagen and with the nearby universities of Oxford and Cambridge. These actions and investment have manifested into jobs and enhanced tax base for the city.

Due to the city's investment and subsequent modernization to its infrastructure through the placement of sensors, cameras, and other data collection tools, it has turned Milton Keynes into a smart city. Today the City tracks data on a variety of items including domestic energy and water consumption, sewer conditions, broadband usage, and various demographic statistics.

Tech and Entrepreneurial Ecosystem

Much of this development was led by various business professionals deciding to bet on Milton Keynes along with consistent support from its city council.

As such, through a series of actions initiated by both the public and private sectors, a comprehensive entrepreneurial ecosystem was formulated. For example, today, there are both public and private business assistance and business accelerators, innovation centers, and a smart city lab.

The NatWest Entrepreneur Accelerator supports and empowers UK entrepreneurs to scale their businesses to the next level. It is a free six-month program for local startups, it specializes in fintech but accepts all different types of startups. It also focuses on helping expand

businesses to larger markets, more talent and providing access to growth funding. Cohorts receive 1:1 coaching by a dedicated acceleration manager and work in an environment or community of like-minded entrepreneurs. Additionally, it provides a network of investor and bank connections.

IM Accelerator, which stands for Intelligent Mobility Accelerator is a partnership between Connected Places Catapult (British Government's innovation agency for the transport industry and autonomous vehicles) and Wayra UK a unit of the Spanish-based ICT accelerator. The accelerator seeks disruptive startups with high-growth potential in transportation tech innovation. Its focus and specialty is in areas such as connectivity/connected vehicles, autonomous vehicles, innovative transport methods, and mobility-focused AI and IoT. The accelerator provides a 6-month program where participants receive access to office space, mentoring, networking, network of investors, government agencies, customers, large corporations, and universities.

InnovateMK is a publicly supported business assistance public effort that uses a third-party management contract with the company (Tech Mahindra) in support of all different forms of tech startups. Participants receive co-working space, access to key entrepreneurial events, mentoring by field professionals, and access to many networking events within the investing community.

The Cranfield Eagle Lab is a partnership between Cranfield University and Barclays Bank. Its focus is on aerospace and the aviation industry. Participants pay to participate; and there are a variety of different levels of memberships, thus the services received vary by memberships. The services provided if a participate paid for all the highest membership including private offices ranging from 4-8 persons, coworking & hot-desking options, access to digital fabrication and IoT hardware equipment, mentoring and coaching from growth specialists, access to Barclays business expertise and connections, professional network links and introductions, potential access to government backed high-growth funding, access to the maker space, and an on-site team to help you scale your business.

The City also has created a **"Smart City Living LabMK,"** a £2.5m innovation hub dedicated to further developing the city's smart technologies, including robotics, virtual reality, digital twinning, sensor technology, intelligent infrastructure, autonomous vehicles, drones, and advanced communications.

In addition, the Spanish – based bank, **Santander** has built its **Innovation Hub** in Milton Keynes. The £150m investment will serve as the Spanish Bank's flagship UK technology hub. The center will house over 5,000 employees. It will also consolidate the bank's existing local office buildings with a world-class working environment specifically designed to encourage collaboration and innovation, supported by cutting-edge technology. The facility is an environmentally sustainable building designed around enhancing the health and well-being of occupants. It includes event space for colleague training as well as community events and classes. A range of health facilities including a fitness center, cycle storage, and rooftop walking and running track; as well as flexible collaboration space offered to local businesses and community organizations. It is anticipated that the building will be ready to open in 2022 (however, this estimate was made before the Covid-19 pandemic).

Adding to the entrepreneurial ecosystem, the community has a variety of programs (meetups) to sustain its economic growth through education and training along with community innovation. They include the following programs:

- Girls Code UK - free coding lessons and workshops for young women
- Entrepreneurs and Professionals MK - afterwork social meetups for entrepreneurs in MK
- Growth Community for MK Tech Startups - group for people in or around MK who have started/or are thinking of starting a tech company
- Internet of Things MK – It is described as the intersection of MK and IoT in the form of social meetups
- Google Developer Group MK – a meetup for discussing and sharing recent Google tech developments
- MKAI - anyone in or around the city who is interested in growing their AI knowledge and experience can meet up and teach or learn various ML-related topics, has hundreds of members
- UXMK – a group dedicated to discussing and learning about all things UX
- Nomad's Social Club – a laidback social events for freelancers, entrepreneurs, and small business owners

The City also presents a variety of hackathons on an annual basis. One is the MK Hackathon. It is a general hackathon focused on opportunities featuring any innovations related to improving and enhancing Milton Keynes. Another is the MKAI Covid-19 Virtual Hackathon. It brought together data scientists, innovators, and idea makers from Milton Keynes to develop solutions for responding to the coronavirus outbreak.

Throughout Milton Keynes, entrepreneurs will find many different forms of co-working spaces available in office buildings, although international based co-working and makerspace operators such as WeWork, or Impact Hub, have not located to Milton Keynes.

Transportation/Mobility Tech

Again, as identified, Milton Keynes is home to the variety of autonomous vehicle companies. These new companies include Starship, the autonomous delivery robot company, and Aurrigo as examples of autonomous vehicle companies that have located to Milton Keynes.

One, such company- Starship claims it has the world's largest fleet of autonomous delivery vehicles located in Milton Keynes. While research was unable to verify this claim, the use of autonomous vehicles however permeates the landscape in Milton Keynes as residents can use the vehicles simply through an app on a smart phone or through their website.

Starship Autonomous Vehicles
Source: Smart Cities World

The City has also made great strides in the support of electric vehicles, providing tax incentives along with infrastructure needed to house a large number of EVs.

In 2014, due to the city's impressive initial investment in artificial intelligence, Milton Keynes was awarded £10 million grant from Innovate UK for developing and testing autonomous cars in the real world. This grant is what jump-started Milton Keynes' endeavors into the mobility tech field, which is currently thriving.

Two years after the initial funding was announced, Milton Keynes had its first self-driving car (courtesy of Transport Systems Catapult) traversing the streets of Milton Keynes. Because this was the first ever instance of an autonomous vehicle on UK public streets, many other transportation and mobility tech companies and projects followed suit locating to Milton Keynes.

Continuing the progress in transportation and mobility technology, Spin Scooters, a subsidiary of Ford's micro-mobility unit, debuted its e-scooters for rent in august 2020 in Milton Keynes. The launch marked the company's first market in the context of the recently legalized e-scooter trials taking place across the United Kingdom.

Spin, headquartered in San Francisco, California, currently operates dock-less *e*-scooters and bikes in campuses and cities in the United States, Germany, and now the United Kingdom. Spin began ramping up its scooter operations in Milton Keynes by placing 100 scooters on the streets for residents to begin using. Spin is planning to steadily increase its fleet size to match demand and is expected to have around 300 scooters on the streets by mid-September 2021.

To take a trip, riders only need to download the Spin App on their smartphone, review a series of rules and safety guidelines, and then start their trip for £0.25 ($0.33) per minute. The scooters can be used on the town's roadways, which provide the local community with convenient connectivity across the town.

Spin at Ford's Micro-Mobility facility in Milton Keynes
Source: TrendiLife Magazine

Spin is working closely with the Milton Keynes City Council to roll out its fleet responsibly, with public safety being the top priority. Spin has held a series of public education events for residents to come out and learn more about how to ride safely. Spin has and continues to hold safety and public engagement events to raise awareness and ensure full compliance with local and national rules. The company also offers all riders access to Spin Safe Digital, an online learning resource that teaches riders how to start and end trips properly, parking techniques to ensure walkways are free of scooter clutter, and safe riding practices during COVID-19.

Riders can receive a £5 ($6.53) discount if they take an online quiz testing them on the rules. "We're excited to bring the first of our UK e-scooter fleets to Milton Keynes and had a highly-encouraging response from local riders at our Spin Safe events," said Felix Petersen, Head of Europe at Spin. "This is the first of many e-scooters trials across the UK as part of our firm commitment to fully integrate e-

scooters into local transportation ecosystems. We hope to responsibly fulfill the need UK residents have for this new transport mode as an accessible, inexpensive, and greener alternative to cars and public transport."

Red Bull MK 7 Esports
Source: Red Bull

Meanwhile, Red Bull Racing of the Formula 1 Racing Circuit has unveiled a new meeting and events space, entitled MK-7, located alongside its Milton Keynes headquarters. It is a fully equipped brand-new Red Bull Racing E-rena in Milton Keynes, the Team's new home of gaming excellence.

The space focuses around an auditorium which houses the team's collection of race cars, dating back to 2005.

The Red Bull *E*-rena
Source: Red Bull Racing

It also has a boardroom and a 40 seat-capacity mezzanine area, decorated with details of the team's history. The space can be configured to suit client requirements, with a maximum capacity of 450 guests for receptions.

Tech and Business Ecosystem

Because Milton Keynes is a high-tech hub constantly pushing the boundaries, they have developed a program named MK:5G in which they plan on deploying dedicated 5G infrastructure throughout the entire city. While this is not currently in place, the broadband currently being provided is already excellent since Milton Keynes has fiber. Milton Keynes also has an abundance of co-working spaces available for companies ranging from startups to large corporations. This is thanks to the rapid business expansion the city is trying to sustain with large amounts of space being dedicated for office buildings.

Access and Availability to Capital

The investment capital scene in Milton Keynes provides yet another positive for startups along with small to medium sized businesses. There are many investors and investment groups located in Milton Keynes. They include groups such as the Investors Network, the Midlands Engine Investment Fund, BGF, Crowdcube and SEMLEP's and Growth Hub. Additionally, these groups provide consulting services for new businesses as well.

Labor Force

Milton Keynes labor force is estimated to be 145,600. It is comprised of a highly educated labor force, with 11,000 of these workers being software engineers. Businesses draw talent from a variety of higher educational and technical institutions. Entrepreneurs find themselves coming to Milton Keynes from these institutes creating an unequalled opportunity.

INSTITUTIONS OF HIGHED EDUCATION THAT SERVE MILTON KEYNES

- **Open University**
- **University of Oxford**
- **University of Cambridge**
- **Institute of Technology**
- **Cranfield University**
- **University College London**

Quality of Life

The quality of life in Milton Keynes boasts the following:

- Most cost-efficient place to live in the UK, since average income is well above living costs
- Extensive and modern transport links, making it a commuter hotspot
- Great public education (Of the 86 primary education institutions in Milton Keynes, around 23 were recently rated outstanding by Ofsted, with a number of its secondary schools rated as good or outstanding.)
- Wide variety of leisure activities available to everyone. Center:MK is one of the top shopping destinations in the UK, many highly-rated parks with small detention ponds, various museums and exhibits, even Europe's largest indoor snow resort)

The only one obvious negative is the lack of affordable housing for individuals, as many not employed in the tech sector struggle to find affordable housing.

Summary

As can be seen, Milton Keynes has done an excellent job in engaging their community by making infrastructure investments and branding their City, while remaining politically committed to the task at hand of growing a tech -based economy.

As such, it provides a great example of a city that has developed an impressive reputation as an emerging tech hub through clear steps of investment in infrastructure and helping formulate and create a strong entrepreneurial ecosystem. Because of this, the City is now reaping the benefits of its early and consistent efforts, ranking third in gross value added (gross value added is the value of the output from the worker, minus the costs of intermediate consumption, or what it took to make the output) per worker in all of England.

Milton Keynes specifically has made great strides in tech growth over the last 20 years, and more specifically in the transportation and mobility tech field. These efforts include autonomous vehicle development and testing, attraction of an autonomous robot delivery fleet, and other mobility technologies listed in the case study.

The City's future is set to get even bigger, with Milton Keynes Council's plans for 31,000 new homes and 28,000-32,000 more jobs by 2031, and 500,000 residents by 2050. The number of businesses, employees, and customers in the area are significantly expanding, especially given its proximity to and lower living costs than London.

Finally, access to homegrown talent is also about to improve significantly. MK:U, a new world-class technology university (in partnership with Cranfield University), is scheduled to open its doors in 2023. As a city built to inspire, invoke, and nurture innovation, it's no wonder that businesses are locating and succeeding there.

Milton Keynes can provide an important guide as to how Federal Way can become an economic powerhouse and emerging tech hub.

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SUMMARY

Keys for Success

Upon review of the nine case studies, there appears to be three key entities and/or actors that initiated the effort to successfully transform their cities into tech hubs. These include the following:

- Local Political Leadership
- Strong Private Sector Actions
- Collaboration with University(ies)

It is important to note that research did not identify any ideal order in which the three key factors needed to appear for success. Rather, each case study identified the different entities and actors highlighted unique sequences achieved to become a tech hub.

However, it appears that local political leadership is a critical component in each case study. In the chart, Composition of Initiators and Sustainers of tech hubs, it appears from the case studies that nearly two-thirds of the time, it was local elected officials that lead the effort.

Two-thirds of the Time, the effort to become a Tech Hub was initiated by Local Political Leadership

For example, Alpharetta, Georgia's case study illustrates that without it, the community may have never deployed broadband. That action led to the subsequent and quick growth in Fintech growing and diversifying their economy by embracing an emerging technology. Political leadership also occurred in Lancaster, California to address the need to diversify their economy while simultaneously fulfilling the state environmental regulation to meet alternative energy requirements. The political leadership accomplished both in a unified strategy. It was apparent through the case studies that political leadership was a key component for success providing strategic direction coupled with public policies to ensure the growth of the tech hub.

Additionally, universities play a crucial role in communities becoming a tech hub. Similar to Federal Way, none of the cities reviewed in the case studies had a university. They all reached out to ask for help and assistance. For example, in Greenville, South Carolina, they were 35 miles away from Clemson University. Milton-Keynes is located between the campuses of Oxford and Cambridge. South San Francisco was also midway from the University of California at San Francisco and Stanford. No community had a gown-to-town relationship by having a major research university in their backyard. However, without a university's participation in providing training, specialized workforce education in business and technology, and applied and advanced research in these communities, they may not have succeeded.

In addition, the private sector actions provided a key component by creating entrepreneurial tools to support startups from business accelerators to incubators. It also provided the needed capital from angel investors, venture capitalists, equity capital, and institutional investment. For example, as it pertains to Richardson, Texas, the private sector did most of the heavy lifting early in its history of becoming the Telecom Corridor. It started with the attraction of major company Collins Radio along with Texas Instruments and continued with a diversity of companies. Following this, the private sector attracted the University of Texas to locate its Dallas campus to Richardson, a key component in driving the future of its entrepreneurial ecosystem.

Stories of Success

Greenville's manufacturing growth began with the arrival of several large manufacturing firms in the late 1960s. These included General Electric, Michelin, and Lockheed. Additionally, the attraction of Clemson University jumpstarted the City's entrepreneurial ecosystem. As such, the University contributed to the City as a growing advanced manufacturing hub through the establishment of the University's International Center for Automotive Research. Since the location has opened, it has invested over $250 million and created 770 onsite jobs. Clemson also contributed to the City's entrepreneurial ecosystem through the establishment of the Phyfer Innovation Hub. The private sector supported entrepreneurs through NEXT, a local social support network for entrepreneurs. Since NEXT's inception it has raised over $300 million for startup businesses through the formulation of the Upstate Carolina Angel Network; opened three different spaces for entrepreneurs and startups; and launched the non-profit makerspace facility, Synergy Mill. Meanwhile, the City of Greenville and the State of South Carolina formulated public policies designed to attract and support business growth in high-technology industries. These public policies were supplemented by the City of Greenville's business recruitment efforts focused on advanced manufacturing industries among others.

Petersburg's tech-based success was based heavily on collaboration with institutions of higher education and the State of Virginia. Leveraging the presence of industry leaders, the Commonwealth Center for Advanced Manufacturing was established to stimulate R&D in the region by bringing together Virginia's public universities and manufacturers. It also trains the next generation of advanced technology workers through the Advanced Manufacturing Apprentice Academy. Additionally, institutions of higher education offer specialized training and career resources to the region. Meanwhile, the private sector led the subsequent efforts to create an entrepreneurial ecosystem with support from the State of Virginia. GO Virginia was launched as a business-led initiative that provides access to capital (debt or equity). The Virginia Economic Development Partnership runs the Virginia Talent Accelerator Program. Ultimately, the City did not play a major role in its success. However, it offers a variety of local business incentives through its Technology Zone, Enterprise Zone, and Federal Opportunity Zone.

San Leandro's efforts to transform the economy relied on political leadership from its Mayor and City Council. The effort can be traced back to a public-private-partnership for enhanced broadband infrastructure. Meanwhile state officials contributed through the formulation of public policy that placed no restrictions on municipal broadband. The result was the deployment of an 18.5-mile fiber loop that provides San Leandro businesses with one of the fastest internet speeds in the nation at up to 100 Gbps. Meanwhile, private sector actors have contributed to success through the launch of tech campuses that leverage the City's broadband and aim to contribute to the economic growth of the community. While there is no university in San Leandro, the City's development undoubtedly benefitted from its proximity to Bay Area institutions of higher education and the national laboratories, Lawrence Livermore and Berkley Livermore.

Lancaster's journey to become "the Solar Capital of the Universe" started with the leadership of its Mayor and City Council. It materialized into a quick and strategic plan to address the aftermath of the Great Recession and California's Assembly Bill 32. As a solution, leadership created a strategy using its locational advantage of over 300 days of sunshine to transition to solar energy. This in conjunction with private-sector innovation aimed to solve the City's economic plight. Through a series of private-sector partnerships, the City exceeded its goal. Through the strategy they excelled at business attraction efforts and even attracted the University of Ottawa which built a testing facility known as SUNLAB. Meanwhile, although young, the City has created a small entrepreneurial ecosystem with the presence of the Lancaster Small Business Incubator and the Small Business Development Center.

Alpharetta's success in creating a technology-based economy, specifically in financial technology, can be traced back to the City's proactive leadership and commitment to growth. Early on, the City installed hundreds of miles of conduit for broadband development. Along with the presence of established fintech firms, it further incentivized the relocation of technology businesses that had already started locating to Alpharetta as a cheaper alternative to Atlanta. Coupled with the Mayor and City Council's rebranding of the City as the "Technology City of the South," the City offered local incentives for new or growing businesses. Again, as seen in other communities, a mayor and city council contributed to creating a well-established technology and entrepreneurial ecosystem. Today, Fintech is a hub for Financial Technologies. In less than a decade they have 70 financial technology companies that employ 10,000 employees.

Richardson's economic success, specifically in information and communication technology (ICT), is unique. The private sector led the initiative and coupled with the public sector have been committed to their efforts for nearly 70 years in becoming the nation's "Telecom Corridor." In doing so, Richardson invested in nearly 418 miles of fiber. Additionally, the City began rebranding and rezoning through the creation of "The Richardson Innovation Quarter." Today, "The Innovation Quarter" includes much of the City's tools for its entrepreneurial ecosystem. The City's success in creating a technology and entrepreneurial ecosystem came through its recruitment and attraction of the University of Texas, Dallas. The University provides a variety of accelerators, incubators, innovation centers, and networks to help grow startups, including CometX, GalXc, Innovate(her), Blackstone Launchpad, UT Dallas Seed Fund, UT Dallas Design Makerspace, and the Venture Development Center. Meanwhile, the City and the Chamber of Commerce have partnered with the establishment of the Richardson Economic Development Partnerships and is supported by a robust regional venture capital environment.

South San Francisco's success was primarily driven by the private sector, specifically with the organic growth of Genentech. Early on, businesses located to the City primarily due to the availability of relatively inexpensive land and buildings close to the San Francisco and Silicon Valley. Subsequently, businesses in the City stimulated the creation of a technology and entrepreneurial ecosystem with the attraction of California Life Sciences Association, numerous accelerators and incubators, innovation centers, and coworking spaces. Located within the Bay Area, the community was able to access venture capital throughout the region. Despite the fact that success was jumpstarted by the private sector, the City supported the growing biotechnology industry through regular engagement with businesses and the approval of infrastructure and transportation investments. Although there are no university in South San Francisco, the City does benefit from close proximity to world-class Bay Area universities.

Songdo, South Korea's tech-based economy is unique as it was built from scratch under the direction of the federal government. As one of three districts in the Incheon Free Economic Zone (IFEZ), the federal government developed the District with the intent to create a smart and sustainable "city." This was intended to provide a higher quality of life for its residents and attract technology-based companies. Meanwhile, Songdo is also home to campuses of numerous national and international universities that contribute to the economy through research and innovation. The concentration of university locations in Songdo is a direct result of the recruitment efforts of the IFEZ. In regards to its technology and entrepreneurial ecosystem, the national government has also partnered with the private sector to establish the nation's largest startup coworking space and accelerator program, Incheon Startup Park. Additionally, leveraging the sale of data collected from the District's ubiquitous sensor network and fiber networks has made the District an ideal environment for the growth of young and mature firms.

Milton Keynes, England's success, similar to many of the other communities, was achieved by local elected officials providing a vision coupled with business professionals deciding to invest in it as an emerging tech

hub. Following this organic development, the City Council made several long-term infrastructure investments into data collection tools and large data hubs including MK:Smart. City investments also include advanced technologies such as robotics, 5G, artificial intelligence and IoT. Coupled with the commitment by local elected officials, these investments led to the transformation of the community into a smart city. Milton Keynes' proximity to Oxford and Cambridge has helped the city to attract talented and educated people. Through a series of public and private actions, a sophisticated technology and entrepreneurial ecosystem, and university support, Milton Keynes became a city built to invoke, and inspired entrepreneurs, nurture innovation.

Local Political Leadership

Over two-thirds of the time, local elected officials provided the initial leadership towards the effort of becoming a tech hub. These efforts included adopting the public policy followed by the initiation of a strategy. As a component of this, a long-term commitment was made. Additionally, the effort focused on community input and subsequent private sector participation. In addition, local elected officials lead the effort to contact and explore opportunities with a university or universities for the purpose of helping grow an entrepreneurial ecosystem. Also, local elected officials dedicated financial support for infrastructure, specifically broadband deployment, through a public-private partnership. Local officials also formulated and engaged in networking to ensure continuity amongst investors and entrepreneurs. Lastly, they ensured the transformative function of promotion, function of capital investment, entrepreneurial assistance, and convening events.

Private Sector Actions

The other part of the equation of these cities becoming a tech hub was the private sector. Their efforts included purchasing real estate and office space, marketing and promotion of their business sector, creating a network of entrepreneurs and businesses, as well as providing a commitment and continuity to their cause. When businesses relocate to these cities the growth of real estate follows behind them which might develop into business corridors of their own. These areas of concentrated business also facilitate networking between not only other businesses and entrepreneurs, but also universities and local government which has been seen to amplify the entrepreneurial ecosystem within the city. Private companies that believe in the growth of these cities have also prioritized the expansion of the entrepreneurial ecosystem well such as creating incubators, accelerators, meetups, and coworking spaces to help educate and train young professionals.

Collaboration with Universities

Research also identified that universities play a crucial role in a city becoming a tech hub. As previously noted, none of the cities reviewed in the case studies had a university when they began their effort to become a tech hub. The research discerned that each reached out to a university for help and assistance. For example, in Greenville, South Carolina, they were 35 miles away from Clemson University. Milton-Keynes is located between the campuses of Oxford and Cambridge. South San Francisco was also midway from the University of California at San Francisco and Stanford. No community had a prior gown-to-town relationship. Thus, they all reached out to create opportunities to a university in providing training, specialized workforce education in business and technology, and applied and advanced research that was located in these cities.

Commitment to Cause

Among the cities examined, the cities with the oldest efforts were Richardson, Taxes; Milton Keynes, England; and South San Francisco, California.

Richardson's development began in 1956 with a Texas Instruments corporate campus followed a year later by a Collins Radio campus. These companies attracted engineering talent, setting the foundation for Richardson's tech success.

Milton Keynes was established in 1967 and by the early 1970s, Cambridge – Milton Keynes was known as the "growth corridor." Milton Keynes is an example of intentional city-building as ultra-long-term investment.

Finally, South San Francisco's biotech development began in 1976 with the founding of Genentech, the pioneer of the City's biotech industry. By 1987, South San Francisco was home to 112 biotech firms.

Overall, the case studies highlighted that these cities nurtured not only investments but also created entrepreneurial ecosystems to support new business starts and the growth of existing firms. However, none documented much detail on the economic development aspect of their efforts. Alpharetta provided the most detail as it pertains to formulating a corresponding economic development strategy – specifically financial technologies and its outcomes. Additionally, limited evidence illustrates that any incorporated diversity, equity, and inclusion in their efforts.

Research illustrated that, of course, by virtue of their high political and social profiles, mayors and elected officials have often played an effective role in convening different groups of stakeholders, also acting as boosters and, in broader terms, salespersons for their cities. But the effects of mayoral actions were greatly amplified in cities with a highly engaged and capable private sector.

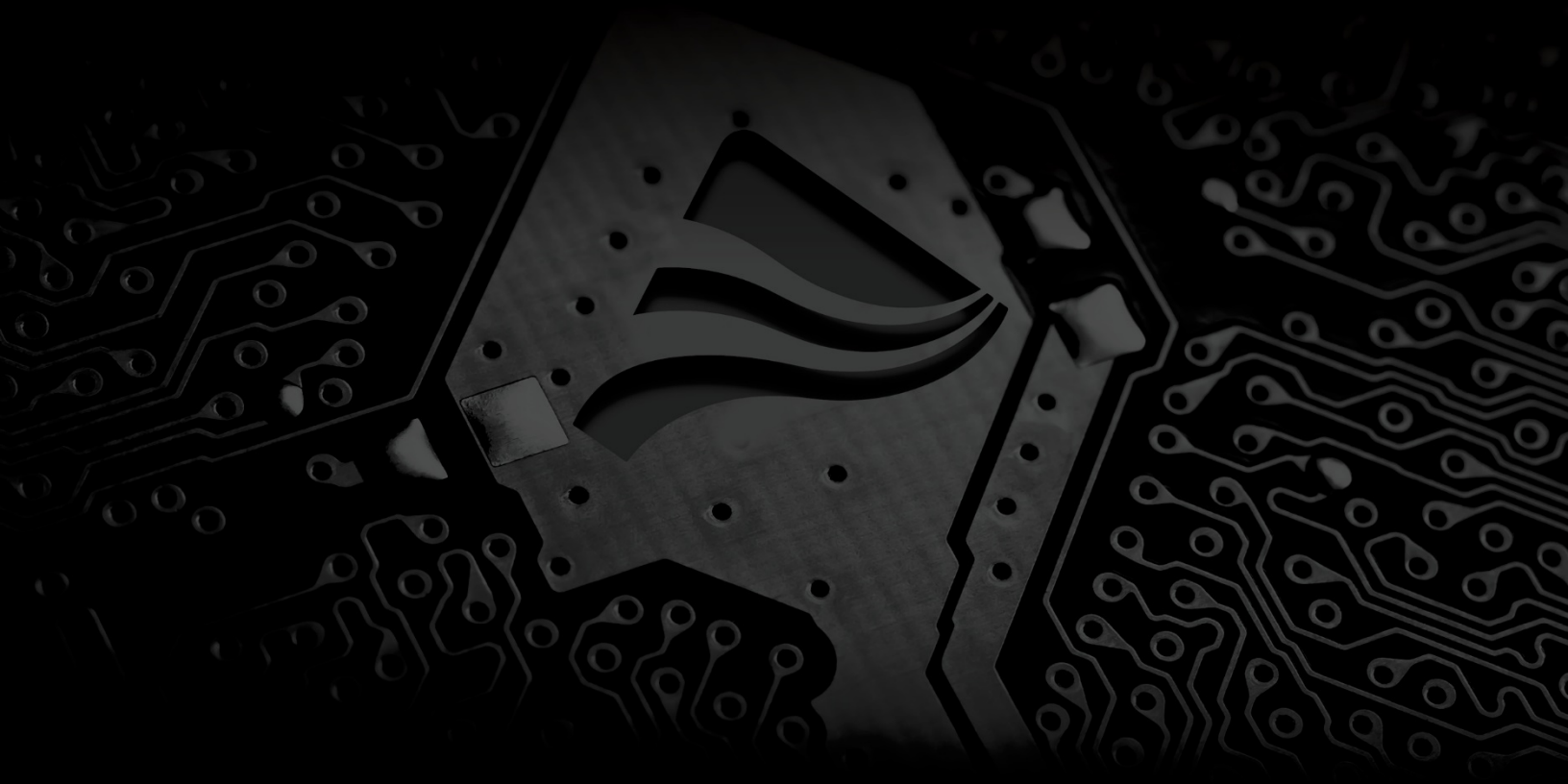
For the majority of the case studies, there are stark differences between which factor initiated the effort to develop a tech-based economy versus who is continuing the effort today. As can be seen in most cases, initial efforts were started by local elected officials. They began the process of working with unversities and the private sector. This is specific case with Alpharetta, Greenville, Lancaster, Milton-Keynes, San Leandro and Songdo. In some cases, the private sector inititiated the such as South San Francisco and Richardson.

So, the largest takeaway from this analysis is that the local government through its elected officials are often the entity that initiates the effort to become a tech-based economy. Once the initiative is deployed by and engaging the private sector and universities, the effort accelerates rapidly in transitioning and growing the local economy. Over time, the efforts of the public and private sector, coupled with the university are often equal.

As it pertains to the type of technology cluster(s) incorporated by cities, it appears that each city focused on sectors where they already had a locational advantage. These include infrastructure; a skilled labor force. Cooperation with nearby research and educational facilities and collaboration with existing tech-firms. These vital assets were identified and further reviewed before any time and resources were poured into sector development initiatives. None of these cities jumped on technology or economic development fads or chased a specific industry because it was fashionable or in vogue. Rather, they targeted a sector because analysis suggested that their city stood a realistic chance of succeeding in growing in that sector.

The research from the case studies clearly demonstrates that no single path leads to a tech hub. Each of the cities pursued its own path to prosperity, accounting either consciously or unconsciously for its particular success. However, the three key factors coupled with the formulation of strong entrepreneurial

ecosystems while correspondingly recruiting or attracting mature firms engaged in the tech-focus of the area comprised the main ingredients for their successes.



TRANSFORMATIVE FUNCTIONS

TRANSFORMATIVE FUNCTIONS

Overview

While understanding the actors or entities (local or state political leadership, private sector and/or university) that initiate the effort is important, the question remains – how effective were these entities at carrying out the key "Transformative Functions," of analytics and promotion, investment facilitation, entrepreneurial assistance and convening events towards becoming a tech hub?

This chapter explains the value of these activities and identifies the creativity used by each city in their deployment illustrating how each city used each category in the effort to become a tech hub.

Analysis

The cities examined in the case studies utilized the four factors to vastly different degrees. Through the use of heuristics, research ranked the transformative functions for each city. For example, Lancaster and San Leandro used Analytics and Promotion the most. Meanwhile, Songdo used Investment Facilitation the most. Richardson used Entrepreneurial Assistance the most, and Greenville used convening events the most.

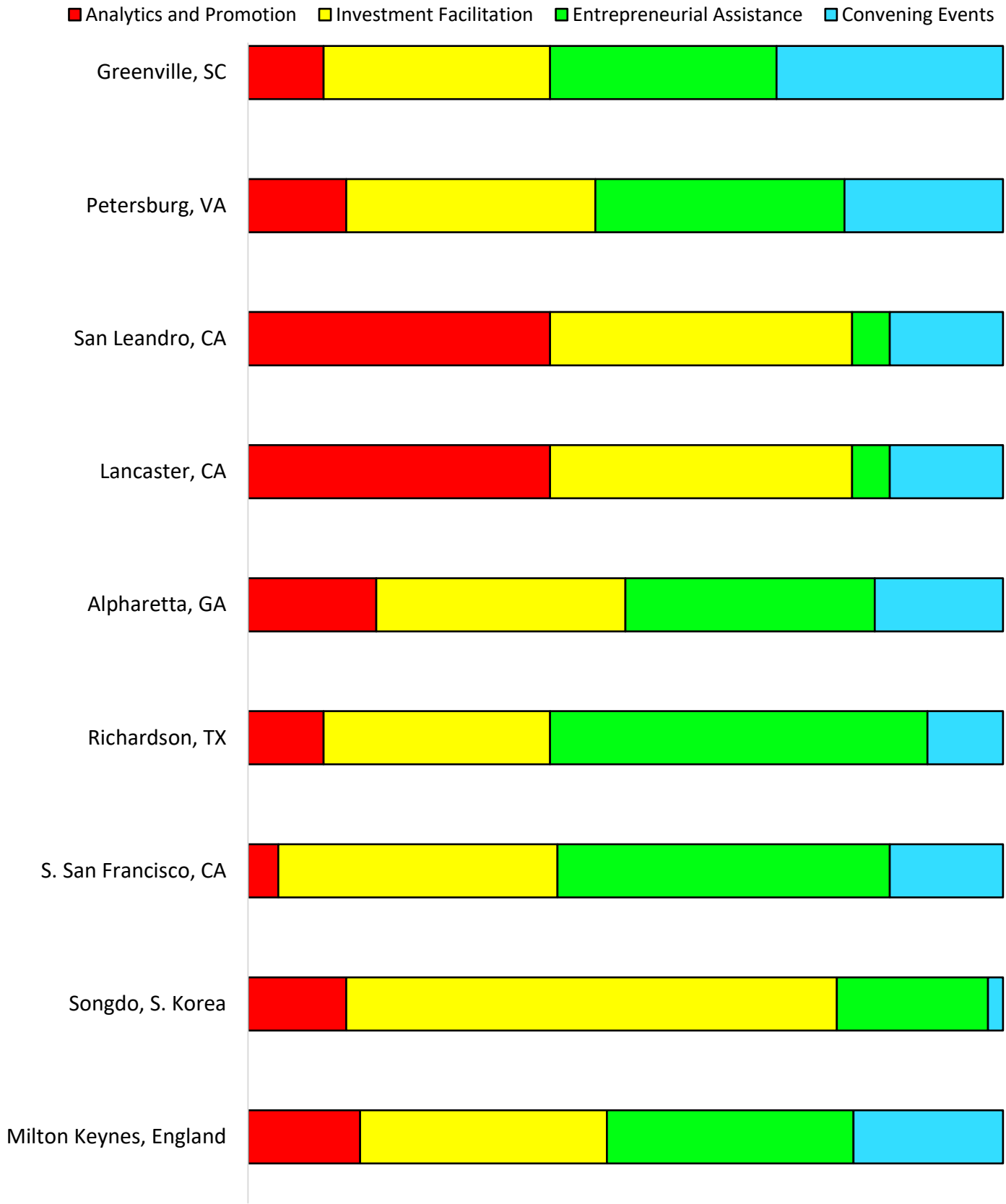
The chart on Transformation Functions of Tech Hubs illustrates the individual city and how each city used the four factors in their development. This is useful either in the attempt to find similarities amongst the Cities and/or to identify a model for success.

Transformative Functions

- **Analytics and Promotion**
  These activities consist of feasibility and competitiveness analysis, planning, and market research, along with branding and marketing.
- **Investment Facilitation**
  These activities include infrastructure such as broadband/internet, zoning, permitting, and business facilities; customized workforce training; business recruitment, expansion, and retention; incentives; and site selection and retention services.
- **Entrepreneurial Assistance**
  This includes activities such as availability and access to capital, technology transfer, business accelerators, incubators, startup studios/foundries, innovation centers, makerspaces, coworking spaces, provision of market information, and export facilitation.
- **Convening Events**
  This function involves engaging stakeholders, fostering industry-academia links, developing partnerships and networks, establishing or leading growth coalitions through meetups, mashups, and hackathons as well as community meetings.

NOTE: The proportions of factors shown for each city were arrived through heuristic estimations, using general rules and experience from analyzing the case studies to determine what factor(s) were more or less significant in the development of each community.

Transformation Functions of Tech Hubs



*Heuristic Estimations

Highlights

Greenville

Analytics and Promotions

Greenville's overall growth was organic. It occurred with its relationship with Clemson University and through their creation of an entrepreneurial ecosystem. Simply, they thrived and were able to do so with little if any promotion.

Investment Facilitation

Greenville's local government along with Clemson University have been able to invest in and facilitate growth in the entrepreneurial ecosystem. One such investment was the development of the University Center of Greenville in 1987, which continues to expand to this day. The private sector also had a hand in the success in Greenville with the development of NEXT, recall the organization focused on growing the startup community in the city.

Entrepreneurial Assistance

NEXT created multiple coworking spaces, launched venture mentoring services, launched a startup incubator and an accelerator, and continues to host an annual event to connect entrepreneurs with venture capitalists. NEXT has been able to raise over $300 million for startup businesses in Greenville since its inception. Additionally, two elements supporting their entrepreneurial ecosystem is the Phyfer Innovation Hub and the Greenville One building. The Innovation Hub provides professionals and students with a place to meet and share innovative thoughts and space, while the Greenville One building houses Clemson University's graduate level programs.

Convening Events

As mentioned, NEXT has helped host a multitude of events. The Annual Advanced Manufacturing and Engineering Night, the South Carolina Manufacturing Conference and Expo, "SOUTHTEC," as well as a host of meetups and hackathons such as the OpenSC are ways that locals stay connected and learn about the future for Advanced Manufacturing.

Petersburg

Analytics and Promotions

Programs such as the Precision Machining Program or the Community College Workforce Alliance have been promoted with the goal of supporting students and young adults to gain skills necessary to work in the Advanced Manufacturing sector in Petersburg. However, it does not seem as though Petersburg has been able to promote much outside of the city and state.

Investments Facilitation

There have been many efforts to facilitate investments such as the Commonwealth Center for Advanced Manufacturing (CCAM) and grants provided by organizations such as GO Virginia. The CCAM itself was a $17 million project that brings entrepreneurs and businesses in the Advanced Manufacturing sector together.

Entrepreneurial Assistance

The entrepreneurial ecosystem has played a significant role in Petersburg's success as well. Through organizations such as the Virginia Commonwealth University (VCU), CCAM, The Advanced Manufacturing Apprentice Academy, and GO Virginia, have provided educational resources, certifications, and job advancement opportunities for citizens of Petersburg. Half of these are business-led that provide private-

sector growth while the other half focus on teaching young adults the skills necessary to excel in advanced manufacturing.

Convening Events
There does not seem to be much in the way of convening events. However, there are some events such as job fairs, employee assessments, and specialized training programs for many Petersburg citizens to advance in their education and careers.

San Leandro
Analytics and Promotions
Lit San Leandro serves as the marketer of the City's broadband to local businesses. City leadership has also been vocal, engaging in events and conferences related to broadband and smart city initiatives.

Investment Facilitation
The City partnered with a private company, Lit San Leandro, to create a fiber network for City operations and local businesses. Lit San Leandro provides hundreds of internet service options for businesses at speeds up to 100 Gbps. This network has also been leveraged to create networks of smart street lights and public Wi-Fi. The fiber network is also connected to the City's two business hubs, Gate510 and the San Leandro Tech Campus, as well as hundreds of businesses outside these hubs in the City and every school in the local school district. Another investment element the City has benefitted from is the PilotCity, an education and internship organization for students from middle school to college.

Entrepreneurial Assistance
Lit San Leandro provides some of the fastest internet speeds in the nation to local businesses, but that is the extent of the entrepreneurial assistance. There are none of the usual element of an entrepreneurial ecosystem in San Leandro such as accelerators, incubators, or innovation centers. Founders Circle Capital is present in the City, operating as a growth-stage investment firm.

Convening Events
There have been few impactful or consistent Convening Events in San Leandro. The Circle, a branch of Founders Circle Capital, is a private leadership community for growth stage executives that provides mentorship and business advising.

Lancaster
Analytics and Promotions
Lancaster has heavily utilized Analytics and Promotions, especially focusing on the City's image as a renewable energy hub. The Mayor labeled Lancaster "the Solar Capital of the Universe" and has been prolific at events and conferences related to clean technologies, especially solar power. The City founded Lancaster Choice Energy, the High Desert Power Authority, and CalChoice all in support of Community Choice Aggregation of power. Lancaster was also the first city to institute a solar mandate, requiring all new construction to have solar panels.

Investment Facilitation
Lancaster has achieved success through business recruitment and attraction. The City attracted an eSolar sun tower then went on to attract the Chinese-based company BYD, an electric bus and battery

manufacturer. Additionally, they attracted a multitude of other solar farms, including an experimental solar farm run by the University of Ottawa, and a hydrogen power company, SGH2.

Entrepreneurial Assistance
Lancaster is without the common elements of Entrepreneurial Assistance such as incubators, accelerators, and innovation centers. The closest the City has is the Lancaster Small Business Incubator which does not act as an incubator but as an office and warehouse with access to business resources from California's Small Business Development Center, Los Angeles Network.

Convening Events
The City worked with the University of Ottawa to create their experimental solar farm and has engaged with private renewable energy firms. The Mayor has collaborated with heads of private renewable energy companies as well as nearby cities to create a renewable energy ecosystem.

Alpharetta
Analytics and Promotion
Alpharetta, Georgia was able to rebrand itself as the "Technology City of the South" after careful strategic planning. The City implemented an initiative called *Grow* Alpharetta to help grow the local entrepreneurial ecosystem, attract new residents and businesses, and reinvent their image. Grow Alpharetta has been able to use data to support this initiative as well, which they have published on their website and update periodically.

Investment Facilitation
The City had an extensive fiber network that supported current businesses, residents, and public facilities while also attracting new businesses and residents to the city. The local government also invested in other infrastructure and economic development projects such as the startup incubator organization *Tech Alpharetta* which has raised millions of dollars for startups.

Entrepreneurial Assistance
Alpharetta worked closely with state and regional economic development agencies to provide tax credits and incentives to complement businesses in the city. Through Tech Alpharetta's Innovation Center, the city has also been able to provide coworking spaces and grow startup companies through a startup incubator. The Technology Association of Georgia (TAG) is also located in Alpharetta which provides networking and assistance programs as well.

Convening Events
Alpharetta has hosted a multitude of events to foster good relationships within the tech community. Meetups, informational events, workshops, coworking spaces and networking events with angel investors are just a few of the events hosted in Alpharetta and usually put on by Tech Alpharetta or TAG.

Richardson
Analytics and Promotion
While not the main factor in Richardson's development, Analytics and Promotions have still played a role. The City self-proclaims itself "the best place in the US to locate your technology company." Part of the city has also earned the name: "The Telecom Corridor." The Richardson Economic Development Partnership also engages in marketing the City to businesses.

Investment Facilitation
Richardson has 418 miles of fiber with FTTC in all residential areas, astonishing accomplishments for a city of 116,000. Tech Titans and the Richardson Economic Development Partnership have both engaged in the business recruitment that was central to the initial growth of Richardson as a telecom city and sustains it today. There are also several venture capital firms in the City.

Entrepreneurial Assistance
Richardson has a large entrepreneurial ecosystem. UT-Dallas's Institute for Innovation and Entrepreneurship is home to the CometX accelerator, GalXc women's accelerator, Innovate(her) STEM education conference, and Blackstone Launchpad network. UT-Dallas also has a startup seed fund, makerspace, and Venture Development Center incubator. These elements of Richardson's entrepreneurial ecosystem all serve to fuel business, using local talent to attract more talent.

Convening Events
The most significant convening event that Richardson has seen was the formation of the Tech Titans, now the largest technology industry association in the southwestern United States. This association has stimulated much of entrepreneurial and investment development the City has achieved.

South San Francisco
Analytics and Promotion
South San Francisco has utilized Analytics and Promotions the least. The City has grown organically with little intentional life science or biotechnology branding. The City of South San Francisco has used digital campaigns for business recruitment, but the City's reputation as a biotech hub has largely been earned due to efforts by the private sector.

Investment Facilitation
The second largest factor toward South San Francisco's development has been investment and development, including Genentech's Futurelab funds K-12 STEM education as well as offering STEM education, competitions, and a biotech lab through Gene Academy, Helix Cup, and Science Garage.

Entrepreneurial Assistance
Entrepreneurial elements have been the largest factor in San Francisco's growth, including the CARB-X accelerator, Johnson & Johnson's JLABS @ SSF incubator, Johnson & Johnson West NA Innovation Center, Eli Lilly's Lilly Gateway Labs, and the nearby Start X community all grow and fuel South San Francisco's tech entrepreneurial ecosystem. These elements provide funding, networking, mentorship, and facilities for local startups to grow and flourish in the City.

Convening Events
Though not to the same degree at the two previously mentioned factors, South San Francisco has utilized Convening Events in its development. The California Life Sciences Association supports California's biotech innovation pipeline through programs in education and competitions, workforce development, and entrepreneurship support, and advocacy. JLABS @ SSF is more than just an incubator; it also offers operational support and educational programs.

Songdo

Analytics and Promotions

Songdo is a smart "city" that utilizes data from its sensors and cameras to improve the District's operations and to create smart city models for other governments. New Songdo International City LLC's marketing of Songdo as an advanced district has contributed to the massive population growth and business attraction.

Investment Facilitation

Large investments have been made by the South Korean government to create and grow Songdo, including smart city infrastructure and a large amount of business recruitment. There are also many satellite campuses in Songdo, including campuses of George Mason University, Ghent University, the University of Utah, the State University of New York, as well as five satellites Korean universities. The Cisco Global Center for Intelligent Urbanization operates as an innovation center for big data and smart city projects as well as a training facility. Two major research centers in Songdo are the District's Technology Research Center and the Stanford Center that, along with its research, collaborates with local universities.

Entrepreneurial Assistance

Though less of a factor than those previously mentioned, Entrepreneurial Assistance has still been critical to Songdo's rapid growth. The District is home to an IoT Accelerator, SparkLabs accelerator, and the business resource and coworking center Incheon Startup Park.

Convening Events

Convening Events have been a relatively minor factor in Songdo's development. The District focused on innovating in big data and smart city technology as well as on attracting universities and top tech companies rather than holding conferences, meetups, or other events. Songdo as a whole could be thought of as one large, convening event considering the degree of orchestration in its creation and growth.

Milton Keynes

Analytics and Promotions

Milton Keynes created an early strategy to transition into a tech hub and they promoted this idea which helped facilitate this idea and attracted large domestic businesses, international corporations, and startups. Through their website, "Invest Milton Keynes," they have been able to analyze data from their MK: Data Hub to find areas of improvement in the city, drive innovation, and support existing residents and businesses as well.

Investment Facilitation

Milton Keynes has had strong private and public investment since the 1970s due to it being viewed as a potential Cambridge-Milton Keynes growth corridor. Milton Keynes was able to invest in a strong broadband infrastructure which helped to support advancement (and more investment) in technologies such as big data, 5G, robotics, artificial intelligence, and IoT. The national government has also provided investment in the city.

Entrepreneurial Assistance

With its close proximity to Cambridge and Oxford, Milton Keynes was able to have an organic entrepreneurial environment and attract educated individuals together. The MK: Data Hub has been able to attract many tech companies from small to midsize to relocate here due to the massive quantity of

data which appeals to many types of businesses. There are many organizations that help these new businesses such as The NatWest Entrepreneur Accelerator, IM Accelerator, and InnovateMK.

Convening Events
There are a plethora of local meetups within the city such as Girls Code UK, Entrepreneurs and Professionals MK, Google Developer Group MK, and many more specific to the city.

Summary

Based on the research, the following highlights the cities that best used a specific category of Transformative Functions to become a tech hub.

Analytics and Promotion

Songdo has one of the most widespread and most recognized smart city networks in the world. On top of using its network to optimize its own operations and services, Songdo also provides consulting services to other up-and-coming smart cities. A great deal of research and development in urban analytics occurs in the District as well.

Milton Keynes has smart city systems compiling data from its IoT technology, sensors and cameras, around the City into its MK Data Hub. The City analyses this data with AI to improve its own operations but also offers the data to local companies to increase innovation. This data has been a large draw for transportation and mobility technology companies.

Top 3 Cities for Analytics and Promotion

1. Songdo, Incheon, South Korea
2. Milton Keynes, England
3. San Leandro, California

San Leandro has leveraged its municipal fiber network into an IoT network of street lights, traffic signals, irrigation clocks, and Wi-Fi hot spots. These smart city technologies have allowed the City to make its own operations more efficient as well as improve the quality of services delivered to residents and businesses.

Investment Facilitation

Milton Keynes has invested heavily in Smart City Big Data, Robotics, 5G, IoT, and AI in the pursuit of economic development and technological innovation. These sector investments along with investment in infrastructure such as city-wide fiber have made the City a leader in transportation and mobility technologies.

Songdo was created entirely through investment from the South Korean Government. Incheon recruited private sector developers to transform what was marshy tidal flats into a modern, smart District. Songdo was also established as part of the Incheon Free Economic Zone to relax policies, reduce business taxes, and attract more foreign investment.

Top 3 Cities for Investment Facilitation

1. Milton Keynes, England
2. Songdo, Incheon, South Korea
3. San Leandro, California

San Leandro's development was initiated and has continued to be centered around broadband development. The City invested in a municipal fiber network which it continues to expand, supplying an increasing number of businesses with ultra-fast internet and enabling investment in smart city technologies and public Wi-Fi.

Entrepreneurial Assistance

Greenville is home to a variety of entrepreneurial elements including the Iron Yard Accelerator, University Center Greenville; Clemson University International Center for Automotive Research; NEXT which has opened three coworking spaces, a venture mentoring service, virtual incubator, and an annual conference to connect entrepreneurs and VCs; the Greenville Technical College Center for Manufacturing Innovation Business Incubator; Phyfer Innovation Hub; and International Transportation Innovation Center.

Milton Keynes has a large network of entrepreneurial ecosystem elements including the NatWest Entrepreneurial Accelerator, IM Accelerator, InnovateMK coworking and mentorship space, Cranfield Eagle Lab, Smart City Living LabMK innovation hub, and Santander Innovation Hub.

Top 3 Cities for Entrepreneurial Assistance

1. Greenville, South Carolina
2. Milton Keynes, England
3. South San Francisco, California

South San Francisco's entrepreneurial ecosystem includes the California Life Sciences Association, the CARB-X accelerator, the JLABS @ SSF incubator, J&J West NA Innovation Center, Lilly Gateway Labs, and the nearby StartX community.

Convening Events

Milton Keynes holds many meetups including Girls Code UK, Entrepreneurs and Professionals MK, Growth Community for MK Tech Startups, Internet of Things MK, Google Developer Group MK, MKAI, UXMK, and Nomad's Social Club. The City also hosts hackathons including the MK Hackathon and the MKAI Virtual Hackathon.

Alpharetta is home to the Technology Association of Georgia, composed of thousands of tech companies. The Association holds an annual fintech conference as well as a world-class fintech summit. Tech

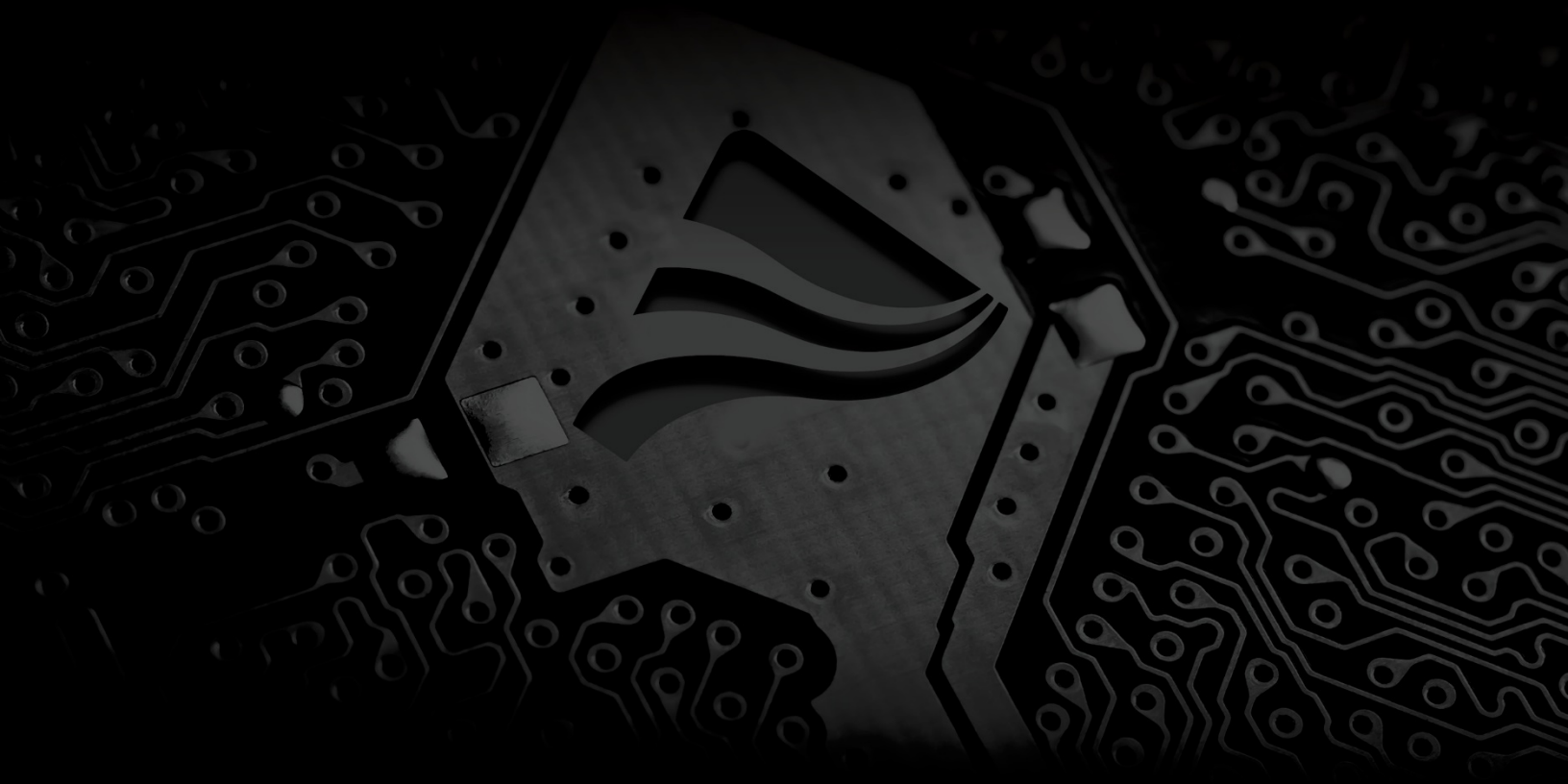
Alpharetta holds recurring tech industry education, networking, and innovation events and has a coworking space where local entrepreneurial and business leaders work together.

Top 3 Cities for Convening Events

1. Milton Keynes, England
2. Alpharetta, Georgia
3. Songdo, Incheon, South Korean

Songdo has an abundance of students with four international university satellite campuses and five domestic university campuses. In conjunction with the many global technology corporations in the District and the District's investment-heavy governance, these students engage with public and private in partnerships and overall collaboration, driving innovation and economic development in Songdo.

Lastly, while there appears to be no one specific use over another that leads to a tech hub, the use of Transformative Functions appear to be required in some percentage of use in order to achieve a tech hub.



ACTIONS FOR SUCCESS

ACTIONS

Overview

This chapter presents the key actions followed by corresponding programs and projects into a corresponding work plan by timeframe. It also provides detail on performance measures for each action along with timeframes for their accomplishment.

ACTIONS FOR SUCCESS -THE PLAN

The Plan identifies 8 actions. They include the following and are highlighted below.

A. Organization & Initiative
B. Broadband Development
C. Entrepreneurial Ecosystem
D. University, College, and School District Relations
E. Diversity, Equity, and Inclusion
F. Technology and Data Analytics
G. Business Attraction
H. Workforce Development

Organization and Initiative

Adoption of city policies that will have catalytic effects on the number and quality of programs, projects and services on the and the potential impact on Federal Way residents, businesses, i.e., startups to mature growth companies.

Broadband Development

Examine citywide internet infrastructure and potential to enhance services to residents, business and visitors. Broadband's key components – Reliability, Speeds, Accessibility, Affordability, Bandwidth, and Symmetry – are what the City must aim for to become a tech hub. It must meet state policy standards, while enhancing education, develop the local economy for its businesses, and improve diversity and inclusion for its residents.

Entrepreneurial Ecosystem

Work to grow an entrepreneurial ecosystem from within the City. An entrepreneurial ecosystem's key drivers include supportive government policies and regulations; strong relationships with local universities; access to financing/capital; quality human capital, and an entrepreneur and business-friendly culture – can all be improved by the public sector directly and indirectly. Under these conditions, business tools – Incubators, Accelerators, Startups Studios/Foundries, Innovation Centers, Tech Transfer Centers, Makerspaces, Coworking Spaces, Meetups, Mashups, and Hackathons – can be utilized to fuel business.

University, College, and School District Relations

Create a beneficial relationship with universities such as University of Washington, both Seattle and Tacoma campuses; community colleges such as Highline; Federal Way School District, and other educational institutions that support STEM education and training, research and innovation and supportive of entrepreneurial success through education, mentoring, and networking.

Diversity, Equity, and Inclusion

Become a community that offers tools and resources designed to help local elected leaders build safe places where people from all racial, ethnic and cultural backgrounds thrive socially, economically, academically and physically.

Technology and Data Analytics

Identification of use of technologies to enhance the organizational effectiveness, efficiency, accountability and transparency of operations of the City of Federal Way. Also, formulation of objectives that identify, analyze and examine the feasibility of use of data collected by the City through its processes of business and determine its use for organization effectiveness and efficiencies and its value to the marketplace potentially leading to subsequent monetization.

Business Attraction

Attract businesses of all sizes and maturities to the City, especially in target technology sectors. Potential businesses include elements of an entrepreneurial ecosystem such as business incubators and business accelerators. Attract businesses that further the City's other aspirations in the areas of smart city technology, connections to academia, broadband development, entrepreneurial ecosystems, and disruptive technologies.

Workforce Development

The Center for Applied Urban Research at the University of Washington Tacoma will work with Federal Way and South Puget Sound communities to understand the characteristics of the region's labor force and identify information to help guide and prepare for the region's economic future. It will conduct a regional study in four phases focused on the following activities for the participating jurisdictions (currently Pierce County, Tacoma, Lakewood, Sumner, and Federal Way). Areas will include all of Pierce County and South King County as a single region. Detailed data at the local level will be provided to participating jurisdictions only.

Organization & Initiative

Action

Formulation of policies that aid the City in achieving a "Tech Hub."

Purpose

Adoption of city policies that will have catalytic effects on the number and quality of projects, programs and services on the potential impact on Federal Way residents, business, startups and entrepreneurs.

Programs and Projects

- Presentation of Action Plan to Community Stakeholders
- Identify how to devise tech-based solutions to City issues by working with local entrepreneurs and innovators.
- Explore and engage private partnerships to achieve a and sustain a "Tech Hub."
- Ensure citizen engagement and use of technologies for transparency and accountability to achieve a "Tech Hub."
- Produce annual review measuring City efforts to achieving and sustaining a "Tech Hub."

Short Term

1. Presentation of Action Plan to Community Stakeholders
2. Complete and present annual reports for years 2021, 2022

Mid Term

1. Identify how to devise tech -based solutions to city issues by working with local entrepreneurs
2. Complete and present annual reports for years 2023, 2024, 2025

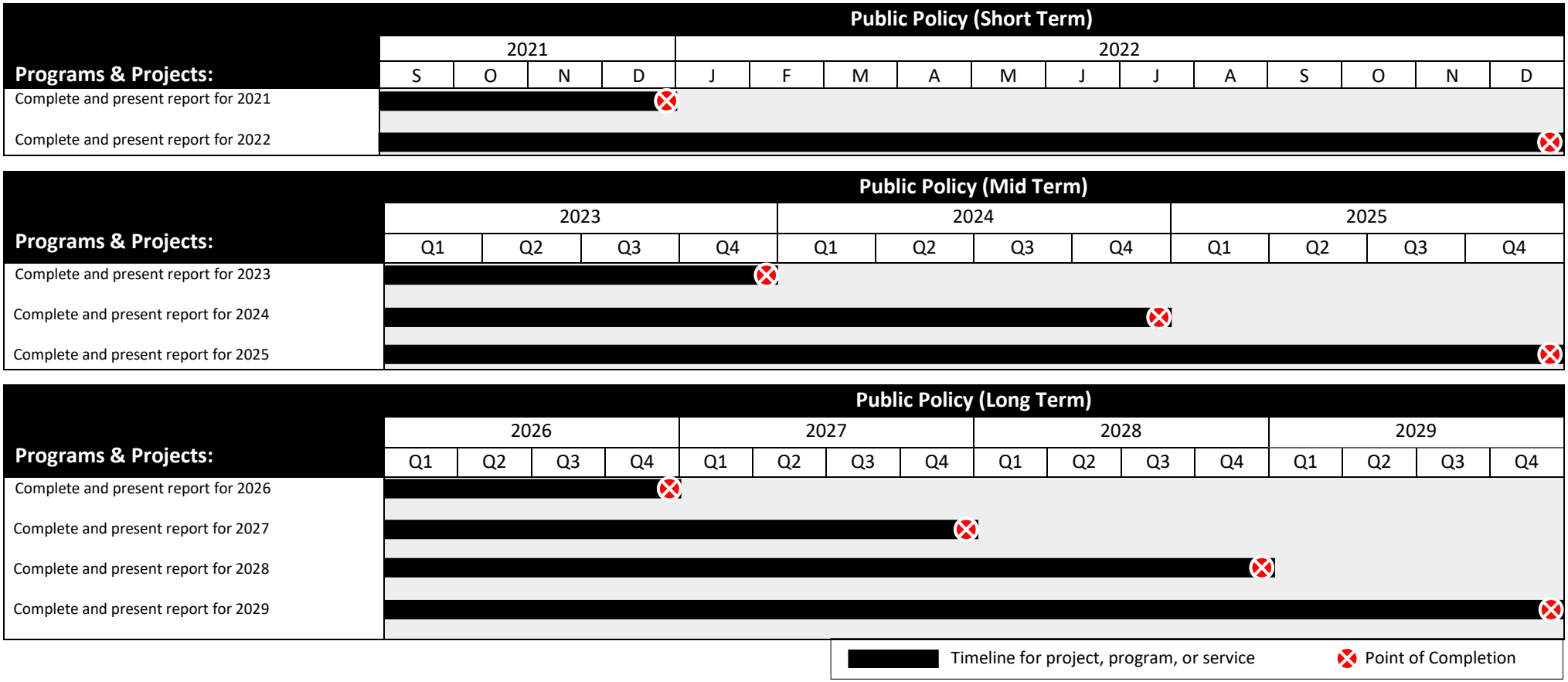
Long Term

1. Complete and present annual reports for years 2026, 2027, 2028, 2029

Performance Measurements

Ensure that programs, projects and services are in alignment with city policies. Budget and timelines are achieved within the identified parameters. Number of jobs created; businesses attracted, retained and expanded; startups supported quality and quantity of programs, for example, numbers of people served, input and subsequent elements implemented.

Organization and Initiative



Infrastructure/Broadband Development

Action

Examine citywide internet infrastructure and potential to enhance services to residents, business and visitors.

Purpose

The purpose of this action would enable Federal Way to achieve multiple goals:

Achieving the City Council Goal – High-speed broadband is essential for attracting, retaining, and growing technology companies.

Meeting State Policy – Local government must meet new state broadband goals:

- By 2024, all Washington businesses and residences should have access to high-speed broadband that provides minimum download speeds of 25 Mbps and minimum upload speeds of 3 Mbps. This State standard aligns with the Federal Communication Commission's (FCC's) current minimum standard for broadband. However, there is currently an effort to increase these federal standards.

- By 2026, all Washington communities should have access to at least 1 Gbps symmetrical broadband service at anchor institutions like schools, hospitals, libraries, and government buildings.

- By 2028, all Washington businesses and residences have access to at least one broadband provider with at least 150 Mbps symmetrical broadband service.

Economic Development – Broadband is an essential infrastructure for business attraction and retention. If capital is the lifeblood of high-tech business, then broadband is the nervous system. A strong broadband network provides a fast, high-volume, and reliable flow of information connecting every aspect of a business and allowing greater efficiency and access to technology developments.

Improving Diversity and Inclusion – Bringing quality broadband to every resident is essential for increasing equity access and economic welfare.

Enhancing Education – Current, high-quality data informs the way educators shape programming for students.

Programs, Projects, and Timeframes

<u>Short Term</u>

1. Align City broadband expansion with new State Policy and broadband goals
2. Approval by City Council for State Grant and Matching Funds
3. Submit State CERB Grant Application
4. Approval by City Council to Issue Broadband Feasibility RFP
5. Issue RFP
6. RFP Awarded and RFP Contract Completed
7. Creation of Broadband Advisory Taskforce (BAT)

If the opportunity presents itself, proceed to the following:

Mid Term

1. Approval by City Council for Funding of Engineering and Design study
2. Approval by City Council to issue RFP
3. RFP Awarded and RFP Contract Completed
4. Selection of Broadband Model (Public, Private, PPP)
5. By 2024, provide minimum download speeds of 25 Mbps and minimum upload speeds of 3 Mbps to businesses and residents within City limits
6. Approval by City Council for Funding of Construction
7. Approval by City Council for RFP Issue
8. RFP Awarded
9. RFP Contract Completed

Long Term

1. Complete construction of broadband network
2. Commence Broadband Operations
3. By 2026, all Washington communities should have access to at least 1 Gbps symmetrical broadband service at anchor institutions like schools, hospitals, libraries, and buildings.
4. By 2028, all Washington businesses and residences have access to at least one broadband provider with at least 150 Mbps symmetrical broadband service

Performance Measurements

Hiring of consultant; completion of study on analysis and feasibility study; acquisition of state Community Economic Revitalization Board grant. Achievement of report/study with timeframes and budget. Identify potential impacts on Federal Way population specifically by income and ethnicity and business served.

Infrastructure/Broadband Services (Short Term)

Programs & Projects:	2021				2022											
	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D
City-wide Broadband Feasibility Study																
Approval by City Council for State Grant and Matching Funds	⊗															
Apply for CERB Grant	⊗															
Approval by City Council for RFP Issue		⊗														
RFP Issued			⊗													
RFP Awarded				⊗												
RFP Contract Completed					⊗											
Broadband Advisory Taskforce (BAT)														⊗		

Broadband Services (Mid Term)

Programs & Projects:	2023				2024				2025			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
If Opportunity,												
Approval by City Council for Funding of Engineering and Design study												
Approval by City Council for RFP Issue												
RFP Awarded and RFP Contract Completed												
Selection of Broadband Model (Public, Private, PPP)												
By 2024, provide minimum download speeds of 25 Mbps and minimum upload speeds of 3 Mbps to businesses and residents within City limits												
Approval by City Council for Funding of Construction												
Approval by City Council for RFP Issue												
RFP Awarded												
RFP Contract Completed												

PENDING DIRECTIONS

Broadband Services (Long Term)

Programs & Projects:	2026				2027				2028				2029			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Complete construction of broadband network																
Commence Broadband Operations																
By 2026, all Washington communities should have access to at least 1 Gbps symmetrical broadband service at anchor institutions																
By 2028, all Washington businesses and residences have access to at least one broadband provider with at least 150 Mbps symmetrical broadband service																

PENDING DIRECTIONS

Timeline for project, program, or service ⊗ Point of Completion

Entrepreneurial Ecosystem

Action

Work to grow an Entrepreneurial Ecosystem in Federal Way.

Purpose

Create an entrepreneurial ecosystem include providing supportive government policies and regulations to allow for ease of starting a business and access to resources; ongoing analysis of how to build and fuel the Federal Way entrepreneurial ecosystem; build strong relationships with local universities to develop entrepreneurial programs, events, and or partnerships; create an environment that allows for ease of access to financing and capital; develop Federal Way's work force and provide access to education in entrepreneurial and tech skills; provide a strong entrepreneurial and business friendly culture through incubators, accelerators, meetups, coworking spaces, mentorship and many others.

Programs, Projects and Timeframe

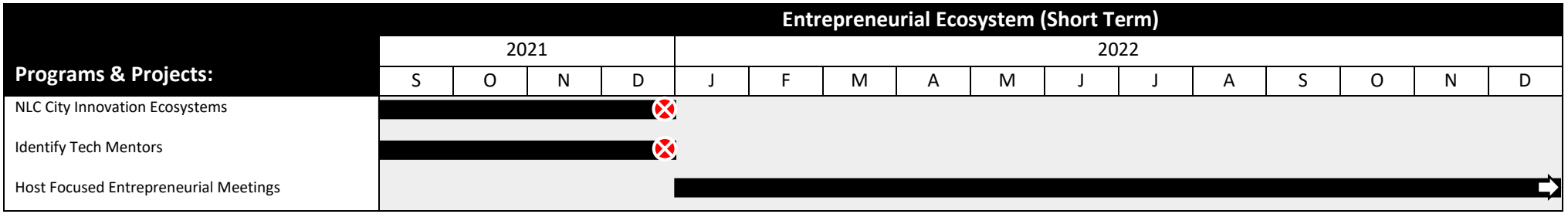
<u>Short Term</u>

1. NLC City Innovation Ecosystems September 2021
2. Identify Tech Mentors
3. Host Focused Entrepreneurial Meetings

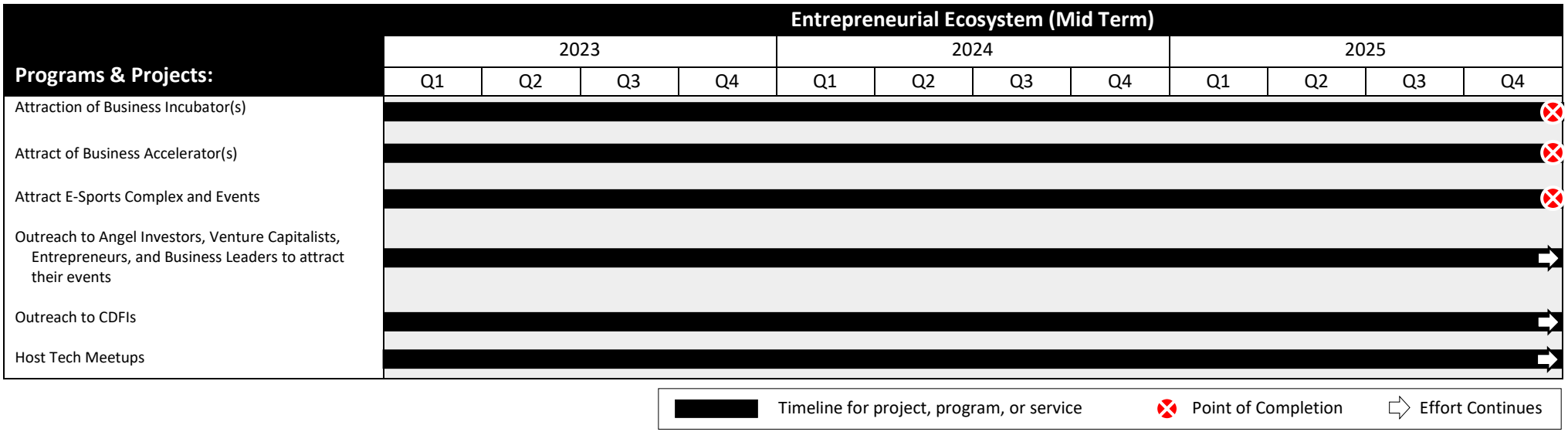
<u>Mid Term</u>

1. Attraction of Business Incubator(s)
2. Attraction of Business Accelerator(s)
3. Attract E-Sports Complex and Events
4. Outreach to Angel Investors, Venture Capitalists, Entrepreneurs, and Business Leaders to attract their events. Potential events include the following:
    - Contact Seattle Angel Conference
    - Contact Startups Meet Angels (Meetup Group)
    - Puget Sound Venture Club
    - Keiretsu Forum Seattle Meetups
    - Seattle Foundation – Neighbor to Neighbor Event
    - TiE Angels Virtual Pitch
    - Microsoft for Startups
    - Venture Out: "Pitch Night"
    - Female Founders Alliance Summit
    - Attraction of Mature Technology Companies
5. Outreach to CDFIs
6. Host Tech Meetups

Entrepreneurial Ecosystem (Short Term)



Entrepreneurial Ecosystem (Mid Term)



University, College, & School District Relations

Action

Create and ensure long term relationship with a research University, college and local school district

Purpose

Create a beneficial relationship with universities such as University of Washington, both Seattle and Tacoma campuses; community colleges such as Highline; Federal Way School District, and other educational institutions that support STEM education and training, research and innovation and supportive of entrepreneurial success through education, mentoring, and networking.

Programs, Projects and Timeframe

Short Term

1. Outreach to ascertain opportunities with the Contact Centers, Facilities, Incubators and Laboratories at the University of Washington to including but not limited to the following:

   UW Seattle CoMotion Labs; UW/BECU FinTech Incubator, Seattle; UW/Sound Credit Union FinTech Incubator, Tacoma; UW Coding Bootcamp; UW FinTech Bootcamp; Technology & Social Change Group; UW Boeing Advanced Research Center, BARC; UW Center for Advanced Materials in Transport Aircraft Structures; UW Autonomous Insect Robotics Laboratory; UW AMP Lab ;UW Washington Nanofabrication Facility, The Mill – UW Makerspace; UW Seattle MetroLab, UW Clean Energy Institute; eScience Institute: Data Science for Social Goo; Tech Policy Lab; Urbanalytics; Urban@UW; Mobility Innovation Center; Autonomous Flight Systems Laboratory; Join Center for Aerospace Technology Innovation; Supply Chain Transportation and Logistics Center

2. Outreach to ascertain opportunities with the Centers, Facilities Incubators and Laboratories at Washington State University including but not limited to the following: WSU Commercialization Tech and Intellectual Property Incubator; WSU Urbanova;

3. Contact to ascertain opportunities with the State Centers of Excellence: Aerospace & Advanced Materials Manufacturing; Agriculture & Natural Resource; Allied Health; Careers in Education; Clean Energy; Construction; Global Trade & Supply Chain Management; Homeland Security-Emergency Management; Information & Computing Technology; Marine Manufacturing & Technology; Semiconductor & Electronics Manufacturing

4. Outreach to Highline College and other community colleges engaged in STEM programming

5. Ensure alignment with Federal Way School District and the Broadband feasibility and analysis

Mid Term

1. Outreach to the following centers, facilities, laboratories:
   A. Pacific Northwest National Laboratory
   B. Outreach to private Coding School(s)
   C. Mayor's Outreach to Universities and Technology Connection

2. Continued Alignment with Federal Way School District and the Broadband feasibility and analysis

Long Term

1. Continued Alignment with Federal Way School District and the Broadband feasibility and analysis

Performance Measures

Achieve outreach within the identified times to the universities and colleges. Ensure alignment with Federal Way School District.

University, College, and School District Relations (Short Term)

Programs & Projects:	2021				2022											
	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D
Outreach to ascertain opportunities with the Contact Centers, Facilities, Incubators, and Laboratories at the University of Washington	⊗															
Outreach to ascertain opportunities with the Centers, Facilities Incubators, and Laboratories at Washington State University	⊗															
Outreach to Highline College and other community colleges engaged in STEM programming	⊗															
Contact to ascertain opportunities with the State Centers of Excellence	⊗															
Ensure alignment with Federal Way School District and the Broadband feasibility and analysis																⊗

University, College, and School District Relations (Mid Term)

Programs & Projects:	2023				2024				2025			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Outreach to the Pacific Northwest National Laboratory	⊗											
Outreach to private Coding School(s)	⊗											
Mayor's Outreach to Universities and Technology Connection	⊗											
Continued Alignment with Federal Way School District and the Broadband feasibility and analysis												⇨

University, College, and School Districts Relations (Long Term)

Programs & Projects:	2026				2027				2028				2029			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Continued Alignment with Federal Way School District and the Broadband feasibility and analysis																⇨

▬ Timeline for project, program, or service ⊗ Point of Completion ⇨ Effort Continues

Diversity, Equity, and Inclusion

Action

Become a community that offers tools and resources designed to help local elected leaders build safe places where people from all racial, ethnic and cultural backgrounds thrive socially, economically, academically and physically.

Purpose

This Direction will enable Federal Way to achieve:

- Foster new and more effective collaboration and cooperation methods amongst our citizens, entrepreneurs, and public partners through hosting meetings or events focused on working together to fix economic systems.

- Invest in People and Education. Every person should have access to resources to elevate their economic situation – from education and career training to capital and technical assistance.

- Provide all businesses and entrepreneurs with equal opportunity to obtain city contracts for businesses owned by women and people of color.

- Provide access to capital and resources for BIPOC and Immigrant Businesses and Entrepreneurs while lowering barriers they face to start, grow, and sustain their businesses.

Programs, Projects and Timeframe

<u>Short Term</u>

1. Broadband Development
2. Focused meetings with Entrepreneurs
    - Meetings with Black Entrepreneurs (Virtual or In Person)
    - Meetings with Latinx Entrepreneurs (Virtual or In Person)
    - Meetings with Asian Entrepreneurs (Virtual or In Person)
    - Meetings with Women Entrepreneurs (Virtual or In Person)
3. Align BIPOC with Regional Entrepreneurs and Institutions

<u>Mid Term</u>

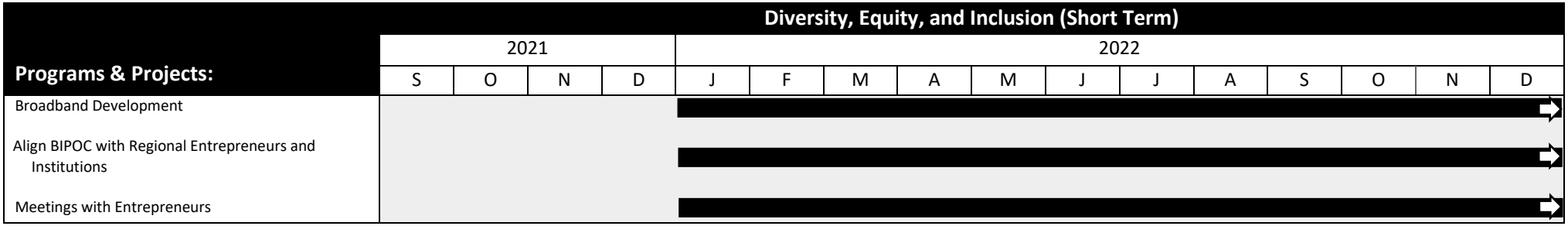
1. National Pilot Program; Seniors and FinTech
2. Work with Angel Investors
3. Work with CDFIs
4. Work with VCs
5. Work with Equity and Institutional Investors

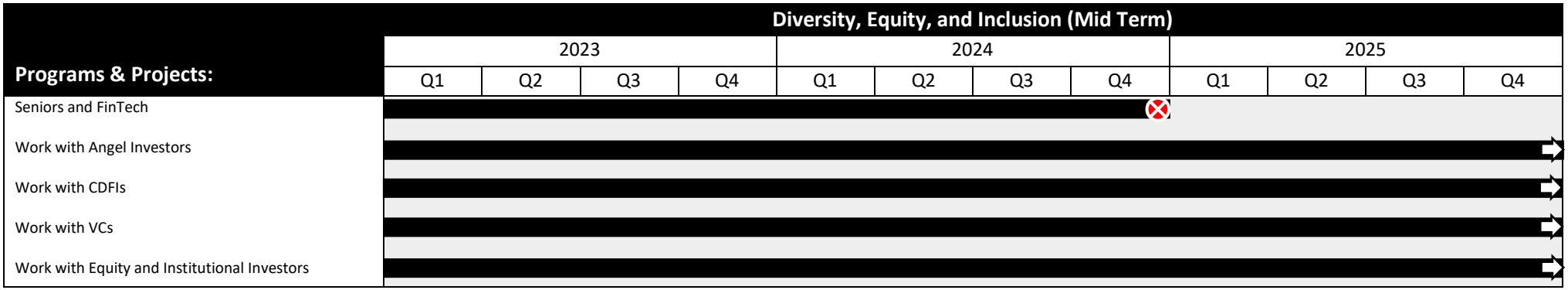
<u>Long Term</u>

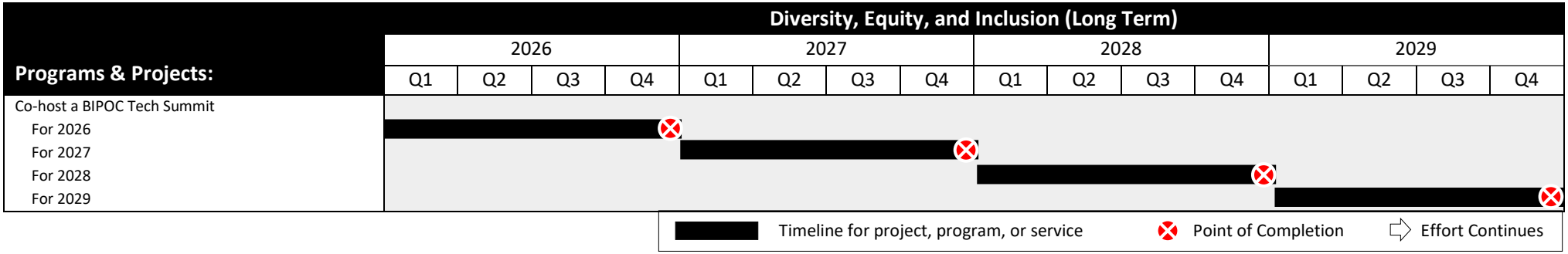
1. Co-host a BIPOC Tech Summit for years 2026, 2027, 2028, 2029

Performance Measurements

Increased number of BIPOC owned businesses; 100% access to broadband network; entrepreneurial meetings; develop broadband to reduce digital divide.


Diversity, Equity, and Inclusion (Short Term)
Programs & Projects:
2021
2022
S O N D J F M A M J J A S O N D
Broadband Development
Align BIPOC with Regional Entrepreneurs and Institutions
Meetings with Entrepreneurs


Diversity, Equity, and Inclusion (Mid Term)
Programs & Projects:
2023
2024
2025
Q1 Q2 Q3 Q4 Q1 Q2 Q3 Q4 Q1 Q2 Q3 Q4
Seniors and FinTech
Work with Angel Investors
Work with CDFIs
Work with VCs
Work with Equity and Institutional Investors


Diversity, Equity, and Inclusion (Long Term)
Programs & Projects:
2026
2027
2028
2029
Q1 Q2 Q3 Q4 Q1 Q2 Q3 Q4 Q1 Q2 Q3 Q4 Q1 Q2 Q3 Q4
Co-host a BIPOC Tech Summit
For 2026
For 2027
For 2028
For 2029
Timeline for project, program, or service
Point of Completion
Effort Continues

Use of Technology & Data Analytics

Purpose

Identification of use of technologies to enhance the organizational effectiveness, efficiency, accountability and transparency of operations of the City of Federal Way**.**

Purpose

Ongoing analysis of how to make Federal Way a Smart City.

Formulation of programs and projects that identify, analyze and examine the feasibility of use of data collected by the City through its processes of business and determine its use for organization effectiveness and efficiencies and its value to the marketplace potentially leading to subsequent monetization.

For example, today the City has a variety of data collection points from its 250 "Safe City" cameras, its community & customer service program "Eyes on Federal Way," which provides citizen interaction and subsequent resolution on issues impacting the community including graffiti abatement, trash clean-up, to grocery carts on public right-a-way. The collection of building permit activity, to its numerous websites, which all provide different data sets that can enhance the organizational effectiveness and efficiency of the operations of the City.

Programs, Projects and Timeframe

Short Term

1. Identify at least one new use of technology/innovation per year to incorporate into the City's process of "Doing Business" for 2022 and beyond
2. Look at other types of Sensors and Monitors towards Furthering the Smart City for 2022

Mid-Term

1. Formulate Data Strategy – Identify what the City wants to achieve by using data collection
2. Identify and recommend how to collect, store and access the existing collection of data
3. Identify and analyze feasibility of how to use existing data collected by City to improve its decision making, customer service and organizational effectiveness and efficiency
4. Look at Other Sensors and Monitors Towards Furthering the Smart City for 2023, 2024, 2025
5. Identify at least one (1) new use of technology/innovation to incorporate into the City's process of "Doing Business" for 2023, 2024, 2025

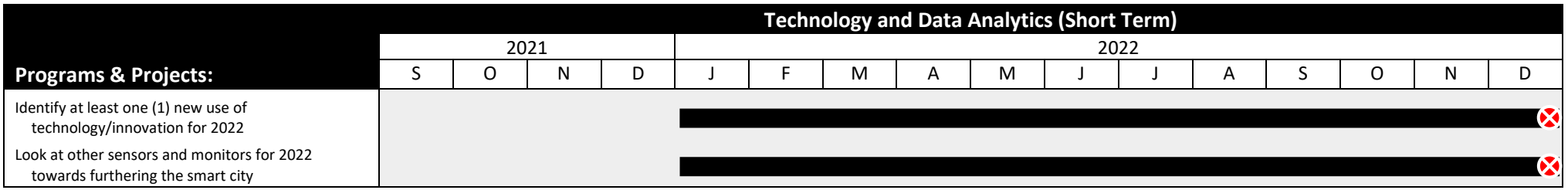
Long Term

1. Identify at least one (1) new use of technology/innovation to incorporate into the City's process of "Doing Business" for 2026, 2027, 2028, 2029
2. Analysis, identification of use of data collected (not impacting civil rights) for the purpose of monetizing data
3. Look at Other Sensors and Monitors Towards Furthering the Smart City for 2026, 2027, 2028, 2029

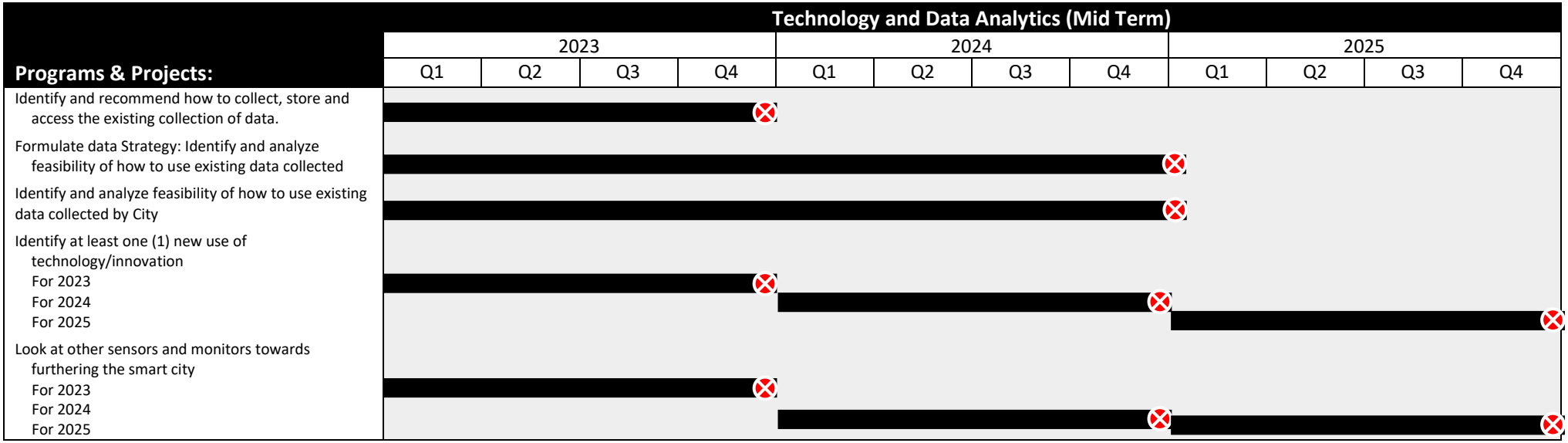
Performance Measurements

Ensure that programs, projects and services are in alignment with city policies. Budget and timelines are achieved within the identified parameters. Enhancement to citywide and departmental budgets. Use and adoption of new technologies and creation of partnerships.

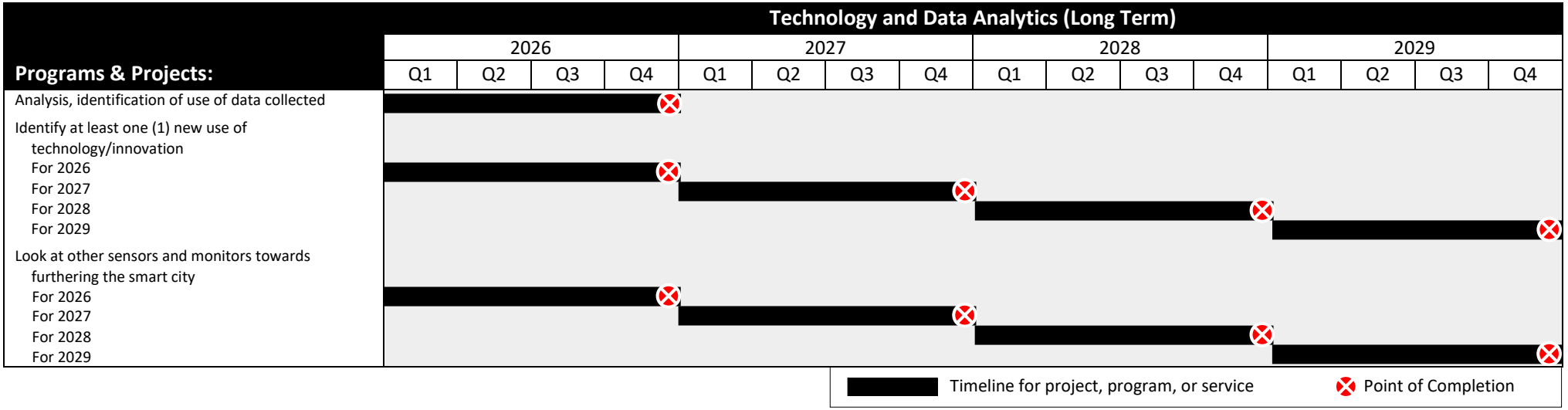
Technology and Data Analytics (Short Term)



Technology and Data Analytics (Mid Term)



Technology and Data Analytics (Long Term)



Business Attraction

Action

Attraction of businesses of all sizes and maturities to the City, especially in target technology sectors.

Purpose

Business attraction round out the City's economic development strategy by providing jobs and needed goods and services. Business attraction is sometimes considered the most well understood of economic development strategies. It involves the inventorying of community assets and goals subsequently leading to the design and implementation of a marketing program geared towards attracting prospective tech-based firms to Federal Way

Short Term

1. Attraction of Mature Tech Companies, especially in Target Sectors:
    a. ICT,
    b. Biotechnology,
    c. Cleantech,
    d. Advanced Manufacturing,
    e. Smart Cities,
    f. FinTech,
    g. Transportation & Mobility tech

Mid-Term

1. Host International Underwater Robotics Competition 2023, 2025
2. Attract Startup Studio or Business Foundry

Long Term

1. Host International Underwater Robotics Competition 2027, 2029
2. Attract Business Incubator(s)
3. Attract Business Accelerator(s)
4. Attract E-Sports Complex and Events
5. Attract Innovation Center(s)

Performance Measures

Key metrics to be incorporated will include the following:

- # & % of Small Businesses Assisted
- Amount of Private Sector Investment
- Net Job Growth
- # of New Jobs
- # of New Businesses Attracted
- Increase in Tax Base

Business Attraction (Short Term)

Programs & Projects:	2021				2022											
	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D
Attract mature technology companies: ICT, Life Sciences, Biotechnology, Cleantech, Advanced Manufacturing, Smart Cities, Fintech, Transportation/Mobility	➡															

Business Attraction (Mid Term)

Programs & Projects:	2023				2024				2025			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Attract Startup Studio or Business Foundry												⊗
Hosting International Robotics Competitions												
For 2023				⊗								
For 2025					⊗							

Business Attraction (Long Term)

Programs & Projects:	2026				2027				2028				2029			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Attract Business Incubator(s)																⊗
Attract Business Accelerator(s)																⊗
Attract E-Sports Complex and Events																⊗
Attract Innovation Center(s)																⊗
Hosting International Robotics Competitions																
For 2027								⊗								
For 2029																⊗

Timeline for project, program, or service ⊗ Point of Completion ⇨ Effort Continues

Workforce Development

Action

Identify the characteristics of the skills and education of the labor force of Federal Way to understand how it aligns with the Tech sector.

Purpose

The Center for Applied Urban Research at the University of Washington Tacoma will work with Federal Way and South Puget Sound communities to understand the characteristics of the region's labor force and identify information to help guide and prepare for the region's economic future. It will conduct a regional study in four phases focused on the following activities for the participating jurisdictions (currently Pierce County, Tacoma, Lakewood, Sumner and Federal Way). Areas will include all of Pierce County and South King County as a single region. Detailed data at the local level will be provided to participating jurisdictions only.

This Direction will enable Federal way to understand the following in in order to achieve a tech hub:

- Analyze the most recent resident occupations/skills centered on the occupational structure and net differential between resident workers, their skills and local jobs by each of the occupation categories listed below.
    A. Construction,
    B. Biotechnology,
    C. Energy,
    D. Transportation,
    E. Manufacturing, and
    F. Information/Communication.

- Determine the overall capacity of the region to attract and grow related industries based on its residents' skills.

- Conduct an online survey of resident workers to identify their place of work by zip code and their occupation benchmarked against economic data and analyze the results. Participating jurisdictions will provide input into questionnaire design and will perform outreach to encourage residents to complete the survey.

- Provide an estimate of the growth potential for the region and individual jurisdictions to expand and attract companies here based on resident skills.

- Identify the work-at-home potential for various job categories and ascertain how this can benefit the participating jurisdictions.

Programs, Projects and Timeframe

<u>Short Term</u>

1. Phase One draft analysis: October 30, 2021
2. Phase Two draft survey results: January 15, 2022
3. Phases Three and Four analyses: February 28, 2022
4. Final Report: March 31, 2022

Mid Term

1. Tech Workforce Reports for Mid Term

Long Term

1. Tech Workforce Reports for Long Term

Performance Measures

A report will be written that describes the research conducted, survey results and data analyzed as well as provides strategies to guide economic development in the region and in each of the participating jurisdictions.

More specifically, the report will:

- Gauge the extent of the potential for continued remote work for the region's residents and how this can support the local economy,
- Ascertain the need to create co-working and other flexible office environments,
- Identify how to attract and support small firms and satellite offices,
- Recognize potential hubs and spokes for a diversity of employment centers (at the regional level),
- Define policy and other strategies to grow and sustain the economy in the South Puget Sound region.

Workforce (Short Term)

Programs & Projects:	2021				2022											
	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D
Phase One draft analysis																
Phase Two draft survey results																
Phases Three and Four analyses																
Final Report																

Workforce (Mid Term)

Programs & Projects:	2023				2024				2025			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Tech Workforce Reports for Mid Term	PENDING PHASE 2											

Workforce (Long Term)

Programs & Projects:	2026				2027				2028				2029			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Tech Workforce Reports for Long Term	PENDING PHASE 2															

Timeline for project, program, or service

Point of Completion

Master Chart – Programs and Projects by Timeframe

Short Term

A-i Complete and Present Annual Tech Hub Reports for years 2021, 2022
B-i Align City broadband expansion with new State Policy and broadband goals
B-ii Approval by City Council for State Grant and Matching Funds
B-iii Submit State CERB Grant Application
B-iv Approval by City Council to Issue RFP
B-v Issue RFP
B-vi RFP Awarded and RFP Contract Completed
B-vii Creation of Broadband Advisory Taskforce (BAT)
C-i NLC City Innovation Ecosystems
C-ii Identify Tech Mentors
C-iii/E-ii Host Focused Entrepreneurial Meetings with Black, Latinx, Asian, and Women Entrepreneurs
D-i Outreach to ascertain opportunities with the University of Washington including but not limited to Laboratories, Contact Centers, Facilities, and Incubators
D-ii Outreach to ascertain opportunities with Washington State University including but not limited to Laboratories, Contact Centers, Facilities, and Incubators
D-iii Outreach to Highline College and other community colleges engaged in STEM programming
D-iv Outreach to ascertain opportunities with the Washington State Centers of Excellence
D-v Ensure alignment with Federal Way School District and the Broadband feasibility and analysis
E-i Broadband Development
E-iii Align BIPOC with Regional Entrepreneurs and Institutions
F-i Identify at least one new use of technology/innovation per year to incorporate into the City's process of "Doing Business" for 2022
F-ii Look at Other Sensors and Monitors Towards Furthering the Smart City for 2022
G-i Attraction of Mature Tech Companies
H-i Initiate a Tech Workforce Report

- Phase One draft analysis
- Phase Two draft survey results
- Phases Three and Four analysis
- Final Report

Mid Term

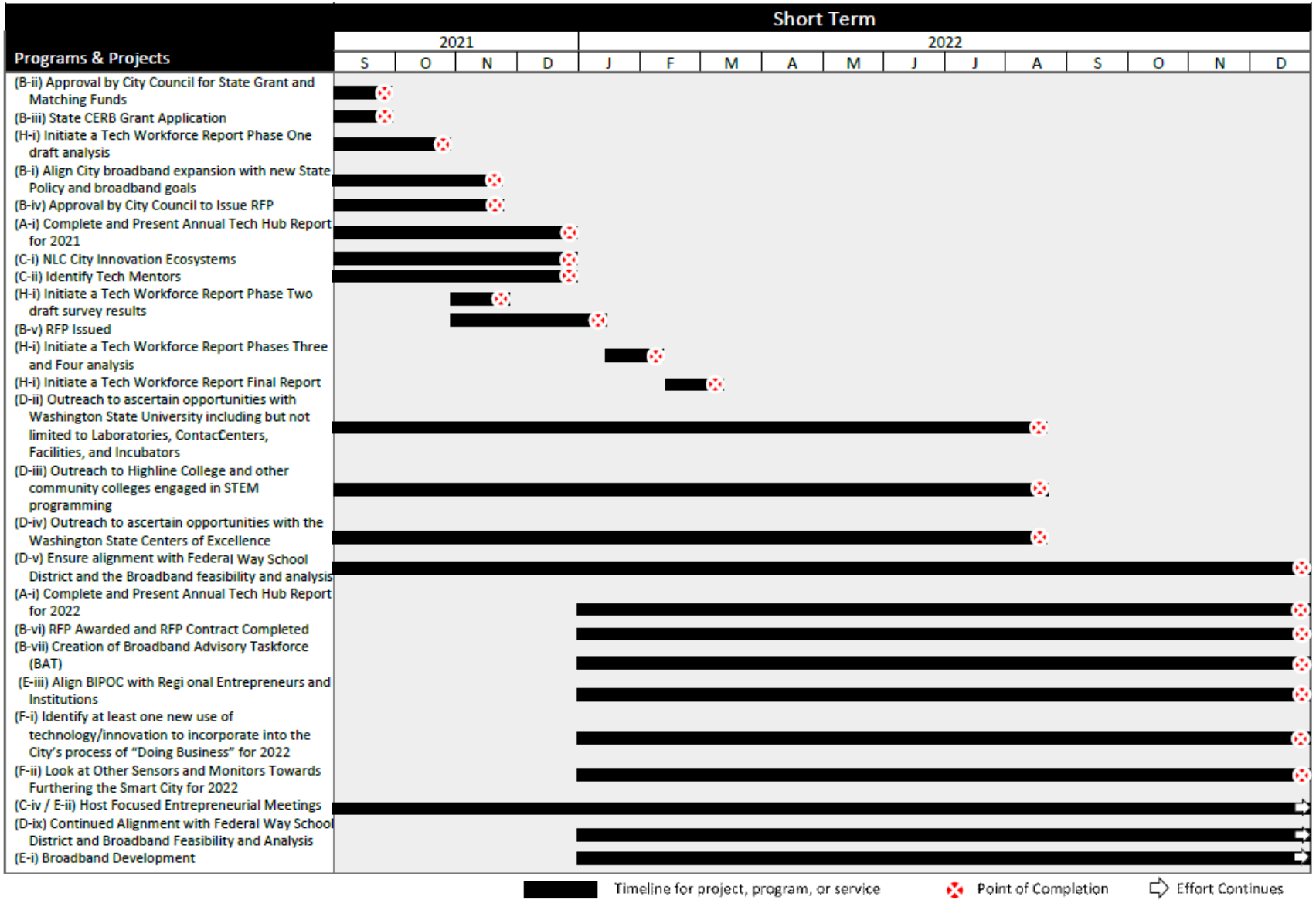
A-ii Complete and present annual Tech Hub reports for years 2023, 2024, 2025
B-viii Approval by City Council for Funding of Engineering and Design study
B-ix Approval by City Council for RFP Issue
B-x RFP Awarded and RFP Contract Completed
B-xi Selection of Broadband Model (Public, Private, PPP)
B-xii By 2024, provide minimum download speeds of 25 Mbps and minimum upload speeds of 3 Mbps to businesses and residents within City limits
B-xiii Approval by City Council for Funding of Construction
B-xiv Approval by City Council for RFP Issue
B-xv RFP Awarded
C-iv Attraction of Business Incubator(s)
C-v Attraction of Business Accelerator(s)
C-vi Attract E-Sports Complex and Events

C-vii Outreach to Angel Investors, Venture Capitalists, Entrepreneurs, and Business Leaders to attract their events
C-viii Outreach to CDFIs
C-ix Host Tech Meetups
D-vi Outreach to the Pacific Northwest National Laboratory
D-vii Mayor's Outreach to Universities and Technology Connection
D-viii Outreach to Private Coding School(s)
D-ix Continued Alignment with Federal Way School District and Broadband Feasibility and Analysis
E-iv National Pilot Program; Seniors and Fintech
E-v Work with Angel Investors
E-vi Work with CDFIs
E-vii Work with VCs
E-viii Work with Equity and Institutional Investors
F-iii Formulate Data Strategy – Identify what the City wants to achieve by using data collection
F-iv Identify and recommend how to collect, store and access the existing collection of data
F-v Identify and analyze feasibility of how to use existing data collected by City to improve its decision making, customer service and organizational effectiveness and efficiency
F-vi Look at Other Sensors and Monitors Towards Furthering the Smart City for 2023, 2024, 2025
F-vii Identify at least one (1) new use of technology/innovation to incorporate into the City's process of "Doing Business" for 2023, 2024, 2025
G-ii Host International Underwater Robotics Competition 2023, 2025
G-iii Attract Startup Studio or Business Foundry
H-ii Tech Workforce Reports for Mid Term

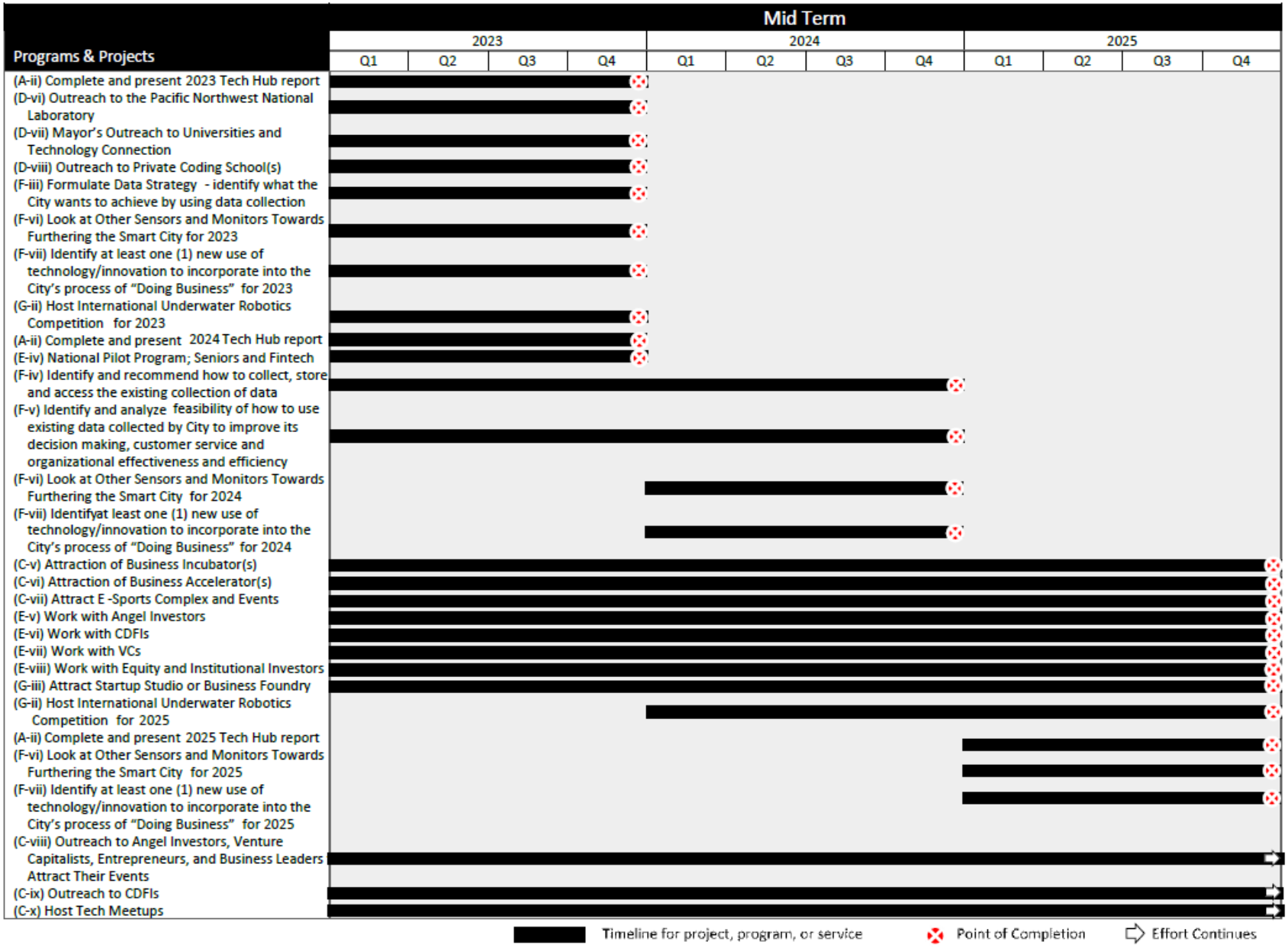
Long Term

A-iii Complete and present annual Tech Hub reports for years 2026, 2027, 2028, 2029
B-xvi Complete construction of broadband network
B-xvii Commence Broadband Operations
B-xviii By 2026, all Washington communities should have access to at least 1 Gbps symmetrical broadband service at anchor institutions like schools, hospitals, libraries, and buildings.
B-xix By 2028, all Washington businesses and residences have access to at least one broadband provider with at least 150 Mbps symmetrical broadband service
E-ix Co-host a BIPOC Tech Summit for years 2026, 2027, 2028, 2029
F-viii Identify at least one (1) new use of technology/innovation to incorporate into the City's process of "Doing Business" for 2026, 2027, 2028, 2029
F-ix Analysis, identification of use of data collected (not impacting civil rights) for the purpose of monetizing data
F-x Look at Other Sensors and Monitors Towards Furthering the Smart City for 2026, 2027, 2028, 2029
G-iv Host International Underwater Robotics Competition 2027, 2029
G-v Attract Business Incubator(s)
G-vi Attract Business Accelerator(s)
G-vii Attract E-Sports Complex and Events
G-viii Attract Innovation Center(s)
H-iii Tech Workforce Reports for Long Ter

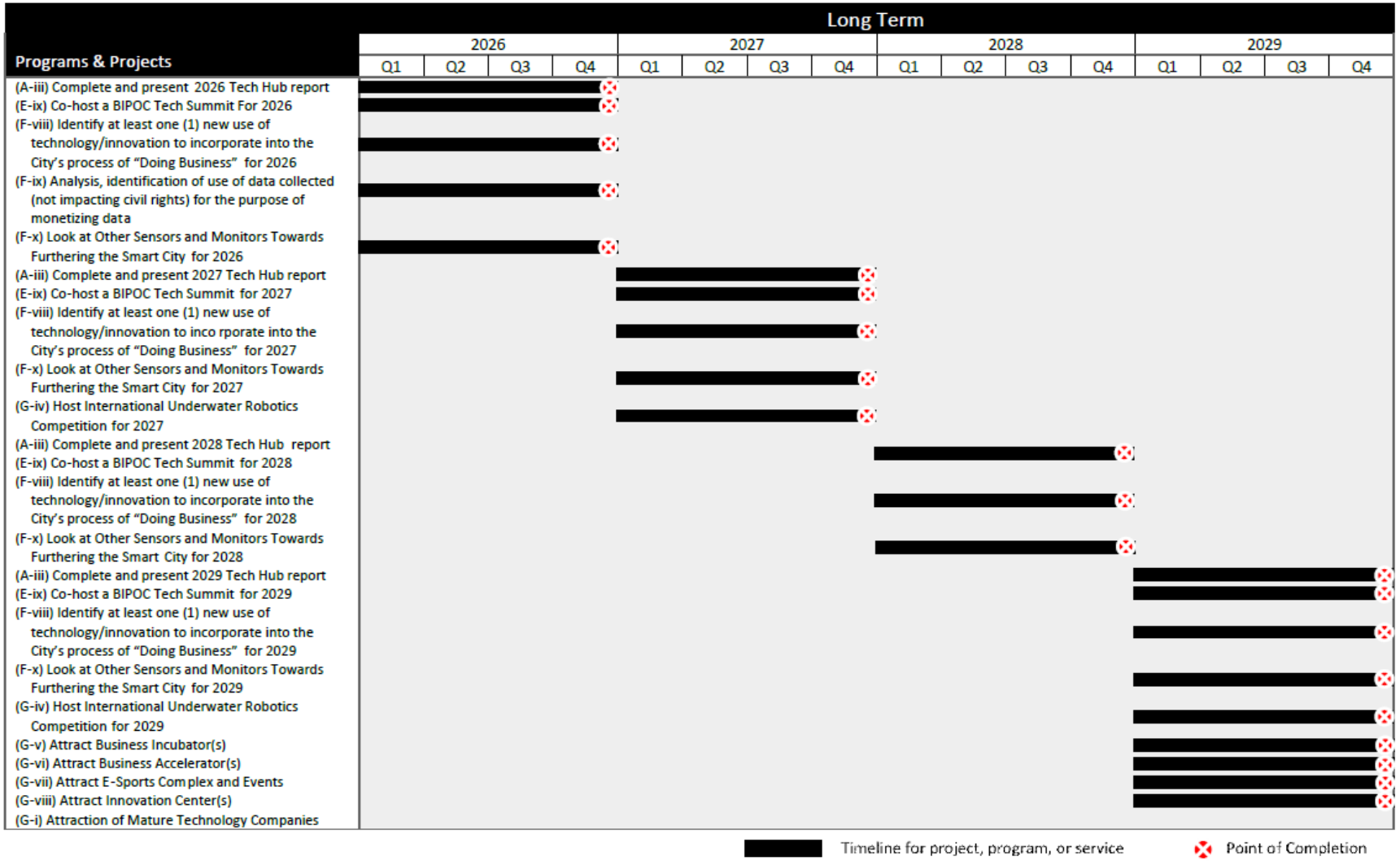
Master Chart – Programs and Projects by Timeframe



Master Chart – Programs and Projects by Timeframe



Master Chart – Programs and Projects by Timeframe



Master Chart – Programs & Projects

A Organization & Initiation of Effort

A-i Complete and present annual Tech Hub reports for years 2021, 2022
A-ii Complete and present annual Tech Hub reports for years 2023, 2024, 2025
A-iii Complete and present annual Tech Hub reports for years 2026, 2027, 2028, 2029

B infrastructure/Broadband

B-i Align City broadband expansion with new State Policy and broadband goals
B-ii Approval by City Council for State Grant and Matching Funds
B-iii State CERB Grant Application
B-iv Approval by City Council to Issue RFP
B-v RFP Issued
B-vi RFP Awarded and RFP Contract Completed
B-vii Creation of Broadband Advisory Taskforce (BAT)
B-viii Approval by City Council for Funding of Engineering and Design study
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B-xii By 2024, provide minimum download speeds of 25 Mbps and minimum upload speeds of 3 Mbps to businesses and residents within City limits
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B-xvii Commence Broadband Operations
B-xviii By 2026, all Washington communities should have access to at least 1 Gbps symmetrical broadband service at anchor institutions like schools, hospitals, libraries, and buildings.
B-xix By 2028, all Washington businesses and residences have access to at least one broadband provider with at least 150 Mbps symmetrical broadband service

C Entrepreneurial Ecosystem

C-i NLC City Innovation Ecosystems
C-ii Identify Tech Mentors
C-iii Host Focused Entrepreneurial Meetings
C-iv Attraction of Business Incubator(s)
C-v Attraction of Business Accelerator(s)
C-vi Attract E-Sports Complex and Events
C-vii Outreach to Angel Investors, Venture Capitalists, Entrepreneurs, and Business Leaders to attract their events
C-viii Outreach to CDFIs
C-ix Host Tech Meetups

D University, College, and School District Relations

D-i Outreach to ascertain opportunities with the University of Washington including but not limited to Laboratories, Contact Centers, Facilities, and Incubators

D-ii Outreach to ascertain opportunities with Washington State University including but not limited to Laboratories, Contact Centers, Facilities, and Incubators

D-iii Outreach to Highline College and other community colleges engaged in STEM programming

D-iv Outreach to ascertain opportunities with the Washington State Centers of Excellence

D-v Ensure alignment with Federal Way School District and the Broadband feasibility and analysis

D-vi Outreach to the Pacific Northwest National Laboratory

D-vii Mayor's Outreach to Universities and Technology Connection

D-viii Outreach to Private Coding School(s)

D-ix Continued Alignment with Federal Way School District and Broadband Feasibility and Analysis

E Diversity, Equity, and Inclusion

E-i Broadband Development

E-ii Focus Meetings with Entrepreneurs

- Black Entrepreneurs
- Latinx Entrepreneurs
- Asian Entrepreneurs
- Women Entrepreneurs

E-iii Align BIPOC with Regional Entrepreneurs and Institutions

E-iv National Pilot Program; Seniors and Fintech

E-v Work with Angel Investors

E-vi Work with CDFIs

E-vii Work with VCs

E-viii Work with Equity and Institutional Investors

E-ix Co-host a BIPOC Tech Summit for years 2026, 2027, 2028, 2029

F Technology and Data Analytics

F-i Identify at least one new use of technology/innovation per year to incorporate into the City's process of "Doing Business" for 2022

F-ii Look at Other Sensors and Monitors Towards Furthering the Smart City for 2022

F-iii Formulate Data Strategy – Identify what the City wants to achieve by using data collection

F-iv Identify and recommend how to collect, store and access the existing collection of data

F-v Identify and analyze feasibility of how to use existing data collected by City to improve its decision making, customer service and organizational effectiveness and efficiency

F-vi Look at Other Sensors and Monitors Towards Furthering Smart City for 2023, 2024, 2025

F-vii Identify at least one new use of technology/innovation per year to incorporate into the City's process of "Doing Business" for 2023, 2024, 2025

F-viii Identify at least one new use of technology/innovation per year to incorporate into the City's process of "Doing Business" for 2026, 2027, 2028, 2029

F-ix Analysis, identification of use of data collected (not impacting civil rights) for the purpose of monetizing the data

F-x Look at Other Sensors and Monitors Towards Furthering Smart City for 2026, 2027, 2028, 2029

G Business Attraction

G-i Attraction of Mature Technology Companies
G-ii Host International Underwater Robotics Competition 2023, 2025
G-iii Attract Startup Studio or Business Foundry
G-iv Host International Underwater Robotics Competition 2027, 2029
G-v Attract Business Incubator(s)
G-vi Attract Business Accelerator(s)
G-vii Attract E-Sports Complex and Events
G-viii Attract Innovation Center(s)

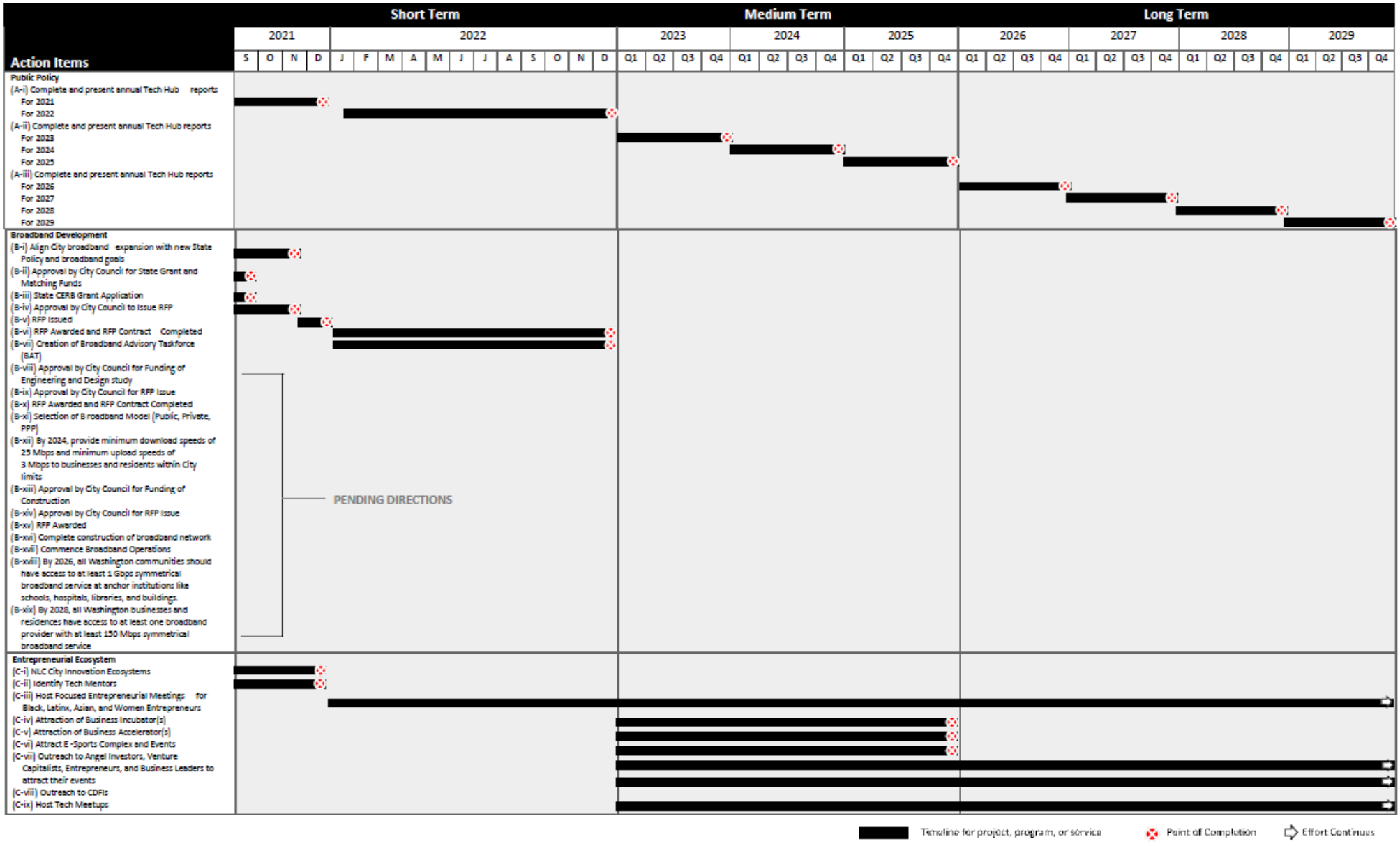
H Workforce Development

H-i Initiate Workforce Report

- Phase One draft analysis
- Phase Two draft survey results
- Phases Three and Four analysis
- Final Report

H-ii Tech Workforce Reports for Mid Term
H-iii Tech Workforce Reports for Long Term

Master Chart – By Action



Master Chart – By Action

Master Chart – By Action

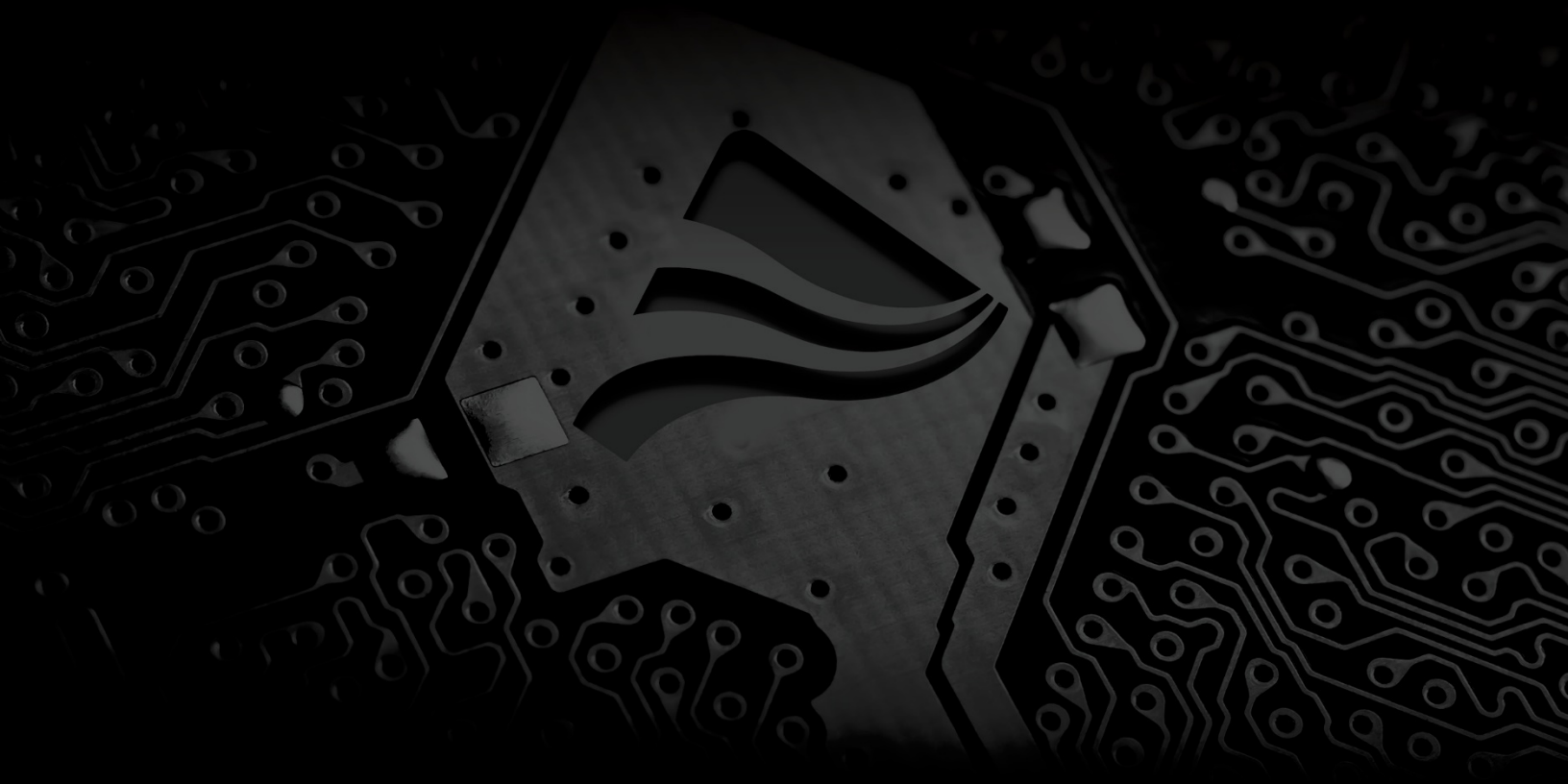
Summary

The vision of this effort is to achieve a prosperous economy and quality of life for all Federal Way residents and businesses by becoming a tech hub.

> ***Achieve a Prosperous Economy and Quality of Life for all Federal Way Residents and Businesses by becoming a Tech Hub.***

Additionally, the research has defined 8 major actions, and a multitude of corresponding programs and projects. The Actions include the following:

- Organization and Initiative
- Infrastructure/Broadband Service
- Building an Entrepreneurial Ecosystem
- University, College, and School District Relations
- Diversity, Equity, and Inclusion
- Use of Technology and Data Analytics
- Business Attraction
- Workforce Development



CONCLUSION

CONCLUSION

If Federal Way succeeds at becoming a tech hub, what does success look like?

A Picture of Success
The research identified that there are many cities of similar size and characteristics to Federal Way that have been successful in becoming a tech hub. Research illustrated a series of actions needed for success. These included the following:

- Local Political Initiative & Organization
- Strong Private Sector Input and Participation
- Active Engagement with a Research University

In addition, research identified that through the case studies, that other transformation functions needed to occur as well such as deployment of an enhanced broadband system along with building a strong entrepreneurial ecosystem.

The research team also had the opportunity to meet with a variety of public officials that have had success with growth as a tech hub. The National League of Cities provided many such introductions. For example, through their generosity, the research team received input from Mayor Eric Garcetti of Los Angeles and Mayor Francis Suarez of Miami.

The mayors identified that technology, while good, can also be a divider of populations and communities. For example, Mayor Garcetti identified that he had seen firsthand how the digital divide has created issues with education, healthcare, and banking. Meanwhile, Mayor Francis Suarez identified that technology has the ability to stratify and gentrify populations and communities specifically at a time when the nation is working towards diversity, equity, and inclusion.

Sage advice was issued by both mayors illustrating that in addition to their individual thoughts on technologies, they both believe that all cities, including Federal Way must take the time to assimilate, into the digital marketplace.

Federal Way must Assimilate into the Digital Marketplace

For example, the mayors identified that in order for Federal Way to assimilate into the digital marketplace, it must work with its youth. As such, research showed that bringing in coding schools and building tech-focused curriculums in K-12 schools are additional elements to achieving success. This would help build a strong foundation for younger populations so they are able to pursue a well-rounded education and be prepared to work in current and emerging technologies.

Additionally, they identified that cities must also educate all residents. As such, senior citizens should also have readily available resources to provide them with basic knowledge about current technologies and how they interact with each other. For example, a 2019 Pew Research Center analysis found that 73% of Americans 65 and older were internet users, and Americans 60 and older spend more than half of their daily leisure time on TVs, computers, tablets, and other electronic devices. The issue is not that they do not have access to technology, it is more that they are not as familiar with tech and tech terminology, which can make adoption of new technology difficult.

Additionally, both mayors recommended that technologists and bureaucrats must engage one another to ensure timely success and inclusion.

The research also identifies that on average it took less than a decade to achieve becoming a tech hub. So, the commitment combined with a strong plan provided the impetus for the cities success.

Ambition without Pretense
Federal Way needs to think big without going overboard. However, beyond the actions city staff has crafted five aspirations to help motivate the City towards its ambition of becoming a tech hub. The aspirations include:

- *Become the City with the "Fastest Internet Speeds" in the State of Washington.*
- *Become the "Most Advanced Smart City" in the Pacific Northwest.*
- *Produce the "Finest Entrepreneurial Ecosystem" in the Pacific Northwest*
- *Become the "Most Connected City to Education" in the nation.*
- *Become the "Global Center for Disruptive Technologies."*

These outcomes reflect Federal Way's desire for shared prosperity and well-being, for unity and inclusion, and to become a community that is *"Centered on Tech Opportunity,"* where the full potential of its citizens can be realized.

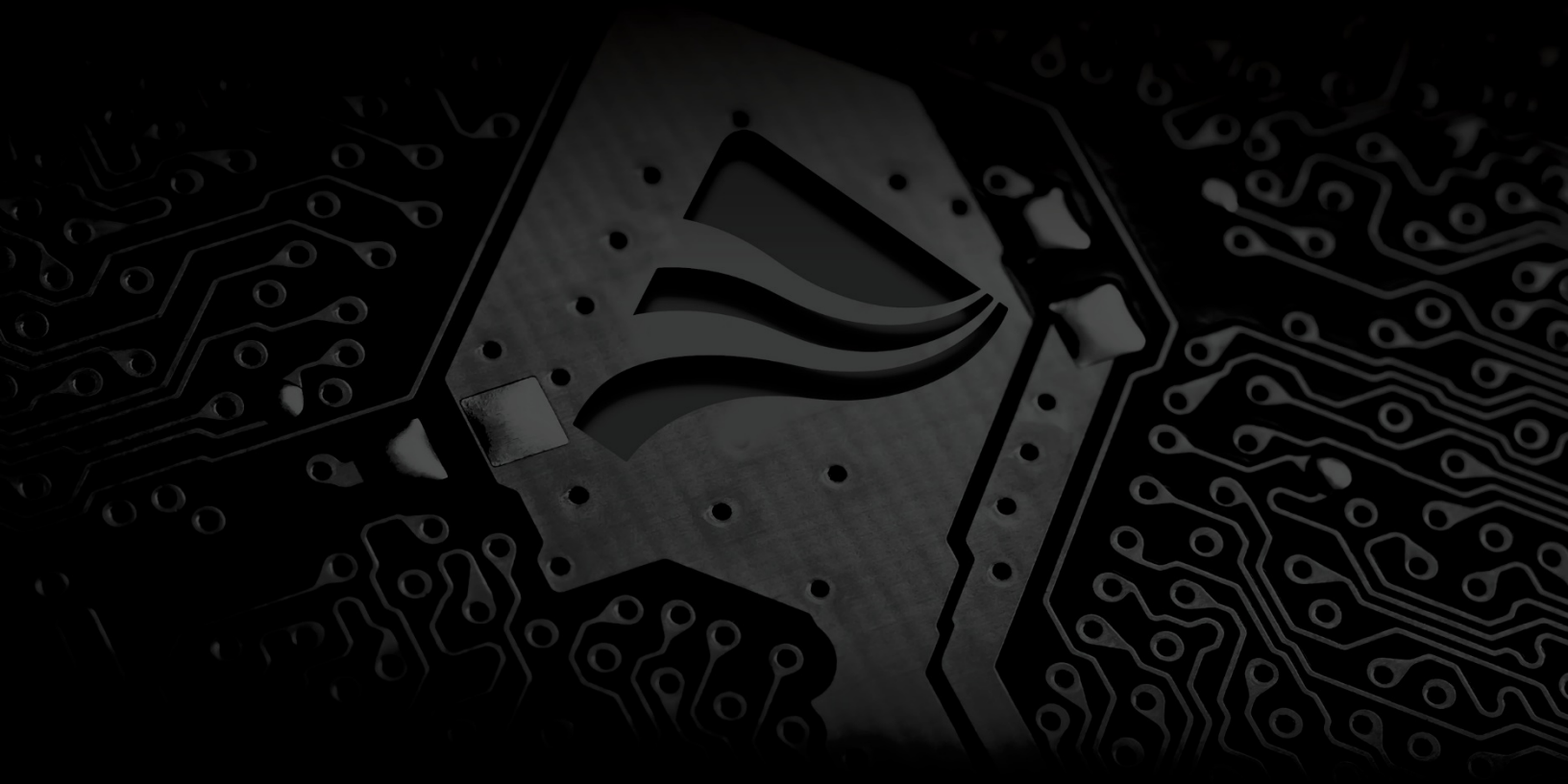
Lastly, the ability of Federal Way to achieve becoming a tech hub should be reached by keeping three key factors in its focus as it progresses towards its ambition.

4. A Need to Develop its Own Tech Credentials
   Federal Way needs to create an unequalled entrepreneurial ecosystem that catches the attention of startups in the tech sector. The vibrancy of a strong entrepreneurial ecosystem should be a source of collective community pride. Federal Way should not hesitate to support the ambition of its entrepreneurs through education, and structured mentorship.

5. Entrepreneurial Parity
   The importance of entrepreneurs to Federal Way's economy plays a significant role in its economic future. Federal Way must focus its attention on supporting its entrepreneurs, not just large corporations.

6. Engineering our Own Center of Tech Gravity
   The City needs to develop a powerful lure and elevate its public image. Federal Way needs to illustrate that it is fashionable and cosmopolitan. Federal Way needs to become a foothold in the Puget Sound for tech- focused entrepreneurs in the effort to become a tech hub.

Recall, this is not a fairy tale but a plan of action based on realty, where its success lies in the journey not a destination, to live happily ever after.



GLOSSARY

GLOSSARY

Accelerator

Business accelerators support early-stage, growth-driven companies through education, mentorship, and financing. Startups enter accelerators for a fixed period of time. They are accepted as part of a cohort of companies on an annual basis or designated timeframe, or by rolling admission. An accelerator works as a holistic business advisory service that provides an intense, rapid, and immersive education aimed at accelerating the life cycle of young, innovative companies, compressing years' worth of learning-by-doing into just a few months. During the program, cohorts of accepted startups are put through rigorous mentoring and training where startups collaborate in their learning and network to set up future capital.

Advanced Manufacturing

Advanced manufacturing is the use of innovative technology to improve products or processes. It can include production activities that depend on information, automation, computation, software, sensing, and networking. However, as these technologies have become more widely used in the manufacturing processes, the line between traditional and advanced manufacturing has become blurred.

Agtech

Agriculture technology or Agtech is the use of technology in agriculture, horticulture, and aquaculture with the aim of improving yield, efficiency and profitability for farm managers and growers. They come up with solutions to improve farming output and quality using drones, sensors, and farm management software. AgTech is also about new farm products, next-generation farms, and urban farming.

Angel Investor

An angel investor (also known as a private investor, seed investor, or angel funder) is a high-net-worth individual who provides financial backing for small startups or entrepreneurs, typically in exchange for ownership equity in the company. Often, angel investors are found among an entrepreneur's family and friends. The funds that angel investors provide may be a one-time investment to help the business get off the ground or an ongoing injection to support and carry the company through its difficult early stages.

Application Program Interface (API)

An application programming interface, or API, enables companies to open up their applications' data and functionality to external third-party developers, business partners, and internal departments within their companies. This allows services and products to communicate with each other and leverage each other's data and functionality through a documented interface. Developers don't need to know how an API is implemented; they simply use the interface to communicate with other products and services.

Artificial Intelligence (AI)

The ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings. The term is frequently applied to the project of developing systems endowed with the intellectual processes characteristic of humans, such as the ability to reason, discover meaning, generalize, or learn from past experience.

Augmented Reality (AR)

AR enhances the real world as it exists with graphical overlays and does not, like VR, create a fully immersive experience. AR is achieved through the use of digital visual elements, sound, or other sensory stimuli delivered via technology.

Autonomous Tech

Automation is about using technology to monitor, control and/or operate any process or function with accuracy and efficiency without human intervention. Autonomous technology is about enriching automated systems with sensors, Artificial Intelligence (AI), and analytical capabilities so they can make independent decisions based on the data they collect.

AUV

AUV stands for autonomous underwater vehicle and is also known as uncrewed underwater vehicle (UUV). AUVs can be used for underwater survey missions such as detecting and mapping submerged wrecks, rocks, and obstructions that can be a hazard to navigation for commercial and recreational vessels. An AUV conducts its survey mission without operator intervention. When a mission is complete, the AUV will return to a pre-programmed location where the data can be downloaded and processed.

UAV

Unmanned aerial vehicles (UAVs) are guided autonomously, by remote control, or both and that carries sensors, target designators, offensive ordnance, or electronic transmitters designed to interfere with or destroy enemy targets. Unencumbered by crew, life-support systems, and the design-safety requirements of manned aircraft, UAVs can be remarkably efficient, offering substantially greater range and endurance than equivalent manned systems.

UGV

Unmanned ground vehicles (UGVs) are robotic systems that operate on land without an onboard human operator. They are used for a wide variety of both civilian and military applications, particularly in environments that are hazardous or unpleasant to humans and for tasks that are difficult, dull, or pose unacceptable risks.

Big Data

Put simply, big data is larger, more complex data sets, especially from new data sources. These data sets are so voluminous that traditional data processing software just can't manage them. Big data contains greater variety, arriving in increasing volumes, and with more velocity than other categories of data. This is also known as the three Vs. Value and veracity are also sometimes included as fourth and fifth Vs. These massive volumes of data can be used to address business problems, especially in tech. A large part of the value today's largest companies offer comes from their data, which they're constantly analyzing to produce more efficiency and develop new products.

Blockchain

A blockchain is a type of database. A blockchain collects information together in groups, also known as blocks, that hold sets of information. Blocks have certain storage capacities and, when filled, are chained onto the previously filled block, forming a chain of data known as the "blockchain." Blockchains are maintained in multiple copies across networks of computers.

Blue Ocean Strategy

Blue Ocean Strategy is a business strategy that creates new demand rather than competing for existing demand (see Red Ocean Strategy). Companies develop uncontested market space rather than fighting over a shrinking profit pool.

Broadband

According to the FCC, the definition of broadband internet is a minimum of 25 Mbps download and 3 Mbps upload speeds. Broadband provides high speed internet access via multiple types of technologies including fiber optics, wireless, cable, DSL, and satellite.

Business Intelligence (BI)

BI leverages software and services to transform data into actionable insights that inform an organization's strategic and tactical business decisions. BI tools access and analyze data sets and present analytical findings in reports, summaries, dashboards, graphs, charts, and maps to provide users with detailed intelligence about the state of the business.

Business-to-Business (B2B)

B2B is a form of transaction between businesses, such as one involving a manufacturer and wholesaler, or a wholesaler and a retailer. B2B refers to business that is conducted between companies, rather than between a company and individual consumer. This is in contrast to business-to-consumer (B2C) and business-to-government (B2G) transactions.

Cable

Cable Internet is a form of connection that uses a cable television infrastructure to provide access to end users. Therefore, the network that provides your cable TV connection is the same one that allows you to access the Internet. Thus, cable Internet access is usually bundled with TV channels, and the service mainly comes from a local cable TV service provider.

CAD

Computer-aided design (CAD) refers to computers being used to assist the design process in a variety of industries. With CAD software, it's possible to build an entire model in an imaginary space, letting you visualize properties like height, width, distance, material, or color before the model is used for a particular application.

Cleantech

Clean technology is any technology that reduces or eliminates pollution, waste, or use of natural resources while seeking to be competitive in cost and performance. Cleantech spans many subsectors, including energy generation, storage, infrastructure, and efficiency; transportation, water and wastewater, air and environment; materials, manufacturing/industrial, agriculture, and recycling and waste.

Cloud Computing (The Cloud)

Cloud computing is the delivery of computing services — including servers, storage, databases, networking, software, analytics, and intelligence — over the Internet to offer faster innovation, flexible resources, and economies of scale. You typically pay only for cloud services you use, helping you lower your operating costs, run your infrastructure more efficiently, and scale as your business needs change. Cloud computing makes it possible to create cloud-native applications; test and build applications; store, backup, and recover data; analyze data; stream audio and video; embed intelligence; and deliver software on demand.

Hybrid Cloud

Hybrid clouds combine public and private clouds, bound together by technology that allows data and applications to be shared between them. By allowing data and applications to move between

private and public clouds, a hybrid cloud gives your business greater flexibility, more deployment options, and helps optimize your existing infrastructure, security, and compliance.

Private Cloud

A private cloud refers to cloud computing resources used exclusively by a single business or organization. A private cloud can be physically located on the company's on-site datacenter. Some companies also pay third-party service providers to host their private cloud.

Public Cloud

Public clouds are owned and operated by a third-party cloud service provider, which delivers their computing resources, like servers and storage, over the Internet. Microsoft Azure is an example of a public cloud. With a public cloud, all hardware, software, and other supporting infrastructure is owned and managed by the cloud provider.

Coding/Programming

Coding, sometimes called computer programming, is how we communicate with computers. Code tells a computer what actions to take, and writing code is like creating a set of instructions. Coding is used to make websites and apps, process data, and perform countless other tasks. Coders/Programmers write code in a variety of coding languages depending on their tasks. For example, someone might use the coding language R to perform statistical analysis while someone might use the coding languages HTML, CSS, and JavaScript to create a website.

Computer Network

Two or more computers that are connected with one another for the purpose of communicating data electronically. Besides physically connecting computer and communication devices, a network system serves the important function of establishing a cohesive architecture that allows a variety of equipment types to transfer information in a near-seamless fashion. Two popular architectures are ISO Open Systems Interconnection (OSI) and IBM's Systems Network Architecture (SNA).

Coworking Space

A coworking space is a shared professional-style working space where you may pay a monthly fee to use a shared workspace. Coworking spaces often foster community, hold networking events, and create ideal solutions for small companies, startups, or remote workers.

Database

A database is an organized collection of structured information, or data, typically stored electronically in a computer system. A database is usually controlled by a database management system (DBMS). Together, the data and the DBMS, along with the applications that are associated with them, are referred to as a database system, often shortened to just database.

Deep Tech

Deep Tech is a term for technologies not focused on end-user services that provide technology solutions based on substantial scientific or engineering challenges. Examples include artificial intelligence, robotics, blockchain, advanced material science, photonics and electronics, biotech, and quantum computing.

Digital Subscriber Line (DSL)

A communications medium used to transfer internet signal over standard telephone lines.

Disruptive Innovation
Disruptive innovation refers to the innovation that transforms expensive or highly sophisticated products or services — previously accessible to a high-end or more-skilled segment of consumers — to those that are more affordable and accessible to a broader population. This transformation disrupts the market by displacing long-standing, established competitors.

Disruptive Technology
Disruptive technology is an innovation that significantly alters the way that consumers, industries, or businesses operate. Recent examples include ride-sharing apps, 3D-printing, and electric vehicles.

Economic Gardening
Economic gardening is an economic development model that embraces the fundamental idea that entrepreneurs drive economies. The model seeks to create jobs by supporting existing companies in a community. Economic gardening connects entrepreneurs to resources, encouraging the development of essential infrastructure and providing entrepreneurs with needed information.

Edtech
Education technology, or EdTech, refers to hardware and software designed to enhance teacher-led learning in classrooms and improve students' education outcomes.

Entrepreneur
An entrepreneur is an individual who creates a new business, bearing most of the risks and enjoying most of the rewards. The process of setting up a business is known as entrepreneurship. The entrepreneur is commonly seen as an innovator, a source of new ideas, goods, services, and is risk tolerant.

Entrepreneurial Network
Entrepreneurial networks (EN) are defined as the patterns of interpersonal relations emerging from entrepreneurial activities. In their everyday activities, entrepreneurs are in contact with a variety of other actors, playing important roles in the functioning of their business.

Equity
Equity, typically referred to as shareholders' equity or owners' equity for privately held companies, represents the amount of money that would be returned to a company’s shareholders if all of the assets were liquidated and all of the company's debt was paid off in the case of liquidation. In the case of acquisition, it is the value of company sale minus any liabilities owed by the company not transferred with the sale.

E-Sports
Competitive tournaments of video games. Esports tournaments usually consist of amateur or professional gamers competing against one another for a cash prize. For example, League of Legends pits five players against another five in a virtual battle arena to destroy each other’s base. Whereas in shooters, like Call of Duty, Overwatch and Counter-Strike, players are pit against each other in various modes to complete an objective.

Fiber Optic

Fiber-optic is capable of the fastest speeds of all the internet transmission methods, using pulses of light through fiber strands to transmit data. An optical fiber carries a signal, commonly internet, from an operator's equipment all the way to a home, business, or curb. The cable is glass or plastic in contrast to traditional copper cables.

Fintech

Fintech is any company using technology to support financial services of any type which can include regulatory tech, lending, payments, saving, investing, insurance, robo-advice, accounting, risk management, claims processing, and underwriting.

Foodtech

The Foodtech sector explores how technology can be leveraged to create efficiency and sustainability in designing, producing, choosing, delivering, and enjoying food.

Geek Culture

At its core, geek culture is maker culture. Geek culture stands out as a place where creativity, imagination, and ingenuity are prized, especially in computer programming circles. The flood of money to the sector is certainly having an impact, but for many, its true attraction is the pleasure, and the power, inherent in creating “worlds” through line by line of code, an endeavor that combines deep intellectual challenges with the pleasures common to other creative activities such as art, cooking, or music.

Graphical User Interface (GUI)

GUI stands for "graphical user interface," known also as simply "interface." An interface is the part of a piece of software that the end user actually sees and interacts with.

Hackathons

A hackathon is a competition that takes place typically within a 48-hour time limit, in which computer programmers and others involved in software development, including graphic designers, interface designers, project managers, domain experts, and others collaborate intensively on software projects to solve technological problems.

Hardware

Computer hardware is the physical components that a computer system requires to function. It encompasses everything with a circuit board that operates within a PC or laptop; including the motherboard, graphics card, CPU (Central Processing Unit), ventilation fans, webcam, power supply, and so on. Although the design of hardware differs between desktop PCs and laptops due to their differences in size, the same core components will be found in both.

Healthtech

Health technology refers to the application of organized knowledge and skills in the form of devices, medicines, vaccines, procedures, and systems developed to solve a health problem and improve quality of lives.

High Tech

High tech is an abbreviation of "high technology," is a broad term describing cutting edge or innovative technology. High tech is more complex, advanced, and/or new compared to lower technology.

ICT

Information and Communication Technologies (ICT) refers to technologies that provide access to information through telecommunications. It is similar to Information Technology (IT), but focuses primarily on communication technologies. This includes the Internet, wireless networks, cell phones, and other communication mediums.

Incubator

A startup incubator is a collaborative program for startup companies — usually physically located in one central workspace — designed to help startups in their infancy succeed by providing workspace, seed funding, mentoring, and training.

Information Technology (IT)

The study or use of systems (especially computers and telecommunications) for storing, retrieving, and sending information.

Infrastructure as a Service (IaaS)

The most basic category of cloud computing services. With IaaS, you rent IT infrastructure — servers and virtual machines (VMs), storage, networks, operating systems — from a cloud provider on a pay-as-you-go basis.

Initial Public Offering (IPO)

An initial public offering (IPO) refers to the process of offering shares of a private corporation to the public in a new stock issuance. Public share issuance allows a company to raise capital from public investors.

Innovation Center

A cross-functional plan that creates a safe haven for new ideas. With opportunities for individual and group collaboration across time zones and continents, it's a place that fosters a culture of innovation through the creation, sharing, and testing of idea.

Innovation District

Innovation districts are dense enclaves that merge the innovation and employment potential of research-oriented anchor institutions, high-growth firms, and tech and creative startups in well-designed, amenity-rich residential and commercial environments. These districts build on and revalue the intrinsic qualities of cities: proximity, density, authenticity, and vibrant places.

Internet Service Provider (ISP)

The term Internet service provider (ISP) refers to a company that provide and charge for access to the Internet to both personal and business customers. ISPs may also provide other services including email services, domain registration, web hosting, and browser packages.

Institutional Investment

An institutional investor is a company or organization that invests money on behalf of other people. Mutual funds, pensions, and insurance companies are examples. Institutional investors often buy and sell substantial blocks of stocks, bonds, or other securities and, for that reason, are considered to be the “whales” on Wall Street. The group is also viewed as more sophisticated than the average retail investor and, in some instances, are subject to less restrictive regulations.

Life Sciences

Life sciences is comprised of sciences that have to do with organisms like plants, animals, and human beings. Life science encompass companies in the fields of biotechnology, pharmaceuticals, biomedical technologies, life systems technologies, nutraceuticals, cosmeceuticals, food processing, environmental, and biomedical devices.

Local Area Network (LAN)

A collection of devices connected together in one physical location, such as a building, office, or home.

Machine Learning (ML)

Computer systems that are able to learn and adapt without following explicit instructions by using algorithms and statistical models to analyze and draw inferences from patterns in data.

Makerspace

A makerspace is a collaborative work space inside a school, library, or separate public/private facility for making, learning, exploring, and sharing that uses high tech to no tech tools. These spaces may be open to everyone or just to entrepreneurs and have a variety of maker equipment including 3D printers, laser cutters, CNC machines, soldering irons, and even sewing machines.

Mashups

A technology mashup mashes together two or more technological innovations in a single device. Examples include smart street lights, microphone headsets, and smart vehicles.

Meetups

A meet-up is defined as an event where a group of like-minded people can listen to experts, share experiences, and simply communicate. Meet-ups can address a variety of subjects. For example, tech-oriented meet-ups cover topics related to technology, such as the future of blockchains, artificial intelligence, software development, and programming languages.

Metadata

Metadata is data about data. There is a wide variety of metadata depending on its purpose, format, quality, and volume. Some of the widely used categories of metadata are: descriptive, structural, administrative, and statistical. One easy example of metadata is everything written on a letter envelope to help the actual content – the letter – get delivered to its recipient. As an internet example, HTML tags instruct web browsers how to layout the pages to make it easier for humans to read them and follow references to other pages.

Metaverse

Metaverse is a combination of the prefix “meta” (meaning beyond) and “universe.” The metaverse is a virtual-reality space in which users can interact with a computer-generated environment and other users.

Open Source Software

Open source software is software with source code that anyone can inspect, modify, and enhance. "Source code" is the part of software that most computer users don't ever see, the code computer programmers can manipulate to change how a piece of software a program or application works.

Platform as a Service (PaaS)
Platform as a service refers to cloud computing services that supply an on-demand environment for developing, testing, delivering, and managing software applications. PaaS is designed to make it easier for developers to quickly create web or mobile apps without worrying about setting up or managing the underlying infrastructure of servers, storage, network, and databases needed for development.

Private Equity
Private equity (PE) is ownership or interest in an entity that is not publicly listed or traded. A source of investment capital, PE comes from high-net-worth individuals and firms that purchase stakes in private companies or acquire control of public companies with plans to take them private and delist them from stock exchanges.

Proptech
Property tech, or Proptech, is an innovative approach to real estate in which technology optimizes the way people research, rent, buy, sell, and manage properties.

Red Ocean Strategy
A Red Ocean Strategy is a business strategy that is about beating out competition. As the market space becomes more crowded, companies compete fiercely for a greater share of limited demand.

Regtech
Regtech, or regulatory technology, is the management of regulatory processes within the financial industry through technology. The main functions of regtech include regulatory monitoring, reporting, and compliance.

Retail Investment
A retail investor, also known as an individual investor, is a non-professional investor who buys and sells securities or funds that contain a basket of securities such as mutual funds and ETFs.

SaaS
Software as a service (SaaS) is a method for delivering software applications over the Internet, on demand, and typically on a subscription basis. With SaaS, cloud providers host and manage the software application and underlying infrastructure, and handle any maintenance, like software upgrades and security patching. Users connect to the application over the Internet, usually with a web browser on their phone, tablet, or PC.

Satellite Internet
Satellite internet is wireless internet beamed down from satellites orbiting the Earth. This is in contrast to land-based internet services like cable or DSL which transmit data through wires. Since it's the only internet service that's available nationwide, satellite internet is often the only way to get online for many rural homes and businesses.

Search Engine
A search engine is a service that allows Internet users to search for content via the World Wide Web (WWW). A user enters keywords or key phrases into a search engine and receives a list of Web content results in the form of websites, images, videos, or other online data that semantically match with the search query. Search engines are becoming more and more intelligent in producing results using machine learning and AI algorithms.

Serverless Computing

Overlapping with PaaS, serverless computing focuses on building app functionality without spending time continually managing the servers and infrastructure required to do so. The cloud provider handles the setup, capacity planning, and server management for you. Serverless architectures are highly scalable and event-driven, only using resources when a specific function or trigger occurs.

Servers

A server is a computer that serves information to other computers. These served computers, called clients, can connect to a server through either a local area network (LAN) or a wide area network (WAN), such as the internet. A server collects and sends information across a network. The Citadel, also owned by Switch, is the largest data center campus in the world. It is located in Tahoe Reno, Nevada, and covers an area of 7.2 million square feet

Smart City

A smart city is a designation given to a city that incorporates information and communication technologies (ICT) to enhance the quality and performance of urban services such as energy, transportation, and utilities in order to reduce resource consumption, wastage, and overall costs. The overarching aim of a smart city is to enhance the quality of living for its citizens through data collection and technology.

Social Media

Social media is a computer-based technology that facilitates the sharing of ideas, thoughts, and information through the building of virtual networks and communities. By design, social media is internet-based and gives users quick electronic communication of content. Content includes personal information, documents, videos, and photos. Users engage with social media via a computer, tablet, or smartphone via web-based software or applications. While social media is ubiquitous in America and Europe, Asian countries like Indonesia lead the list of social media usage. More than 3.8 billion people use social media.

Software

Software is a set of instructions, data, or programs used to operate computers and execute specific tasks. Software is a generic term used to refer to applications, scripts and programs that run on a device. Software works in concert with hardware, which is the physical aspects of a computer.

SPAC

A special purpose acquisition company (SPAC) is a company with no commercial operations that is formed strictly to raise capital through an initial public offering (IPO) for the purpose of acquiring an existing company. SPACs are also known as "blank check companies."

Startup

The term startup refers to a company in the first stages of operations. Startups are founded by one or more entrepreneurs who want to develop a product or service for which they believe there is demand. These companies generally start with high costs and limited revenue, which is why they look for capital from a variety of sources such as venture capitalists.

Startup Studios/Foundries

A startup studio or foundry is a team of engineers and entrepreneurs building startups that transform industries. A key point with a studio or foundry is that you can't outsource the job of turning great ideas into great businesses. So, operators of studios or foundries are real co-founders on projects they pursue. They mentor members of the team to become the future CEOs and CTOs of the companies built.

STEM
An interdisciplinary, applied approach to learning science, technology, engineering and math (STEM).

Tech Ecosystem
An interconnected and interdependent network of diverse entities coming together to spur innovation in the tech environment pertaining to products and services in a sustainable manner. Ecosystems include business incubators and accelerators, universities and colleges, private companies, entrepreneurs, and many other technology players.

Tech Transfer
Technology transfer is the process of transferring (disseminating) technology from the person or organization that owns or holds it to another person or organization. These transfers may occur between universities, businesses (of any size, ranging from small, medium, to large), governments, across geopolitical borders, both formally and informally, and both openly and secretly. Often it occurs by concerted effort to share skills, knowledge, technologies, manufacturing methods, samples, and facilities among the participants.

Transportation and Mobility Tech
Transportation technology is referring to the electronic advancements made for traveling and vehicles. Advances include innovation in autonomous vehicles, hyperloop, maglev trains, and vehicle battery tech.

Two-Factor Authentication (2FA)
2FA is an extra layer of security used to make sure that people trying to gain access to an online account are who they say they are. First, a user will enter their username and a password. Then, instead of immediately gaining access, they will be required to provide another piece of information. This second factor could be something you know, have, or are such as answers to secret questions, a code sent to your email or phone number, or a fingerprint.

User Experience (UX) and User Interface (UI)
"User Experience Design" is often used interchangeably with terms such as "User Interface Design" and "Usability." However, while usability and user interface (UI) design are important aspects of UX design, they are subsets of it – UX design covers a vast array of other areas. A UX designer is concerned with the entire process of acquiring and integrating a product, including aspects of branding, design, usability and function. A UI is a place where interactions between humans and machines occur, allowing users to effectively operate a machine to complete a task or achieve a specific goal, like making a purchase or downloading an app.

Venture Capital (VC)
VC is a form of private equity and a type of financing that investors provide to startup companies and small businesses that are believed to have long-term growth potential. Venture capital generally comes from wealthy investors, investment banks, and any other financial institutions.

Venture Capitalists
A venture capitalist is an investor who provides capital to businesses that exhibit high growth potential in exchange for an equity stake in the business.

Virtual Reality (VR)

VR refers to a computer-generated simulation in which a person can interact within an artificial 3D environment using electronic devices such as special goggles with a screen or gloves fitted with sensors. In this simulated artificial environment, the user is able to have a realistic-feeling experience.

Wi-Fi (Wireless Fidelity)

Wi-Fi is the radio signal sent from a wireless router to a nearby device, which translates the signal into data you can see and use. The device transmits a radio signal back to the router, which connects to the internet by wire or cable. A Wi-Fi network is simply an internet connection that's shared with multiple devices in a home or business via a wireless router. The router is connected directly to your internet modem and acts as a hub to broadcast the internet signal to all your Wi-Fi enabled devices

Wireless Broadband

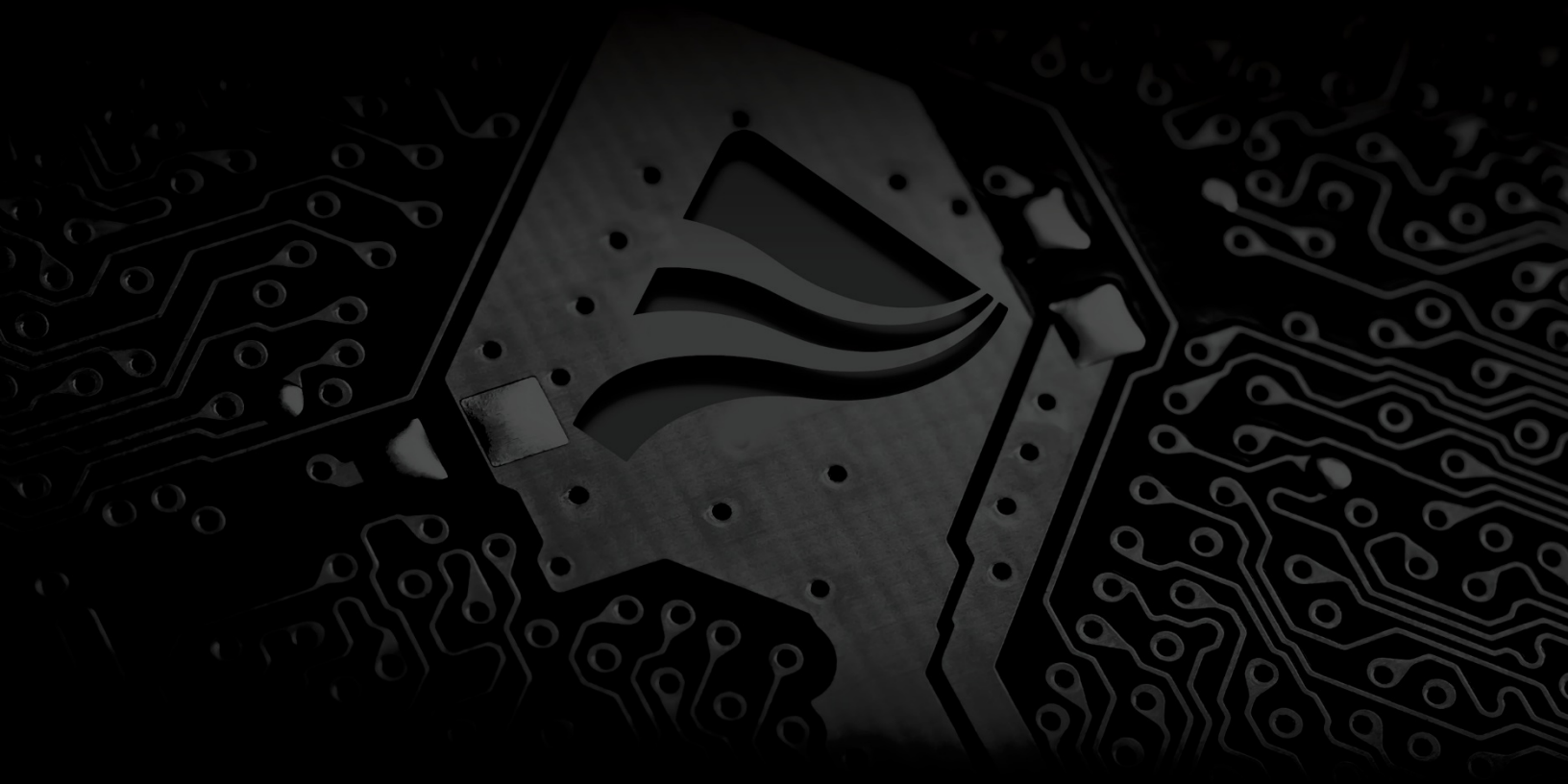
Wireless broadband is a networking technology designed to impart high speed Internet and data service through wireless networks. Wireless broadband may be delivered through wireless local area networks (WLANs) or wide area networks (WWANs). Similar to other wireless services, wireless broadband can be either fixed or mobile.

Wireless Local Area Network (WLAN)

A network that connects two or more devices using a wireless distribution method and provides access to the public Internet.

Wireless Wide Area Network (WWAN)

A wireless wide area network (WWAN) is a specific type of network that sends wireless signals beyond a single building or property.



SOURCES

SOURCES

American Trucking Association
Antelope Valley Press
Association of Washington Business
Biotech Innovation Organization
Bloomberg New Energy Finance
Boston Consulting Group
Business Korea
Business Research Corporation
Catherine Grooms, Director of Small Business Development Center, Los Angeles Network
CB Insights
CBRE Group, Inc.
Center of Excellence for Aerospace & Advanced Manufacturing
Center of Excellence for Information & Computing Technology
Choose Washington
City of Alpharetta
City of Greenville
City of Lancaster
City of Milton Keynes
City of Petersburg
City of Richardson
City of San Leandro
City of Seattle
Cleantech Group
Cleantech.org
CompTIA
Crunchbase
Deloitte
District of Songdo, Incheon, South Korea
Dr. Robert Harmon, Professor Emeritus of Marketing & Service Innovation, Cameron Research Fellow, Portland State University
East Bay Times
ERI Economic Research Institute
Ernst & Young
ESD Washington State
fDi Intelligence
Fintech Atlanta
Forbes
GeekWire: American Technology News Website
Georgia Tech Financial Services Innovation Lab
Grand View Research
Grow Alpharetta
International Data Corporation (IDC)
Kathy Otterson, General Manager, Lit San Leandro
Life Sciences Washington
Lit San Leandro

Lockheed Martin
Magellan Advisors
Massachusetts Institute of Technology
McKinsey & Co.
Metro Atlanta Chamber of Commerce
MIT Management Sloan School
Mordor Intelligence
Nation Association of Manufacturers
National Institute of Standards and Technology
New Tech Northwest
Pacific Northwest Center of Excellence for Clean Energy
Pacific Northwest Laboratories
Pitchbook
Plaid
Port of Seattle
PricewaterhouseCoopers (PWC)
Puget Sound Business Journal
Ren21
ResearchandMarkets
Robotics Business Review
San Leandro NEXT
Site Selectors Guild
Small Business Development Center, Los Angeles Network
Smart City Hub
State of Washington Department of Commerce
Statista: A German-based market and consumer data company
Tech Alpharetta
TechCrunch: American Startup and Technology Newspaper
Technology Association of Georgia
TechRepublic: Online trade publication and social community for IT professionals
Tellus Venture Associates
The Antelope Valley Times
U.S. Bureau of Economic Analysis
University of Washington, Bothell
University of Washington, Seattle
University of Washington, Tacoma
Washington Technology Association
Washington Blockchain Chain Council
Washington State Centers of Excellence
Washington Technology Institute
World Bank

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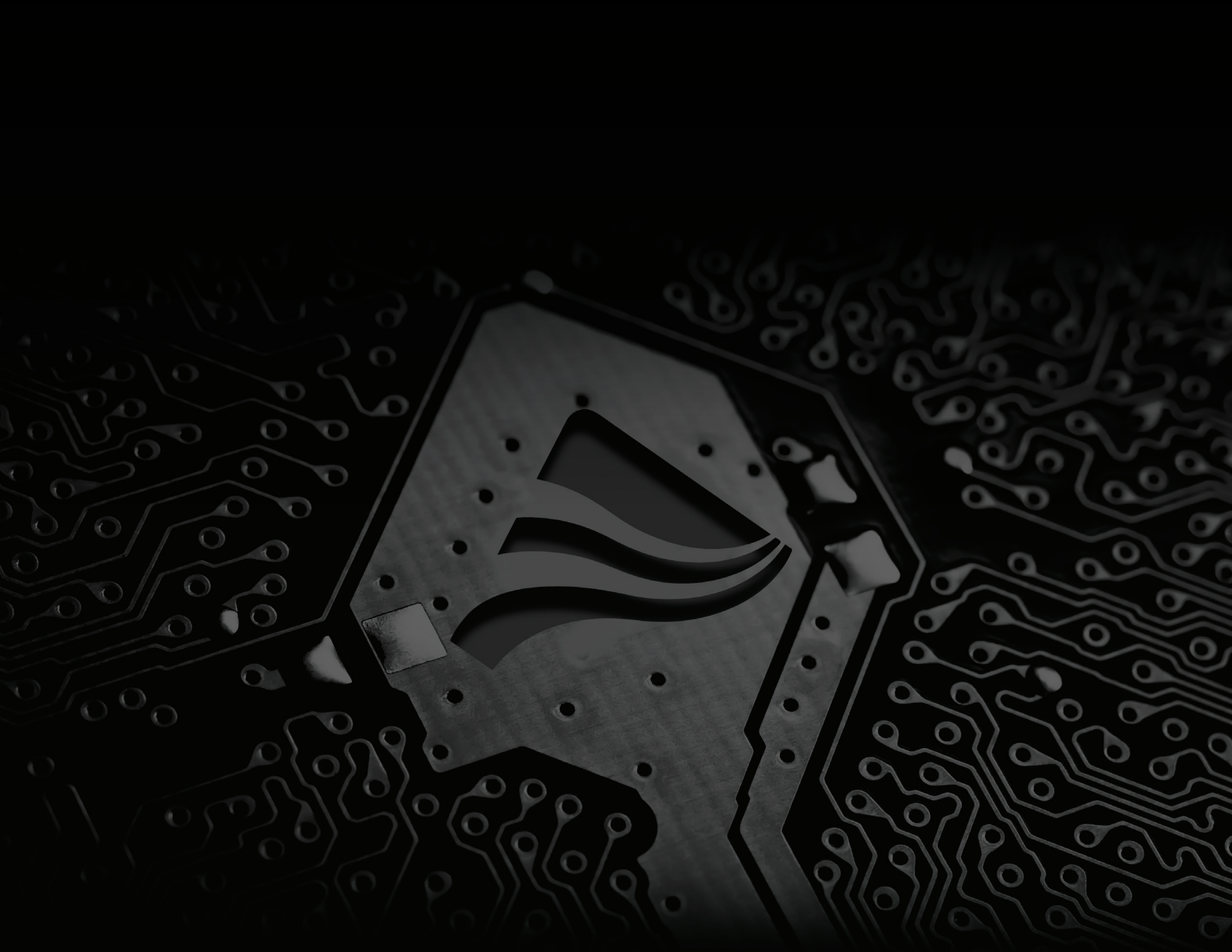
CITY OF Federal Way
Tech
Centered on Opportunity



ADDENDUM

A TECH HUB ACTION PLAN

2021

CITY OF
Federal Way
Tech
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TABLE OF CONTENTS

Introduction 1
Technology and Entrepreneurial Ecosystem 2
- Business Accelerators 3
- Incubators 12
- Innovation Centers 18
- Tech Transfer Centers 23
- Mash-ups, Meet-ups, & Hackathons 30
- Coworking Space & Makerspace 36
- Economic Gardening 40
- Start-Up Studios or Foundries 45

Infographics 52
- Advanced Manufacturing 52
- Clean Tech 54
- Fintech 56
- ICT 58
- Life Sciences/Biotech 60
- Transportation/Mobility Technology 62
- Smart Cities 64

INTRODUCTION

This Addendum contains two elements that supplement the report, *A Tech Hub Action Plan*.

As identified in the report, building a strong innovation-led entrepreneurial ecosystem is crucial in Federal Way's efforts to become a tech hub.

The first component of the Addendum is *Technology and Entrepreneurial Ecosystem – Federal Way*. It is a series of articles on the programs and services that typically comprise an entrepreneurial ecosystem.

The programs and services that typically comprise technology and entrepreneurial ecosystems include the following:

- Business Accelerators
- Incubators
- Innovation Centers
- Tech Transfer Centers
- Mash-ups, Meet-ups, & Hackathons
- Coworking Space & Makerspace
- Start-up Studios or Foundries

These articles examine the opportunities to be considered in the attraction of each element or service to Federal Way.

The second element clusters of tech hubs provides a series of infographics of the technologies sought by the cities in the case studies.

The technologies include the following:

- Advanced Manufacturing
- Clean Tech
- Fintech
- ICT
- Life Sciences/Biotech
- Transportation/Mobility Technology
- Smart Cities

As previously mentioned, research did not seek a specific type(s) of technology for Federal Way, but rather focused on the methodology used to become a tech hub. Therefore, the infographics are purely to provide contextual information on the technologies embraced by the cities highlighted in the case studies. The infographics provide global and national data. They also illustrate statewide and regional resources that exist in the state as a reference for future analysis to understand the opportunities of one or more as the focus of a tech hub for Federal Way.

TECHNOLOGY AND ENTREPRENEURIAL ECOSYSTEM - FEDERAL WAY

During this research, an examination was undertaken to understand the programs and services that typically comprise and support the growth of innovation- focused entrepreneurial ecosystems.

These involved reviewing and understanding the following:

- What is a Business Accelerator?
- What is a Business Incubator?
- What is an Innovation Center?
- What is a Tech Transfer Center?
- What are Mashups, Meetups, and Hackathons?
- What is Makerspace and Coworking Space?
- What is Economic Gardening?
- What is a Startup Studio or Foundry?

As identified in the Recommendations, Federal Way's "Technology and Entrepreneurial Ecosystem" is thin compared to Seattle and Tacoma. However, Federal Way is home to International Paper's Global Cellulose Fibers Innovation Center. The value of having an innovation center of a Fortune 500 firm is of significant value. Specifically, the ability to leverage this facility in marketing to others in the effort towards achieving a tech hub.

Additionally, the research identified these as a starting point - a series of opportunities. Thus, these articles identify ways to formulate a framework to place Federal Way on the right direction to achieve the following:

- Job creation – Creating new employment opportunities for local residents and youth in the participating startups.

- Enhanced image – Building the region's reputation as a center for excellence for growth in the tech-based economies.

- Increased entrepreneurialism – Elevating awareness of entrepreneurs and stimulating confidence in the targeted industries of Federal Way to create new products and services.

- Business development – Sparking new opportunities for established area businesses to develop relationships with early-stage companies.

- Increased tax revenue – Generating a larger, more diverse tax base to support public services and contribute to livability and health.

- Workforce development – Nurturing the next generation of diverse, inclusive, and representative workforces with technological expertise and access to clean, healthy, living-wage jobs.

WHAT IS A BUSINESS ACCELERATOR?

Alex Forte
City of Federal Way, Washington
Economic Development
April 2021

It is tough to start a business. It is even more so during a pandemic. So, where can a young, struggling startup business find help and financing?

Accelerators Defined

Business accelerators support early-stage, growth-driven business engaged in a product or service. Accelerators support these businesses through education, mentorship, and financing.

An accelerator works as a holistic business advisory service that provides an intense, rapid, and immersive education aimed at accelerating the life cycle of young, innovative companies, compressing years' worth of learning-by-doing into just a few months. During the program, cohorts of accepted startups are put through rigorous mentoring and training where startups collaborate in their learning and network to set up future capital.

According to research, there are four attributes that make accelerators unique. Accelerators are Fixed-Term, Cohort-Based, Mentorship and Education-Driven, and they culminate in a graduation or Demo Day. This demo day is when the startups present their products to investors.

ATTRIBUTES OF ACCELERATORS

1. Fixed-Term
2. Cohort-Based
3. Mentorship and Education-Driven
4. Demo Day

Cohen & Hochberg (2014)

Startups in an accelerator are generally provided funding in exchange for equity. The initial funding is mainly meant to cover company and living expenses. Accelerators may provide office space for the participating startups or encourage the startups to find their own spaces.

> ***A useful analogy is that an incubator provides a longer-term, nurturing "childhood" for a startup while an accelerator puts a startup through its "teenage" years via rapid development and increasing maturity.***
>
> Sepulveda (2012)

It should be noted that there are a variety of nuances in describing the tools and organizations that comprise an entrepreneurial ecosystem. For example, the terms start-up or seed accelerator are synonymous. They are privately run entities that offer financial support but also provide training and mentorship that provide a startup opportunity to networks with other businesses and specifically to the marketplace.

Accelerators aim to speed up the growth of new business helping the founders of businesses scale them to the market.

Accelerators specialize in funding startups during their seed stage. The seed stage is the earliest round for funding and is often referred to as the angel investor round Accelerators support startups during their initial steps by helping them create something significant that can achieve even further growth.

However, the definition of an accelerator has evolved and continues to evolve, hence not all accelerators offer exactly the same services. So, this means that what is expected of an accelerator today, may alter in years to come. Thus, the definition of an accelerator is not simple either. This is because not all accelerators are the same. Accelerator involve varying features. These include:

- Overall goals
- Specialization of types of startups
- Funding structure
- Funding levels
- Experience of Advisors/Mentors
- Allotted time to startup
- Opportunities for future investment
- Regions where they operate

Understanding these features, a startup needs be aware of the landscape of accelerators and look for the accelerator that best addresses or serves their needs. Thus, a startup needs to be well read on accelerators.

For example, a start-up must apply through a formal process to be accepted into any and all accelerators. As such, accelerators allow anyone to apply. However, that does not mean the startup will be accepted. The startup needs to understand the features of the accelerator. As the competition is fierce. Wasting time to get into an accelerator that will not serve a startups needs can be a significant exercise of futility. Accelerators do not care about elitism, they care about working ideas that are scalable. The number of startups accepted are miniscule. Research identifies that less than 10 percent of businesses that apply are accepted and with the larger more sophisticated accelerators only 1 to 2 percent of those who apply are accepted.

Accelerators do not care about elitism, they care about working ideas that are scalable.

Startup accelerator programs will sometimes offer pre-seed stage investment. This is usually under $50,000 and is given in exchange for equity. Seed stage investment is also given, and that amount can be upwards of several hundred thousand dollars.

Startup accelerators invest their money, time, and advising in the earliest stages of a business. In turn, they expect to see quick results. This is the reason why accelerators usually offer a limited time of support from 3 to 6 months after acceptance into their program. This does not mean the accelerator exits from that point. They maintain their equity, but take a back seat after the intense period of guidance, subsequently turning their attention towards new applicants. Their aspirations for those businesses which have gone through their accelerator program will increase in value over time without further interaction. Accelerators tend to focus on founders who impress them. However, the guidance given is usually focused on the larger team at a startup rather than an individual. The reason is that most startups are a collaborative process. Research shows that accelerators/investors are more likely to invest in businesses with more than one founder because the burden of running a startup is shared, as is the ideas. Accelerators want to see a strong team and a great product. Yet, in a major analysis from NYU and the Wharton School, entrepreneurs who start a business on their own are likelier to success than those who do so with one or more partners. The 2018 report, "Sole Survivors: Solo Ventures Versus Founding Teams by Jason Greenberg & Ethan Mollick identified that popular consensus suggests that new venture perform better when launched by teams rather than by individuals.

This seems to fly in the face of logic however, consistent with investors' bias toward teams rather than solo founders (and perhaps the fact that multiple people have more friends and family members than one), they found that companies with multiple founders were able to raise more money than those headed by a solo entrepreneur. You might think this would give founding teams an advantage over single founders, but you'd be wrong. Despite starting off with a smaller stake, companies with a single founder were more likely to still be in business than those with two or more. And though teams may have been able to raise more money initially than single entrepreneurs, companies with only one founder also saw higher revenue than those with two or more.

In either case, accelerators tend to specialize in niches. They invest their time and resources in startups that are operating either in similar innovations or have similarities. This maximizes the time and impact of the advisors focusing relevant advice specific to a niche. Additionally, this increases the chance of success.

Accelerators unlike their cousins' incubators, accelerators provide both advising/mentoring and funding.

Types of Accelerators

There are many ways to categorize accelerators. Some accelerators are defined by the type of product or sector such as biotech, artificial intelligence, advanced manufacturing and software applications. Other accelerators are defined by the stage of a business, anywhere from pre-seed to beyond series C.

There are three overarching categories for accelerators. They include:

1. Venture Accelerators
2. Corporate Accelerators
3. Government Accelerators

A Venture Startup Accelerator is financially backed by venture capitalist(s). They look to invest in businesses that will yield a significant return on investment over a short period.

An example of a Venture Accelerator is the organization known as Y Combinator. It has been identified as the first accelerator in the world and continues to be the most prominent venture accelerator in the world. Y Combinator now has over 2,000 alumni and has helped startups raise over $34 billion.

A Corporate Accelerator is hosted or sponsored by a large corporation. The sponsoring company of the accelerator has complete control over the direction and research. It often provides its startups or cohort of startups access to its own resources such as advertisement, software, or equipment, either discounted or for free. This accelerator may funnel to the startups access to its R&D and/or access to its customer base.

Corporate accelerators come in many different shapes and sizes depending on the goals of the sponsoring company. For example, M12, formerly known as Microsoft Ventures, is an accelerator focused on enterprise-level tech startups that benefit the customer base of Microsoft. Each startup or "portfolio company" is assigned a relationship manager. The relationship manager is a well-connected expert that explores the needs of the portfolio company and connects them with the right people, to help accelerate the project's growth.

Sponsored Microsoft startups receive assistance from a team of professionals that have in-depth knowledge of marketing, operations, investing, finance, and engineering. All portfolio companies receive access to Microsoft technologies (Office, Azure, Github Enterprise, and others).

Again, as previously identified, models differ and not to confuse the entities but this model is different from corporate venture studios in which a company partners with an outside specialist organization to develop a homegrown spin-off venture, meaning the resources of the parent corporation and studio are pulled together in the process.

A Government accelerator is operated and supported by the public sector. These accelerators have significantly different goals compared to venture and corporate accelerators. Their focus is not primarily profit. Instead, it may involve competing globally on innovation, creating jobs, and creating a stronger talent pool.

Government accelerators are typically found outside the United States. For example, the Scottish Government runs CivTech, the Chilean Government runs Start-Up Chile, and the Korean Government runs In2Korea to name a few.

The closest thing to a government accelerator in the United States are accelerators run by public universities.

For example, the Jones + Foster Accelerator program at the University of Washington helps early-stage, student-led businesses through the first six months of business development. Businesses receive mentoring from a committee of entrepreneurs and investors, a variety of workshops, access to up to $1,000 in expense funding, and guidance in creating a list of milestones with up to $25,000 in follow-up funding if the milestones are met. The program does not take equity in participating businesses.

An example of a foreign government accelerator is CivTech. It was created by the Scottish Government in 2016. This program harnesses entrepreneurial tech innovation – typically from start-ups and Small to Medium Enterprises, (SME's) – to solve identified public sector challenges.

Moving beyond these three types of accelerators, there are pre-accelerator and post-accelerator programs.

A Pre-accelerator is similar to accelerators, except they typically do not have equity-based funding and are focused on earlier-stage companies. Pre-accelerators have a competitive application process, have mandatory attendance, and include a culminating event just like accelerators. Pre-accelerators take startups from concept to a minimum viable product. Pre-Accelerator are typically a three to six-month program.

For example, FoundersBoost, (previously Startup Boost), runs 6-week, mentorship-driven, pre-accelerator programs in the US and across the globe. The program is designed to help pre-seed/seed stage startup companies. The program runs like a lean accelerator and takes no equity or fees from startups in the program.

Over the past 4 years, FoundersBoost has pre-accelerated 300+ startups which have gone on to receive $85+ million dollars in investment.

A Post-accelerator extends the value of the accelerator program service to the startup. It also provides lasting value through continued access to mentorship and investor networks, peer support, member-exclusive discounts, and additional programming.

An example, is the post- accelerator Bold Growth. It is run by the Minneapolis/St. Paul regional economic development organization. Its focus is on food and agriculture.

Accelerators: The Marketplace

As previously identified, any startup can apply to an accelerator program, but selection can be extremely competitive. Two of the top accelerator companies in the world, Y Combinator and Techstars have acceptance rates of between 1% and 2%, for example. Some accelerators are generalists, such as being open to any sector or any tech startup, while other accelerators focus on narrow sectors such as artificial intelligence, healthcare tech, or cleantech to name a few.

One of the largest accelerator programs, Techstars, annually selects over 300 companies to join its three-month, mentorship-driven program. Techstars invests $120K in each startup and provides hands-on mentorship and access to their network for life. Techstars hosts dozens of accelerator programs across different cities and industries, including in the Seattle region.

According to report from Chalmers University of Technology, Sweden; a startup goes through five distinct steps in its journey through an accelerator program (except for a corporate accelerator):

1. The Awareness Step when a startup becomes knowledgeable about a certain accelerator.

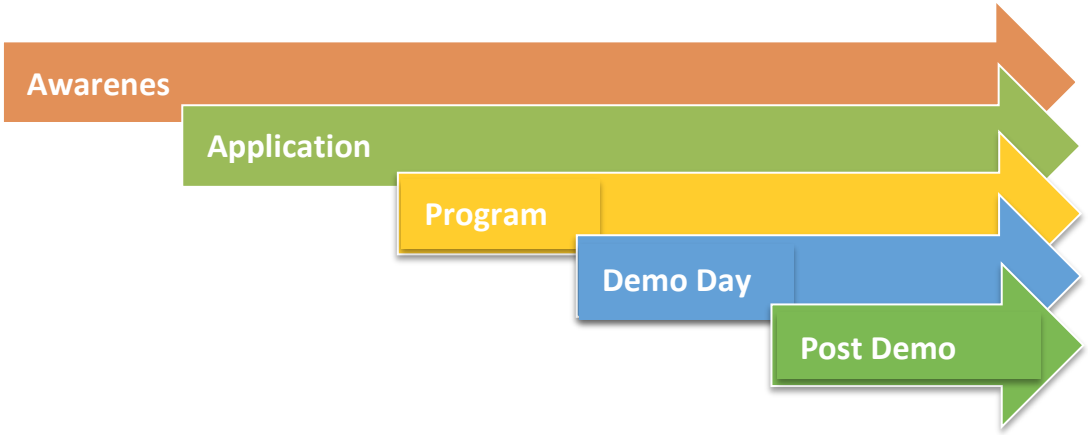
2. The Application Step is when a startup applies to the accelerator program.

3. The Program Step is the accelerator itself, where startup teams interact with mentors, attend lectures, collaborate with peers, practice presenting and pitching, and work on their products for an intense period of often no more than three months.

4. The Demo Day Step serves as graduation for the startups of the accelerator program. On demo day, startups give pitches to VCs and business angles in the hopes of receiving additional funding to continue the development of their companies.

5. Finally, the Post Demo Day Step is when startups go out on their own as alumni from the accelerator. If the accelerator retains equity in the company after the program, the accelerator may or may not continue to help the startup raise money.

THE 5 STEPS OF AN ACCELERATOR



Chalmers University of Technology (2012)

Local Accelerators

The Seattle region is home to many accelerators that help startups grow. Examples of accelerators in the region include, as previously identified, Techstars, along with the global accelerator network Fledge and Reactor.

Courtesy: Port of Seattle

Rendering of the Maritime Innovation Center scheduled to open in 2023 in the Ballard neighborhood of Seattle. It will be 12,000 square feet in size with a mix of working space for incubators, accelerators, and anchor tenants along with fabrication and event space.

Maritime Blue is a statewide organization of maritime stakeholders, including the State's Department of Commerce, and the Port of Seattle. It has launched its second group (cohort) of startup companies focused on maritime innovation at its accelerator. The goal of the program is to spur new technology and investment in the maritime sector in the state. Washington Maritime Blue is a non-profit. It is a strategic alliance formed to accelerate innovation and sustainability in support of an inclusive "Blue Economy." The Maritime Innovation Center in Ballard will become the home of the Maritime Blue Accelerator upon its completion in 2023.

Another example of an accelerator is the Port of Seattle's PortGen Accelerator. It supports the economic vitality of the Puget Sound by introducing new small businesses to the Port and improving their ability to land future business opportunities. The PortGen Accelerator is one of the first business accelerators in the region to focus on women and minority-owned firms. Recognizing that diversity, equity, and inclusion are vital to the success of any organization and necessary to strengthen the Puget Sound region and specifically King County, the Port of Seattle contracted with the newly formed accelerator operator, Find Ventures, to select and engage businesses in the environmental, travel, and airport-related industries, specifically those owned by women and people of color, to assist the Port in its mission to advance economic development through the operation of the international airport and the Port's maritime trade and travel gateways.

Value of Accelerators

In the article, *Do Startup Accelerators Deliver Value? The Economics of Creating Companies*, by Andy Wu of the Harvard Business School provides insight on the value of an accelerator. He identifies that accelerators provide four key values. These include, education through mentorship and training; networking with peers, experts and investors. Money for expenses or development and the help or assistance in raising money and finally, signaling to investors that the start-up is credible.

Accelerators have a highly competitive application process and will invest equity capital in selected companies. Accelerator programs are typically short-term, ranging from three to six months. Accelerator programs accept companies in cohorts and provide learning opportunities and intense mentoring to each company, with a culminating graduation event where companies pitch their concept or product in front of investors and other key stakeholders. Space is usually provided to companies for the duration of the program, although some seed accelerators run cohorts virtually with required attendance at educational or networking events.

VALUE OF AN ACCELERATOR

- Education through mentorship and training
- Networking with peers, experts, and investors
- Money for expenses or development and help raising money

Federal Way

Unfortunately, Federal Way has no accelerators for business. However, it does provide an economic development opportunity to help diversify and strengthen the business climate. The following illustrates the opportunities to be considered in examining the attraction of an accelerator to Federal Way in the effort to create a technology hub.

- Job creation – Creating new employment opportunities for local residents and youth in the participating startups.

- Enhanced image – Building the region's reputation as a center for excellence for growth in the tech-based economies.

- Increased entrepreneurialism – Elevating awareness of entrepreneurs and stimulating confidence in the targeted industries of Federal Way to create new products and services.

- Business development – Sparking new opportunities for established area businesses to develop relationships with early-stage companies.

- Increased tax revenue – Generating a larger, more diverse tax base to support public services and contribute to livability and health.

- Workforce development – Nurturing the next generation of diverse, inclusive, and representative workforces with technological expertise and access to clean, healthy, living-wage jobs.

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Organizations:
FoundersBoost

Geekwire

Port of Seattle

Washington Maritime Blue

Y Combinator

About the Author

Alex Forte is a member of the Economic Development Internship Program at the City of Federal Way. He is majoring in Economics with a minor in Applied Mathematics at UW-Seattle. He is pending graduation at the end of this year. He grew up in Lakewood, WA, and has a strong connection to the Puget Sound. His career goal is to work towards improving communities and fueling business. Alex's research is on technologies associated with the clean, green, and construction sectors. His research involves the examination of several cities that have shown both success and disappointment in their efforts to attract and grow technology-based companies in this sector. Alex's motivation will enable Federal Way to understand opportunities in becoming a tech hub in this field

WHAT IS AN INCUBATOR?

Dominic Minichillo
City of Federal Way, Washington
Economic Development
April 2021

Creating and growing a business is difficult, just ask anyone that has started one. Startups require collaboration, mentoring, capital, product, and customers all of which are supported by the missions of business incubators.

Business Incubator Defined

Business incubators support entrepreneurs through a plethora of resources and services that often include office and/or lab space, access to capital, networking, mentoring, and training.

A business incubator is simply a workplace created to provide new ventures and startups access to growth-related resources they require, found all under one roof.

Business incubators support entrepreneurs through a plethora of resources and services such as office-related resources including office and/or lab space, access to advisors, training, administrative support, potential investors, and mentorship and training programs.

Many incubators serve as a temporary launching pad for new ventures, expecting that participants (the startups) will eventually grow and move out. Unfortunately, not all companies achieve this objective as some do not realize their business ideas finding out that they are not viable and forego further progress.

Incubators typically charge monthly program fees or membership dues in exchange for office and or lab space and access to program offerings. Start-ups usually are required to apply as part of a cohort (class) annually to ensure they meet the incubator's operator criteria or mission (industry, stage of company, founder demographics, etc.). Thus, the application process is one of membership. Start-ups become members upon approval of an application typically on an annual or rolling basis and are typically able to reside in the incubator for 1-3 years as incubators usually have graduation policies that are typically based on achievement of agreed-upon milestones, growth metrics, or time-based stipulations.

According to a 2016 survey by the International Business Innovation Association, (InBIA), there are an estimated 1,400 incubators serving entrepreneurs nationwide. Operations of incubators can be publicly and or privately owned.

Many highly successful and well-known companies began as startups that used incubators to spur growth, including current industry giants such as Reddit, Dropbox, and Airbnb.

An estimated 1,400 incubators serve entrepreneurs nationwide

InBIA

Incubators vs. Accelerators

Sometimes, there is confusion between the terms, business incubator and accelerator. To clarify this the terms incubator and accelerator are often used interchangeably, but the two programs are different.

For example, incubators allow companies to use the service for 1-3 years (sometimes indefinitely) and either take a fee or equity position for the right to use the incubator resources.

Additionally, business incubators vary and often exist to help business founders determine if a business idea is viable. They also help put startups on an aggressive growth trajectory. Again, some incubators set a time limit. During this time the start-up is charged a fee to be in the incubator's space, or the incubator's operator may take an ownership stake.

Business incubators are typically operated by partnerships or corporations. Additionally, they can be operated by universities, colleges, government entities, for-profit ventures, and economic development organizations.

Accelerators are typically operated by the private sector. Accelerators require a competitive application process. Nationally, less than 10% of applicants are accepted. Most limit the number of participating businesses and accelerator operators limit the number of startups. Additionally, business accelerators provide equity financing to the startup.

Service and Value

A significant service that Incubators provide is access to industry expertise providing mentors for the startup. These mentors are able to provide insights and lessons that are not widely available to startups.

Additionally, incubators provide affordability and flexibility as it pertains to space at a significantly discounted rate than the general marketplace. Also, incubators provide office equipment at a discounted rate for startups. Incubators have access to expensive equipment that small companies are usually unable to afford.

Networking is a prerequisite of business; incubators allow startups to excel in this requirement. Companies will be working beside dozens of other startups. Sharing experiences is significant in the incubator experience as entrepreneurs gain exposure to information, business contacts, and exposure to new industries.

Business incubators provide professional resources through administrative and technical support. Administrative support includes business and market planning, while technical support gives software and business tools.

Meanwhile, the value of a business incubator is captured in a 2020 article entitled "9 Reasons why Startups Need Incubators," from University Lab Partners, an independent, non-profit program founded in partnership with the University of California, Irvine. The article identifies that the value of an incubator provides an entrepreneur the means of growing a business while eliminating many of the inherent risks of starting a business. They include the following: Accelerate the Growth of your Startup, Structure to Maintain Business Focus, Access to Funding, Exposure to Industry Leaders and Mentorship, Low Cost Space & Flexibility, Low Cost Access to Expensive & Sophisticated Equipment, Partnerships, Access to Professional Resources, and Networking Resources.

9 REASONS STARTUPS NEED INCUBATORS

1. **Accelerate Growth**
2. **Maintain Business Focus**
3. **Access to Funding**
4. **Exposure to Industry Leaders and Mentors**
5. **Low Cost Space**
6. **Access to Sophisticated and Expensive Equipment**
7. **Partnerships**
8. **Access to Professional Resources**
9. **Networking**

University Lab Partners: 2020

Sometimes, operators of a business incubator will provide funding to the startups. However, incubators typically help startups financially by linking them to investors.

Types of Business Incubators

There are two types of incubators. They include, traditional where the entrepreneur(s) are located in a physical facility and online or virtual incubators where entrepreneurs work on line. In either case, some they may specialize in a specific industry and only provide service to those types of businesses.

Traditional Business Incubator

The traditional company incubator model requires a startup to set up a physical shop at the incubator's location (site). Thus, start-ups can find private offices, or a dedicated desk or what is called a float desk. This means the startup uses a desk that is available when the startup is at the incubator. Additionally, startups can participate in a larger room, sometimes referred to in the industry as a café, where numerous start-ups are located and working in a larger space. Examples of traditional business incubators in Washington include the SURF Incubator and Madrona Venture Labs.

Virtual Business Incubator

A virtual business incubator is also known as an online business incubator. This type of incubator offers support for startups that need capital and advice to achieve their business concepts. The virtual incubator offers similar services. Additionally, their cost structure and time span for assistance also differ. Also, different than traditional incubators it is a rolling admission rather than an annual enrollment.

Types of Operators of Business Incubators

As previously identified, different operators create incubators with distinct interests. The National Business Incubation Association (NBIA) categorizes incubators into five types be they traditional or virtual. They include incubators from academic institutions, non-profit development corporations, for-profit property development ventures, VC firms, and some combination of the above.

TYPES OF OPERATORS OF INCUBATORS

1. **Academic Institution**
2. **Non-Profit**
3. **For-Profit**
4. **Venture Capital Firm**

Incubators in Washington and the Puget Sound

In the Puget Sound region, there are a myriad of incubators that cover a variety of sectors.

For example, Capria (Downtown Seattle) works to develop solutions to world problems by investing in emerging market startups. Capria is one of the most successful incubators for global impact startups.

Another example, is the AI2 incubator at the Allen Institute for Artificial Intelligence. In 2019 it moved to accommodate growth in this sector. The incubator is now located in over 7,000 square feet. It has workstations for entrepreneurs in residence and chief technology officers in addition to numerous team pods, phone booths, conference rooms, a classroom, lounge, and a large deck overlooking Lake Union. The incubator plans to have 20 to 25 incubatees at any given time in the facility.

CoMotion is yet another site for incubator space. CoMotion Labs provides a multi-industry incubation environment for early stage startups with a focus on University of Washington spinoffs. CoMotion provides desks, dry benches, wet lab space, and prototyping capabilities. Labs are currently home to 60 startups and operate in three locations on the UW Seattle campus; two in Fluke Hall and one in Startup Hall. Each lab focuses on a particular industry sector: life sciences and hardware in Fluke Hall, and technology in Startup Hall.

Another example, is the new Tacoma Maritime Innovation Incubator. It is part of Washington Maritime Blue Initiative. The organization announced the formulation of an incubator focused on clean water and alternative energies and logistics and other initiatives addressing marine and maritime innovation. The incubator will support at least seven companies each year with expert mentorship and free rent, and access to funders, and opportunities to collaborate. The incubator will be housed at the Centro for Urban Waters on the Thea Foss Waterway.

The William M. Factory Small Business Incubator located in Tacoma helps new businesses in as a way to improve living and working conditions. They house over 40 companies in specialty trade construction, applied technologies, and business services.

Federal Way

Federal Way currently has no business incubators. However, it is an opportunity that might yield great outcomes. Incubators can create new employment opportunities for local residents and specifically youth through participating in startups. Along with increased entrepreneurialism, it can create new products and services in Federal Way including the following:

- Job creation – Creating new employment opportunities for local residents and youth in the participating startups.

- Enhanced image – Building the region's reputation as a center for excellence for growth in the tech-based economies.

- Increased entrepreneurialism – Elevating awareness of entrepreneurs and stimulating confidence in the targeted industries of Federal Way to create new products and services.

- Business development – Sparking new opportunities for established area businesses to develop relationships with early-stage companies.

- Increased tax revenue – Generating a larger, more diverse tax base to support public services and contribute to livability and health.

- Workforce development – Nurturing the next generation of diverse, inclusive, and representative workforces with technological expertise and access to clean, healthy, living-wage jobs.

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Allen Institute of Artificial intelligence

CoMotion Labs, University of Washington

About the Arthur

Dominic Minicello is a member of the economic development internship program at the City of Federal Way. Dominic is junior at UW-Seattle majoring in economics (specifically microeconomics along with minoring in Entrepreneurship and Data Science. His collegiate work has involved housing and consumer concerns on food pricing. His academic work is on entrepreneurship in the life sciences sector and opportunities, and tech tools that can be incorporated into the strategy achieving the goal of Federal Way becoming a tech hub. He is examining the suburbs of Boston; (Braintree, Cambridge, Quincy and Somerville); San Francisco (S. San Francisco) and San Diego, (Carlsbad, Vista and San Marcos) as case studies of how they became life science hubs.

WHAT IS AN INNOVATION CENTER?

Quinn Zilly
City of Federal Way, Washington
Economic Development
April 2021

Technological innovation even in a pandemic has accelerated at an incredibly fast pace in today's business landscape. A comprehensive report completed in 2015 surveyed the world's 200 largest corporations and found that 90% of the companies believe that they are too slow in bringing new products to market.

Corporations have known for decades that to remain competitive, they must be quick to the marketplace. Thus, many corporations have opened innovation centers to help them respond faster and effectively.

Innovation Center Defined

Innovation centers are facilities where a specified team of employees is located and dedicated to innovation. It allows companies to experiment with disruptive technologies, new business models, and to test new products.

They also tend to be physically separated from the parent company and are setup more like a startup than a traditional R&D department. The centers tend to be located in communities known for being tech hubs where they can take advantage of resources specifically from universities and local technology ecosystems.

Innovation centers represent a variety of sectors from manufacturing, telecom, automotive, consumer retail, financial, and insurance services. Their main mission is to gain access and exposure to latest technologies and a deeper understanding of customer needs.

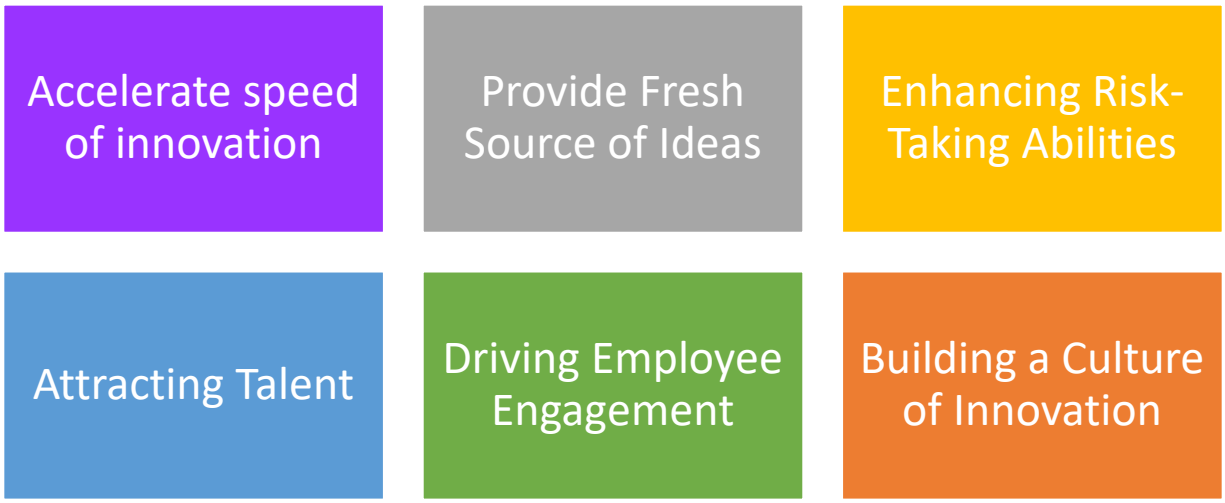
The areas of focus for centers cover a variety of products and services. They may include how to use or enhance operations through some of the following areas: Mobility, Big Data/Analytics, Internet of Things, Robotics, Virtual Reality, and 3D printing.

Purpose

Corporate innovation centers are setup to accomplish several key goals. These include: Accelerate the Speed of Innovation; Provide fresh source of ideas; Enhance risk-taking abilities; Attracting talent; Driving employee engagement; and Building a culture of innovation.

> ***"The mission of our innovation centers is to pioneer and test disruptive solutions for our customers, testing new business models and technologies that advance our position in the market. . . "***
>
> ***Debbie Bracken, Citi Ventures***

PURPOSE OF INNOVATION CENTERS



The Innovation Game: Why and How Business Are Investing in Innovation Centers
Capgemini Consulting 2015

Types of Centers

There are several different categories of innovation centers. Each have different business models for achieving their goals. The categories are in-house innovation labs, university residences, community anchors, and innovation outposts. Further insight illustrates that they can also be privately and publicly operated.

In-house Innovation Labs

These type of innovation centers carry out large-scale innovation projects in-house for the parent company and are generally working in tandem with a specific department within the company. They are typically larger in size than the other categories, often with hundreds of employees. For example, Walmart Labs was setup as part of their global ecommerce team to innovate with respect to their website. Coca-Cola has KOLab, where they develop new products from beverages to digital features.

University Residences

Companies will invest in a university research center to innovate through collaboration with university researchers. An example of this type of innovation center is the Volkswagen Automotive Innovation Lab (VAIL) at Stanford University.

Community Anchors

These innovation centers strengthen the local startup ecosystem through mentoring and investment in startups. They offer opportunities for startups to work alongside the company to design and test new products, sometimes taking an equity position in startups. Allianz Digital Labs is an example of a

community anchor that helps students, innovators, and early-stage startups in the innovation process, giving them the potential for future licensing or commercialization of their product.

Innovation Outposts

This type of innovation center consists of a small team with a low level of investment from a parent company. They work in collaboration with the local startup ecosystem, gathering ideas and new technology trends that the parent company can then use to create an innovative new product or service. Nissan opened an innovation outpost in Silicon Valley in 2013 that was initially focused on vehicle IT and advanced engineering, but is now specializing in autonomous driving and connected vehicles.

Location of Centers

While there is a high concentration of innovation centers in Silicon Valley, 61% of companies with innovation centers have a presence in the Silicon Valley, they are not however, unique to this region. In fact, innovation centers are emerging across the United States, with the top ten cities for innovation centers comprising only 35% of the total.

TOP TEN GLOBAL CITIES FOR INNOVATION CENTERS, 2015

San Jose/Silicon Valley	53
London	10
Paris	9
Singapore	7
Tokyo	6
Shanghai	6
Berlin	5
Munich	5
Tel Aviv	4
Chicago	4

Seattle Regional

An example of an innovation center in the region is the Mobility Innovation Center, (MIC), housed at the CoMotion Lab at the University of Washington in Seattle.

The MIC is a university residence innovation center partnership between 17 Seattle area firms focused on researching transportation technology to create a sustainable and equitable transportation system for the region. With funding from project specific grants, the Center works with community partners to find technology and policy-driven solutions to address the transportation needs of the greater Seattle area.

CoMotion Lab at the University of Washington, Seattle

Another example is the Joint Center for Aerospace Technology Innovation (JCATI). The Center is an initiative to stimulate aerospace economic development and job creation in Washington state by funding collaborations between the aerospace industry in the state and academic researchers at the state's public 4-year institutions of higher education. Their offices are housed at the University of Washington.

The Center has funded UAV's, aerospace materials such as composites and metals, composite recycling, advanced manufacturing and 3D printing, fuels along with guidance and controls, software, and sensors. The Center helps companies when they need to look deeper into projects because they do not have the bandwidth, therefore, they come to the Center to find the expertise. The Center enables these companies to approach the faculty and personnel at the state's institutions to make this connection. JCARTI can through its process help fund innovations. Thus, JCARTI is a technology innovation center that pushes the frontier of aerospace while embracing economic development.

Meanwhile, local corporations like Amazon and Microsoft both have innovation centers or labs. The Microsoft Research Lab is involved in artificial intelligence, computer vision, human-computer interaction, security, privacy, and cryptography, along with systems and networking.

Federal Way

Innovation centers can add a lot of value to a growing tech hub. Centers like the MIC and JCATI can improve the economy and overall quality of life of a region by upgrading its technological infrastructure. Community anchors and innovation outposts help cultivate a startup ecosystem through investment and collaboration with local startups and attracting talent to a community.

Federal Way was home to an innovation center when the Weyerhaeuser Corporation was headquartered here. Today, that facility is now home to International Paper's Global Cellulose Fibers Innovation Center.

If Federal Way establishes itself as a tech hub in the region, recruiting innovation centers could help to grow the city's tech ambition.

Federal Way needs to understand that companies set up innovation centers to be close to tech, business, and customers. So, the methodology and decision making to site an innovation center can be based on a multitude of reasons.

Federal Way also needs to understand how to market its position in the region, geographically, and by specific business clusters to illustrate the opportunity to attract more centers.

Photo by Tim Johnson

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CB insights (a consultancy practice)

About the Author

Quinn Zilly is a member of the economic development internship program at the City of Federal Way. Quinn is a senior at the UW-Seattle studying Economics. His focus has been on macroeconomics specifically labor economics along with work in data analysis and visualization. Quinn is examining opportunities upon graduation by either attending graduate or law school. In the meantime, his work is focused on the manufacturing sector and corresponding advanced technologies. Federal Way has found Quinn's research invaluable on so many fronts specifically in the examination of advance manufacturing in helping understand how Federal Way can become a tech hub along with understanding innovation centers.

WHAT IS A TECH TRANSFER CENTER?

Ankit Bhat
City of Federal Way, Washington
Economic Development
April 2021

The concept of a technology transfer center or a center of commercialization is the process of transferring scientific findings from one organization to another for the purpose of further development and commercialization. In a 2012 report issued by the Washington Clean Technology Alliance, it defined technology transfer as "the process by which research, knowledge and technology move from the laboratory to the marketplace and society."

The terms tech transfer center and center of commercialization are often interchangeable since they provide similar services. Technology transfer centers deal more with the research process while centers of commercialization deal with already researched products and their path to becoming a public service or product. The main intention of both is to connect with inventors and help commercialize new products and innovations.

From Concept to Implementation

Tech Transfer centers are often research spaces funded by institutions, corporations, federally funded, and/or by private funding to give entrepreneurs and driven minds the ability to collaborate on ideas and share knowledge. The main purpose is to allow more accessibility to projects and to further widen the range of groups who can develop the technology into new products, applications, or services.

The concept of tech transfer has a long history in the nation. Universities such as MIT and Stanford began efforts in the 1930's. However, most have been around since the middle of the 20th century, most notably during World War II. Large private-public partnership research projects led to several advancements in technology and weaponry. An example of tech transfer in this time period was the Manhattan Project, where combined research and development from various parties created the world's first nuclear weapon. Following World War II, it became apparent that researching and developing new technologies would expand the boundaries of technology and create new jobs which could eventually spur economic growth.

Subsequently, the history of federal funding for research has been a driving force for the growth of research at universities, thus many American universities became hot spots for research and innovation.

The Bayh-Dole Act, which came into law in 1980, enabled universities rather than federal funding agencies to claim the intellectual property generated by research funded by the U.S. government. Retaining title created a larger incentive for universities and innovators to pursue patents and licensing. The Act explicitly encouraged collaboration with the private sector, and aimed to reduce the administrative costs of pursuing commercialization by moving control out of government bureaucracy to universities. In the decade after the passing of the Act in 1980, the number of university Technology Transfer Offices, (TTO's) in the U.S. increased from 25 to 200. The number of TTO-tracked patents issued has almost quadrupled from 1995 and 2015.

Today, most major U.S. research universities support technology transfer offices that actively seeks, registers, and patents Intellectual Property (IP), and manages the commercialization of their discoveries.

Tech transfer is accomplished through licensing IP to companies/organizations that have both the resources and desire to develop and produce the technology by monetizing it for commercial applications.

Tech transfer centers help pave the way for future innovations to be developed and used. Many technology experts believe these tech centers are mainly used to impact the information and communication sector but in reality, they impact virtually every scientific field and every part of life.

As an example, innovations have occurred in the following areas:

- Life-saving Drugs and Medical Devices
- Alternative Energy Solutions
- Computer Hardware and Software
- New Modes of Transportation
- Blockchain Technologies
- Artificial Intelligence
- Vaccines
- Robotics
- Cybersecurity
- Environmental solutions
- Agricultural innovations
- Aerospace

Over the last 20 years, technology transfer has supported the creation of nearly 4 million jobs according to the Association of University Technology Managers. Additionally, the organization also identifies that since the approval of Bayh-Dole Act over 200 drugs and vaccines have been developed, nearly 1,100 companies have been formed nationally from technology transfer and technology transfer has contributed to nearly $1 trillion to the U.S. Gross Domestic Product from 1996 to 2017.

ECONOMIC IMPACTS FROM TECH TRANSFER

4 million Jobs
1,100 Companies
$1 Trillion to US GDP

Association of University Technology Managers, 2017

An example of an innovation, a technology transfer that occurred in Washington State is Bellwether Bio.

Bellwether Bio was a Seattle-based biology startup focusing on developing next generation cancer diagnostics using cell-free DNA. Bellwether Bio was developed at the UW's CoMotion Labs. It emerged and though the University of Washington's own technology transfer center, it was acquired by Guardant Health in April of 2019. It is an example of one of hundreds of startups that have been successful in the state.

The Process of Tech Transfer

The Massachusetts Institute of Technology has summarized in a chart the steps to a technology transfer process. The common core elements in the process of tech transfer are laid out in three phases. These include:

- Submitting a Technology Disclosure
- Patenting an Invention
- Commercializing an Invention

Submitting a Technology Disclosure

The first stage in the process of tech transfer research is submitting a technology disclosure. Submitting an invention involves providing a description of your invention or development to the Technology License Office. The submission (also called a technology disclosure) should also list all sponsors of the research and should include any other information necessary to begin pursuing protection and commercialization activities.

Patenting an Invention

The second stage is patenting an invention. Shortly after submitting a technology disclosure, an assessment takes place of the disclosure. For example, each disclosure is then evaluated for commercial potential and patentability Factors considered in evaluating the commercial potential of an invention include:

- Problems solved or unmet needs addressed by the technology
- Potential applications
- Market Size
- Potential competitors/partners
- Potential challenges to patenting and commercialization

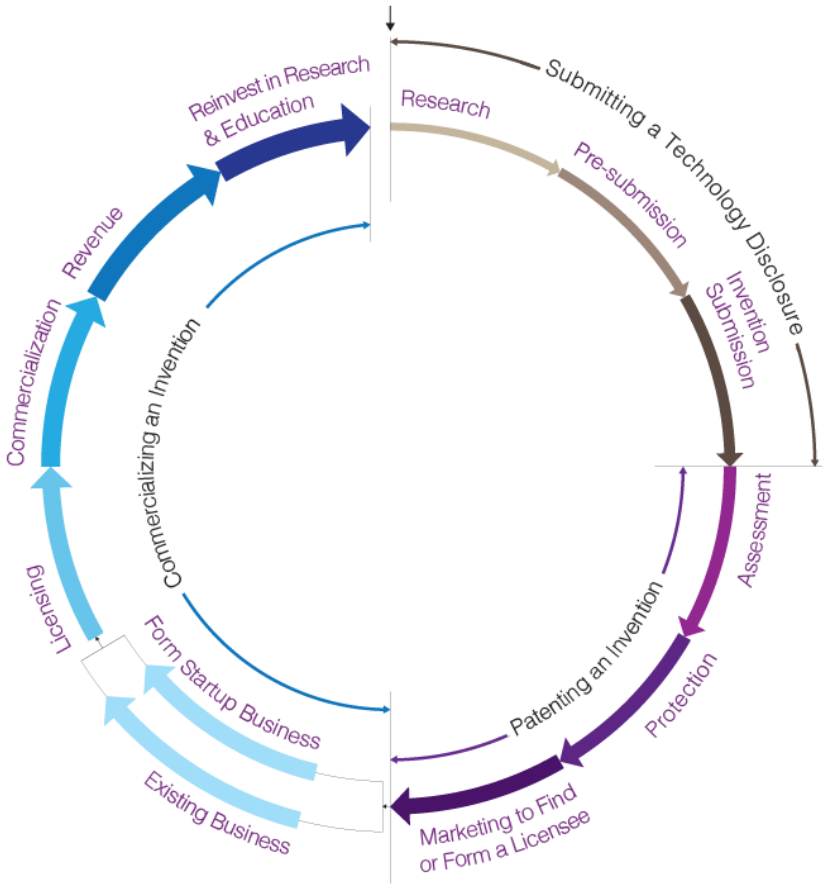
With input from inventors, assessment of patentability is performed by technology licensing officers in consultation with patent attorneys and/or literature search specialists. Typically, a technology licensing officer will contact the inventor(s) to identify next steps, whether that is filing a patent or not. This can take over 2 years to have a patent granted.

Commercializing an Invention

The third stage is to commercializing the invention. This process involves the Licensing, Product Development Revenue, Reinvestment and Relationships. Revenue from the product is split between the center and the creators with varying rates depending on the center. See chart next page.

The revenue that makes its way to the center is then reinvested into new ideas creating a cycle of invention and innovation. The process can take 1-3 years depending on the service or product and the speed of which the U.S Patent and Trademark office can produce the necessary protection for the product.

Technology Transfer Process, MIT



Many companies, universities, and government organizations have created Offices of Technology Transfer (TTO) which are dedicated to identifying research and ideas which have potential to be commercialized. Inside these TTO's is space (incubators) built for collaboration coupled with an environment that breeds growth.

National Impact

Today, there are over 700 tech transfer centers according to the Federal Laboratory Consortium for Technology Transfer (FLC). The largest and most successful tech transfer are scattered across the United States in universities and national laboratories.

In 2017, the Milken Institute ranked more than 200 universities for their technology transfer activities. The rankings are based on a weighted average of four data points as self-reported by universities to the Association of University Technology Managers. The data points included patents issued, licenses issues, licensing income, and startups formed. The report ranked University of Utah, Columbia University, University of Florida, Brigham Young University, and Stanford University as having the top centers of tech transfer and commercialization. The University of Washington was ranked 7th and Washington State University was ranked 73rd.

The U.S government funds over $100 billion in research and development, which leads to a continuous cycle of new inventions and technology within government research facilities.

The process of tech transfer varies from center to center due to differences in commercial and patent rights. Different institutions require varying amounts of ownership of the product or service they are developing.

Additionally, there are several national associations devoted to tech transfer research and development. These associations help fund as well as support innovators in the quest of creating an innovate product that can be commercialized. These associations accomplish this through the process of tech transfer that occurs in centers of commercialization and tech transfer centers. They include the following:

- Association of University Technology Managers (AUTM)
- Alliance of Technology Transfer Professionals (ATTP)
- The National Council of Entrepreneurial Tech Transfer
- Federal Laboratory Consortium for Technology Transfer (FLC)

State Impact

Currently there are four centers of commercialization or tech transfer centers in the state.

The Fred Hutchinson Cancer Research Center is located Seattle. The Business Development & Strategy group works to commercialize intellectual property generated by Fred Hutch researchers so their discoveries and innovations can contribute to better cancer care at the clinic level. They seek industry partners who share the organization's drive to innovate, collaborate and accelerate development of new inventions and approaches. They also cultivate partnerships with philanthropic supporters, research colleagues, and policymakers to extend the reach of their scientific advances.

The areas of focus of their research includes:

- Blood stem cell transplantation and immunotherapy
- Cancer risk factors and causes, prevention and treatment outcomes
- Vaccine development and virus-associated cancers
- Molecular underpinnings of cancer and Tumor-specific translational research

The Pacific Northwest National Laboratory, (PNNL) is a U.S. Department of Entergy national laboratory in Seattle. It focuses on global security, computational and data analytics, coastal sciences, energy, and environmental research. Specifically, the Laboratory taps into the region's innovation hotbed at the nexus of science and technology. The PNNL Seattle Research Center is one of three PNNL offices in Washington. The state is also home to PNNL's headquarters in Richland and the Marine Coastal Research Laboratory in Sequim.

Washington State University's, Office of Commercialization is responsible for the evaluation, protection and commercialization of innovations and discoveries for the expansion of WSU's public impact. Spokane/Pullman works with many departments and academic programs to help license technologies and launch businesses in fields such as food safety and preservation, nutrition, agriculture, animal health, hydrogen fueling stations, and lithium battery safety.

TECH TRANSFER CENTERS IN WASHINGTON

- **Fred Hutchinson Cancer Research Center - Seattle**
- **Pacific Northwest National Laboratory - Seattle**
- **University of Washington CoMotion, previously Center of Commercialization-Seattle**
- **Washington State University**

The University of Washington, Seattle has a variety of tech transfer activities. For example, at the College of Engineering, their industry Relations Officer facilitates the commercialization of new innovations arising from University of Washington research by managing intellectual property. The formal process captures the intellectual output of UW research, protects it through patents and copyrights, and licenses it to industry for the ultimate benefit of society. The University has three incubators. They include:

- Life Sciences Incubator, focused on Life Sciences, Biotech, and Medtech
- Hardware Incubator focused on Consumer, Health, Materials, and Medtech
- Startups Incubator focused on Fintech, IT, and Software

UW- Seattle has been ranked:

#1 Most Innovative public university in the US, Reuters
#7 Best University for Technology Transfer, Milken Institute
#9 in Nation for Number of Startups Launched AUTM 2017

Federal Way & the State

Photo by: We are Engineers

Federal Way currently does not have any Tech Transfer Centers, but it's location within the Greater Seattle area is prime. With talent surrounding the city and a highly respected research university in the University of Washington -Seattle (not more than 40 minutes away in distance) and the campus of the University of Washington- Tacoma located only 20 minutes away, locating a Tech Transfer Office would be an excellent economic development opportunity to help strengthen and shift the city's economy.

The commercialization of technology has a clear potential to benefit universities, but also businesses, industries, communities, consumers, along with the state's and Federal Way's economies through new products, partnerships and collaborations. Technology transfer activities can further the growth of the state's and city's economies.

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Pacific Northwest National Laboratory, Seattle Research Center

Washington State University

About the Author

Ankit Bhat is a member of the economic development internship program at the city of Federal Way. He is sophomore at UW-Seattle, majoring in economics and specifically focused on globalization, health and development through the Department of Geography. He has strong interest in working in the public sector with the goal of improving economies of cities across the globe. His work is on the information, communications and technology sector. He is examining Richardson, Texas and its long history and future as a communications corridor to determine opportunities in the way of tools, programs and projects to support Federal Way's efforts. His global enthusiasm is well placed to serve Federal Way.

WHAT ARE MASH-UPS, MEET-UPS, AND HACKATHONS?

Jason Golliet
City of Federal Way, Washington
Economic Development
April 2021

Mash-ups, meet-ups, and hackathons are words that define new techniques of how individuals and organizations communicate. Through the sharing of knowledge and advice in formal and informal networks, people, organizations, and even communities can engage on topics and issues through an impactful manner with successful outcomes using these tools.

Mash-ups

A mash-up (sometimes spelled as one word, *mashup*) is a web page or application that integrates complementary elements from two or more sources.

There are many types of mash-ups, such as technology, business, consumer, and data mash-ups.

The most common type of mash-up is the *consumer mash-up* which is an innovation used by the general public. An example of a consumer mashup is Google Maps. It has spawned hundreds, if not thousands, of mash-up applications. These include applications that use Google Maps to highlight information, such as rating areas in a city or identify points of interest. It can also show the current speed limit in an area or show roads that are undergoing construction. These applications take some of the functionality and data from Google Maps and combine it with their own programming to create a new application, hence the name mash-up.

Another type of mash-up is a technology mash-up. It is the merging of two or more technologies into a single device or system. For example, a technology mash-up might have some combination of mechanical, electronic, and/or digital innovations included in it. An example is a cellphone that can record videos and streams music.

Yet another type of mash-up is a data mash-up. The City of Amsterdam is an example of a municipality that has embraced the idea of data mash-ups. The City shares their traffic and transportation data with interested parties, like developers, who then create mapping apps connected to the city's transport systems for use by the public and private sectors.

Furthermore, municipalities may utilize other data from energy distribution, electric vehicle charging, and smart waste management solutions, such as the data collected from the use of sensors at city waste bins that alert crews to when they need attention.

In any of these types of mash-ups, the use and application of the Internet of Things, or data collected that is acquired, can allow an organization to be more efficient and effective. Simply, mash-ups drive the ease of use for consumers, companies, and communities.

Meet-ups

A meet-up is defined as an event where a group of like-minded people can listen to experts, share experiences, and simply communicate. Meet-ups can address a variety of subjects. For example, tech-oriented meet-ups cover topics related to technology, such as the future of blockchains, artificial intelligence, software development, and programming languages.

ELEMENTS OF MEET-UPS

- Talks – a speaker comes and talks about a particular topic, and the audience can ask questions at the end.
- Lightning Talks – multiple speakers come and speak for 5-10 minutes on a common topic.
- Workshops – hands-on learning.
- Social – people can hang out and have food and drinks while talking about core topic.

Meet-ups allow opportunities for experts and newcomers, startups to Fortune 500 companies, and everyone in between to learn and share. Through in person or online events, meet-ups connect people to a variety of activities from business, technology, education, legal, medicine, and social innovations to grow their networks, and even create employment opportunities.

Photo by NeONBRAND

The use of this entrepreneurial tool can provide a myriad of opportunities. Meet-ups can support an entrepreneurial culture, by assisting startup businesses, identifying and explaining new technologies, presenting new product design, and access to finance and human capital all in the effort to create and grow a tech hub.

Hackathons

A hackathon is a relatively new entrepreneurial tool used to drive sustained innovation through crowdsource solutions addressing real-life business problems and/or social issues. A definition by researchers, Maria Angarita of the University of Tartu and Alex Nolte of Carnegie Mellon, identify that a hackathon is typically a time-bound competitive event where participants collaborate in person to build proofs of concept and minimum viable products for a specific pre-defined problem and to facilitate innovation.

Photo by Alex Kotliarskyi

According to the 2017 Global Hackathon Report conducted by HackerEarth, the common goals of hackathons include the following: Explore New Technologies; Drive Business Innovation; Create Potential Startups; Reward Innovative Thinking; Create Solutions for Social Causes; Fuel Connectivity in Tech Community; Branding; and Create Opportunities for Participants.

Hackathons have some very clear advantages over traditional management processes that are top-down, linear-structured and static. Simply, they are inclusive, affect positive change, promote multidisciplinary collaboration, and have shorter innovation cycles that are better suited to addressing the fast-changing world of technology.

Of interest, before undertaking the dynamics of a hackathon, beginners have the opportunity to become oriented to the process through what is called "Learn-a-thons." As its name suggests, it is focused on learning. Its design is to help future participants in a hackathon to understand the basic and how to prepare for a hackathon.

Prior to the pandemic there were a multitude of hackathons in the Greater Seattle region. Companies such as Amazon and Expedia have hosted hackathons on items such as e-commerce to creating apps to revolutionizing travel. Additionally, national organizations have come to the Seattle region to host hackathons, such as Neurodata Without Borders.

COMMON GOALS OF HACKATHONS

Source: HackerEarth: Global Hackathon Report 2017

Many of the local hackathons that have been presented in the Seattle region are generally sponsored by private businesses and universities. For example, founded in 2017, Tech Together is the nation's largest initiative to address gender inequities in the hackathon community. The Tech Together Chapter serving Seattle constantly provides local hackathons.

Additionally, the UW-Seattle Foster Technology Club and Entrepreneurship and Venture Capital Club (EVCC) will present the inaugural event, "The Husky Hackathon 2021," in November at the Buerk Center for Entrepreneurship at the UW – Seattle Campus. The event will be an exciting 48 hours filled with building prototypes, pitching prototypes to investors, and actually creating a business. Through this event, teams will experience the entire lifecycle of building a product. Workshops and speaker series from local technology leaders will precede the event. There are cash awards for the 1st, 2nd, and 3rd place teams as high as $5,000.

Federal Way

As technology improves, the communities that enable these activities may see an improvement by way of a more cohesive and engaged community. Through hackathons, meet-ups, and mash-ups, there are an abundance of opportunities that not only challenge society, business, and education, but may also strengthen individual growth and help the grow of the community in ways that will lead to the creation or improvements of innovation positively impacting the world and specifically Federal Way.

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About the Arthur

Jason Golliet is a member of the Economic Development Internship program with the City of Federal Way. He is a junior at UW-Tacoma majoring in Public Policy and Economics. His focus is on global economics and international politics and he is also pursuing a minor in Data Analytics. Jason has had the opportunity to study abroad specifically Japan where he attended Kogakuin University of Technology and Engineering. Jason's work involves examining Financial Technologies (Fintech) as it pertains to opportunities to diversify Federal Way's economy as a potential tech hub. He is examining a variety of similar sized global cities to Federal Way identifying their efforts to attract and or grow this dynamic and fast growth sector. Jason's previous overseas education experience bodes well for Federal Way.

WHAT IS COWORKING SPACE AND MAKERSPACE?

by Taylor Jackson
City of Federal Way, Washington
Economic Development
April 2021

With more than one-third of the US population being part of the gig economy, it can safely be said that freelancing and remote working are on the rise. In fact, experts predict that this number will double by 2028 as freelancing becomes the new norm in the workforce. So, what does this group of independent workers value, and how does this affect the future of work spaces for startups and freelancers alike?

Above all else, freelancers prioritize flexibility. And to meet this newfound need for flexible working conditions, two new types of work places were born: Coworking spaces and Makerspaces. While both cater to the same core audience (freelancers), they differ in their main features and target two very different crowds of people. This paper defines the types of uses found at coworking and maker spaces.

Freelancing and Remote Work Will Double by 2028

Coworking Space (CWS)

Coworking spaces or hubs are membership-based shared office spaces, generally boasting an open-concept layout that encourages networking and professional collaborations. These workspaces come with flexible opening times. People utilizing them do not belong to the same organization, but come from diverse companies or run their own businesses.

The goal of coworking offices is to help freelancers combat the loneliness and isolation of a solopreneur life. They also provide a fantastic community atmosphere, where like-minded people can gather, exchange ideas and perhaps even form business partnerships. With a modern, chic interior and fantastic on-site amenities, coworking offices spark creativity in a professional, well-equipped environment.

Coworking spaces are a critical piece in the development of startup companies for three main reasons:

1. Affordable workspace option in many high-cost locations.
2. Ease of networking/collaboration, as compared to traditional office spaces.
3. Environment leads to more creativity and innovation.

US Chamber of Commerce 2019

As office space costs increase throughout America's tech hubs, price efficiency is highly prioritized. Due to requiring less overall space and amenities, co-working spaces often cost less than their traditional office counterparts.

Networking is another key aspect of CSWs, as many host competitions where various organizations working out of the same space can show off their products to potential investors or clients. Even without explicit events, the constant interactions CWSs create means networking between peers is inevitability.

By creating an environment where conversations are occurring between peers, workers are constantly introduced to fresh perspectives and ideas. This helps spark creativity and innovation, as individuals get the opportunity to pick the brains of other working professionals in their field and bounce ideas off of each other.

Makerspace

Makerspaces are non-program-based community work spaces that provide equipment for specific sectors of manufacturing (e.g., technology, 3D printing, woodworking, food) and provide the tools, training, and specialized equipment necessary for creating and building physical products.

These spaces are an entirely new realm outside of traditional office spaces. They are developed in an effort to foster a community that likes to build, invent, create, and learn. Rather than creating a networking space that encourages like-minded computer dwellers to collaborate, makerspaces encourage the handcrafters to gather together.

These niched workspaces directly target handcrafters and are equipped to cater to their professional needs. They often come with 3D printers, various tools, instruments, models, machines, software, and electronic devices that can come in handy in the "making" process. But besides these features, another extremely important element of makerspaces is their lifelong-learning mindset.

Makerspaces are typically privately-owned businesses, made with the intention of driving a profit. They generate funds through a variety of ways, including membership fees, renting out areas of the space, paid classes, and any grants or donations they can collect. Most makerspaces provide different levels of membership, with the more expensive options providing access to more of the tools in the space.

While physical spaces and tools are vital to the creation of a makerspace, the most integral piece is an interested maker community. The strength of the community in this endeavor outweighs all other aspects, as the community is the focal point of the makerspace being created in the first place. An actively engaged maker community ensures that the local makerspace can become a hotspot for collaboratively developing new crafts and concepts.

Photo by Benjamin Thomas

Washington state features several makerspaces, with a majority located in Seattle. These include Seattle Makers, UW's Area 01, ALTSpace, and many more.

The makerspaces range from more tool-focused workshops to computer software and program-focused spaces, with most providing a variety of tools for either focus.

It's ever-important for designers to keep in mind that while makerspaces are still in their infancy, they will only grow from here. Discovering the right space to innovate, development along with the freedom to collaborate is imperative for today's designers.

Regional Impacts

Thanks to the large success of coworking spaces, there are many large companies that operate facilities across the nation and the world. These include companies such as Impact Hub, Spaces, and WeWork. All three businesses operate globally, with large footprints in the US. Seattle hosts many of these coworking spaces with the intention of attracting startups in order to maintain the city's status as a flourishing technology hub. The largest coworking spaces in Washington include Ballard Labs, Thinkspace, and WeWork's Seattle locations.

Photo by Eloise Armbursley

According to the publisher Statista, while new coworking spaces seem to pop up on a daily basis, makerspaces are still quite a rarity, with only an estimated 500 of them found in the US. But their numbers are increasing, and they are definitely an exceptional way to learn the ins and outs of a handcraft business. While both spaces are slightly different and used for different types of work, they both help usher in a new era of developing the freedom to work away from your specific office.

Federal Way has coworking space dedicated to startup and established companies. Additionally, there is a makerspace planned for the Federal Way Library. Unfortunately, pandemic-related issues have postponed its opening. However, with the proper implementation of these features they can provide a key stepping stone for the City to achieve its goal as a tech hub.

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Geekwire

Impact Washington

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Taylor Jackson is a member of the economic development internship program at the City of Federal Way. Taylor is a junior at UW-Seattle majoring both in Economics and Information. His academic focus has covered a spectrum of topics including data science and work in programming and coding. Taylor's goal is to leverage the academic world into improving situations of marginalized communities using information technology and economic analysis to make more informed and equitable decisions. Taylor's work is focused on the dynamic area of transportation and mobility technologies. Additionally, Federal Way has found Taylor's research invaluable on so many fronts not only the examination of transportation and mobility technologies but also understanding makerspaces and coworking space in the effort to make Federal Way a tech hub.

WHAT IS ECONOMIC GARDENING?

By Tim Johnson
City of Federal Way
April 2021

Ever plant a garden? In short that is what economic gardening is but only with businesses, not vegetables. Simply, economic gardening connects entrepreneurs to resources, specifically capital and information.

Some in the economic development profession identify it as an alternative way of doing economic development that is focused on growing local stage-two companies as opposed to attracting outside companies. It can be characterized as a "grow from within strategy," in contrast to traditional business assistance. Economic gardening focuses on strategic growth challenges, such as development of new markets, refining business models, and gaining access to competitive intelligence.

The strategy of economic gardening focuses on providing market research to so-called second stage companies – growth-oriented businesses with external market potential that have moved beyond the startup stage. Typically, second stage companies employ 10-99 people and bring in at least $1 million in revenue each year.

While economic gardening is an appealing idea, as with any economic development tool, it's no silver bullet. Challenges with economic gardening include overcoming a lack of trust in the resources the effort can bring to the table, building awareness about programs, and actually identifying second stage companies.

These challenges are common with many government-led business assistance programs. However, working with trusted partners, like the chambers of commerce, universities, small business development centers, or trade groups can help overcome some of these barriers.

Also, an economic gardening program is not a quick fix. It is important for cities to understand that economic gardening is not a quick solution. A city cannot expect silver bullet solutions to sudden economic woes. Economic gardening takes time to put into place and time to reach a critical mass of growing companies.

According to the International City and County Managers Association (ICMA), economic gardening is an entrepreneurial approach to economic development that seeks to grow the local economy from within one's community. Its premise is that local entrepreneurs create the companies that bring new wealth and economic growth to a region in the form of jobs, increased revenues, and a vibrant local business sector. Economic gardening seeks to focus on growing and nurturing local businesses rather than hunting for "big game," i.e., large companies outside the area.

Preparing a strategy for an economic gardening program can be complicated as there are many elements that must be developed first, taking into consideration unique community needs and available resources. ICMA identifies seven steps to developing an implementation strategy for an economic gardening program to succeed. These include the following:

1. Gain the support of local officials and other stakeholders
2. Identify your community's assets
3. Develop a collaborative effort among resource partners
4. Create a system-wide operating agreement
5. Determine the target audience for services
6. Develop a delivery system to provide services to the target audience
7. Develop a communication system to gain community support and buy-in

The following details the seven steps for economic gardening:

1. Gain the support of local officials and other stakeholders

Governing bodies generally do not like to be handed a program and asked to vote on it. It takes time and effort to develop the support of elected officials for an economic gardening approach. The first step is to sit down with each official and other key stakeholders and listen to their concerns about economic development. Inadvertently or deliberately excluding a key stakeholder or someone on the governing board from providing input or participating in the decision-making process can lead to opposition and future undermining of your efforts.

2. Identify your community's assets

Develop an inventory of community and business assets available to you. Frame the discussion with the following questions:

- What human capital exists in my community? Human capital refers to "the unique capabilities and expertise of individuals that are productive in some economic context," generally linked to formal education and experience.
- What skills and expertise are available?
- Opportunities to partner with other organizations?
- What systems and organizations already exist to support entrepreneurs?
- Who is already motivated and passionate to make something happen?
- What cultural, recreational, and other quality-of-life amenities are available?
- What assets can be leveraged outside our community?

Additionally, identify a list of assets including the usual suspects such as economic development organizations, chambers of commerce, small business development centers (SBDCs), SCORE, workforce centers, universities and community colleges, financial institutions, and civic and service organizations along with the following:

- Public and university libraries
- Professional business associations and groups
- Community foundations and loan funds
- Microfinance organizations
- Elected and appointed officials
- Utility companies
- Successful entrepreneurs
- Council of governments
- Arts and cultural entities
- Consultants

- Policymakers
- Healthcare agencies
- Immigrant and citizenship initiatives

Look for individuals who have skills and expertise in areas such as business coaching and mentoring, finance, employment/workforce development, research, marketing, meeting facilitation, organizing and managing projects, public speaking, legal support, and fundraising.

Identify individuals who can become champions and advocates for the economic gardening project.

3. Develop a collaborative effort among resource partners

Explore which entities and individuals are likely to become resource partners in moving your economic gardening venture forward. Set up a steering committee that can guide and implement the project. Bring key resource partners together to reach common agreement on goals and directions for the project, as well as to identify who will take responsibility to carry out each piece. If key resource partners are not willing to take ownership of the project, then your community may not be ready to launch an economic gardening project. You may have to step back and address the political and community development issues that are driving your locality.

4. Create a system-wide operating agreement

Because an economic gardening project generally involves multiple entities, it is important for the steering committee to develop a formal or informal operating agreement that addresses key operational and long-term planning issues. Questions that must be addressed include:

- Which entity or group will make program-level decisions?
- How will the program be funded?
- Who will serve as fiscal agent?
- Who will oversee and coordinate delivery of services?
- What role will each partner fulfill in the overall project?
- What resources will each partner contribute to the effort?
- How will the program be tracked and evaluated?
- How will a sustainable capitalization plan be developed?

5. Determine the target audience for services

One of the most important questions an economic gardening project needs to answer is, "Who will we serve?" Economic gardening programs around the nation take a variety of approaches, depending on their identified goals and community expectations. Some programs support all types of small businesses; others work only with growth-oriented companies. The important thing is to know which group of businesses you are targeting and why.

The first step in determining your target audience(s) is to inventory the available entrepreneurial talent in your community. What kinds of businesses are located there? What is their level of growth or maturity? Small local businesses generally fall into three categories: start-ups, lifestyle businesses (local consumer-based ventures), and growth businesses (offering goods and services to external markets).

According to current research, growth-oriented companies produce the greatest economic impact on a region because they bring money into the community from outside markets. Many of these are so-called

Stage 2 businesses, those that have between 10 and 99 employees and at least $1 million in revenue, although growth-oriented ventures can be found across the entire universe of companies. Growth companies also tend to be relatively young (less than five years old). They can range from companies with a local market that want to expand and reach external markets (often through e-commerce) to companies that have specialized expertise or knowledge and primarily sell to external markets.

Rapidly expanding, high-growth companies, sometimes referred to as gazelles, are a rarity in many communities, making up only 3 to 5 percent of all businesses. Most small localities do not have the technical resources to support the needs of high-growth businesses. These companies typically have the connections and technical assistance they need to grow, and they will do it with or without an economic gardening program.

Stage 1 companies—those with one to nine employees and less than $1 million per year in revenue—are by far the largest group of businesses in the United States. They collectively represent 28 percent of all employment nationally. Most of these firms are start-ups and lifestyle businesses, but some are growth-oriented companies too.

Lifestyle businesses (the so-called mom 'n' pops) are the small retail and service businesses in every community. They do not usually "grow" the local economy by bringing in new wealth, but they recirculate the wealth throughout the local community. They are essential to what makes a local community a vibrant and desirable place to live and work. They can also provide significant political capital for an economic gardening program through their testimonials and support.

The "sweet spot" for most local and regional economic gardening programs is to target its entrepreneurs who have started a venture that is between one and five years old and who want to grow it, regardless of its size. These ventures aren't necessarily high-tech, but they have developed some sort of innovation in their product, process, or delivery method. They also have a potential or actual market outside the local economic region and create high-quality, living-wage jobs.

These nascent growth-oriented companies can provide significant economic impact and can benefit greatly from the services an economic gardening program typically provides. To focus on this target audience, it is recommended to find companies that meet the following criteria:

- Firmly established (in business for one to three years)
- Have financial statements that include profit and loss and cash flow numbers
- Have a clearly defined market
- Demonstrate revenue growth over time (even if the company has not yet reached the break-even point)
- Clearly intend to grow (as expressed in the desire to hire employees, expand operations or market area, or seek capital investment)
- Have a product or service that is scalable and preferably unique (i.e., cannot be easily imitated)
- Have a potential or actual market outside the local region

These growth businesses will sometimes look like secondary businesses—local retail and service companies. The key is that they have both the desire and the ability to sell their goods and services outside the local area. For example, a local producer of specialized jams and jellies can sell their products over the internet; a local coffee roaster using solar technology can wholesale their organic beans to coffee shops

throughout the region; or a local printing company can provide on-demand printing and graphic services through their website.

You can also consider offering tiered services to different audiences. You might, for example, provide basic services to your lifestyle businesses and more comprehensive support to growth-oriented companies.

6. Develop a delivery system to provide services to the target audience

Steps involved in creating a viable delivery system include finding or developing qualified business coaches, providing or linking to technical assistance resources, locating entrepreneurs within one's target audience, offering market research services, identifying financial resources, and partnering with other providers within and outside the local area. A local referral network of small business professionals and service providers is a crucial element at this juncture.

7. Develop a communication system to gain community support and buy-in

Make public presentations explaining the economic gardening program and gain the support of local media. Use entrepreneurs and your local referral network as advocates to deliver your message to funders, prospective clients, and the public. Build regular reporting functions into your ongoing activities.

The ability to help second stage companies may not be in Federal Way's future given the supply. However, the awareness of economic gardening specifically as a tool for creating an entrepreneurial ecosystem in Federal Way is important to the City's efforts.

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About the Author

Tim Johnson is director of Economic Development & Redevelopment for the City of Federal Way. He has nearly 40 years of experience working in both the public and private sectors in the formulation and implementation of programs and services in support of business growth. Tim's efforts have been on creating conducive business environments across the nation for the growth of entrepreneurial endeavors. His work includes the deployment of the "MainStreet" program supporting small business through the National Trust for Historic Preservation across the western U.S. to the initial formulation of the Small Business Development Centers in California. Additionally, Tim serves on the Board of Advisors for the nationwide Community Development Financial Institution – The Clearinghouse CDFI. The program provides financial assistance to underserved micro and small businesses specifically in communities of color.

WHAT IS A START UP STUDIO OR FOUNDRY?

By Tim Johnson
City of Federal Way
April 2021

Still searching for help and assistance with your startup? Many are familiar with startup incubators and accelerators, but there's an emerging model for building high-growth-potential startup companies.

A startup studio, also known as a startup factory, or a startup foundry. A Studio or Foundry is aimed at building several companies in succession. This style of business building is referred to as "parallel entrepreneurship". Whereby they produce multiple businesses at the same time.

Historically, the first was created in 1996. It was known as the Idealab. It has created over 150 companies whose products and services change the way people think, live and work. Some of the most notable are: Picasa, acquired by Google; NetZero, merged with Juno; Tickers.com and Coinbase.

Studios and Foundries Defined

However, unlike an accelerator or incubator, a Startup Studio or Foundry is different.

Elaborating further, a startup studio or foundry is a team of engineers and entrepreneurs building startups that transform industries. A key point with a studio or foundry is that you can't outsource the job of turning great ideas into great businesses. So, operators of studios or foundries are real co-founders on projects they pursue. They mentor members of the team to become the future CEOs and CTOs of the companies built.

A Studio or Foundry is to build multiple companies in parallel. This concept has led to the emergence of startup studios, foundries which focus on creating companies from the ground up.

A startup studio/foundry fosters the creation of multiple companies per year, following a repetitive process that leverages an infrastructure of resources, multi-disciplinary team members, and collaborative methods.

Instead of working on one company at a time, startup studios work on multiple projects simultaneously. They dedicate a team to the continued development of a project, with the hope of that project turning into a startup and then into a scalable, full-fledged company.

The idea is to build multiple startups, getting involved at the ground floor and helping recruit a team to carry the company forward.

To understand the right model for one's needs, it is important to understand the differences between start-up studios and accelerators and incubators.

Professionals Help Innovators Create and Grow Startups

Studios and Foundries don't actually select startups and they don't actually provide assistance. In fact, the technical term for the model is "startup studio," as opposed to "startup incubator." Rather than selecting pre-existing startups, funding them, mentoring them, and then releasing them to the wild over a period of three to six months like most incubators, this model involves starting companies by the operators of the studios or foundries themselves. As such, they are starting from scratch -- taking an idea, vetting it, funding it, staffing a team to it, and growing it internally. A candidate startup may stay within the Foundry for one to two years before gaining the team and traction needed to spin it out as its own autonomous company.

Differences & Similarities: Studios/Foundries vs. Accelerators/Incubators

To understand the differences of a studio/foundry versus an accelerator/incubator presents the analysis by the following categories: Support, Program Length, Grouping, Price, Operators, Investments, Stage and Location.

Support

Startup Studios/Foundries

Startup Studios and foundries help build a company from the ground up and often provide hands-on support for everything from marketing to logistics to technology to talent and beyond.

Accelerators/Incubator

Accelerators and incubators, on the other hand, are more focused on education, networking, co-working spaces, mentorship, and fundraising.

Program Length

Startup Studios/Foundries

Startup studios and foundries typically don't have fixed program lengths.

Accelerators/Incubator

Accelerators typically offer programs that run for a finite length of time (typically three to six months) and culminate in an investor pitch demonstration and graduation to follow. An incubator can run longer in timespan depending upon the nature of the incubator.

Grouping

Startup Studios/Foundries

Startup studios, on the other hand, can typically offer more scalability by staggering the start dates of their ventures to balance the resource demands that are required to build companies. However, the industry is starting to see more venture builders group cohorts together by quarter or year, but stagger the actual start dates within that time period.

Accelerators/Incubators

Most accelerators group cohorts together and run group sessions in which everyone starts and ends at the same time, which makes accelerator operations scalable. Although accelerators and incubators have begun to take rolling admissions

Price

Startup Studios/Foundries

Both startup studios and accelerators charge “fees” for their services, often in the discount they receive on the equity for their investment. Sometimes, startup studios charge cash fees in addition to an equity position to help with the much larger resource commitment to each startup, and some will take more equity in lieu of cash.

Accelerators/Incubators

Accelerators often end up with anywhere from two to 10 percent equity, and many have a program fee that they claw back from the investment to help cover their overhead.

Operators

Startup Studios/Foundries

Startup studios vary in how they approach this, as such some work with external operators and some with internal operators to run their companies. Similarly, some build their own ideas while others source them from elsewhere.

Accelerators/Incubators

Startup accelerators work with outside operators that bring their own business.

Investments

Startup Studios/Foundries

Startup studios typically fund their own internal ideas, but usually don’t make a cash investment when ideas come from outside. Similar to startup accelerators, venture builders often have continuity funds to invest in the later stages in their own companies.

Accelerators/Incubators

Almost all accelerators make investments in the companies they select, usually ranging from $50,000 to $150,000, and some have continuity funds that they continue to invest in later-stage capital raises.

Stage

Startup Studios/Foundries

The purpose of the startup studio model is usually to start companies from an idea-on-a-napkin and take them to (at least) the seed stage. Many support the concepts from idea to exit.

Accelerators/Incubators

Accelerators typically work with post-product and early-revenue tech companies and speed up the ramp to the seed stage, which can mean $10,000 a month in revenue — or sometimes even up to $100,000 a month.

Location

Startup Studios/Foundries
Startup studios typically don't require a location, and tend to be more open to remote operators.

Accelerators/Incubators
Most accelerators require you to be in person during the program to co-work with the rest of the selected cohort of businesses.

DIFFERENCES & SIMILARITIES: STUDIOS/FOUNDARIES vs. ACCELERATORS/INCUBATORS

CATEGORIES	STUDIOS/FOUNDRIES	ACCELERATORS/INCUBATORS
Support	**Broad**	**Focused**
Program Length	**No Lengths**	**Finite Length**
Grouping	**Flexible**	**More Structured**
Price	**Charge fees**	**Charge fees**
Operators	**Internal & External Operators**	**External Operators**
Investments	**Funding Support**	**Funding Support**
Stage	**Start up**	**Start up**
Location	**In person or remote**	**In person**

The Startup Studio is the key that helps corporations and entrepreneurs unlock innovative and financial success with a model that is sustainable and proven. However, the Startup Studio model remains — much like the accelerator was a decade ago — mystery to many. How is it different, and what are the benefits?

Startup Studios don't put the focus on creating cohorts to compete on a "Demonstration Day," rather they focus on creating the best possible outcomes with as little waste as possible. Startup Studios are intimately familiar with the idea and problem stages of creating a startup or corporate spinout. This means Studios have a lot of experience with generating creative and impactful business models, building thoughtful user flows for the customer and product journey, as well developing roadmaps for talent acquisition and securing funding.

The real secret of the Startup Studio? They don't just build the technology, they help founders build their business — going along the journey every step of the way. From early-stage startups, to Fortune 500 companies, the purpose of the Startup Studio is to validate and build business models that disrupt a particular industry, accelerate innovation, and remain competitive long-term. Startup Studios provide an opportunity for shared learning and leadership. Startup Studios don't just build the technology, they help founders build their business — going along the journey every step of the way.

At the same time, it's extremely difficult for corporations to quickly and efficiently validate and commercialize transformative innovation internally. To address this challenge, many organizations are trying open innovation, looking outside their core business to innovate. Unfortunately, most consultancies and similar organizations are unable to provide sustainable solutions that generate lasting results, leading to inevitable innovation fatigue.

To fight this fatigue, one company has positioned itself in the marketplace on fatigue avoidance. Coplex is a corporate startup studio that helps companies increase the speed of decision making, creating innovative technical solutions, building strong teams, and attracting revenue for new startups. The Coplex Startup Studio combines an experienced team and a proven methodology for rapidly commercializing the best ideas and maximizing the chance of spinout success.

In other words: We turn the art of designing businesses into a repeatable science. The process is broken down into five phases:

- Business Design
- Validation
- Go-to-Market
- Revenue
- Growth

From ideation, marketing and branding strategies, talent acquisition, and revenue generation, the startup studio is designed to propel companies forward with a low-risk, high reward mindset.

The startup studio concept is not new as previously identified. Idealab was one of the first to find success, more than 25 years ago. But more and more are cropping up as the global total is near 600, according to Enhance Ventures.

Washington

Local Studios and Foundries in the state include the following:

- Seattle's Kernel Labs
- Pioneer Square labs
- Madrone Venture labs
- Perch Partners

Federal Way

Unfortunately, Federal Way has no start up studio or business foundry. However, it is an economic development opportunity in the effort to help diversify and strengthen the business climate. The following illustrates the opportunities to be considered in examining the strategy to make Federal Way a technology hub.

- Job creation – Creating new employment opportunities for local residents and youth in the participating startups.
- Enhanced image – Building the region's reputation as a center for excellence for growth in the blue economy.
- Increased entrepreneurialism – Elevating awareness of entrepreneurs and stimulating confidence in the maritime industry to create new products and services.
- Business development – Sparking new opportunities for established area businesses to develop relationships with early-stage companies.
- Increased tax revenue – Generating a larger, more diverse tax base to support public services and contribute to livability and health.
- Workforce development – Nurturing the next generation of diverse, inclusive, and representative maritime workforces with technological expertise and access to clean, healthy, living-wage jobs.

Sources

So, what is a startup foundry anyway? Julia Austin September 2014

Geekwire

Global Startup Studio Network; *"The Rise of Startup Studios "March* 2019 Denver

Startup Studios, A new Path to AI Growth, by Annie Brown, Forbes magazine online edition, December 17, 2019.

About the Author

Tim Johnson is director of Economic Development & Redevelopment for the City of Federal Way. He has nearly 40 years of experience working in both the public and private sectors in the formulation and implementation of programs and services in support of business growth. Tim's efforts have been on creating conducive business environments across the nation for the growth of entrepreneurial endeavors. His work includes the deployment of the "MainStreet" program supporting small business through the National Trust for Historic Preservation across the western U.S. to the initial formulation of the Small Business Development Centers in California. Additionally, Tim serves on the Board of Advisors for the nationwide Community Development Financial Institution – The Clearinghouse CDFI. The program provides financial assistance to underserved micro and small businesses specifically in communities of color.

ADVANCED MANUFACTURING

Use of innovative technologies to create existing products and the creation of new products. Advanced manufacturing can include production activities that depend on information, automation, computation, software, sensing, and networking.

Global
Total Manufacturing Market Size 2019

- $13.672 trillion
- 15.391% of total GDP

World Bank

Smart Manufacturing Market Size

- 2019 $215.8 billion
- Projected CAGR 11.8% from 2020 to 2027

Grandview Research

Select Major Global Companies

- Siemens AG
- GE
- Rockwell Automation
- Schneider Electric
- Honeywell International
- Emerson Electric
- Fanuc Corporation

US Manufacturing Sector

- Over 625,000 manufacturers
- Employs over 12 M
- Average Salary $88,406 annually including wage and benefits

Contributed $2,359.9 billion to U.S. GDP amounting to 11.0% of total GDP.

National Institute of Standards and Technology

For every $1.00 spent in manufacturing, another $2.79 is added to the economy.

National Association of Manufacturers

It is forecasted over the decade of the 2020's, an additional 4.6 million manufacturing jobs will be needed.

National Association of Manufacturers

Manufacturers in the United States perform 61.8% of all private-sector R&D in the nation, driving more innovation than any other sector.

Bureau of Economic Analysis

In 2017, $1 billion worth of VC money was invested in smart manufacturing startups in the U.S.

Deloitte

Largest U.S. Manufacturers

- Exxon Mobil
- Apple
- General Motors
- Ford
- General Electric

Elements of Advanced Manufacturing

WASHINGTON'S ECOSYSTEM FOR ADVANCED MANUFACTURING

Washington Manufacturing

- Home to Firms: 75,000 (2019)
- Employs over 305,000 (2019)
- Average Salary $81,200
- 11.21% of State GDP

Assn of Washington Businesses
National Association of Manufacturers

Key Areas of Advanced Manufacturing in Washington

- Aerospace
- Maritime Manufacturing
- Food Processing
- Semiconductors and Electronic Manufacturing

Advance Manufacturing Seattle Region

- # of Firms; 2,460
- 105,400 employed

Select Advance Manufacturers — Seattle

- Boeing
- Honeywell Aerospace
- Paccar
- Crane Aerospace
- Hexcel Corp.
- Blue Origin

Puget Sound Business Journal

Washington's Thriving Ecosystem

Manufacturing has long been a leading driver of job creation in Washington. There are a variety of institutes and research centers in support of advanced manufacturing. These include facilities at Washington State University and University of Washington involved in business incubation and technology transfer.

State Centers of Excellence:

- Aerospace and Advanced Materials Manufacturing — Edmonds Community College
- NW Center for Marine Manufacturing and Technology — Skagit Valley College
- Semiconductor and Electronics Manufacturing- Everett Community college

Washington State University

- WSU Advanced Manufacturing Initiative, School of Mechanical and Materials Engineering
- WSU Composite Materials & Engineering Center

University of Washington

- UW Boeing Advanced Research Center, BARC
- UW Paccar Advanced Research Center
- UW Washington Clean Energy Testbeds
- UW Center for Advanced Materials in Transport Aircraft Structures
- UW Autonomous Insect Robotics Laboratory
- UW AMP Lab
- UW CoMotion Hardware Incubator
- UW CoMotion Labs Technology Incubator (Startup Hall)
- UW CoMotion Tech Transfer
- UW Washington Nanofabrication Facility
- UW Makerspaces — The Mill

Other

- Toolbox Manufacturing Incubator, Spokane, WA
- Joint Center for Aerospace Innovation Technology
- Washington Aerospace Training & Resource Center
- Advance manufacturing Training and Education Center (AMTEC)
- Boeing: Manufacturing Technology Integration. Research Center
- Center for Advanced Manufacturing Puget Sound (CAMPS)

"If I have a thousand ideas and only one turns out to be good, I am satisfied."
Alfred Nobel

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CLEAN TECH

Cleantech is any product or service that uses technology to compete on price and performance while reducing pollution, waste, and use of natural resources. Cleantech is at the center of the green economy.

EVERYONE IS IN ON CLEANTECH

Estimated Market Size 2020

Global: $7.87 T

US: $1.3 T

Georgeson, Maslin (2020)

Public & Private Investment in Clean Energy

China	US	Japan	India	Brazil
$83.4 B	$55.5 B	$16.5 B	$9.3 B	$6.5 B

Bloomberg New Energy Finance (2020)

Globally, venture capital invested $17 B across 1,000 deals in 2020, 49% was US-led.

Bloomberg New Energy Finance (2021)

600+ cities worldwide have set 100% renewable energy targets.

REN21 (2021)

50 of the 100 most promising clean tech start ups are in the U.S.

Cleantech Group (2021)

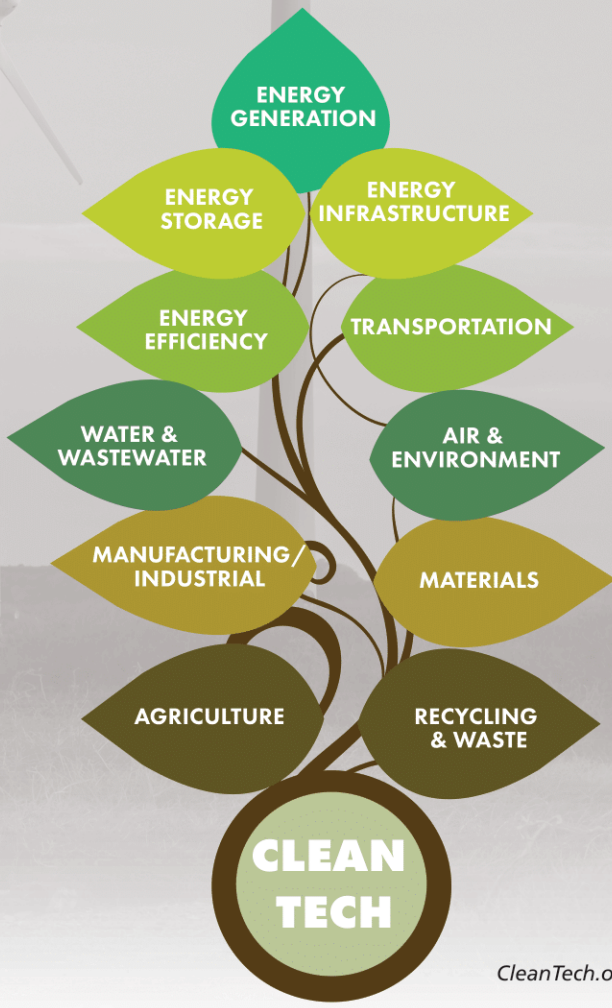
US Clean Energy Industry

- Employs over 3.3 million
- Median wage of $23.89

E2 (2020)

Select Cleantech Companies in the US

- GE
- Iberdrola
- Vestas
- NextEra Energy
- Renewable Energy Group
- First Solar
- Turner Construction
- Tesla
- Waste Management
- GM
- Republic Services



CleanTech.org

WASHINGTON'S ECOSYSTEM FOR CLEAN TECHNOLOGIES

Washington Clean Energy Industry 2019

- Employs over 85,000
- Median hourly wage of $25.39

E2 (2020)

Select Cleantech Companies in Washington

- McKinstry
- PACCAR
- Gly Construction
- Itron
- Helion Energy
- Parametrix
- PlanLED – Federal Way

Incentives & Tax Exemptions

To attract investment, Washington offers an array of business incentives and tax exemptions for clean technology.

- Business and occupation tax reductions for manufacturers of solar energy systems
- Components or semiconductor materials
- Sales and tax exemptions for semiconductor gases and chemical purchases
- Sales and tax credits for equipment that generates electricity using renewable energy
- Business and occupation tax credit and sales tax exemption for forest-derived biomass harvesters

73% of Washington's energy is from renewable sources, mostly hydropower.

Choose Washington

Investment Capital

- Microsoft Climate Innovation Fund: $1 B for Climate Change Mitigation Technologies
- Amazon Climate Pledge Fund: $2 B into Cleantech

Private Research Centers & Organizations

- Cleantech Alliance (largest state trade association of cleantech businesses in US)
  - Cascadia Cleantech Accelerator
- Pacific Northwest Center for Excellence for Clean Energy
- Northwest Institute for Material Physics, Chemistry, & Technology
- Pacific Northwest National Laboratory
- Northwest Energy Efficiency Council
- Lighting Design Lab
- GridForward
- Spark Northwest
- E8 Angels
- Energy Foundry

Major Public Research Centers

University of Washington

- UW Clean Energy Institute
  - Clean Energy Testbeds
  - Center for Advanced Materials and Clean Technologies (CAMCET)
- Burke Center for Entrepreneurship
  - Jones + Foster Accelerator
  - Alaska Airlines Environmental Innovation Challenge

Washington State University Office of Clean Technology

- Energy Systems Innovation Center
- Institute for Sustainable Design
- Composite Materials & Engineering Center
- State of Washington Water Research Center
- Center of Excellence for Alternative Jet Fuels & the Environment
- Biproducts, Sciences, and Engineering Laboratory
- Composite Materials and Engineering Center
- Center for Materials Research
- Center for Sustainable Agriculture and Natural Resources
- Center for Precision & Automated Agricultural Systems

"Clean tech is growing up."

Dan Reicher, Former US Assistant Secretary of Energy,
Currently, Senior Research Scholar, Stanford University, Woods Institute for Environment

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FINTECH

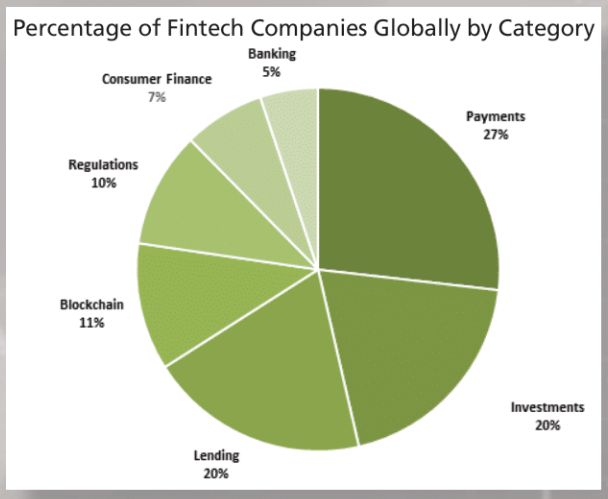
Fintech is a term used to describe financial technology, an industry encompassing any kind of technology in financial services - from businesses to consumers. Fintech describes any company that provides financial services through software or other technology and includes anything from mobile payment apps to cryptocurrency.

Fintech by the Services

- Banking
- Insurance
- Loans
- Personal Finance
- Digital Payments
- Wealth Management
- E-commerce

Fintech by Technology

- API
- Artificial Intelligence
- Blockchain Technology
- Distributed Computing



90% of people in 2021 will make a mobile payment using a smartphone.

(Business Research Corporation 2020)

By 2022, 78% of millennial population in the U.S. will become a digital banking users.

(Business Research Corporation 2020)

73% of Americans view fintech as the future (new normal) because of COVID-19.

(Plaid 2020)

VALUE OF GLOBAL FINANCIAL SERVICES MARKET:

$26.5 trillion in 2022 with CAGR at 6%

(Business Research Corporation 2020–2028)

Global Fintech Market $128 billion in 2018 25% AGR through 2022.

(The Business Research Company)

3 out of 4 People Worldwide have used Fintech Services.

(Ernst & Young 2019)

There were over 26,000 Fintech Startups Globally in 2020.

(Statist 2021)

Select Major U.S. Companies in Fintech

- Ant Financial
- PayPal
- Venmo
- Square Stripe
- Affirm
- Credit Karma
- Sofi
- Kabbage
- Lendio
- Robinhood

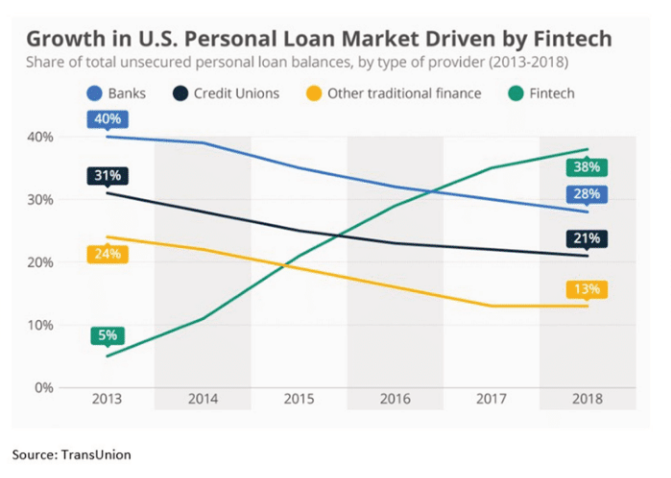
As of 2021, there are 8,775 Fintech startups in North America.

(Statista, 2021)

In 2018, the US accounted for 57% of the Global Fintech Market.

(Mordor Intelligence)

WASHINGTON'S ECOSYSTEM FOR FINTECH



Business Recources State/Seattle Region

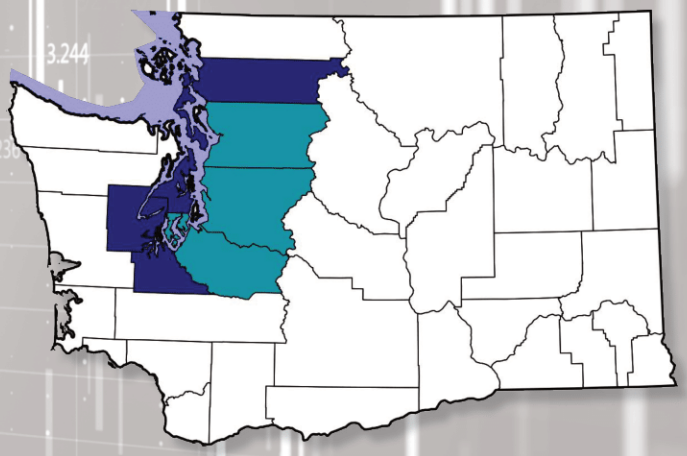
- BECU/UW — Seattle CoMotion, Fintech Incubator
- Sound Credit Union/UW — Tacoma, Fintech Incubator
- Fintech Boot Camp, UW — Professional and Continuing Education

Over 101,000 employees in Washington State are employed in 52 Financial Activities related to FinTech.

(ESD Washington State March 2021)

67,300 (66.5%) of financial sector employees in Washington are located in the Seattle-Tacoma Metropolitan Area.

(ESD Washington State March 2021)



Select Fintechs — Seattle

- Tracxn
- Remitly
- Transparent Systems
- PitchBook
- Gravity Payments
- Possible Finance
- Carta
- Alogent
- Doxo
- Tomorrow

Nation Wide Venture Capital Deals & Investments (2020)

- Total: $21.9 Billion
- Deals: 1,040

(Pitchbook-NVCA Venture Monitor 2021)

Seattle Region Venture Capital Deals & Investments (2014–2019)

- Total: $128 Million Annual Average Investments
- Deals: 12.3 Average Deals per Year

(Pitchbook)

"The world needs banking but it does not need banks"
Bill Gates

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ICT

ICT is an abbreviation for the term information and communication technology, an industry defined as a diverse set of technological tools and resources used to communicate and to create, disseminate, store, and manage information.

Key Categories of ICT include: Software, Devices & Infrastructure, IT & Business Services, Emerging Tech, and Telecom Services.

Services that surround ICT include: Cloud Computing, Software, Hardware, Digital Transactions, Internet, and Digital Communications.

Global

The ICT market was valued at $4.3 trillion in 2020 with a CAGR or 11.1% through 2023. However, a post pandemic forecast for the Value of Global ICT Services Market will near $5 Trillion in 2021 with an estimated 5% CAGR through 2024.

IDC

Worldwide full-time employment in the ICT sector is to reach 55 million in 2020 (pre-corona estimate) a 3.9% increase over 2019

Statista

US

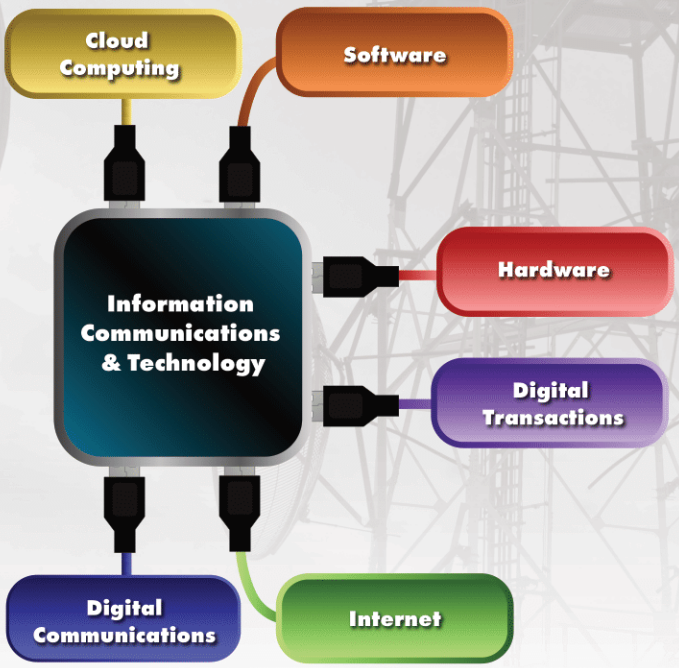
Largest tech market in the world representing 32% of total global market

- 12.4 million jobs
- 7.9% of Workforce is in tech
- Economic impact of Tech industry almost $2 Trillion
- Representing 10.5% of the nation's GDP
- 585,000 tech businesses

CompTIA

Select Major ICT companies in the United States

- Apple
- Microsoft
- Amazon
- Alphabet/Google
- Facebook
- Verizon
- Intel
- Adobe
- AT&T
- Cisco
- T-Mobile
- Oracle
- IBM



WASHINGTON'S ECOSYSTEM FOR ICT

ICT is among Washington's largest sectors. It is fast growing, pays strong wages, and is an increasingly important driver of other industries in the state.

ICT in Washington

- 405,816 total in Tech Occupations
- 278,000 jobs in Tech Industry
- 11.2% of workforce in tech
- 16,547 Tech Establishments
- Annual Economic Impact of Tech Industry, $121 billion

CompTIA

Annual Average wages including benefits of $188,700 per worker.

Association of Washington Business 2021

An estimated 15% of employees in the space sector work in ICT occupations, such as software engineering. Within biotech, nearly 9% of the workforce are ICT workers, and 6% in energy systems.

Association of Washington Business 2021

Seattle-Tacoma-Bellevue

- 326,000 total in Tech Occupations
- 231,000 jobs in Tech Industry
- 11,239 tech Establishments
- Annual Economic Impact of Tech Industry, $113 billion

CompTIA

80% of ICT employment is concentrated in the Seattle Region

Association of Washington Business 2021

Select Major ICT companies in the Seattle Region:

- Amazon
- Google
- Expedia
- Tableau
- Facebook
- Microsoft
- Boeing
- Adobe

Washington is Ranked #1 in the Nation for Tech Economic Impact as percent of State Economy.

Washington is identified as one of the Top Ten Cyberstates States by Employment Size and Employment Gains in ICT.

Washington has the second highest concentration of tech workers relative to its overall employment base at 11.2%.

CompTIA

Business Resources in State/Seattle Region:

There are a variety of accelerators, and incubators throughout the Seattle region for ICT. Additionally, there are multiple research centers at both the University of Washington and Washington State University helping support innovation through collaboration.

- Washington State University Office of Commercialization Tech and Intellectual Property Incubator.
- 5G Open Innovation Lab.
- A12 Incubator at the Allen Institute for Artificial Intelligence.
- Pacific Northwest National Laboratory
- Schweitzer Engineering Laboratories
- Center of Excellence for Information & Computing Technology — Bellevue College
- UW Center for Information Assurance and Cybersecurity-UW Bothell
- UW Center for Technology and Disability Studies
- UW Law Technology and the Arts
- UW Technology & Social Change Group
- Center for Design of Analog-Digital Integrated Circuits
- Center for Game Science
- Center for Research and Education on Accessible Technology
- UW CoMotion
- DigiPen Institute of Technology

Associations

- Association of Washington Business
- Tech Alliance
- Washington Technology Alliance
- New Tech Northwest

"The number one benefit of information technology is that it empowers people to do what they want to do. It lets people be creative. It lets people be productive. It lets people learn things they didn't think they could learn before, and so in a sense it is all about potential."

Steve Ballmer, Former CEO, Microsoft

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LIFE SCIENCES/BIOTECH

Life science is the study of living organisms and life processes; biotechnology uses the understanding of natural science to develop technologies to help improve lives and the health of the planet.

LIFE SCIENCES INCLUDES:

- Research, Testing, and Medical Laboratories
- Drugs and Pharmaceuticals
- Medical Devices and Equipment
- Agricultural and Industrial Biotech
- Bioscience Distribution

BIO Report 2020

The Global Biotechnology Market has been valued at $445 to nearly $500 billion in 2019. The CAGR is forecasted from 6% to 8% through 2028.

Global Market Insights & Fior Markets

Global Industry

Top 5 Life Science Countries (# of Firms)

1. US
2. China
3. Japan
4. Switzerland
5. UK

fDi Intelligence (2020) & Forbes (2020)

Globally, Venture Capital invested in nearly $295 billion in USD in nearly 33,000 deals across the spectrum in 2019.

Crunchbase

66% of Biotech Venture Capital is raised by US companies.

BIO, Biotech VC by the numbers

US Life Sciences/Biotech Industry

- Over 100,000 firms
- Employs 9.4M
- Average Salary $107,601

BIO Report 2020

Biotech raised $13.9B in Venture Capital in 2020 with 326 deals.

PWC, Healthcare Moneytree Report

Select Life Science & Biotech Companies in the US

- Amgen
- Gilead Sciences
- Biogen
- Illumina
- Johnson & Johnson
- Pfizer (Pharmaceutical)
- Bristol-Myers Squibb (Pharmaceutical)

Nationally, Venture Capital invested in nearly $137 billion across nearly 11,000 deals in 2019.

PitchBook and NVCA

WASHINGTON'S ECOSYSTEM FOR LIFE SCIENCES/BIOTECH

Washington Life Sciences/Biotech Industry

- Home to 1,150 life science firms
- Employs over 94,000
- Average Salary $93,000
- Creates $11.5B to state GDP

Life Sciences Washington

Life Science employment jumped 17.4 percent in Seattle from 2014 to 2017.

Geekwire

TOP LIFE SCIENCES/ BIOTECH MARKETS IN THE U.S.

Seattle Metro Area ranks #9.

CBRE, US Life Sciences Report 2020
JLL, Life Sciences Real Estate Outlook, 2020

University of Washington ranks #8 Best Global Universities.

US News & World Report's (2020)

University of Washington, School of Medicine #1 in primary care and #2 in family medicine.

US News Rankings (2021)

Select Life Sciences and Biotech in Washington including Federal Way

- Seagen
- Juno Therapeutics
- Nanostring Tech
- Bio-Rad Laboratories
- Adaptive Biotechnologies
- DiagnosTechs — Federal Way
- Sagemax — Federal Way

Washington's Thriving Startup Ecosystem

Variety of accelerators and incubators focused on life sciences/biotech such as Cambia Grove, IntuitiveX, Icogenix bioincubator, SP3NW, Bothell Biomedical Investment Zone, and UW CoMotion along with WIN Mentoring program. Fred Hutch, UW, and WSU are home to Tech Transfers Centers driving innovation through collaboration

Major Life Science/Biotech Research Centers & Organizations

- Fred Hutch Cancer Research Center
- University of Washington, School of Medicine
- Bill and Melinda Gates Foundation
- Allen Institute for Brain Science
- Washington State University
- LFW Institute
- Benaroya Research Institute at Virginia Mason/Franciscan Health
- Seattle Children's Hospital
- Seattle Cancer Care Alliance
- The Hope Heart Institute
- Institute for Systems Biology
- PATH
- Pacific Northwest Pacific Institute
- Infectious Disease Research Institute
- Puget Sound Blood Center
- Seattle Biomed
- Heart Clinics Northwest

Industry Trade Associations

- Life Science Washington
- Washington Global Health Alliance

Washington State Department Commerce

"I think the biggest innovations of the 21st century will be at the intersection of biology and technology. A new era is beginning."

Steve Jobs

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TRANSPORTATION/ MOBILITY TECHNOLOGY

Transportation and mobility technology encompasses all technologies that make the process of moving people and goods more seamless, sustainable, accessible, affordable, and safe.

The global transportation market was valued at $5.938 trillion in 2020, and is expected to grow to $7.5 trillion by 2023 with a CAGR of 9%.

ResearchandMarkets

20% of websites of large ocean carriers and steamship companies take more than 20 seconds to load.

McKinsey

Intelligent Transport Systems (ITS) will reach more than $72.3 billion globally by 2022.

PwC 2019

US Transportation/Mobility Industry

- Employs 5.6 million
- Generates $895 billion
- Over 180,000 firms total

Select Transportation/Mobility Companies in the US

- Tesla
- Lyft
- Uber
- Toyota
- Waymo
- Ford

Even with advancements, 80% of warehouses today are manually operated.

Robotics Business Review

The annual sale of autonomous trucks is estimated to reach 600,000 units per year by 2035.

Trucks.com

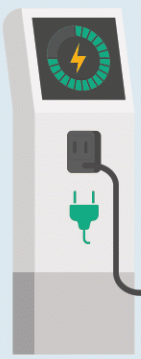
Value of Autonomous Technology, it is estimated that by 2026 driver shortage in the US will be 174,000.

American Trucking Association

In 2019, venture capitalists invested over $33.5 billion across more than 750 deals into mobility technology.

Pitchbook

Transportation/Mobility Tech Includes:



- **Autonomous Vehicles**
- **Aerospace Transportation**
- **Electric Vehicles**
- **First and Last Mile Transportation**
- **Mobility as a Service**
- **Smart Transportation/Connectivity**

WASHINGTON'S ECOSYSTEM FOR TRANSPORTATION/MOBILITY TECHNOLOGY

Washington Transportation/Mobility Tech Industry

- Home to 4,657 firms
- Employs over 100,000
- Total $5.5 billion in Annual Wages

ESD State of Washington

Select Transportation/Mobility in Washington

- Boeing
- Expeditors
- Alaska Airlines
- Carrix
- Saltchuk
- Tideworks
- Convoy
- Flexports
- ioCurrents
- Blue Origin

Initiatives

The state has multiple initiatives including transition to an electric fleet of vehicles to an EV infrastructure pilot program. Also, the state participates in the West Coast Green Highway program providing fast charging stations for electric vehicles along major interstates.

Investments

Billions of dollars are being invested globally. A variety of venture capital groups such as Tandem Capital, Fontinalis Capital Partners are specifically focused on transportation and mobility. Toyota, BMW, and Samsung Automotive Innovation Fund are focused exclusively on the automotive market.

Incentives

The State offers a variety of incentives for business. Selected examples include; The Governor's Strategic Reserve Fund, Community Economic Revitalization Board Funds, Clean Energy Fund, and Targeted Employment investments.

Washington's Research & Business Ecosystem

- University of Washington Mobility Innovation Center
- University of Washington Urban Freight Lab
- Washington State Freight Mobility Strategic Investment Board

Public and Private Research Institutes and Centers

- Washington State Freight Mobility Strategic Investment Board
- Washington State Autonomous Vehicle Working Group, State of Washington
- Pacific Northwest Transportation Consortium
- Washington State Transportation Center
- Center for Advanced Transportation and Energy Solutions
- Aces Northwest Network
- University of Washington Mobility Innovation Center
- University of Washington Urban Freight Lab
- Autonomous Flight Systems Laboratory, UW
- Boeing Advanced Research Center, UW
- Joint Center for Aerospace Technology Innovation, UW
- Supply Chain Transportation and Logistics Center, UW
- Kernel Labs Incubator

"If I had asked what people wanted they would have said faster horses"
Henry Ford

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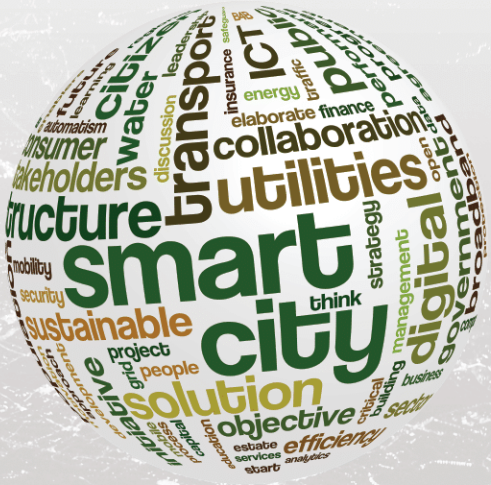
SMART CITIES

Cities that use information communication technology and data analysis to increase the efficiency and effectiveness of core systems that are central to a city's operations and development.



Smart City Solutions Include:

- Smart Governance
- Smart Mobility & Transportation
- Smart Public Safety
- Smart Buildings
- Smart Utilities
- Smart Healthcare

Global Smart City Market Size 2021:

- $111.70 billion

Source: Grand View Research

Global Smart City Market Size 2028:

- $676.01 billion

Source: Grand View Research

Smart City Market Growth

- Projected CAGR of 29.3% from 2016-2028

Source: Grand View Research

Globally Leading Smart Cities 2020

1. Singapore
2. Seoul
3. London
4. Barcelona
5. Helsinki
6. New York City
7. Montreal
8. Shanghai
9. Vienna
10. Amsterdam

Source: Statista

High Growth Jobs in Smart City Services

- Machine Learning Scientist
- Cybersecurity Analyst
- Industrial Network Engineer
- Virtual Reality Specialist
- Network Reliability Engineer
- Integration Engineer
- Data Scientist
- Cloud Architect
- Alliance/Partnership Manager
- Chief City Experience Officer
- Urban Informatics Analyst

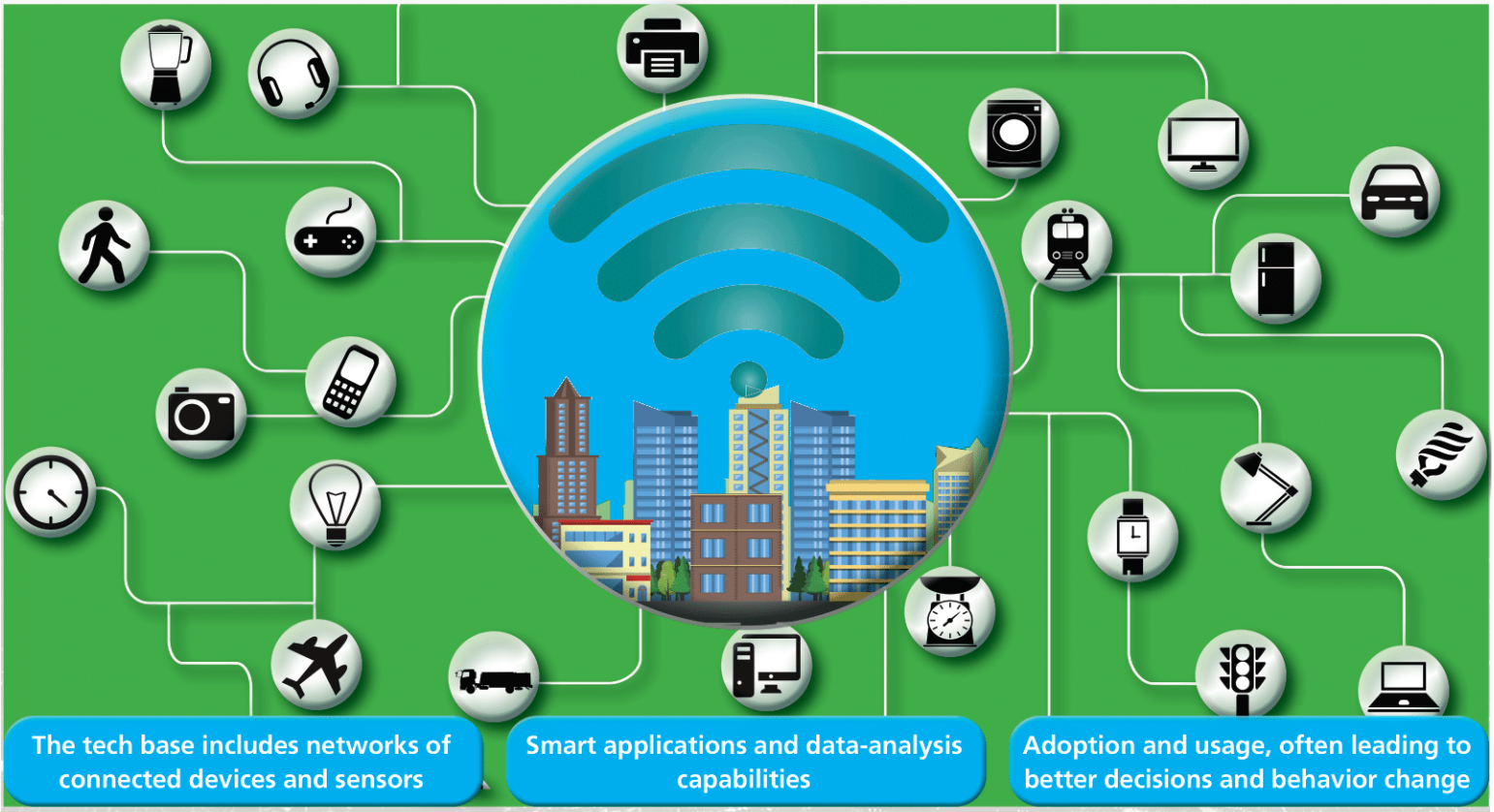
Source: TechRepublic

Major Global Companies

- Cisco
- IBM
- Schneider Electric
- Siemens
- Microsoft
- Hitachi
- Huawei
- Ericsson
- Toshiba
- Oracle

Source: Smart City Hub

SMART CITIES



The tech base includes networks of connected devices and sensors

Smart applications and data-analysis capabilities

Adoption and usage, often leading to better decisions and behavior change

Elements of Smart Cities

- Technology Base
- Smart Applications and Data Analysis
- Adoption and Usage

Source: McKinsey & Company

WASHINGTON'S ECOSYSTEM FOR SMART CITIES

Major Global Companies in Washington

- Cisco
- Ericsson
- IBM
- Itron
- Microsoft
- Oracle
- Schneider Electric
- Siemens AG
- Verizon

University of Washington

- Seattle MetroLab
- eScience Institute Data Science for Social Good
- Tech Policy Lab
- Urbanalytics
- Mobility Innovation Center
- Urban@UW

Washington State University

- Urbanova

Northeastern University-Seattle

- Master of Science in Urban Informatics

Other

- Huawei Tech Research Center

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